SATA 1D: Along-Track Ramps

Targets Close Together and Separated, with an Ambiguity Metric that Reads the Ambiguities

Lucas Ayub Rodrigues Tuler
DLR–DAAD Fellowship No. 734 — Microwaves and Radar Institute (DLR–HR)

September 2026

Setup. SATA 1D, $N_{rx} = 4$, PRF = 2000 Hz, $\mathrm{PRF}_{\mathrm{ch}} = 500$ Hz, DPCA along-track baselines ($\Delta b_{at} = 7.69$ m), $b_{xt} = 63.7, 27.0, 4.1, 1.7$ m ($U(0, 100)$, seed 0), $r_0 = 766.2$ km, $\theta = 20°$, $\lambda = 0.25$ m. Along-track ramp: $N \in \{2, 3, 5, 7, 9, 13\}$ iso-range targets with heights $h_j = j \cdot 200\,\mathrm{m} \cdot \tan\alpha$, $\alpha \in \{0.1, 1, 1.5, 2, 3, 4, 5, 10, 15, 20, 30, 40, 50\}°$. Two spacings with the *same heights*: 200 m (78 cases) and 2.6 km (39 cases, $N \leq 5$). Each case is reconstructed without SATA, with SATA whole band, with SATA per sub-band, and with the oracle (each target rotated by its own ΔC_0 before the channels are summed).

1 Why the heatmap showed zero at small slopes

The previous heatmaps used the *worst ambiguity*: the largest value of the focused response outside ±40 samples (±154 m) of every target. At small slopes that maximum is not an ambiguity. With SATA it is the pedestal around the main lobe (the skirt at $|t| \lesssim 0.3$ s, about −36 dB), which does not depend on the correction; without SATA it is the ambiguity at ±0.81 s, which at $\alpha = 5°$, $N = 2$ happens to be at the same −36.5 dB. The difference of the two maxima is therefore 0 dB although the ambiguities differ by 12 dB (Fig. 1). Of the 36 cells with $1° \leq \alpha \leq 5°$ at 200 m, 15 read exactly 0.0 with the old metric.

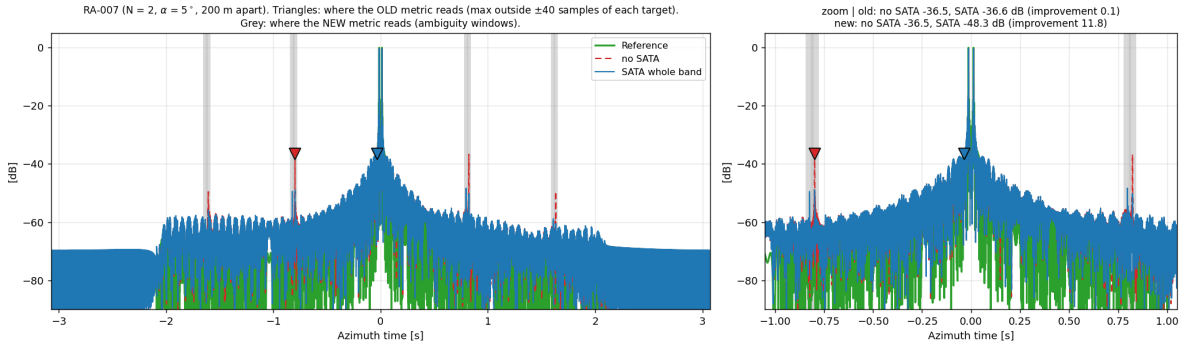


Figure 1: Case RA-007 ($N = 2$, $\alpha = 5°$, 200 m apart). Triangles: where the old metric reads for no SATA (the ambiguity at +0.81 s) and for SATA (the main-lobe pedestal). Grey bands: the windows of the new metric.

New metric: ambiguity peak. A Doppler shift of $m\,\mathrm{PRF}_{\mathrm{ch}}$ moves the focused copy of a target by $m\,\Delta t_{amb}$, with

$$\Delta t_{amb} = \frac{\mathrm{PRF}_{\mathrm{ch}}}{K_a} = \frac{\mathrm{PRF}_{\mathrm{ch}}\,\lambda\,r_0}{2\,v^2} = 0.810\ \mathrm{s} \quad (6.23\ \mathrm{km}). \tag{1}$$

The ambiguity peak is the largest |IRF| within ±20 ms of $x_j/v + m\,\Delta t_{amb}$, $m = \pm 1, \pm 2$, over all targets j, leaving out ±20 ms around every target's own main lobe, in dB relative to the

reference peak. It reads the ambiguities and nothing else: for the reference (no ambiguities) it gives −52 to −63 dB, the level of the IRF floor there (and, with targets 2.6 km apart, of the skirt of a neighbour's main lobe). The same number is now in every figure title, and the grey bands in every IRF page show where it is read. Improvement = ambiguity peak (no SATA) − ambiguity peak (SATA); gap = ambiguity peak (SATA) − ambiguity peak (oracle).

2 Results

	cases	improvement median (max)	improvement, $1^\circ \le \alpha \le 5^\circ$ median (min)	gap to oracle median (max)	|sub − whole| max
200 m apart	78	18.8 (30.7) dB	18.8 (7.0) dB	6.7 (31.1) dB	0.09 dB
2.6 km apart	39	20.2 (44.2) dB	12.8 (1.1) dB	0.0 (0.9) dB	0.39 dB

Table 1: SATA whole band, new metric. With the old metric the improvement at $1^\circ \le \alpha \le 5^\circ$ had a median of 2.1 dB.

- **Small slopes.** SATA reduces the ambiguities already at $\alpha = 1^\circ$ (heights up to 42 m for $N = 13$): 7.0 dB for $N = 2$, 20 dB for $N = 13$ at 200 m. Only $\alpha = 0.1^\circ$ (heights below 5 m) stays at 0–3 dB, where there is nothing to correct.
- **Close targets (200 m).** SATA follows the oracle up to $\alpha \approx 2^\circ$ and then falls behind it (gap 2–9 dB at 5°, up to 31 dB at 50°): neighbours 200 m apart share one sub-aperture cell ($\delta x = 309$ m) and receive one correction, so the more their heights differ, the more is left. The old metric hid this because the oracle, too, was read on the main-lobe pedestal.
- **Separated targets (2.6 km).** SATA is at the oracle within 0.9 dB in every cell, whole band and per sub-band, up to $\alpha = 50^\circ$: each target has its own block in the ΔC_0 map and its own correction. For $N = 3$ and 5 the ambiguity windows of a target also see the skirt of the main lobe of the target 2.6 km away (−46 and −43 dB, identical for all methods), which is why the improvement at small α is lower there.
- **Spacing sweep** (Fig. 2). With the heights fixed, SATA approaches the oracle as the spacing grows, except at 1.6 and 2 km, where one pair of targets lies within 0.63 km of a multiple of $\Delta t_{amb}\, v = 6.23$ km (4×1.6 and 3×2 km) and the two targets share Doppler bins (the Doppler-alias collision).

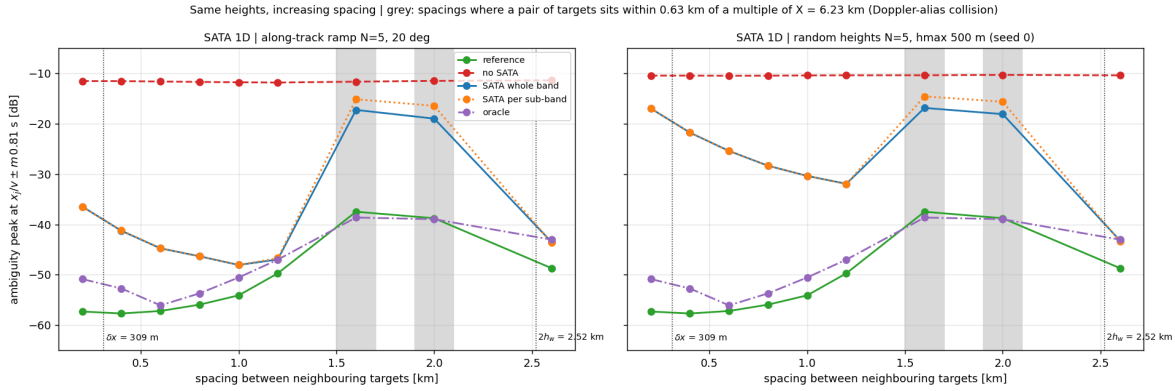


Figure 2: Ambiguity peak against the spacing between neighbouring targets, same heights. Left: along-track ramp $N = 5$, $\alpha = 20^\circ$ (heights 0–291 m). Right: random heights $N = 5$, $h_{max} = 500$ m (seed 0). Grey: collision spacings; there the ambiguity of one target falls 20–30 ms from the main lobe of another target, so even the reference and the oracle read −38 dB in the windows.

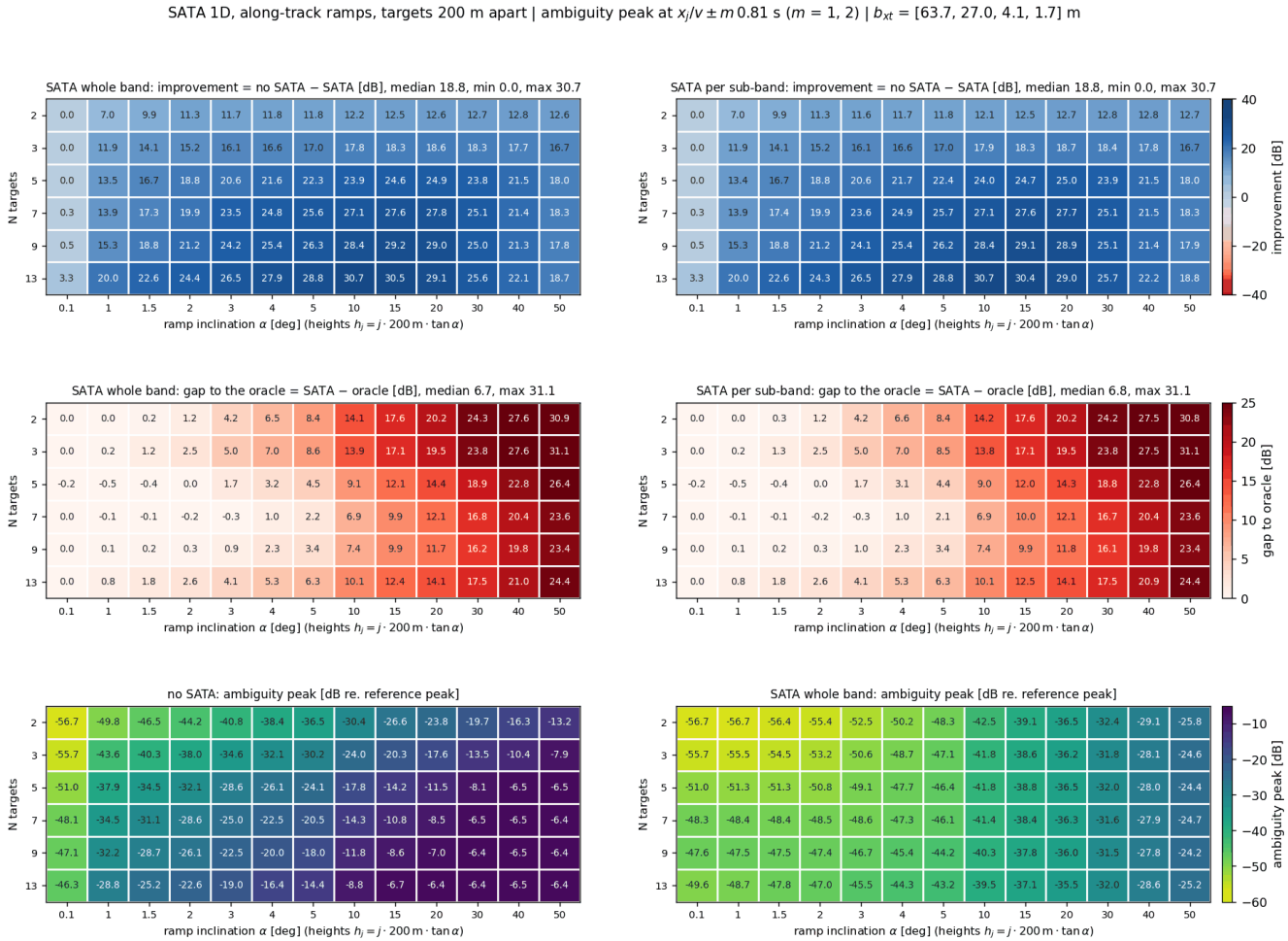


Figure 3: Targets 200 m apart. Top: improvement over no SATA, whole band and per sub-band. Middle: gap to the oracle. Bottom: the ambiguity peak itself, no SATA and SATA whole band.

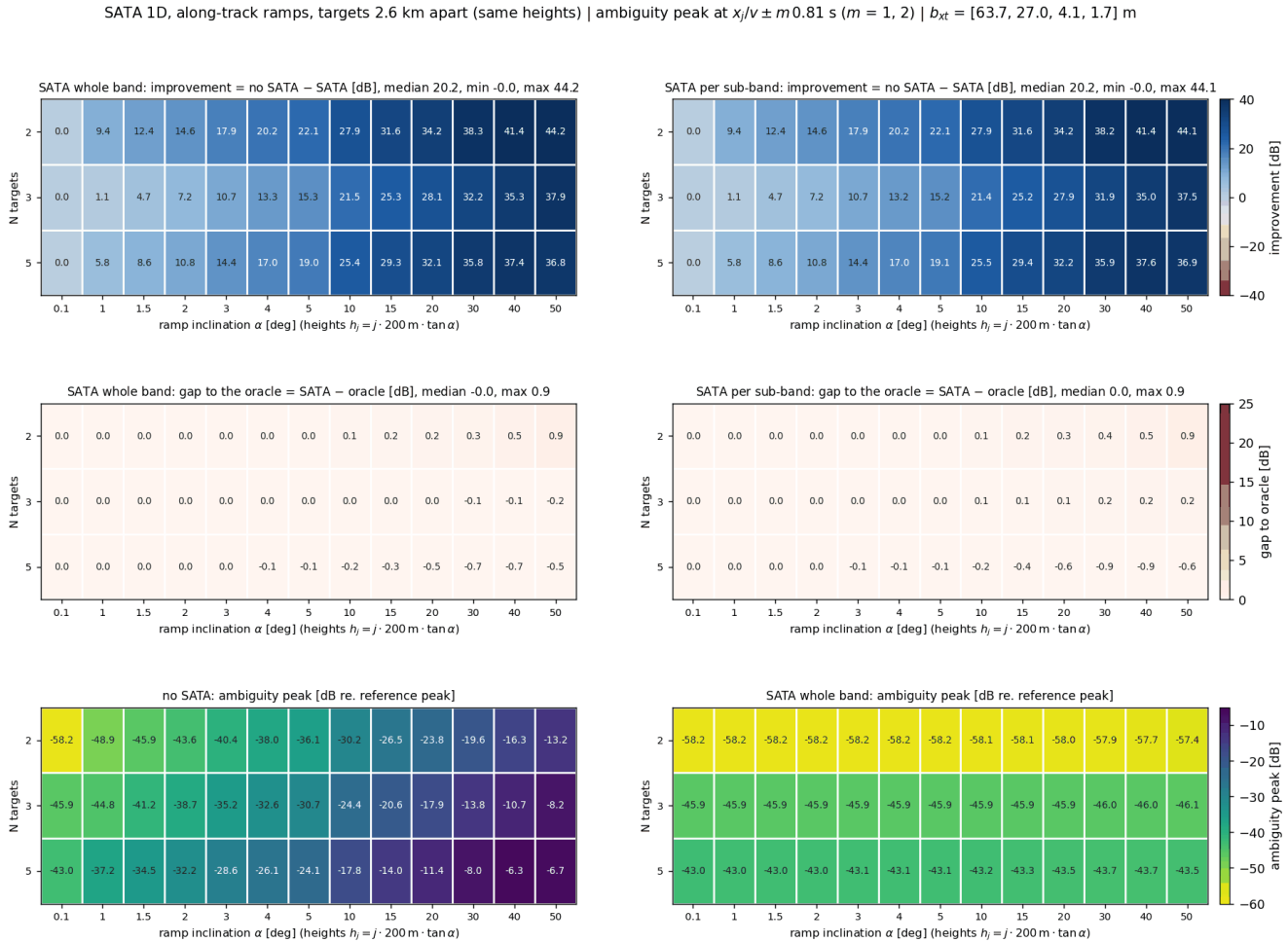


Figure 4: Targets 2.6 km apart, same heights. Same panels as above.

3 IRFs, targets 200 m apart

Top: reference, SATA whole band and SATA per sub-band. Bottom: reference and no SATA. Grey bands: the ambiguity windows. Right: target geometry in 2D and 3D (iso-range targets sit further out in ground range the higher they are).

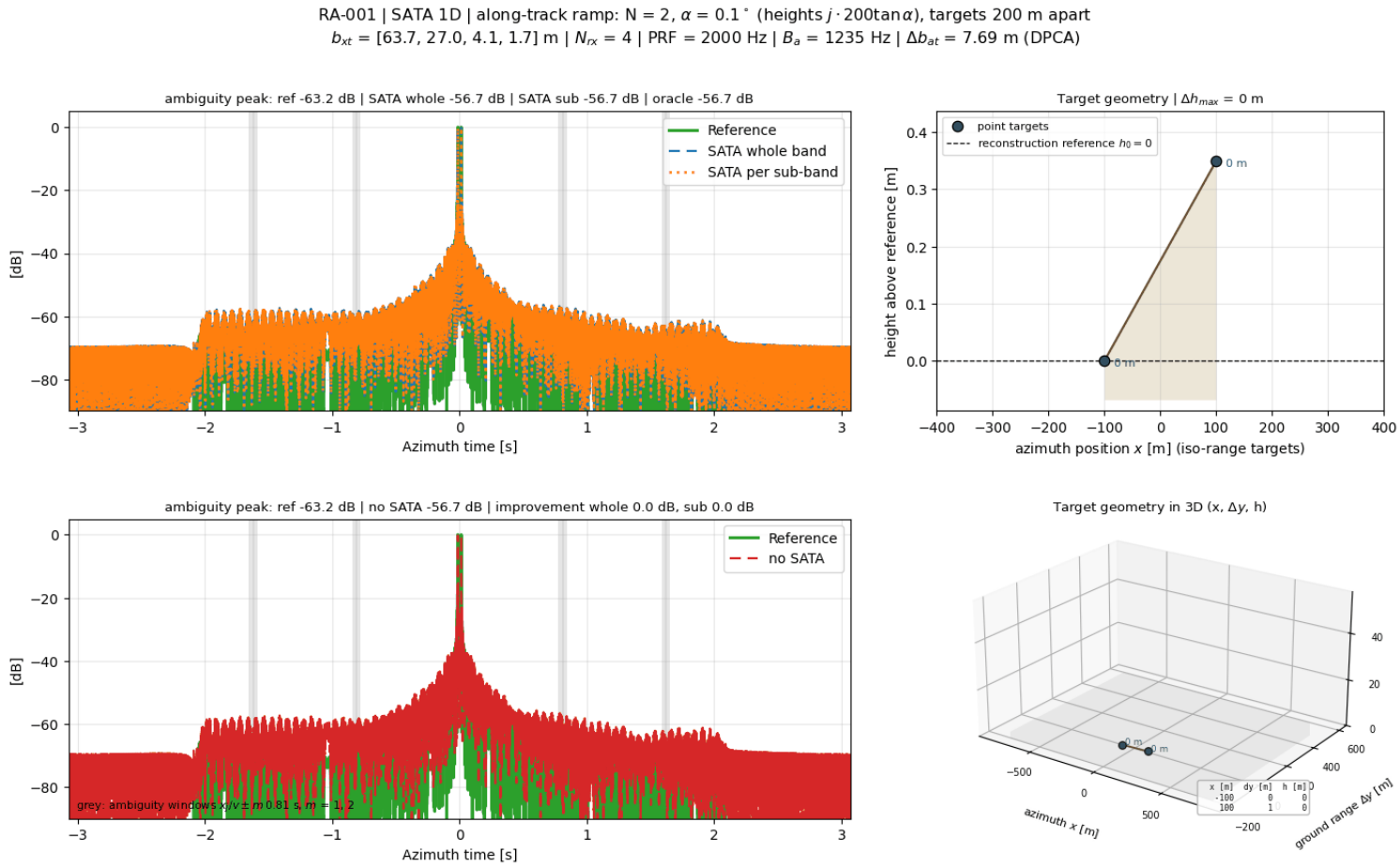


Figure 5: Targets 200 m apart: RA-001, $N = 2$, $\alpha = 0.1°$. Ambiguity peak: no SATA −56.7, SATA whole −56.7, per sub-band −56.7, oracle −56.7 dB (improvement 0.0 dB).

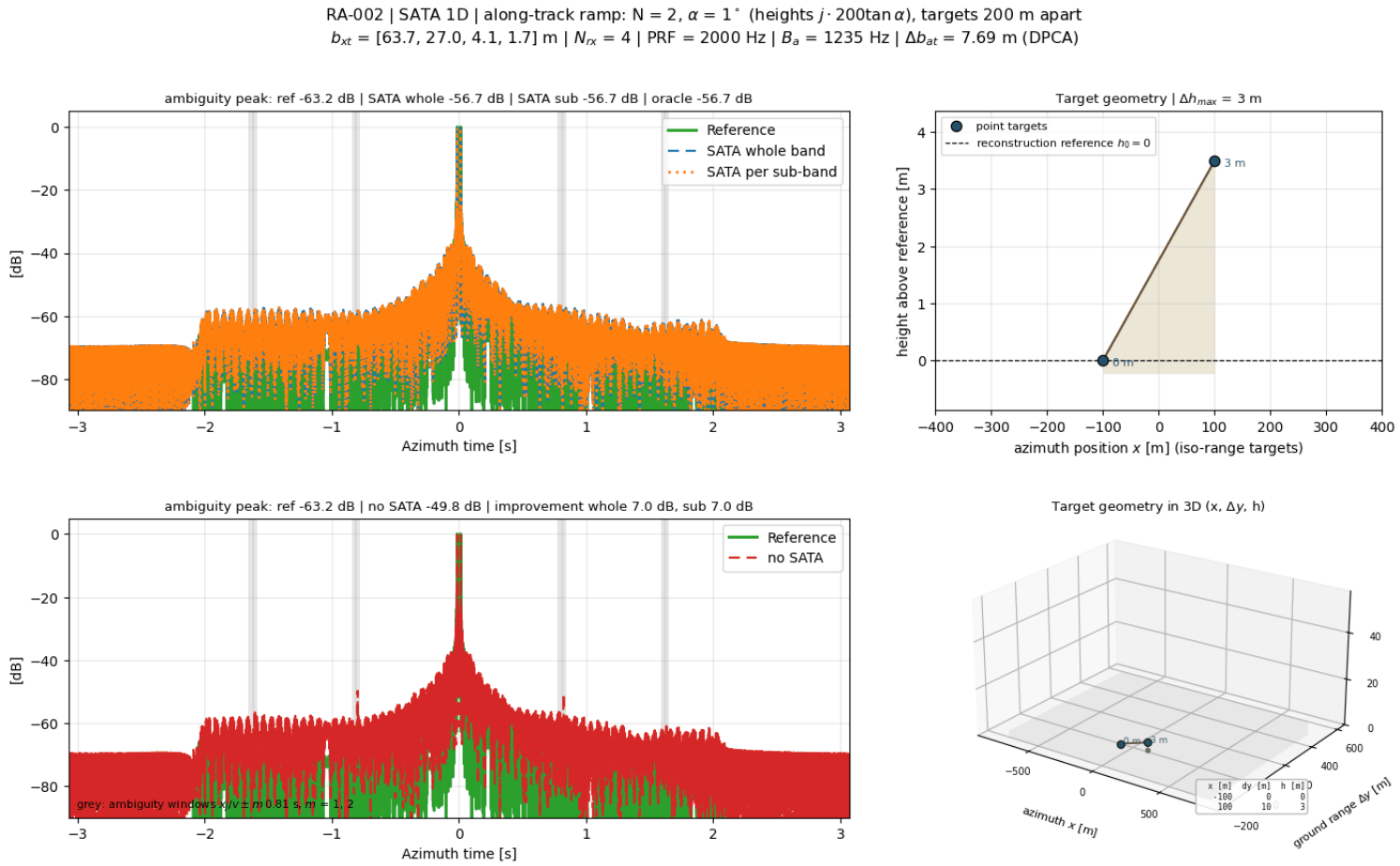


Figure 6: Targets 200 m apart: RA-002, $N = 2$, $\alpha = 1°$. Ambiguity peak: no SATA -49.8, SATA whole -56.7, per sub-band -56.7, oracle -56.7 dB (improvement 7.0 dB).

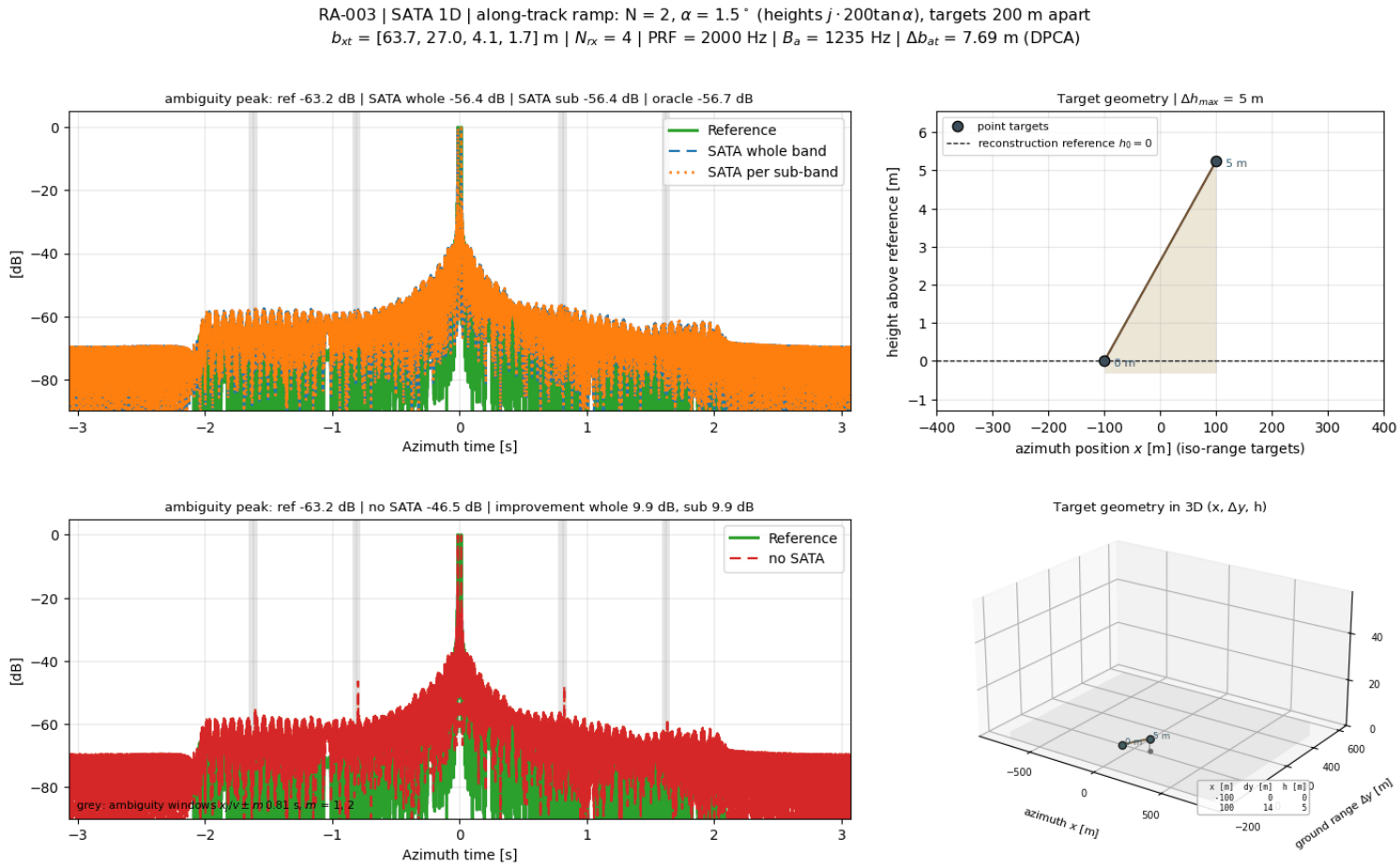


Figure 7: Targets 200 m apart: RA-003, $N = 2$, $\alpha = 1.5°$. Ambiguity peak: no SATA -46.5, SATA whole -56.4, per sub-band -56.4, oracle -56.7 dB (improvement 9.9 dB).

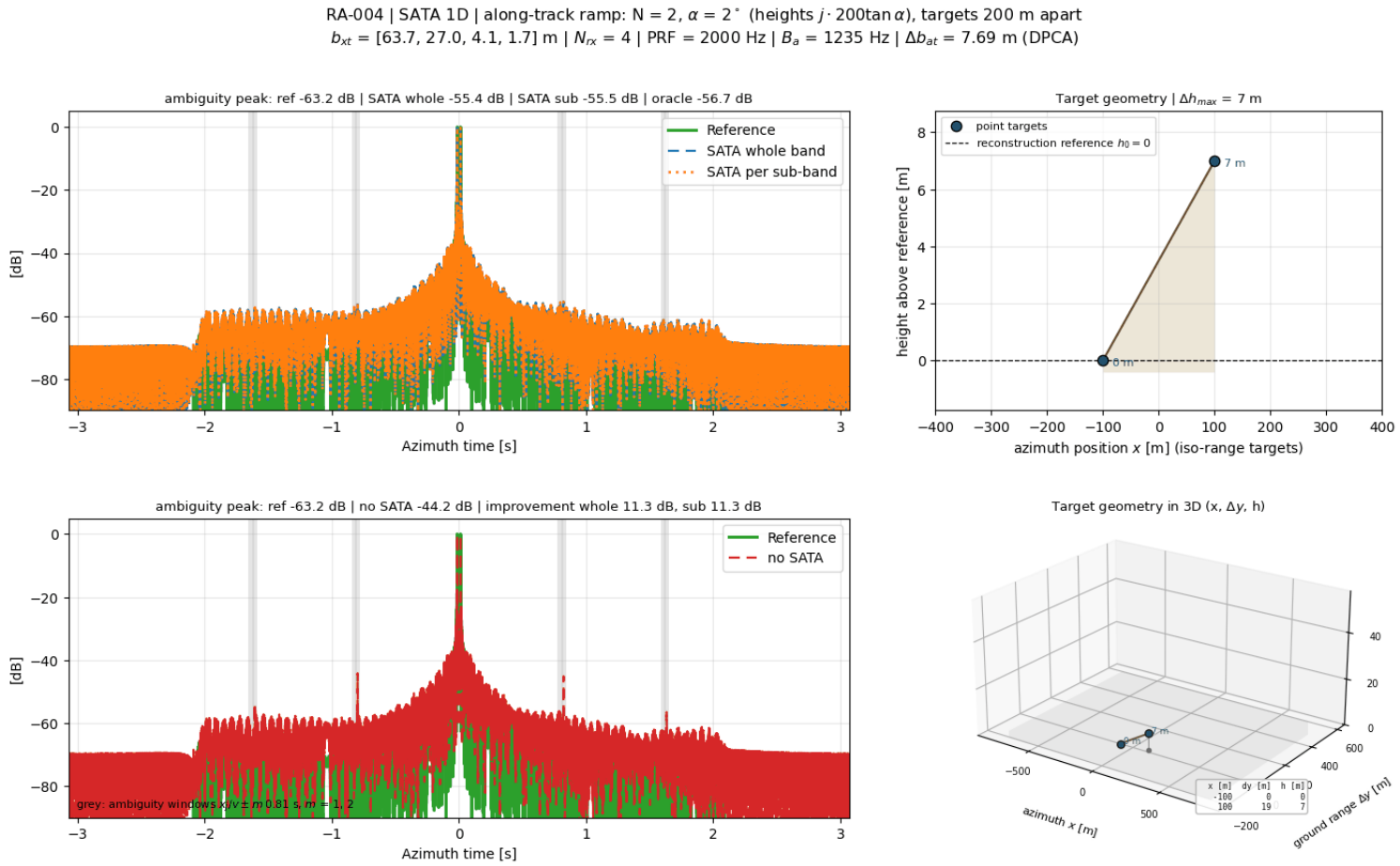


Figure 8: Targets 200 m apart: RA-004, $N = 2$, $\alpha = 2°$. Ambiguity peak: no SATA -44.2, SATA whole -55.4, per sub-band -55.5, oracle -56.7 dB (improvement 11.3 dB).

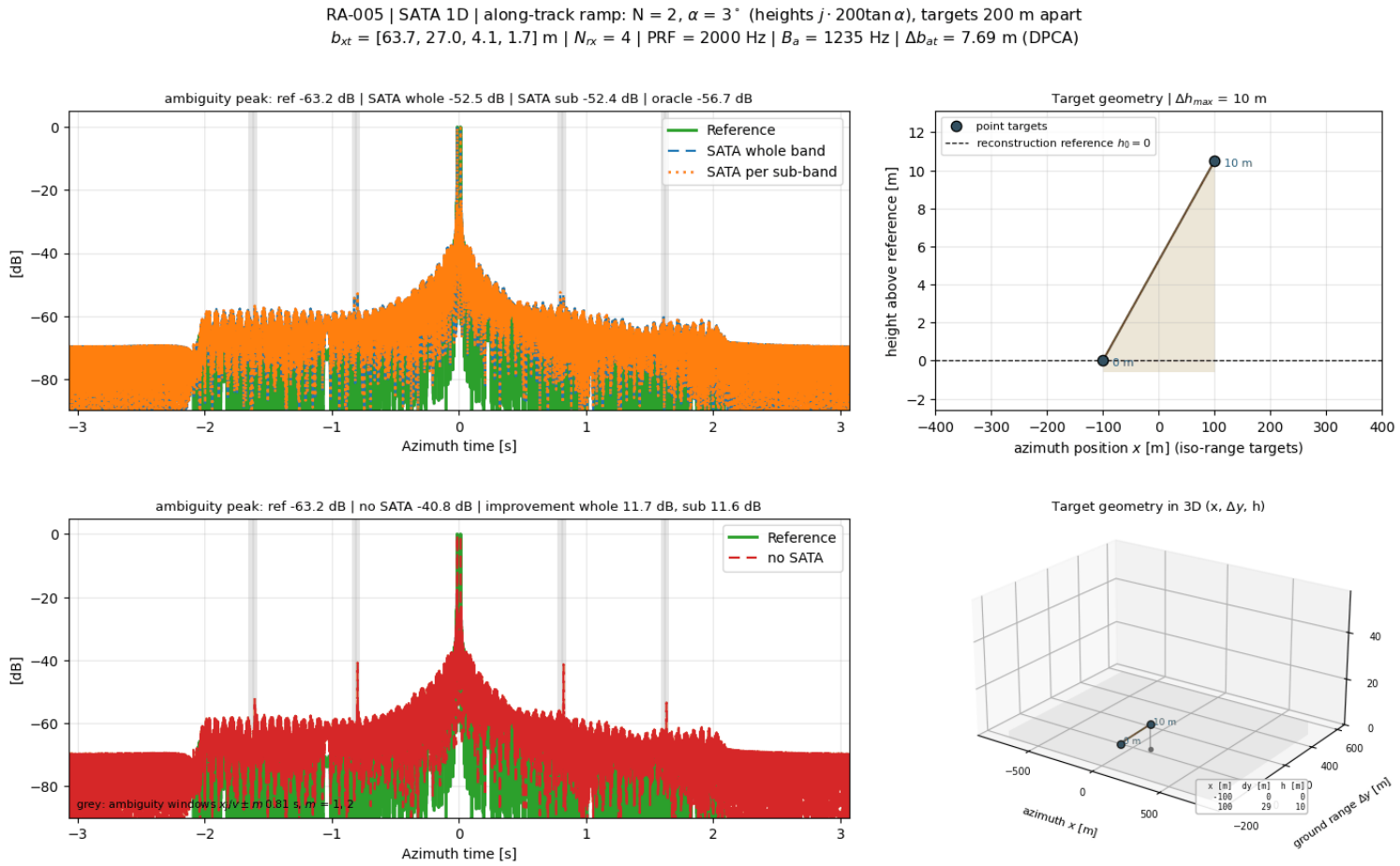


Figure 9: Targets 200 m apart: RA-005, $N = 2$, $\alpha = 3°$. Ambiguity peak: no SATA -40.8, SATA whole -52.5, per sub-band -52.4, oracle -56.7 dB (improvement 11.7 dB).

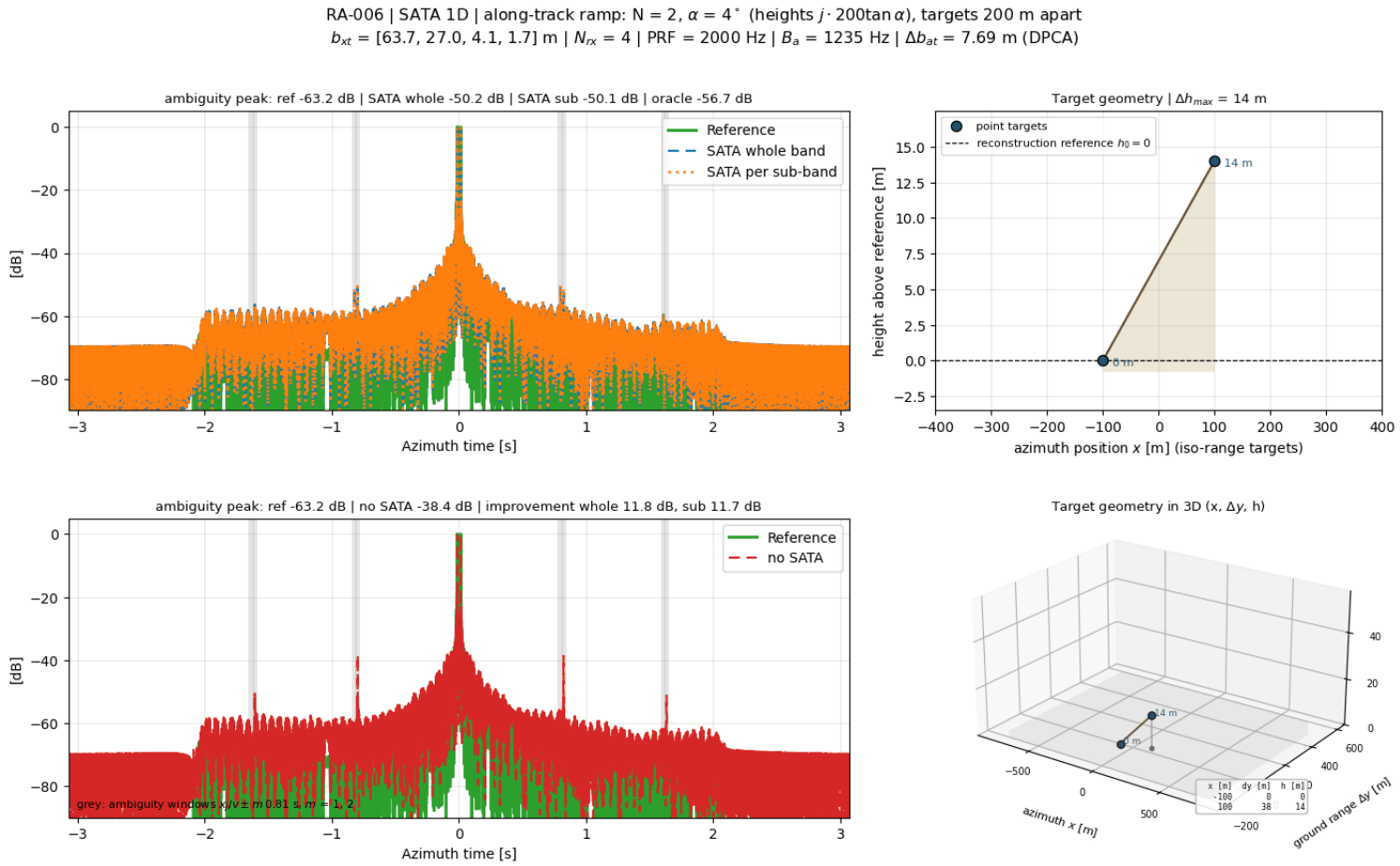


Figure 10: Targets 200 m apart: RA-006, $N = 2$, $\alpha = 4°$. Ambiguity peak: no SATA -38.4, SATA whole -50.2, per sub-band -50.1, oracle -56.7 dB (improvement 11.8 dB).

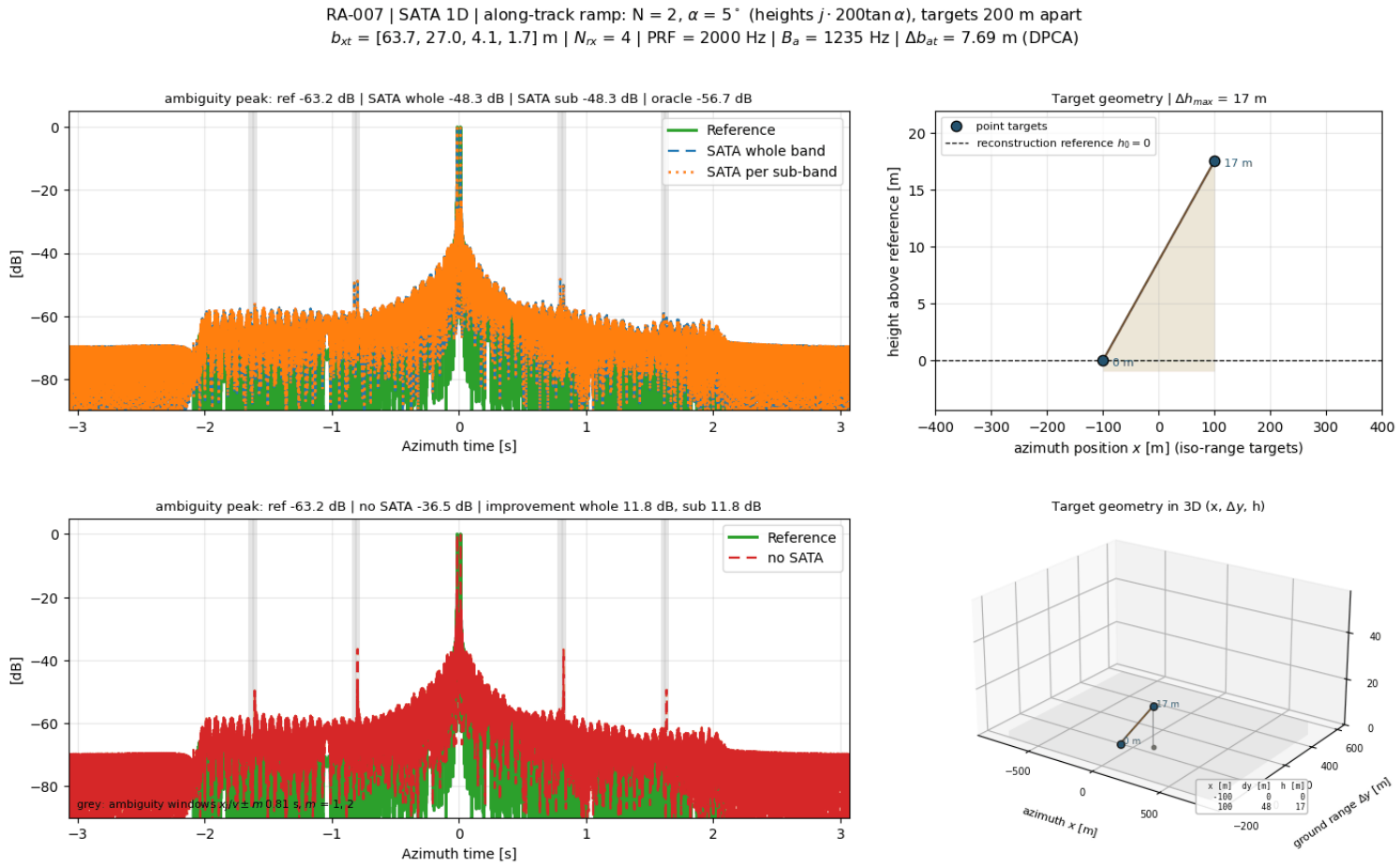


Figure 11: Targets 200 m apart: RA-007, $N = 2$, $\alpha = 5°$. Ambiguity peak: no SATA -36.5, SATA whole -48.3, per sub-band -48.3, oracle -56.7 dB (improvement 11.8 dB).

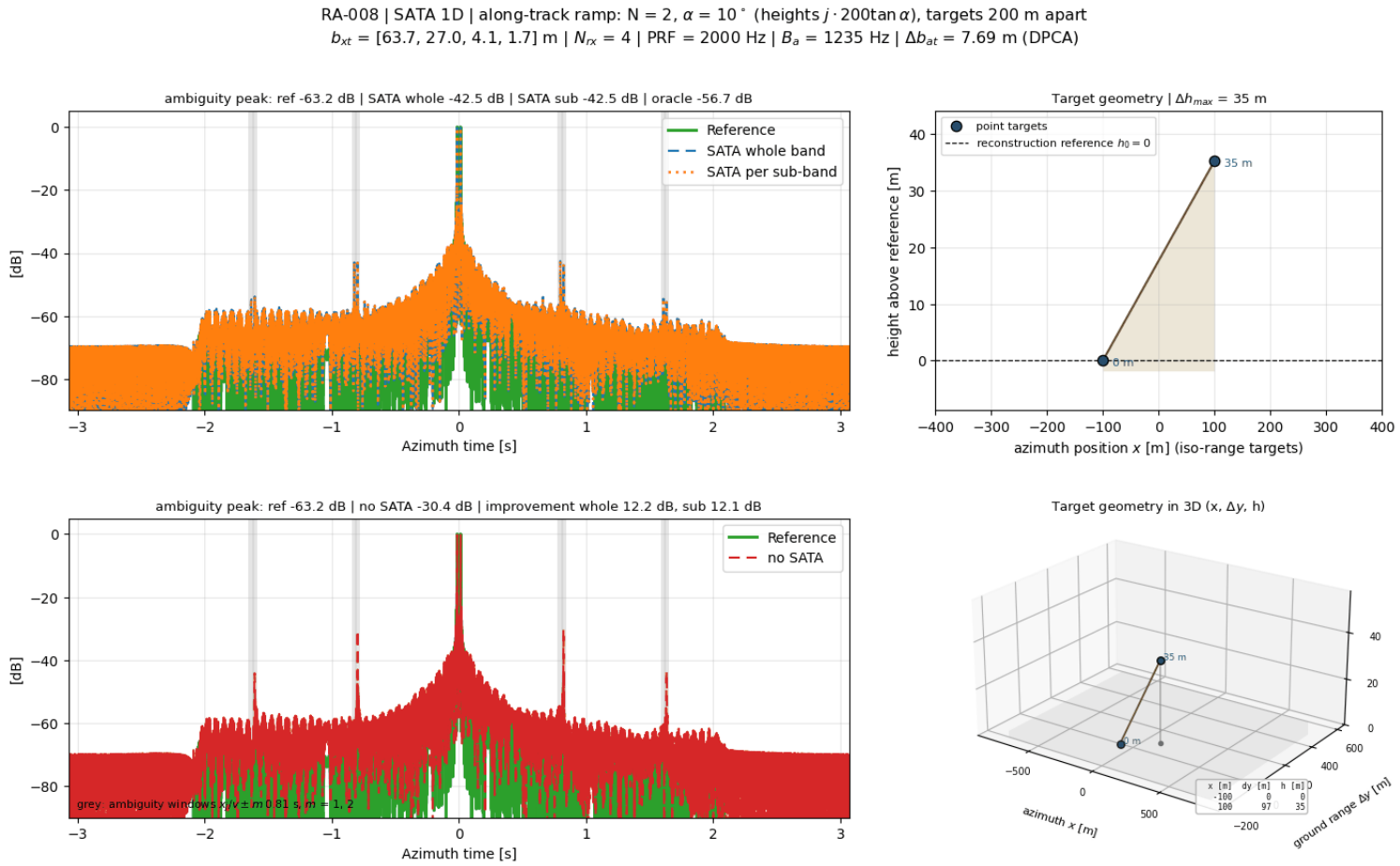


Figure 12: Targets 200 m apart: RA-008, $N = 2$, $\alpha = 10°$. Ambiguity peak: no SATA -30.4, SATA whole -42.5, per sub-band -42.5, oracle -56.7 dB (improvement 12.2 dB).

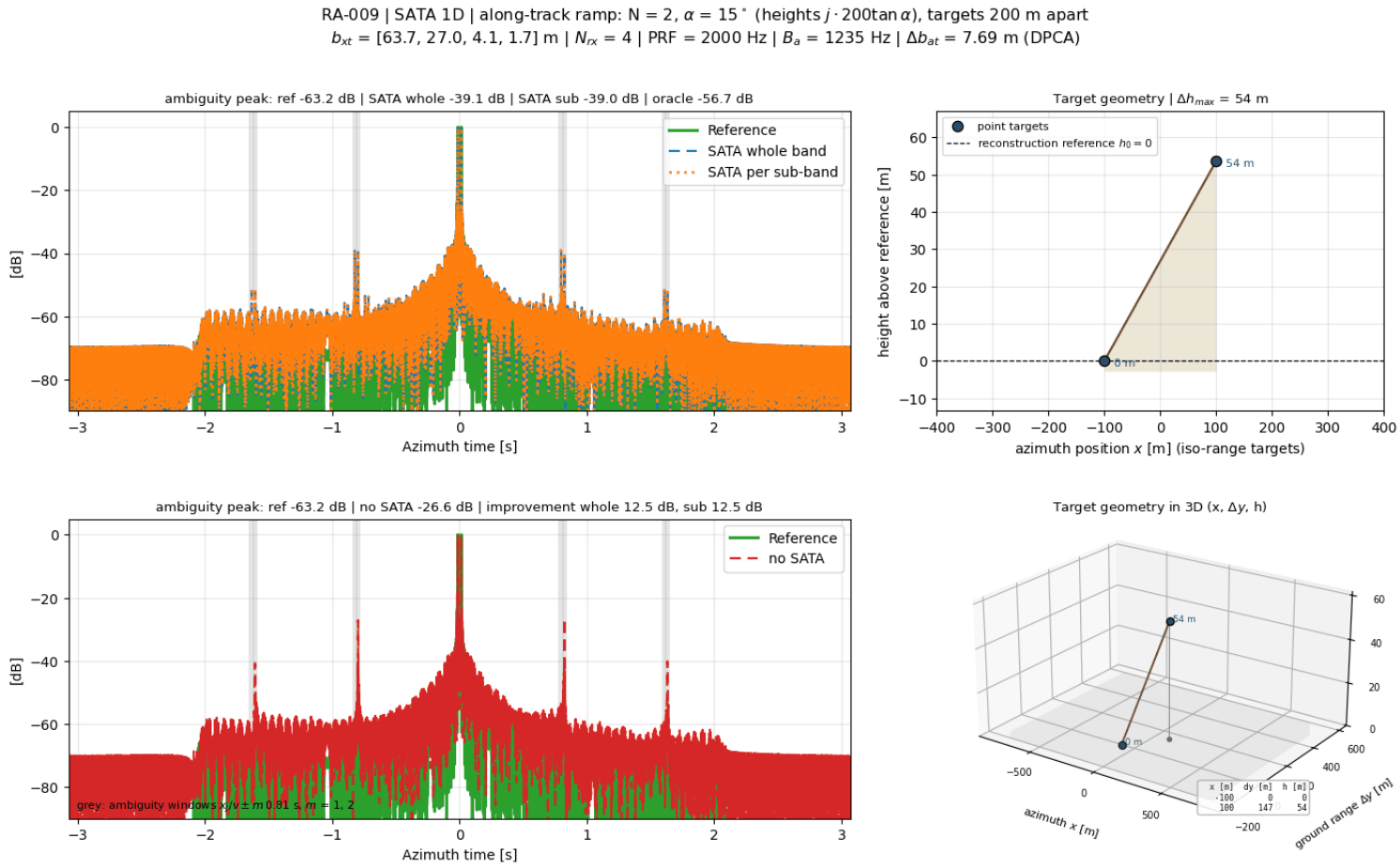


Figure 13: Targets 200 m apart: RA-009, $N = 2$, $\alpha = 15°$. Ambiguity peak: no SATA −26.6, SATA whole −39.1, per sub-band −39.0, oracle −56.7 dB (improvement 12.5 dB).

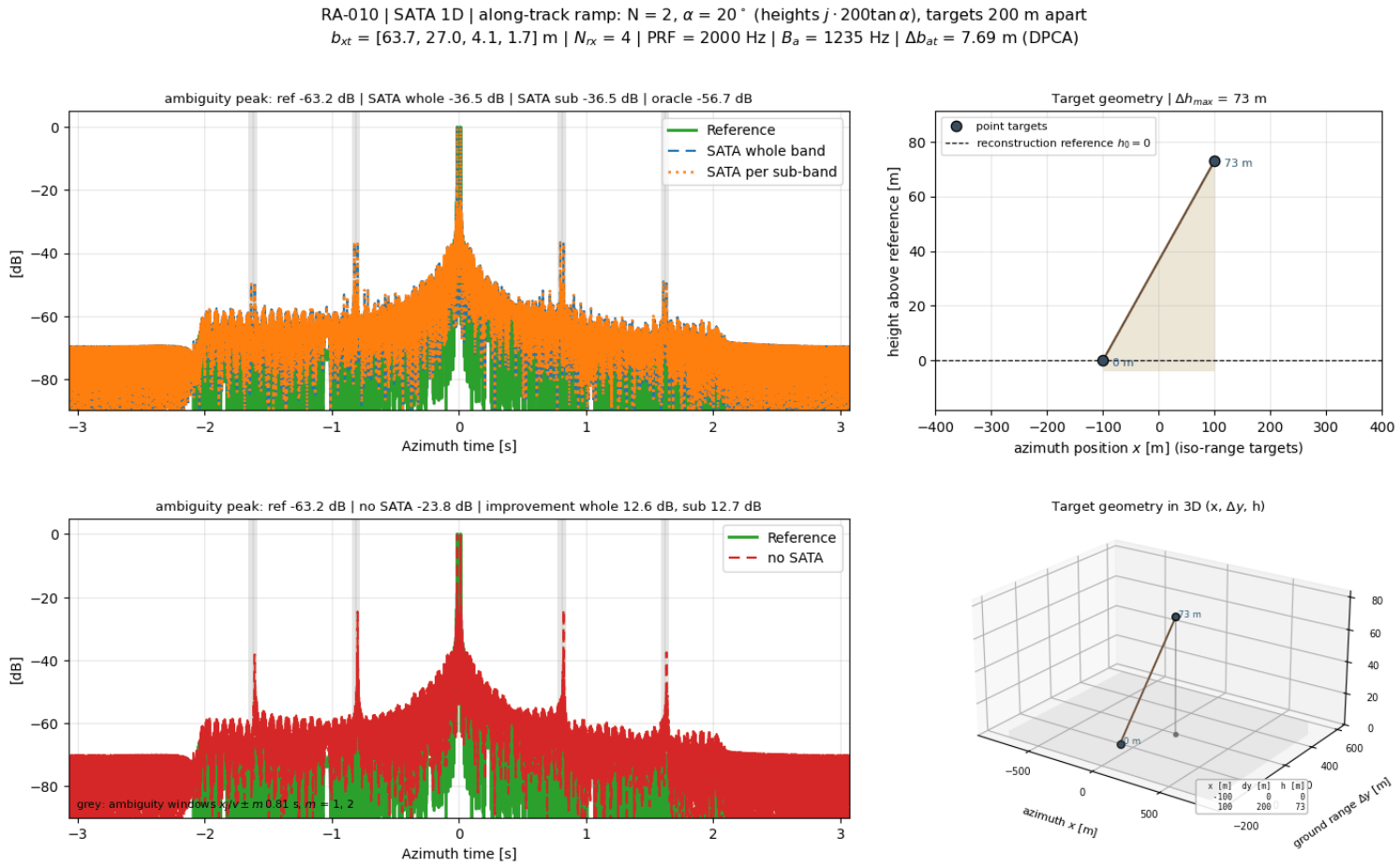


Figure 14: Targets 200 m apart: RA-010, $N = 2$, $\alpha = 20°$. Ambiguity peak: no SATA -23.8, SATA whole -36.5, per sub-band -36.5, oracle -56.7 dB (improvement 12.6 dB).

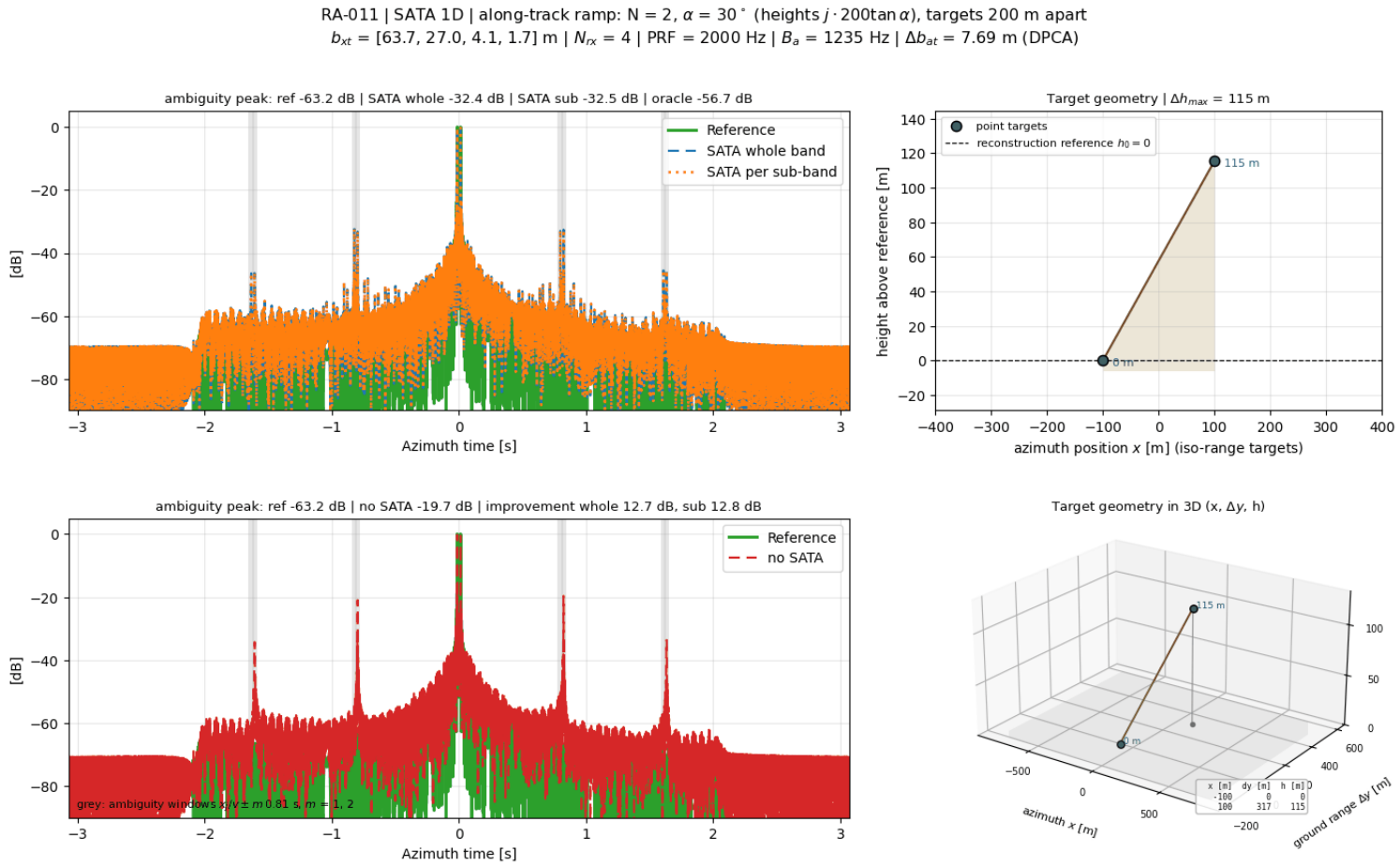


Figure 15: Targets 200 m apart: RA-011, $N = 2$, $\alpha = 30°$. Ambiguity peak: no SATA −19.7, SATA whole −32.4, per sub-band −32.5, oracle −56.7 dB (improvement 12.7 dB).

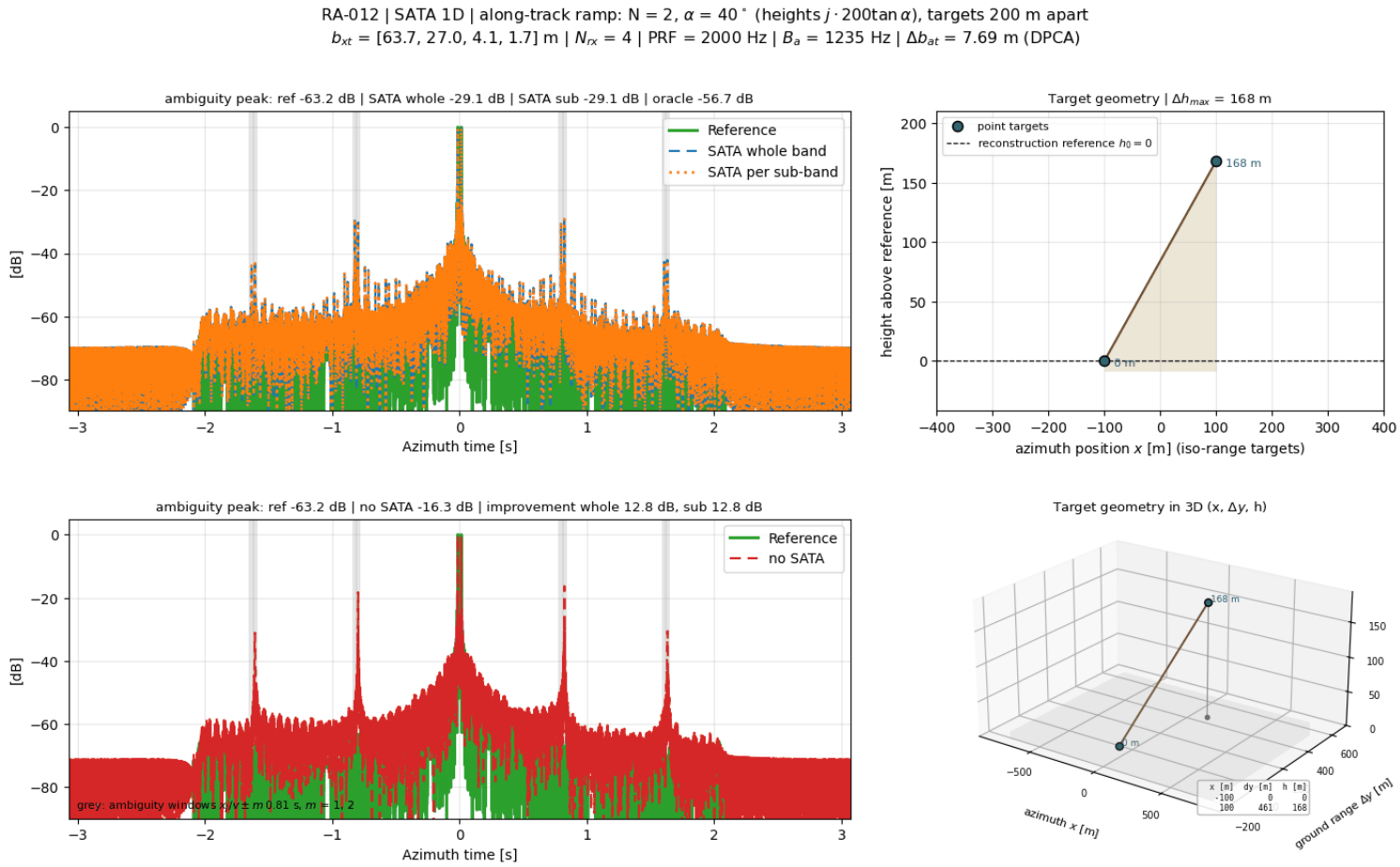


Figure 16: Targets 200 m apart: RA-012, $N = 2$, $\alpha = 40°$. Ambiguity peak: no SATA −16.3, SATA whole −29.1, per sub-band −29.1, oracle −56.7 dB (improvement 12.8 dB).

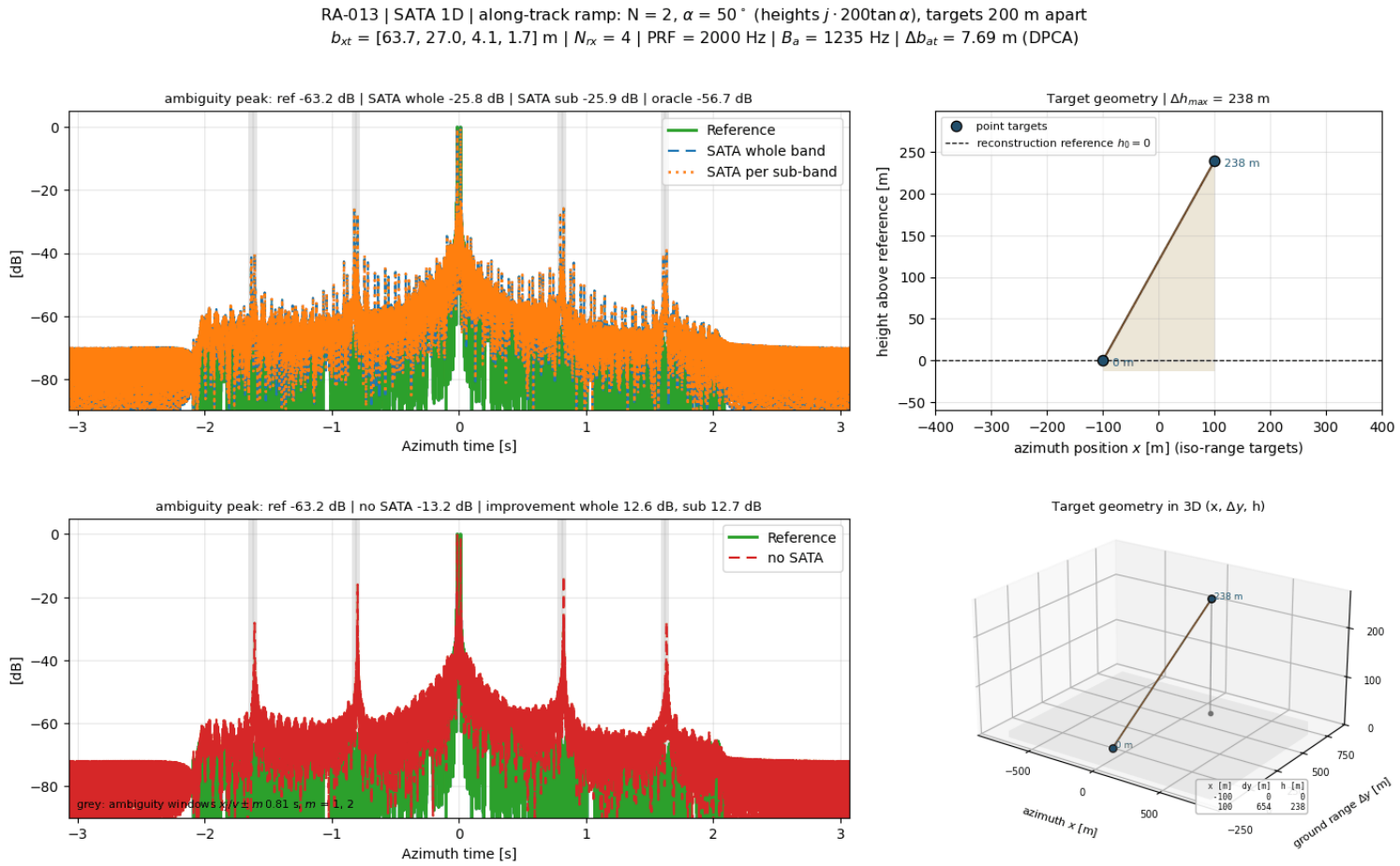


Figure 17: Targets 200 m apart: RA-013, $N = 2$, $\alpha = 50°$. Ambiguity peak: no SATA −13.2, SATA whole −25.8, per sub-band −25.9, oracle −56.7 dB (improvement 12.6 dB).

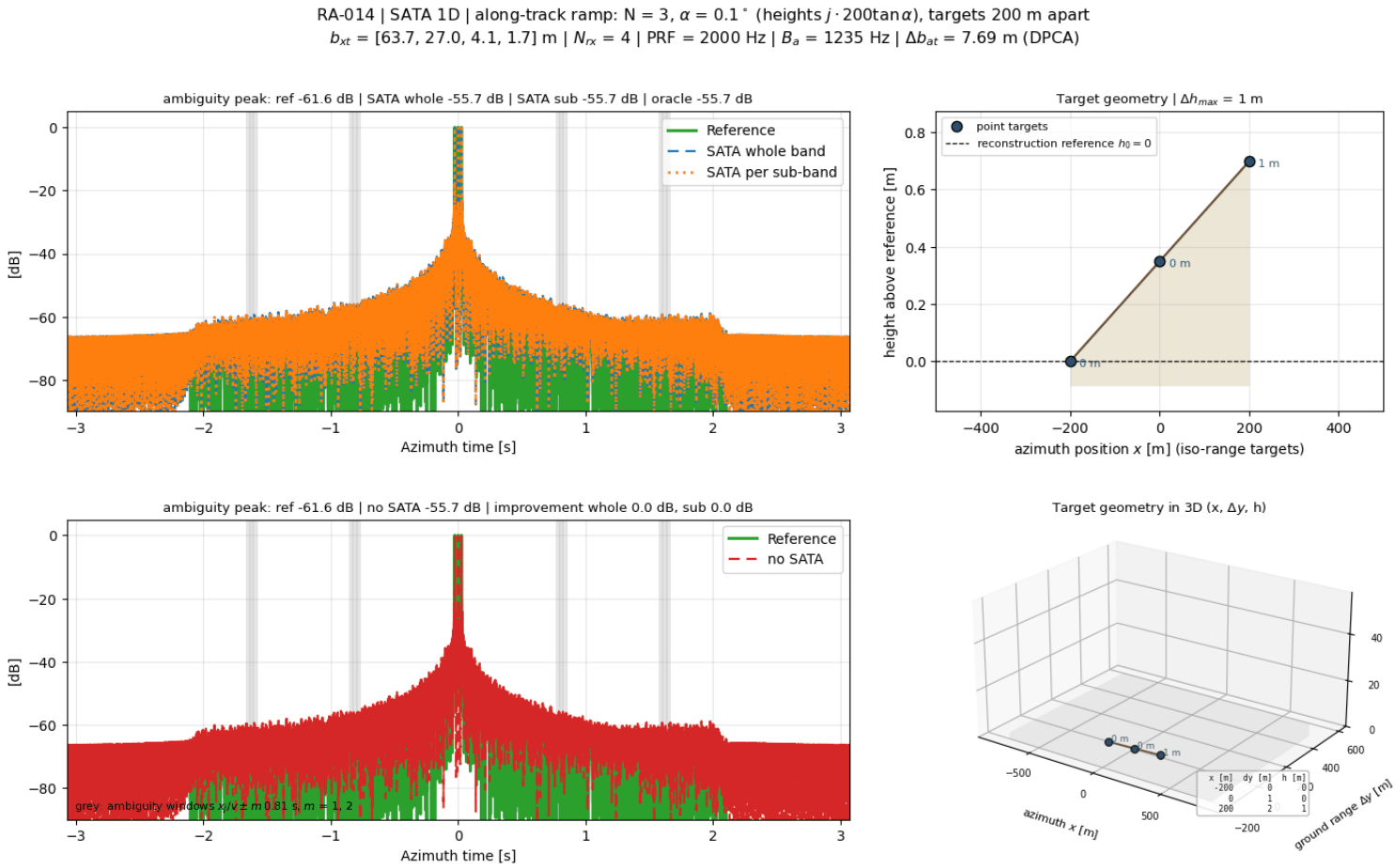


Figure 18: Targets 200 m apart: RA-014, $N = 3$, $\alpha = 0.1°$. Ambiguity peak: no SATA −55.7, SATA whole −55.7, per sub-band −55.7, oracle −55.7 dB (improvement 0.0 dB).

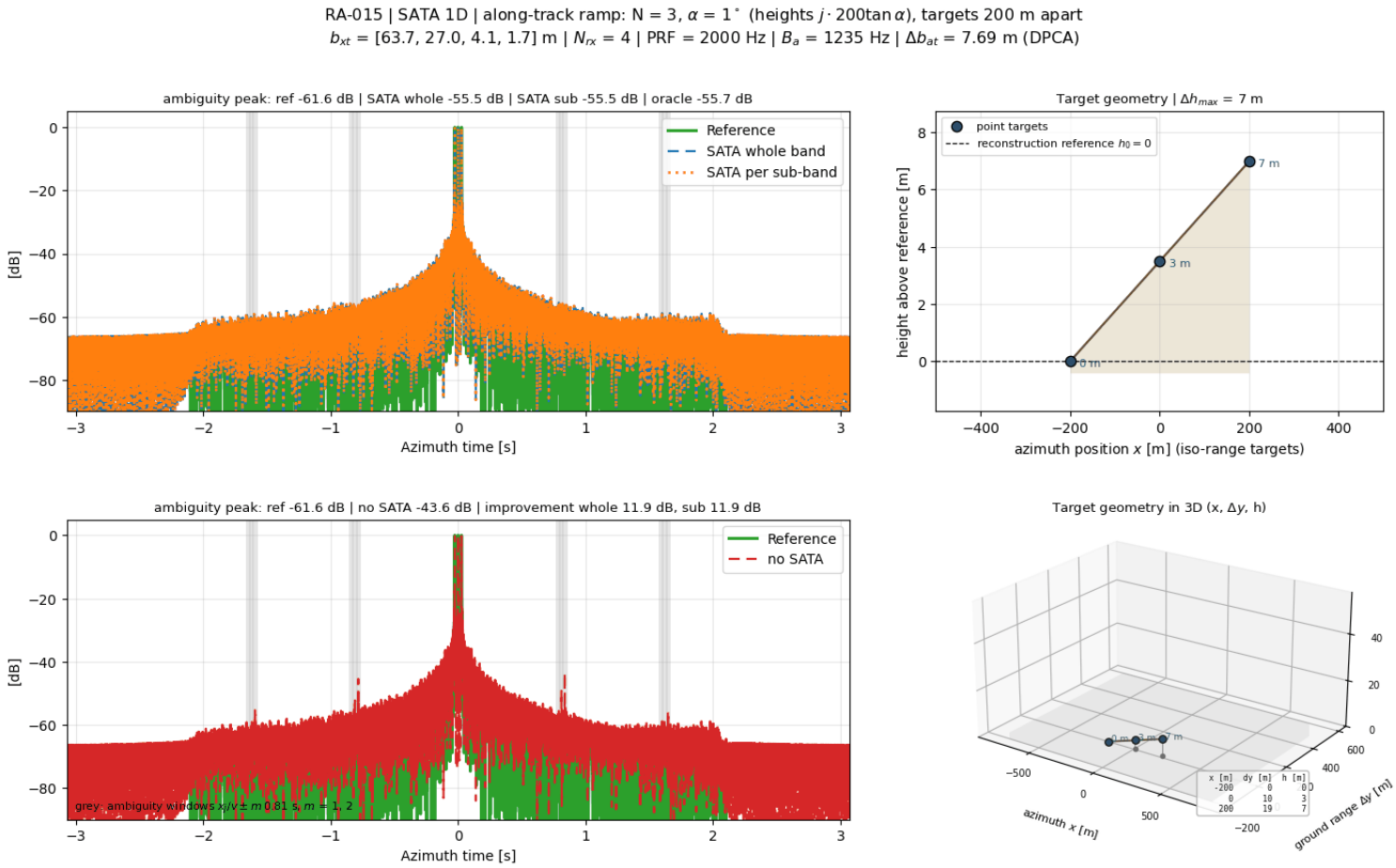


Figure 19: Targets 200 m apart: RA-015, $N = 3$, $\alpha = 1°$. Ambiguity peak: no SATA -43.6, SATA whole -55.5, per sub-band -55.5, oracle -55.7 dB (improvement 11.9 dB).

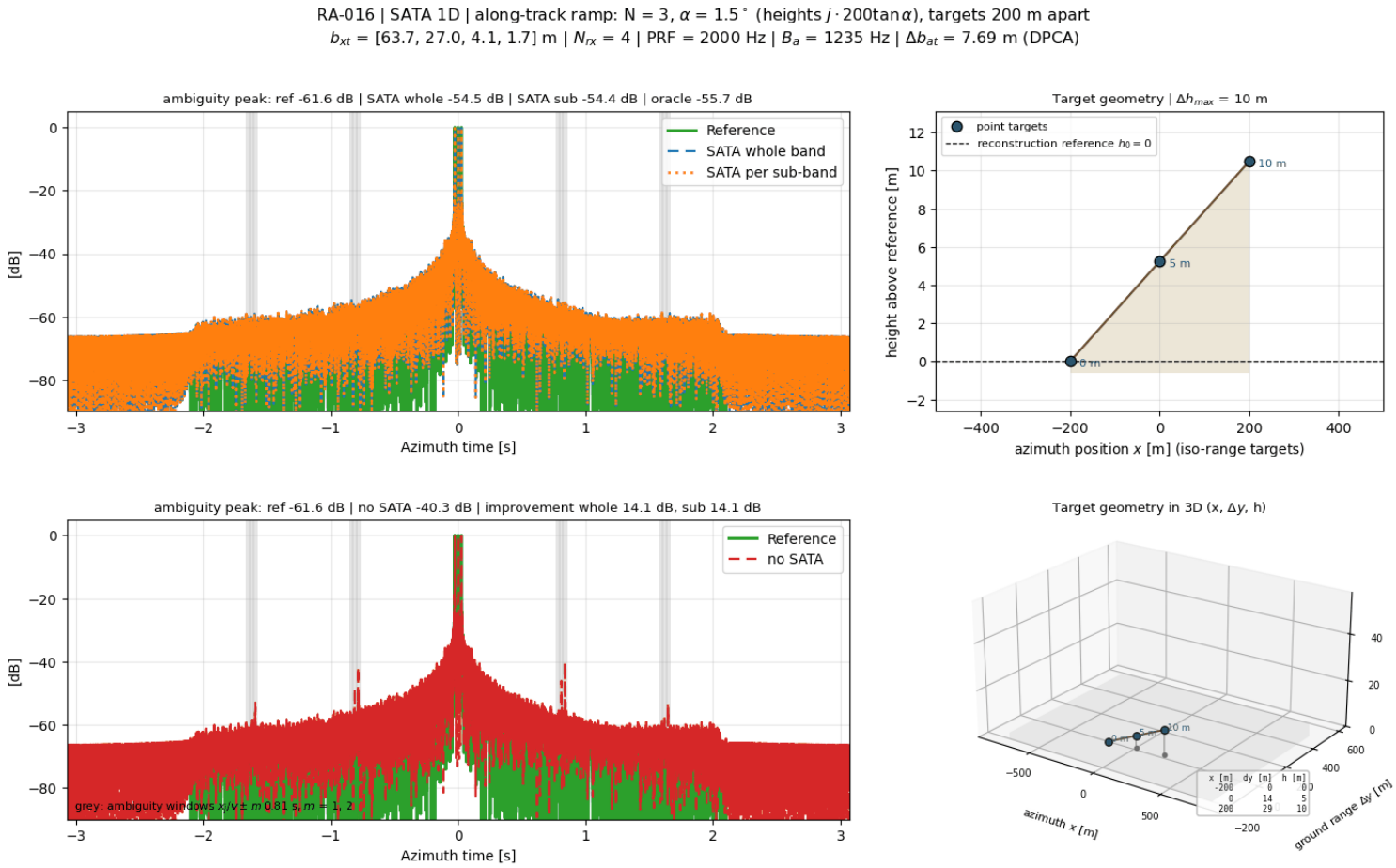


Figure 20: Targets 200 m apart: RA-016, $N = 3$, $\alpha = 1.5°$. Ambiguity peak: no SATA -40.3, SATA whole -54.5, per sub-band -54.4, oracle -55.7 dB (improvement 14.1 dB).

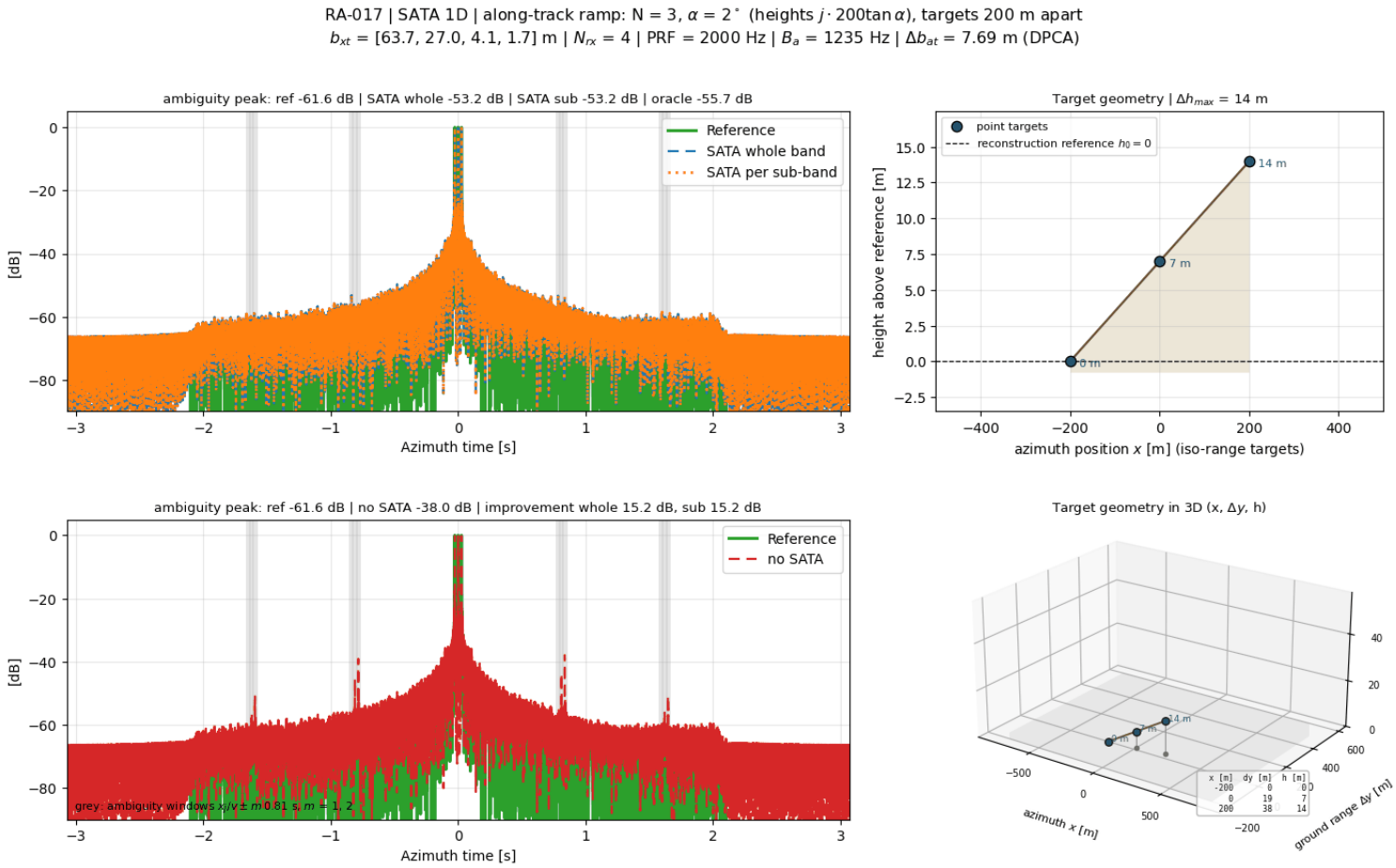


Figure 21: Targets 200 m apart: RA-017, $N = 3$, $\alpha = 2°$. Ambiguity peak: no SATA -38.0, SATA whole -53.2, per sub-band -53.2, oracle -55.7 dB (improvement 15.2 dB).

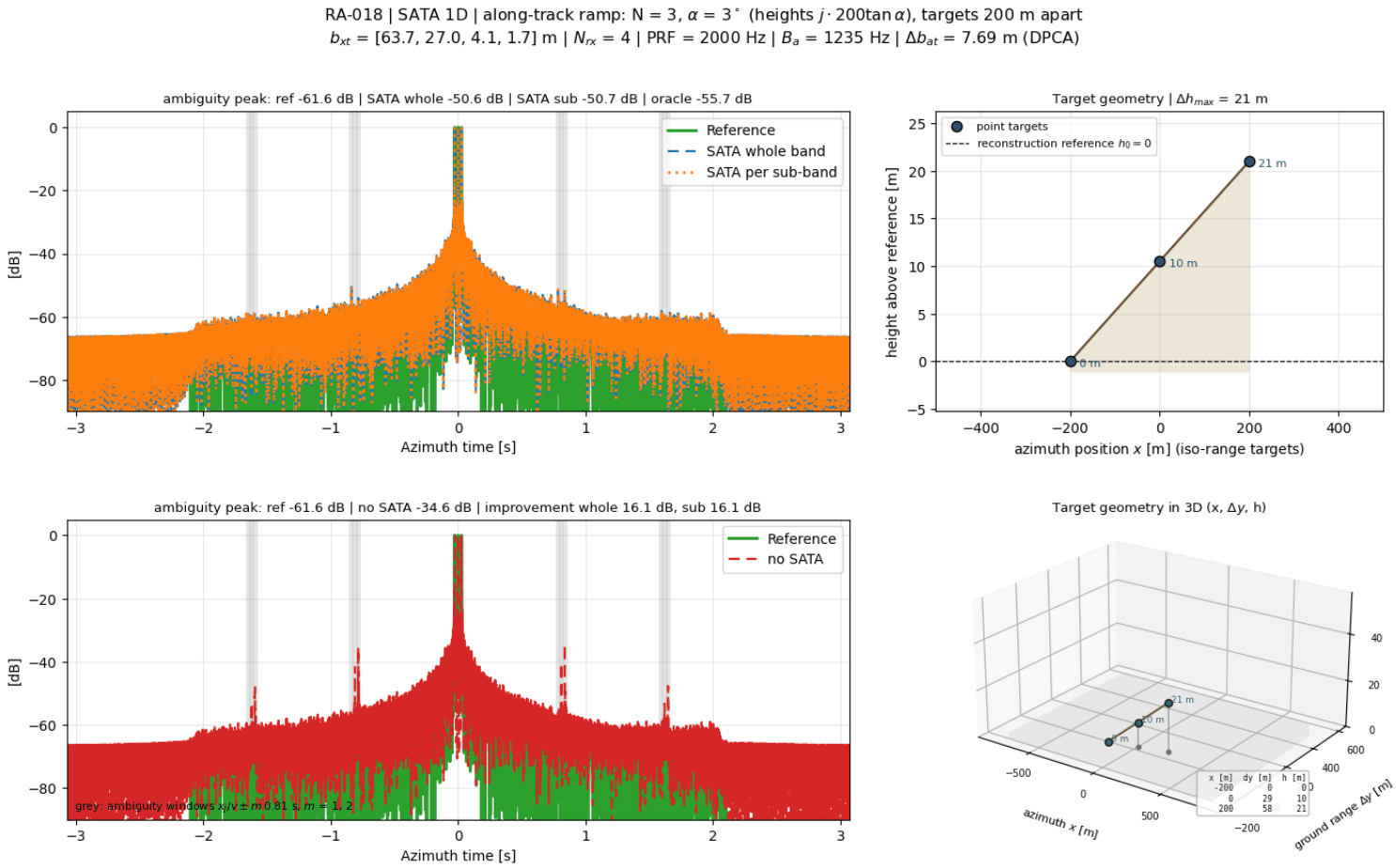


Figure 22: Targets 200 m apart: RA-018, $N = 3$, $\alpha = 3°$. Ambiguity peak: no SATA -34.6, SATA whole -50.6, per sub-band -50.7, oracle -55.7 dB (improvement 16.1 dB).

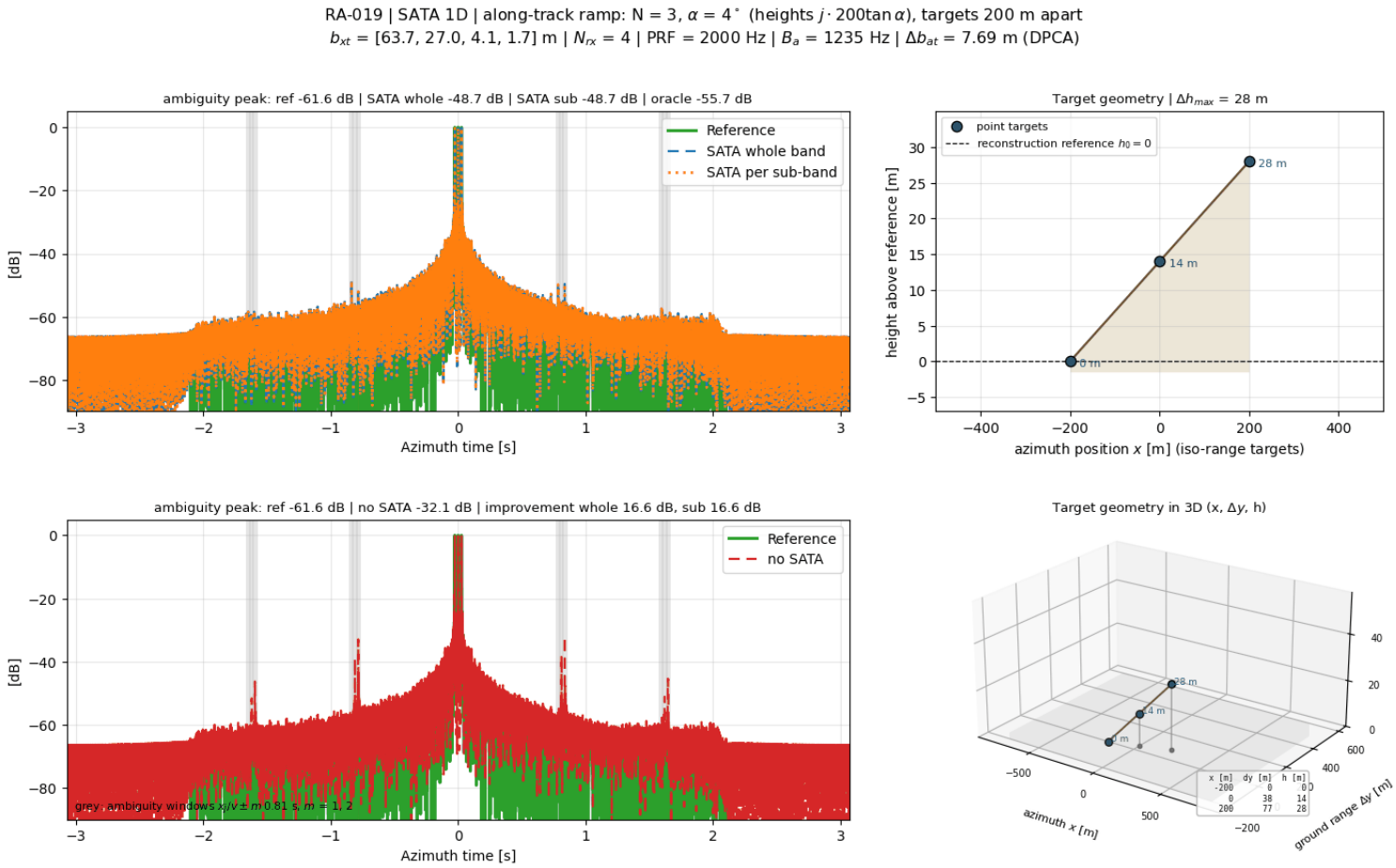


Figure 23: Targets 200 m apart: RA-019, $N = 3$, $\alpha = 4°$. Ambiguity peak: no SATA -32.1, SATA whole -48.7, per sub-band -48.7, oracle -55.7 dB (improvement 16.6 dB).

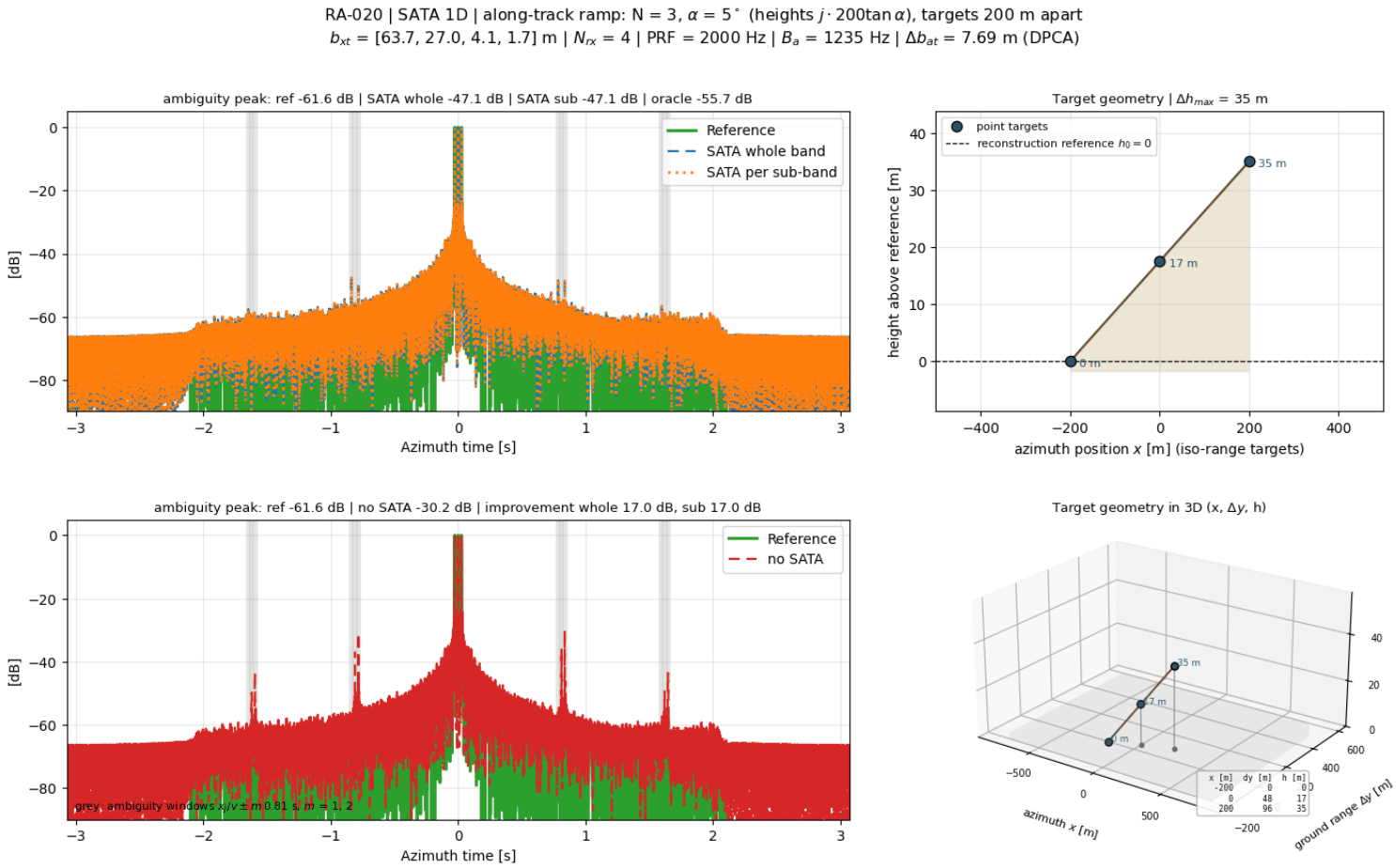


Figure 24: Targets 200 m apart: RA-020, $N = 3$, $\alpha = 5°$. Ambiguity peak: no SATA -30.2, SATA whole -47.1, per sub-band -47.1, oracle -55.7 dB (improvement 17.0 dB).

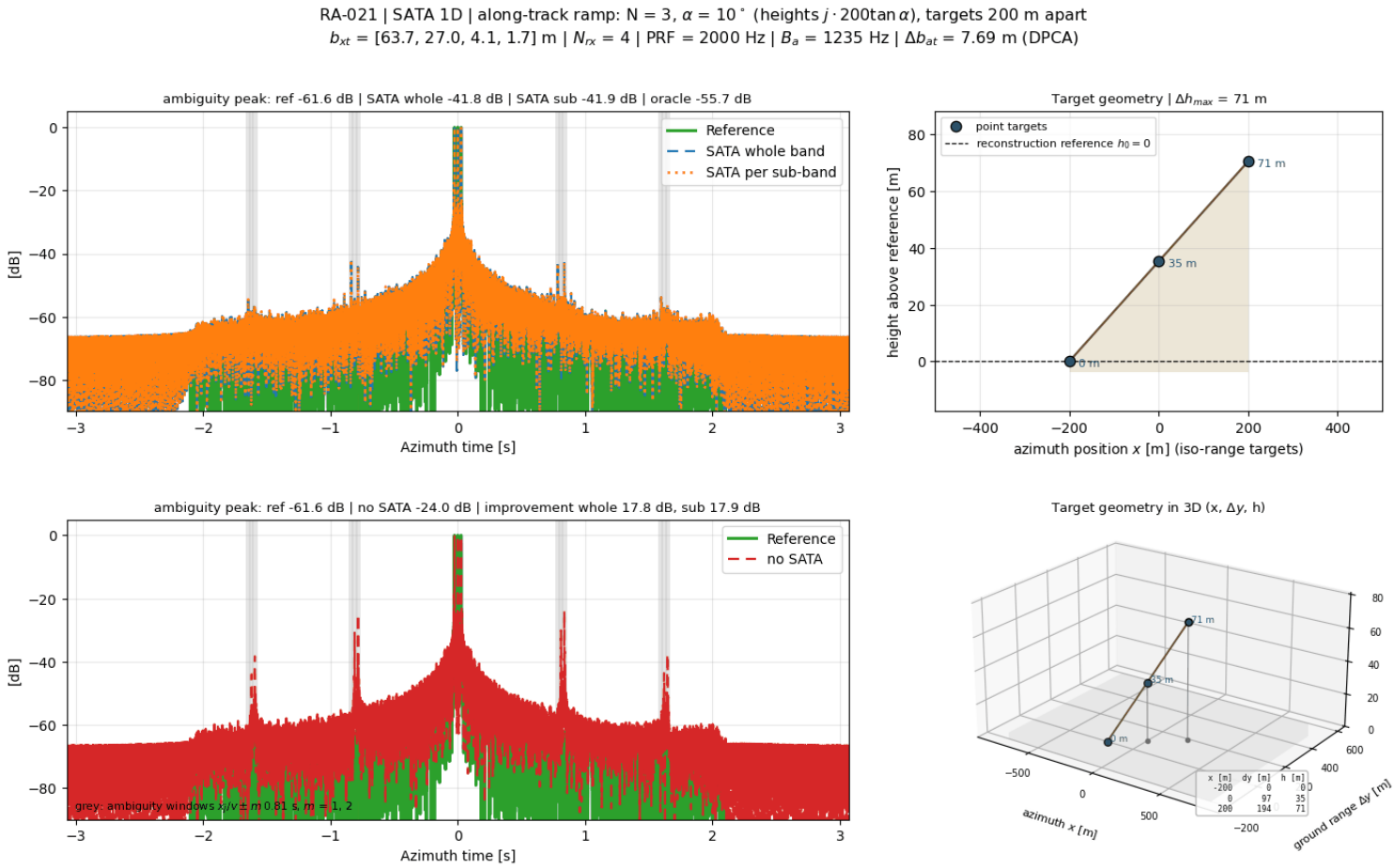


Figure 25: Targets 200 m apart: RA-021, $N = 3$, $\alpha = 10°$. Ambiguity peak: no SATA -24.0, SATA whole -41.8, per sub-band -41.9, oracle -55.7 dB (improvement 17.8 dB).

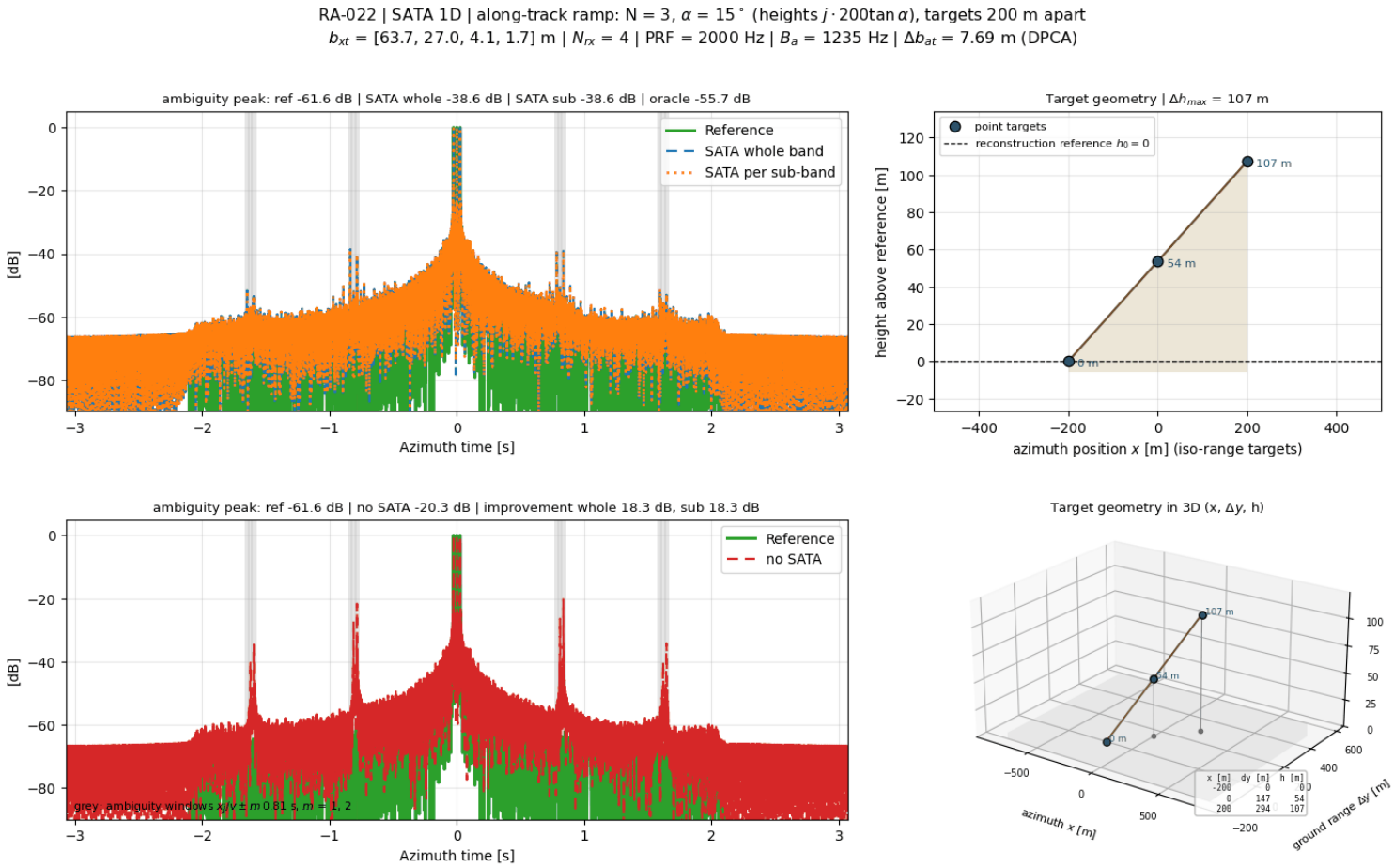


Figure 26: Targets 200 m apart: RA-022, $N = 3$, $\alpha = 15°$. Ambiguity peak: no SATA -20.3, SATA whole -38.6, per sub-band -38.6, oracle -55.7 dB (improvement 18.3 dB).

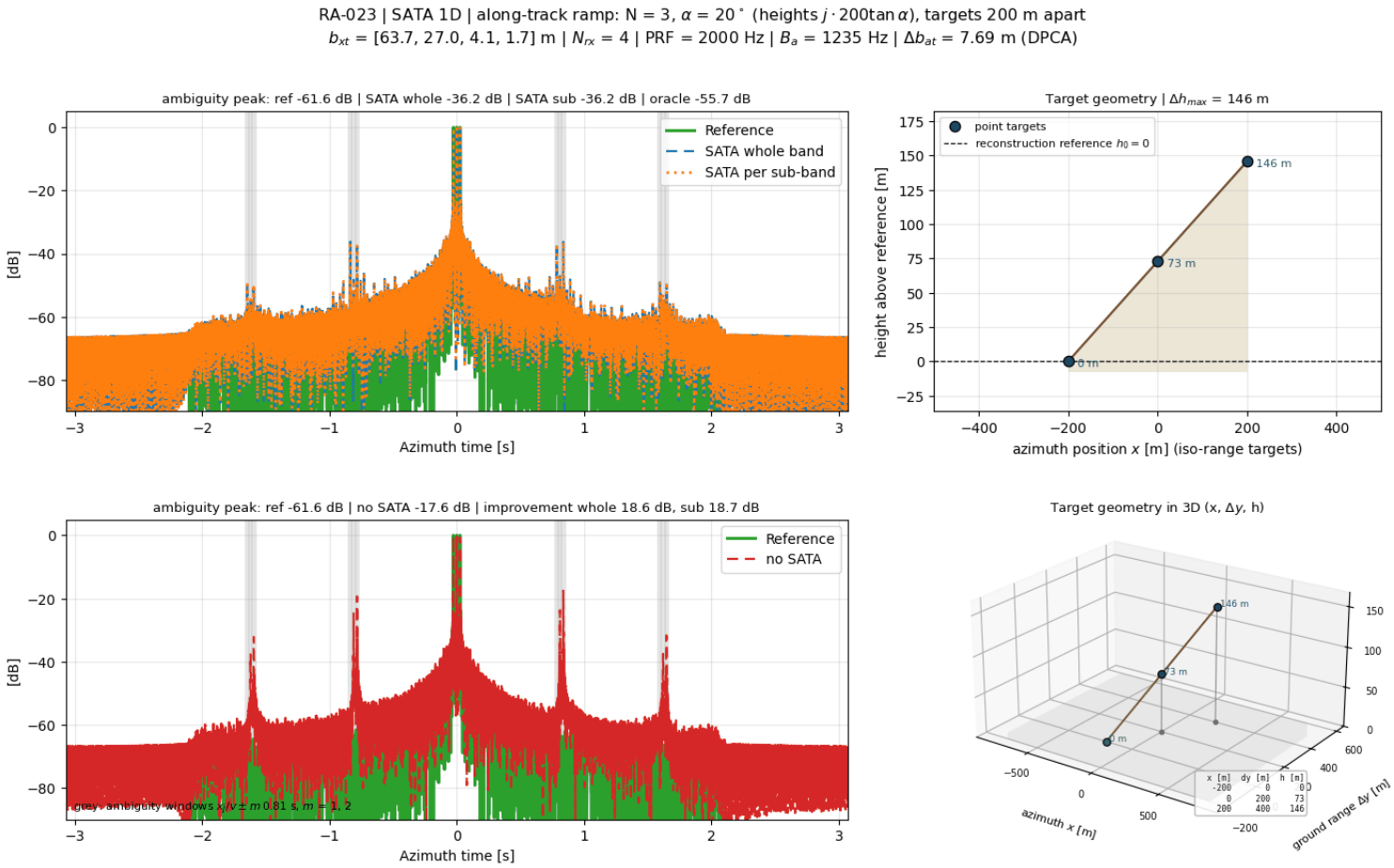


Figure 27: Targets 200 m apart: RA-023, $N = 3$, $\alpha = 20°$. Ambiguity peak: no SATA -17.6, SATA whole -36.2, per sub-band -36.2, oracle -55.7 dB (improvement 18.6 dB).

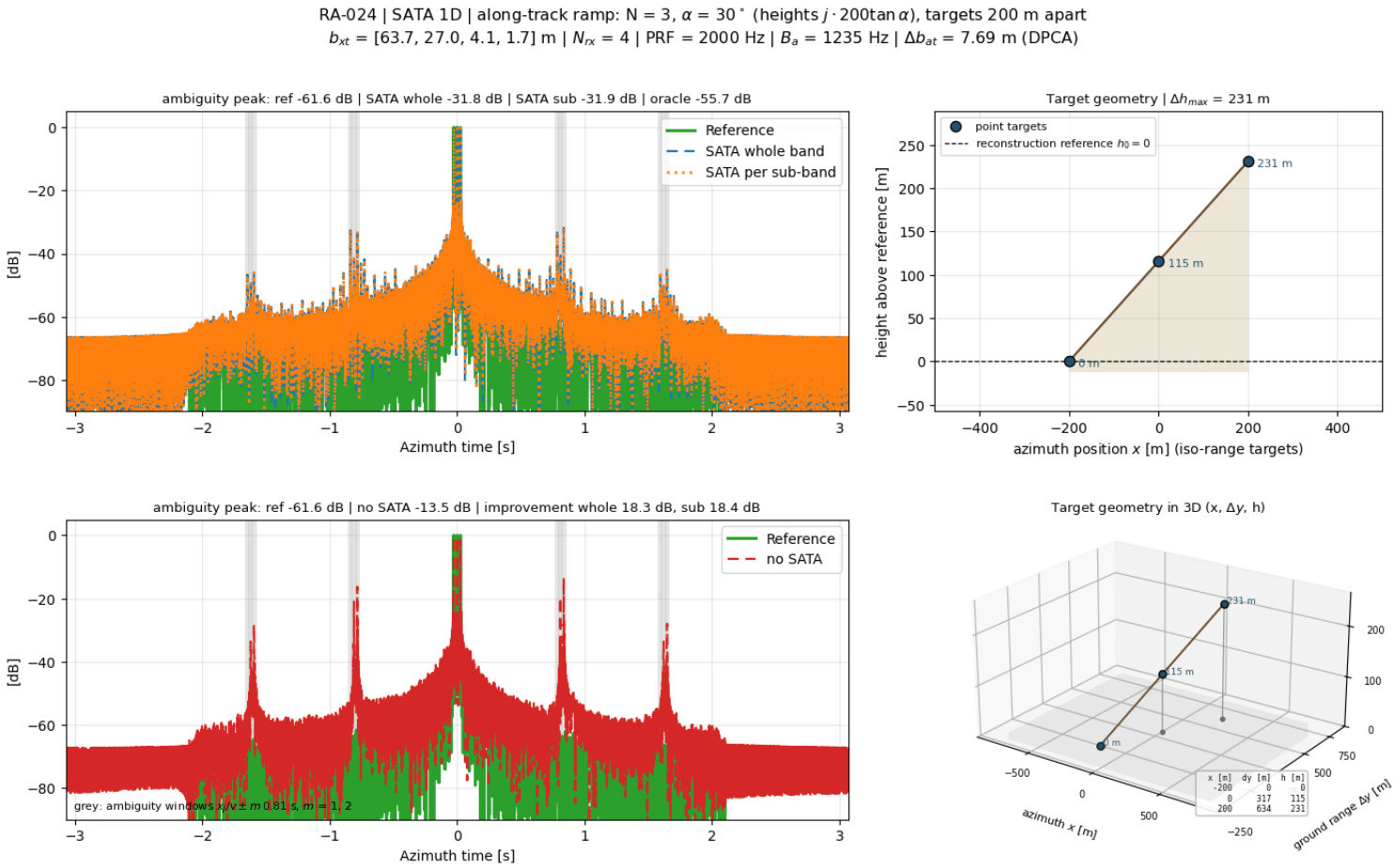


Figure 28: Targets 200 m apart: RA-024, $N = 3$, $\alpha = 30°$. Ambiguity peak: no SATA −13.5, SATA whole −31.8, per sub-band −31.9, oracle −55.7 dB (improvement 18.3 dB).

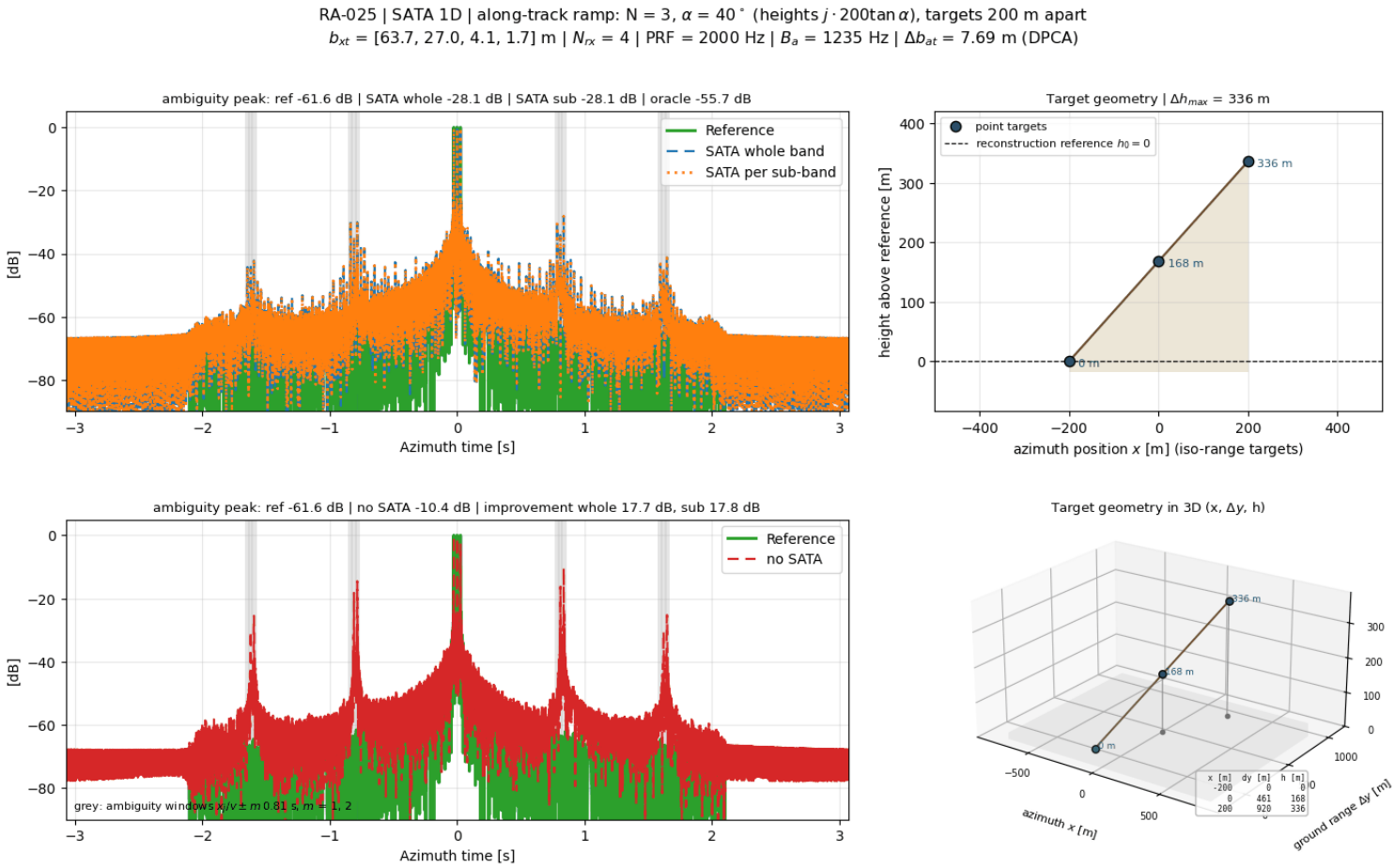


Figure 29: Targets 200 m apart: RA-025, $N = 3$, $\alpha = 40°$. Ambiguity peak: no SATA -10.4, SATA whole -28.1, per sub-band -28.1, oracle -55.7 dB (improvement 17.7 dB).

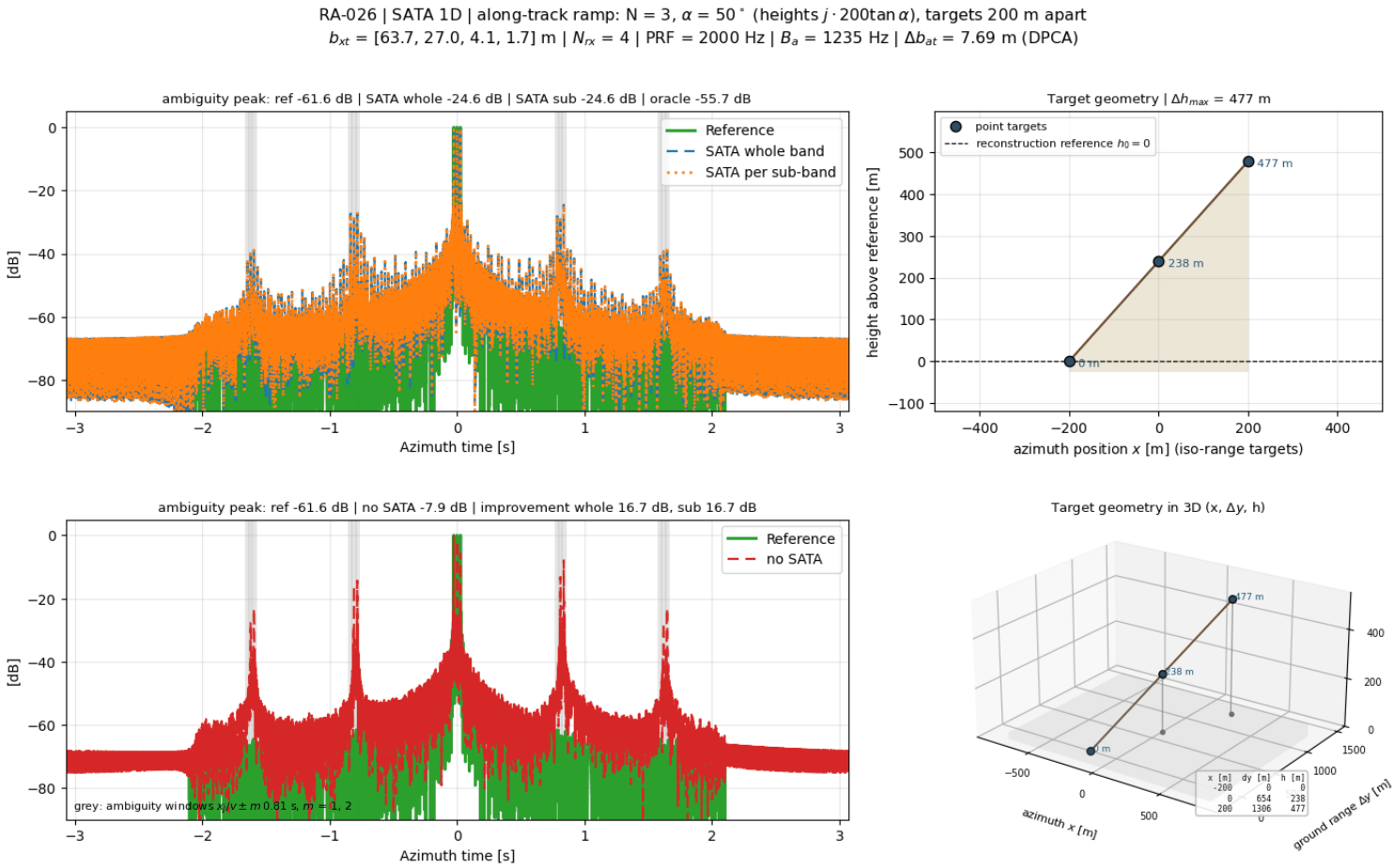


Figure 30: Targets 200 m apart: RA-026, $N = 3$, $\alpha = 50°$. Ambiguity peak: no SATA -7.9, SATA whole -24.6, per sub-band -24.6, oracle -55.7 dB (improvement 16.7 dB).

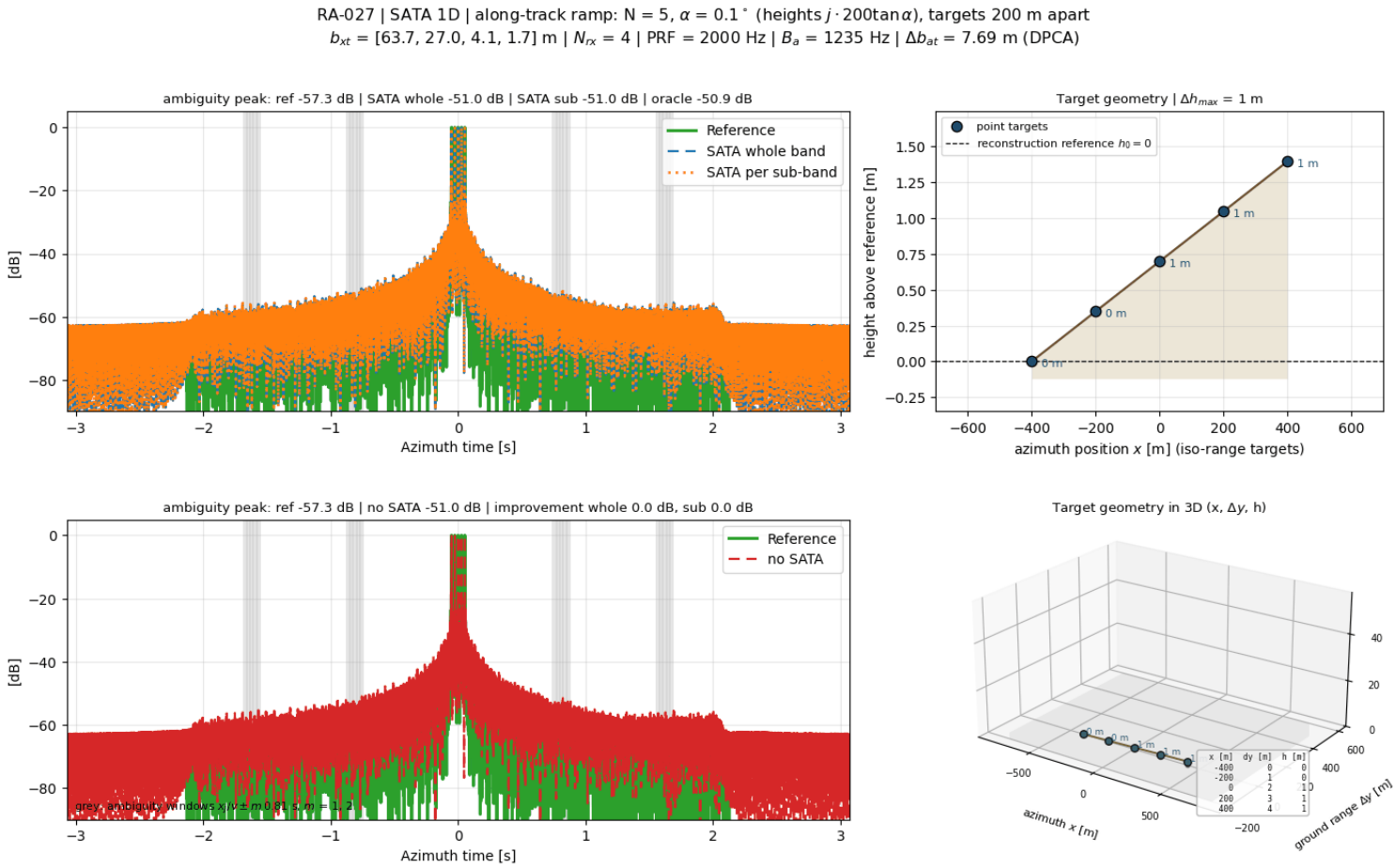


Figure 31: Targets 200 m apart: RA-027, $N = 5$, $\alpha = 0.1°$. Ambiguity peak: no SATA −51.0, SATA whole −51.0, per sub-band −51.0, oracle −50.9 dB (improvement 0.0 dB).

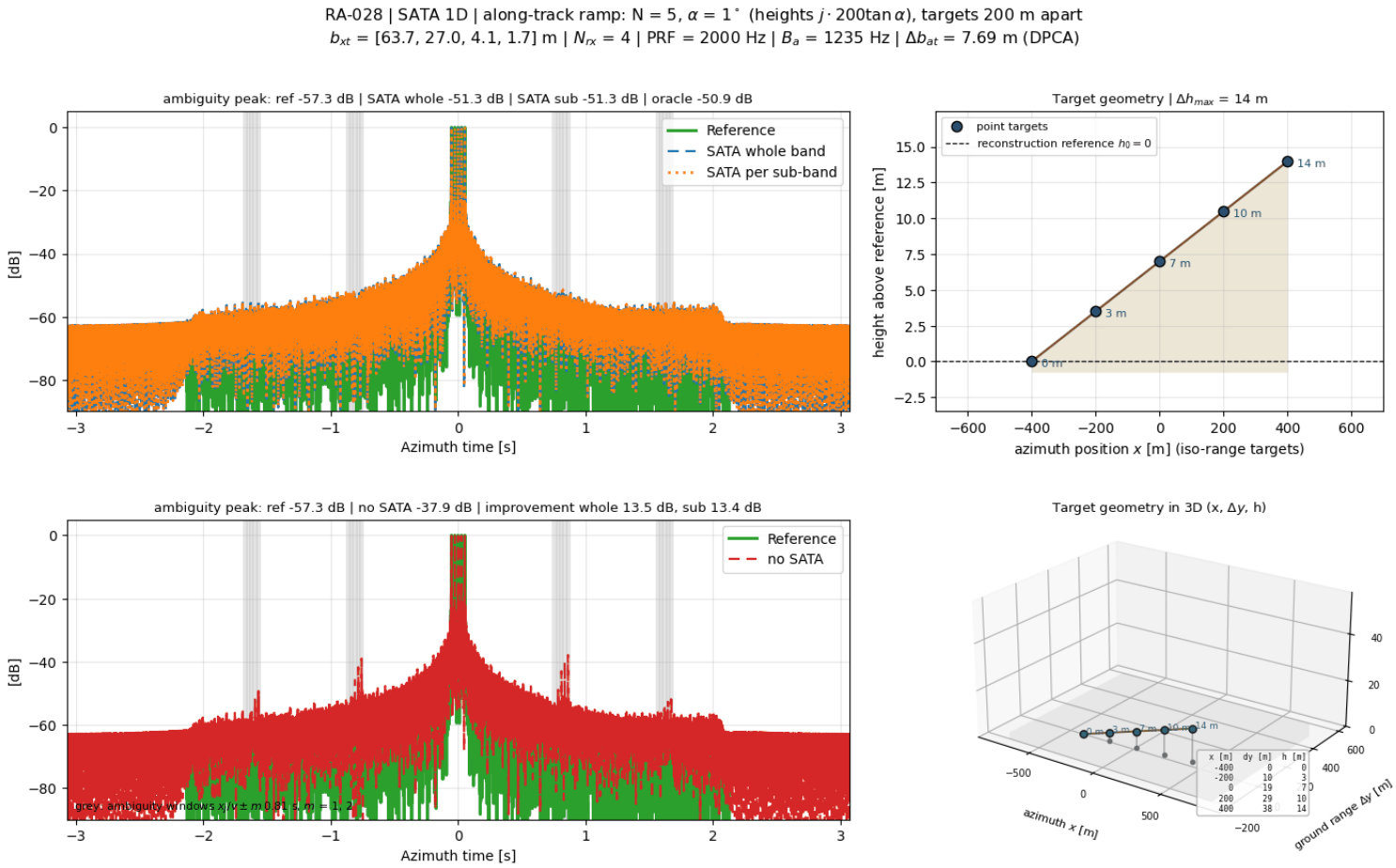


Figure 32: Targets 200 m apart: RA-028, $N = 5$, $\alpha = 1°$. Ambiguity peak: no SATA -37.9, SATA whole -51.3, per sub-band -51.3, oracle -50.9 dB (improvement 13.5 dB).

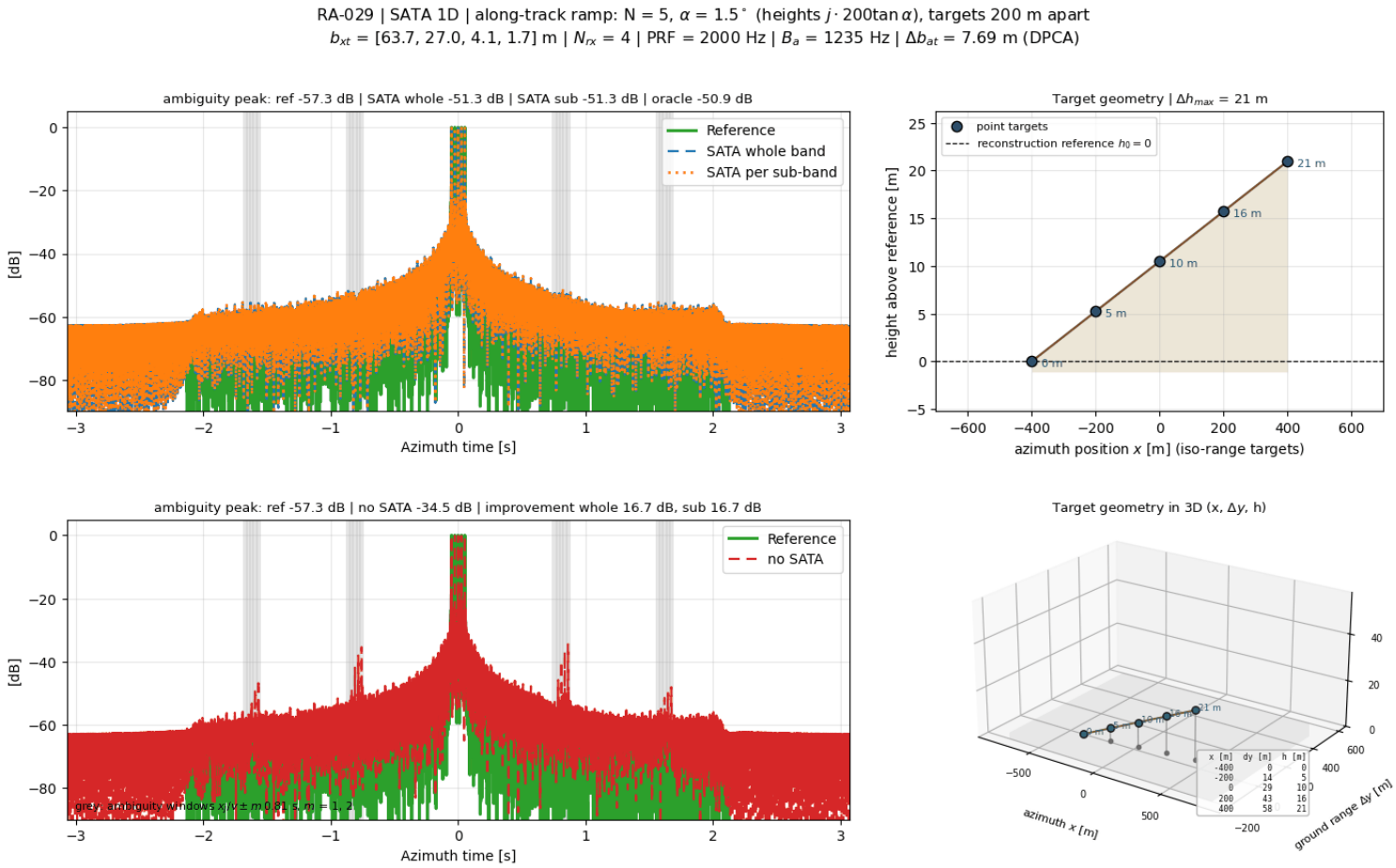


Figure 33: Targets 200 m apart: RA-029, $N = 5$, $\alpha = 1.5°$. Ambiguity peak: no SATA -34.5, SATA whole -51.3, per sub-band -51.3, oracle -50.9 dB (improvement 16.7 dB).

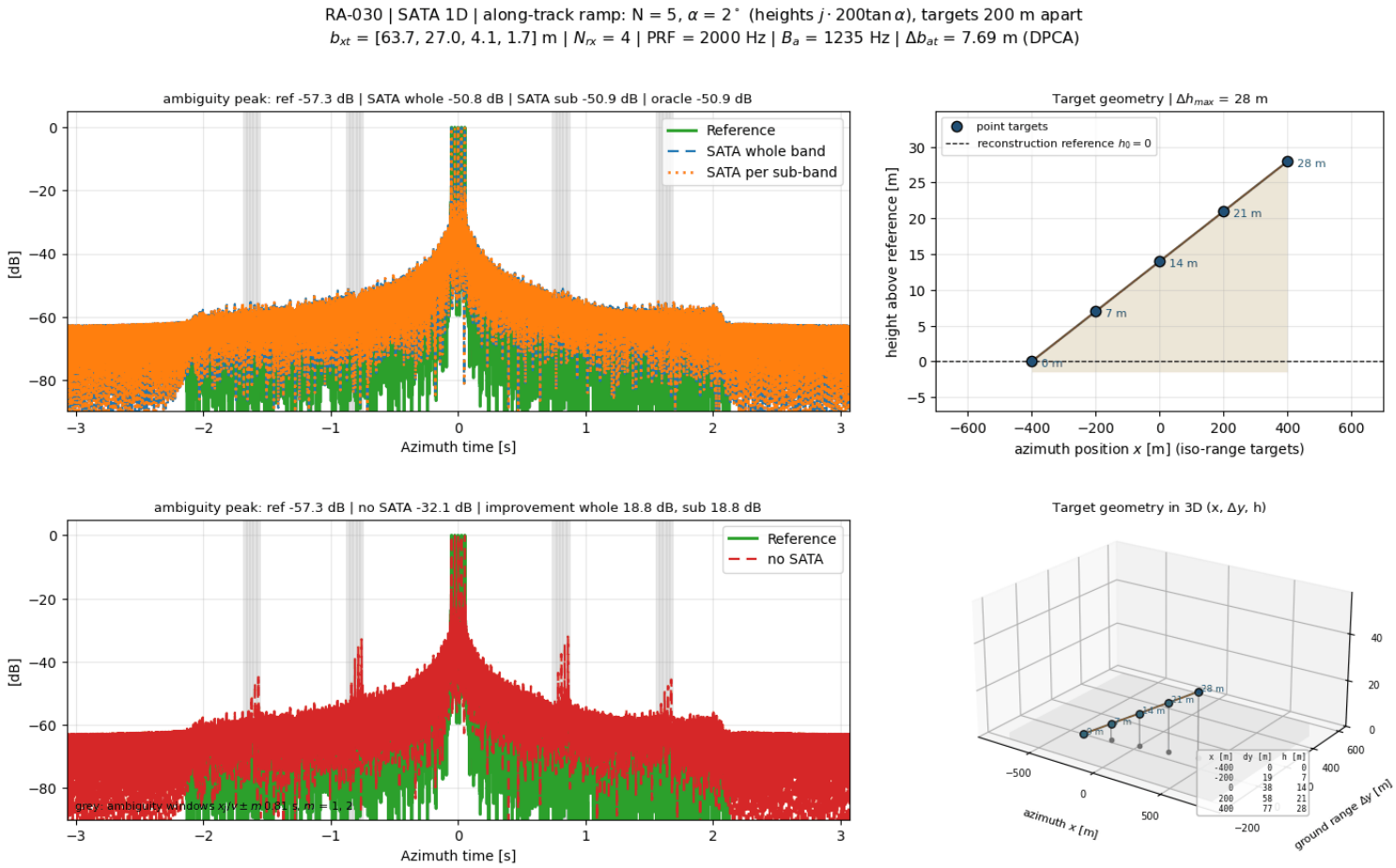


Figure 34: Targets 200 m apart: RA-030, $N = 5$, $\alpha = 2°$. Ambiguity peak: no SATA -32.1, SATA whole -50.8, per sub-band -50.9, oracle -50.9 dB (improvement 18.8 dB).

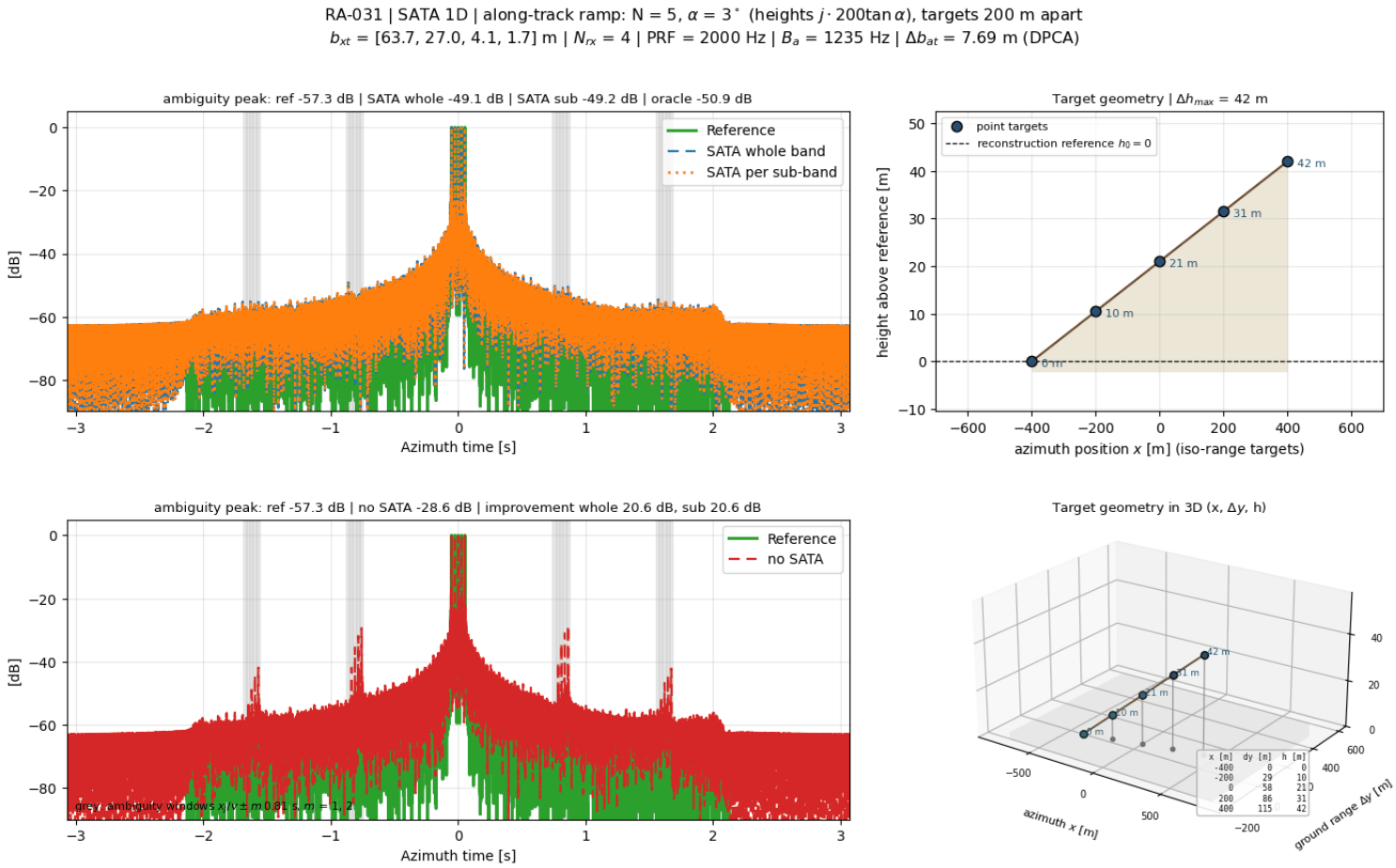


Figure 35: Targets 200 m apart: RA-031, $N = 5$, $\alpha = 3°$. Ambiguity peak: no SATA -28.6, SATA whole -49.1, per sub-band -49.2, oracle -50.9 dB (improvement 20.6 dB).

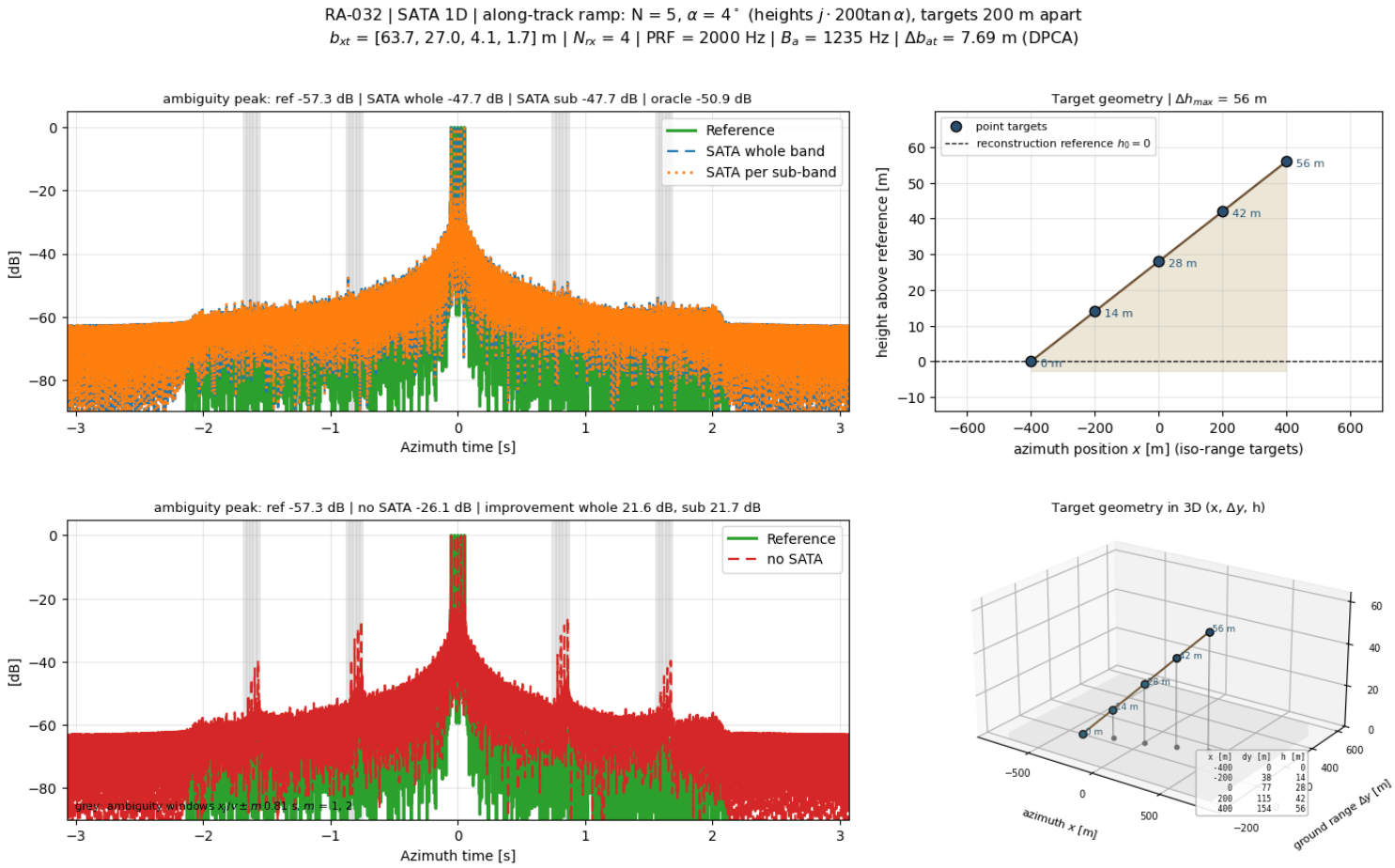


Figure 36: Targets 200 m apart: RA-032, $N = 5$, $\alpha = 4°$. Ambiguity peak: no SATA -26.1, SATA whole -47.7, per sub-band -47.7, oracle -50.9 dB (improvement 21.6 dB).

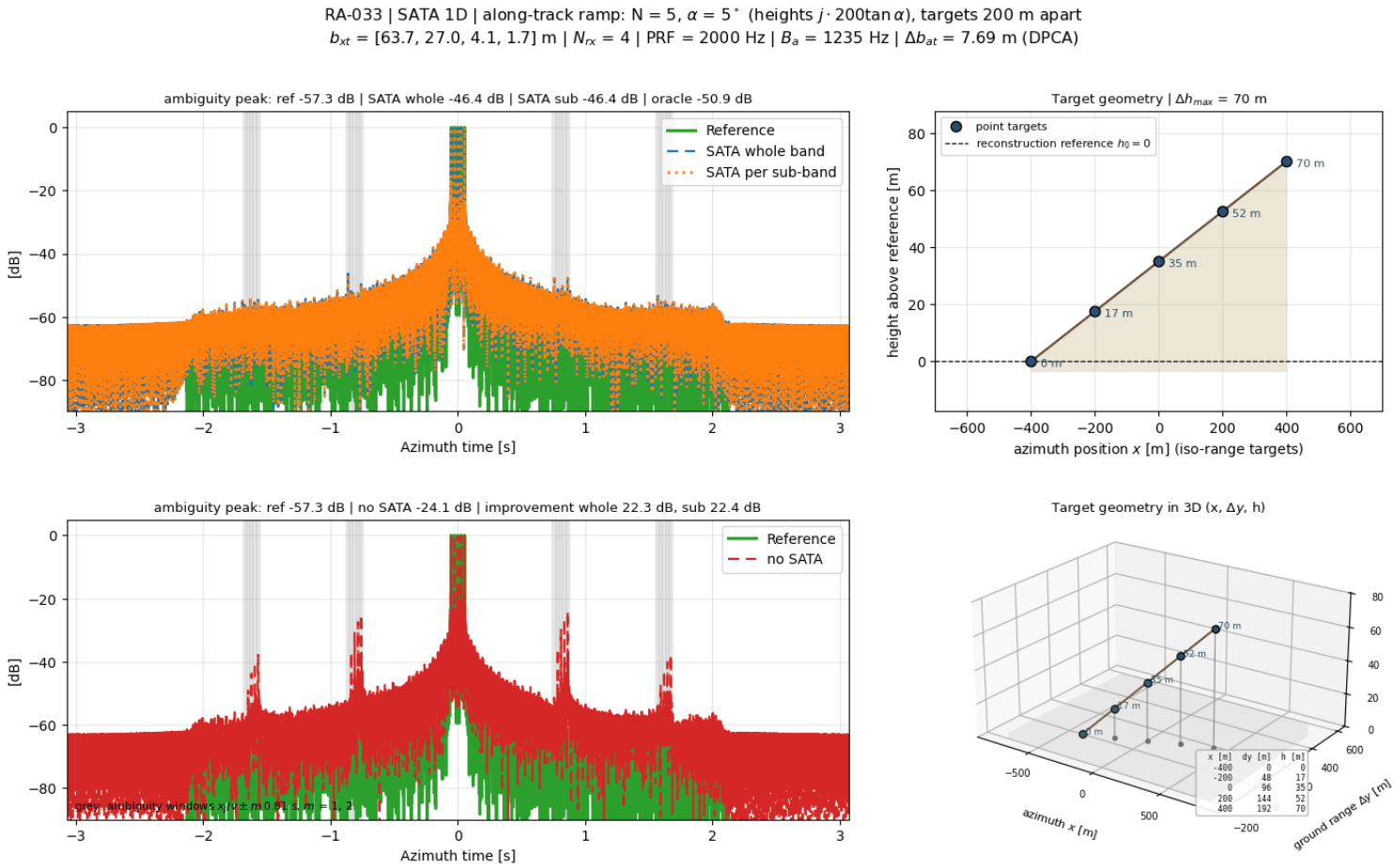


Figure 37: Targets 200 m apart: RA-033, $N = 5$, $\alpha = 5°$. Ambiguity peak: no SATA -24.1, SATA whole -46.4, per sub-band -46.4, oracle -50.9 dB (improvement 22.3 dB).

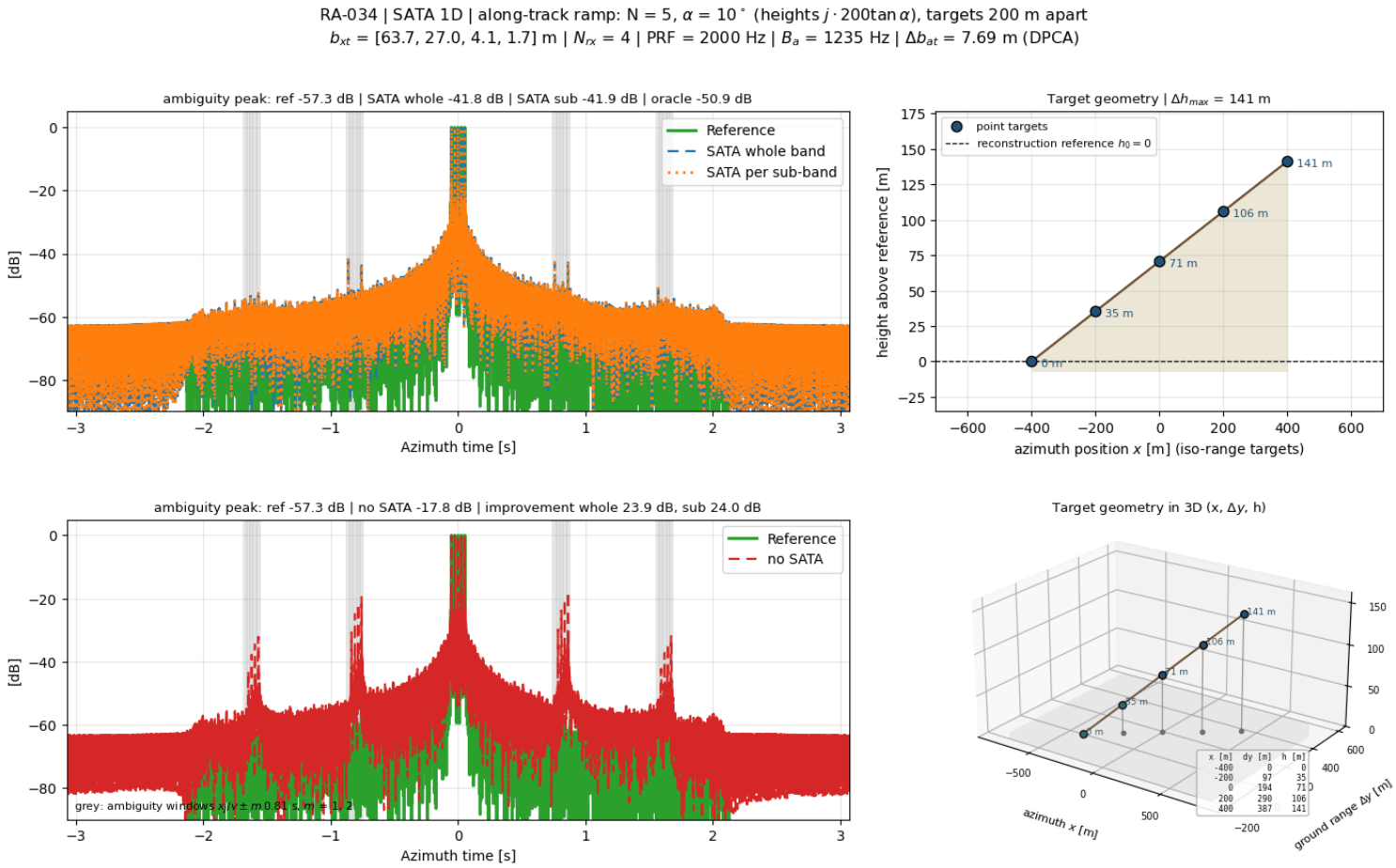


Figure 38: Targets 200 m apart: RA-034, $N = 5$, $\alpha = 10°$. Ambiguity peak: no SATA -17.8, SATA whole -41.8, per sub-band -41.9, oracle -50.9 dB (improvement 23.9 dB).

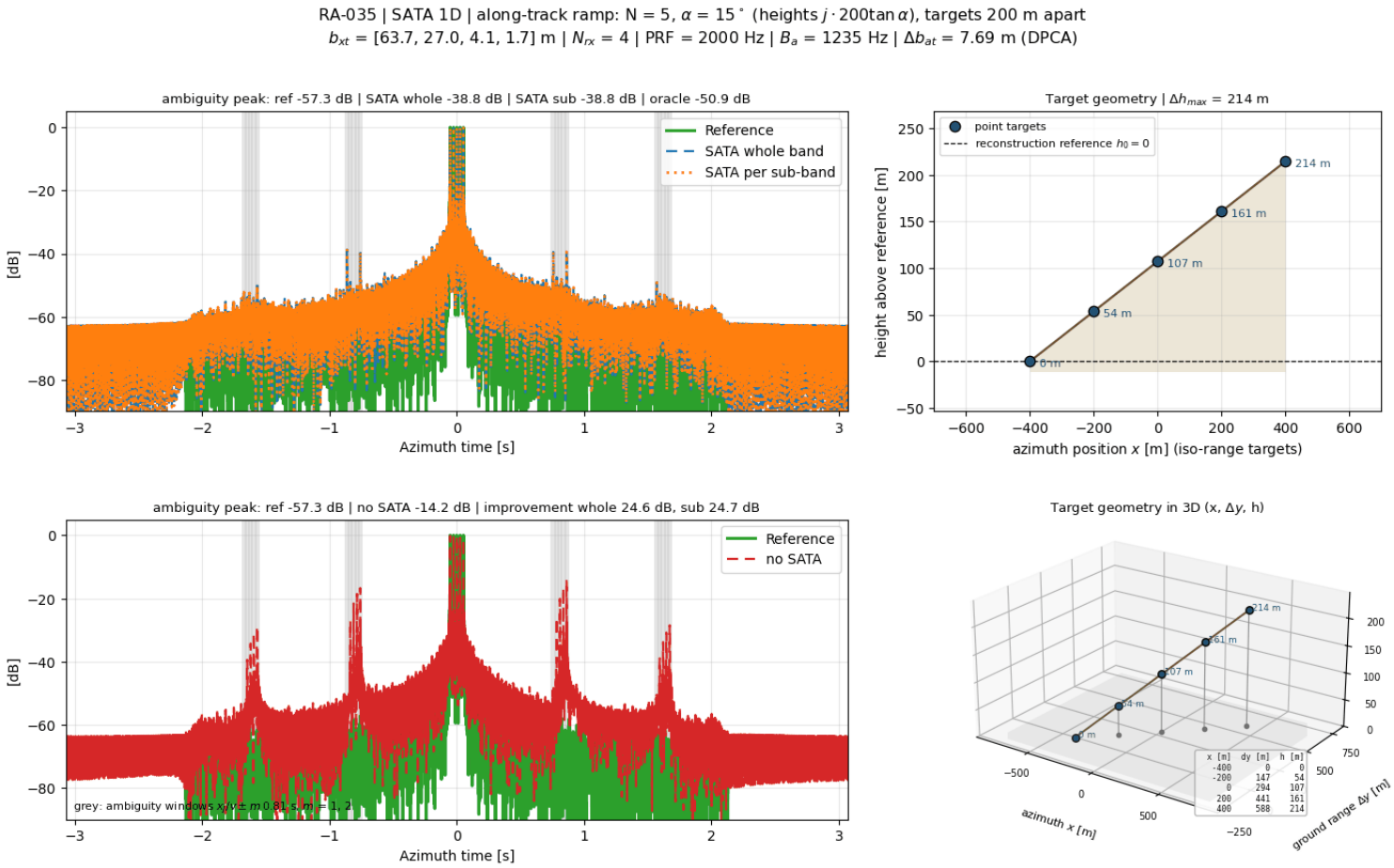


Figure 39: Targets 200 m apart: RA-035, $N = 5$, $\alpha = 15°$. Ambiguity peak: no SATA −14.2, SATA whole −38.8, per sub-band −38.8, oracle −50.9 dB (improvement 24.6 dB).

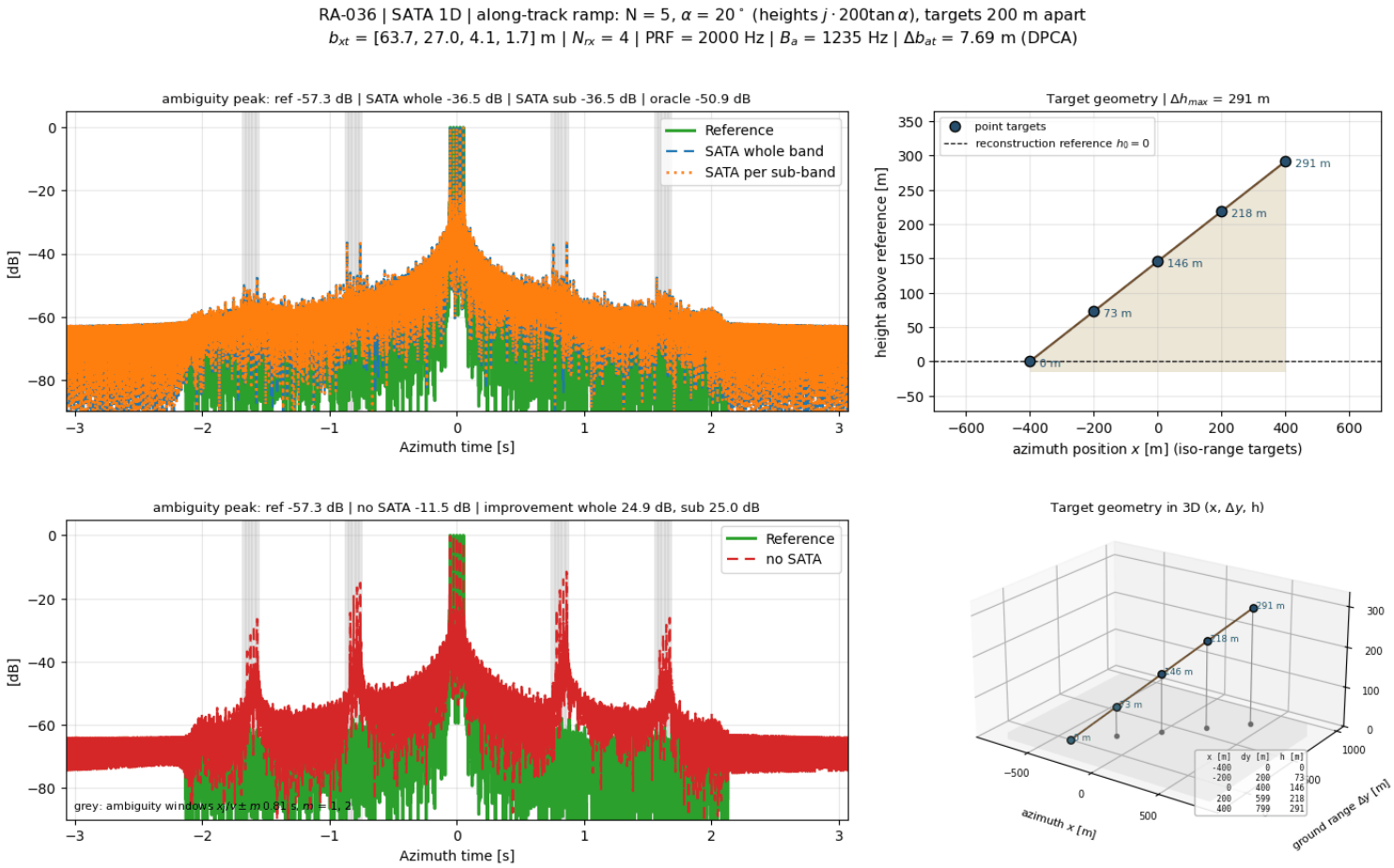


Figure 40: Targets 200 m apart: RA-036, $N = 5$, $\alpha = 20°$. Ambiguity peak: no SATA -11.5, SATA whole -36.5, per sub-band -36.5, oracle -50.9 dB (improvement 24.9 dB).

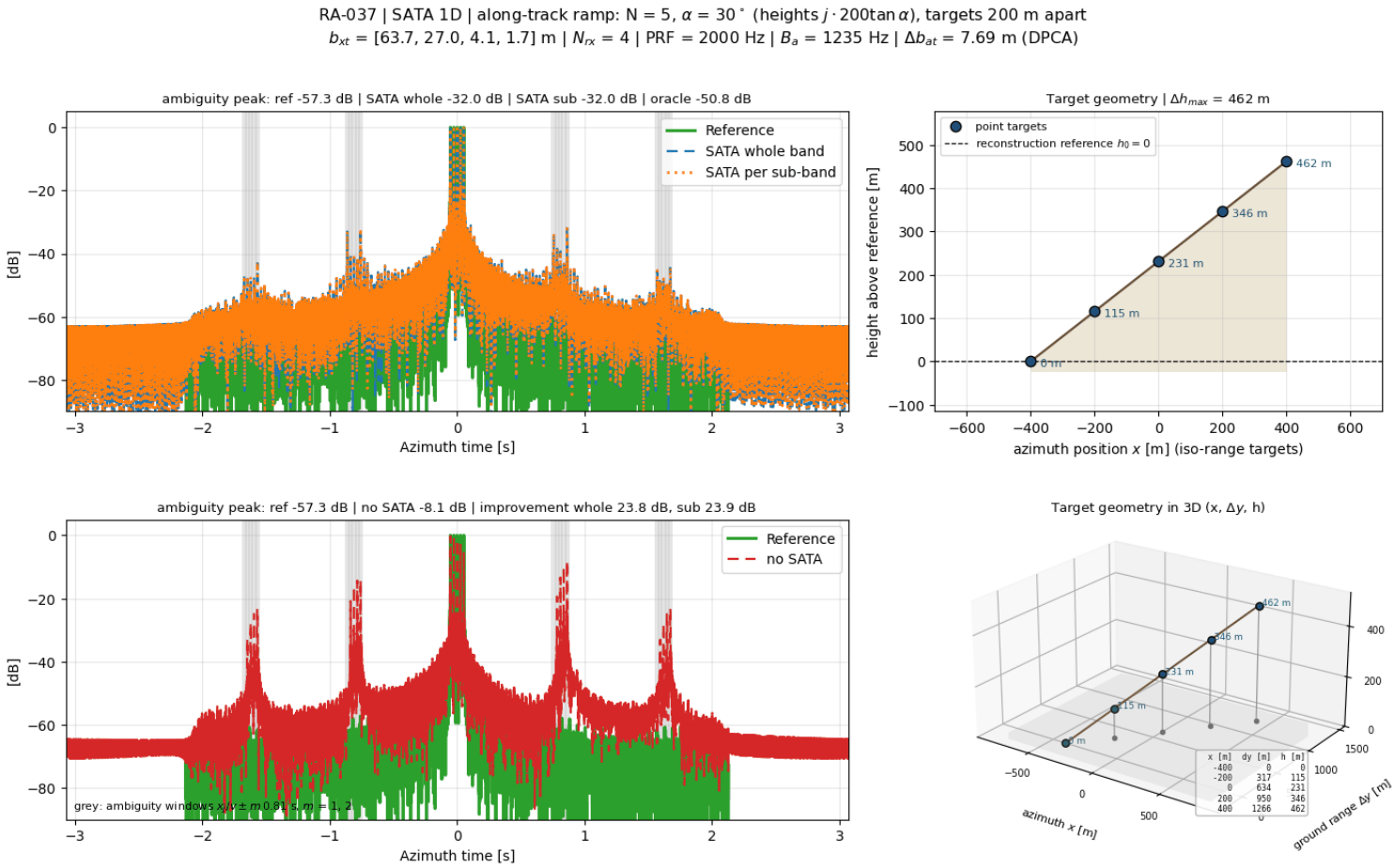


Figure 41: Targets 200 m apart: RA-037, $N = 5$, $\alpha = 30°$. Ambiguity peak: no SATA -8.1, SATA whole -32.0, per sub-band -32.0, oracle -50.8 dB (improvement 23.8 dB).

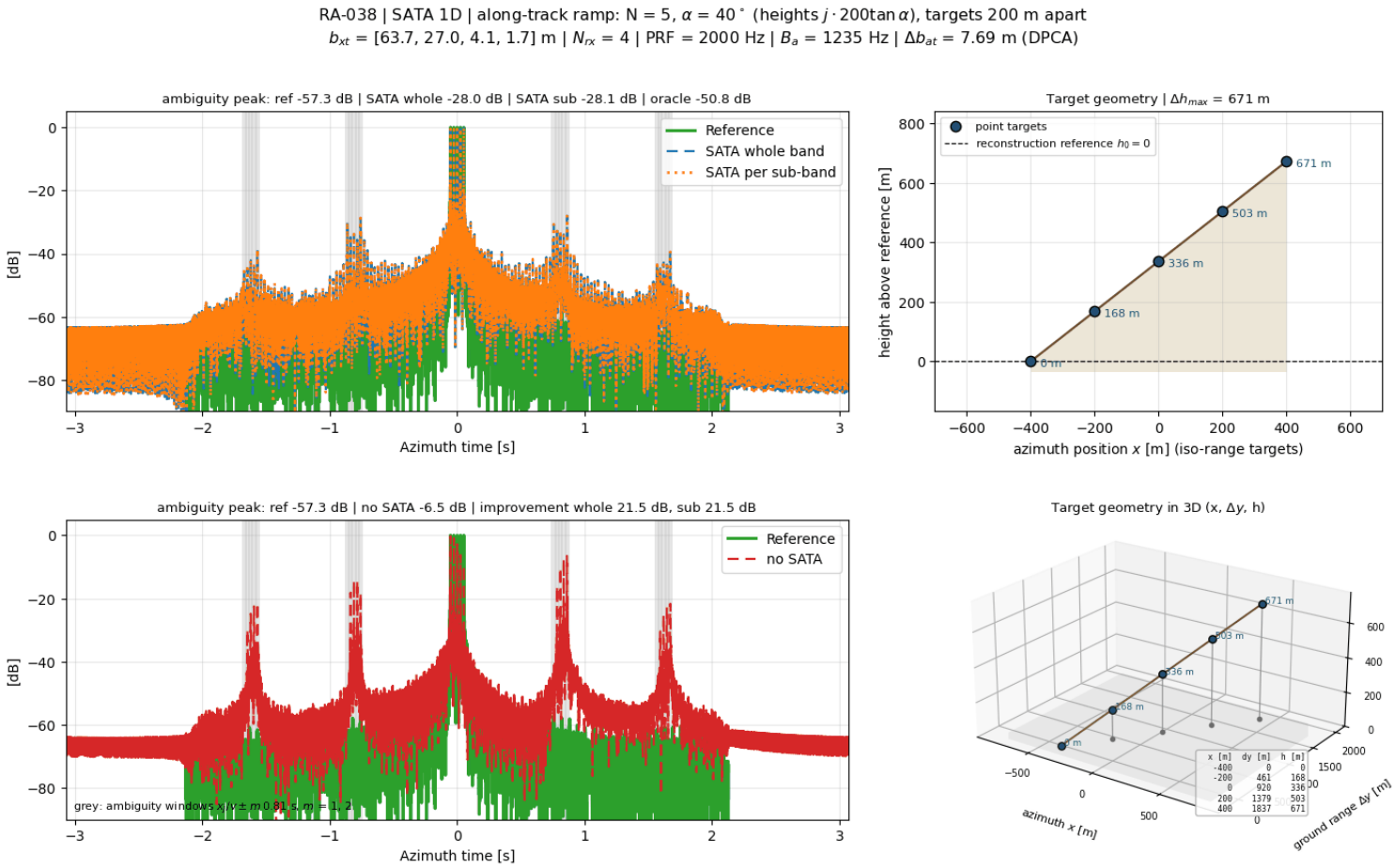


Figure 42: Targets 200 m apart: RA-038, $N = 5$, $\alpha = 40°$. Ambiguity peak: no SATA -6.5, SATA whole -28.0, per sub-band -28.1, oracle -50.8 dB (improvement 21.5 dB).

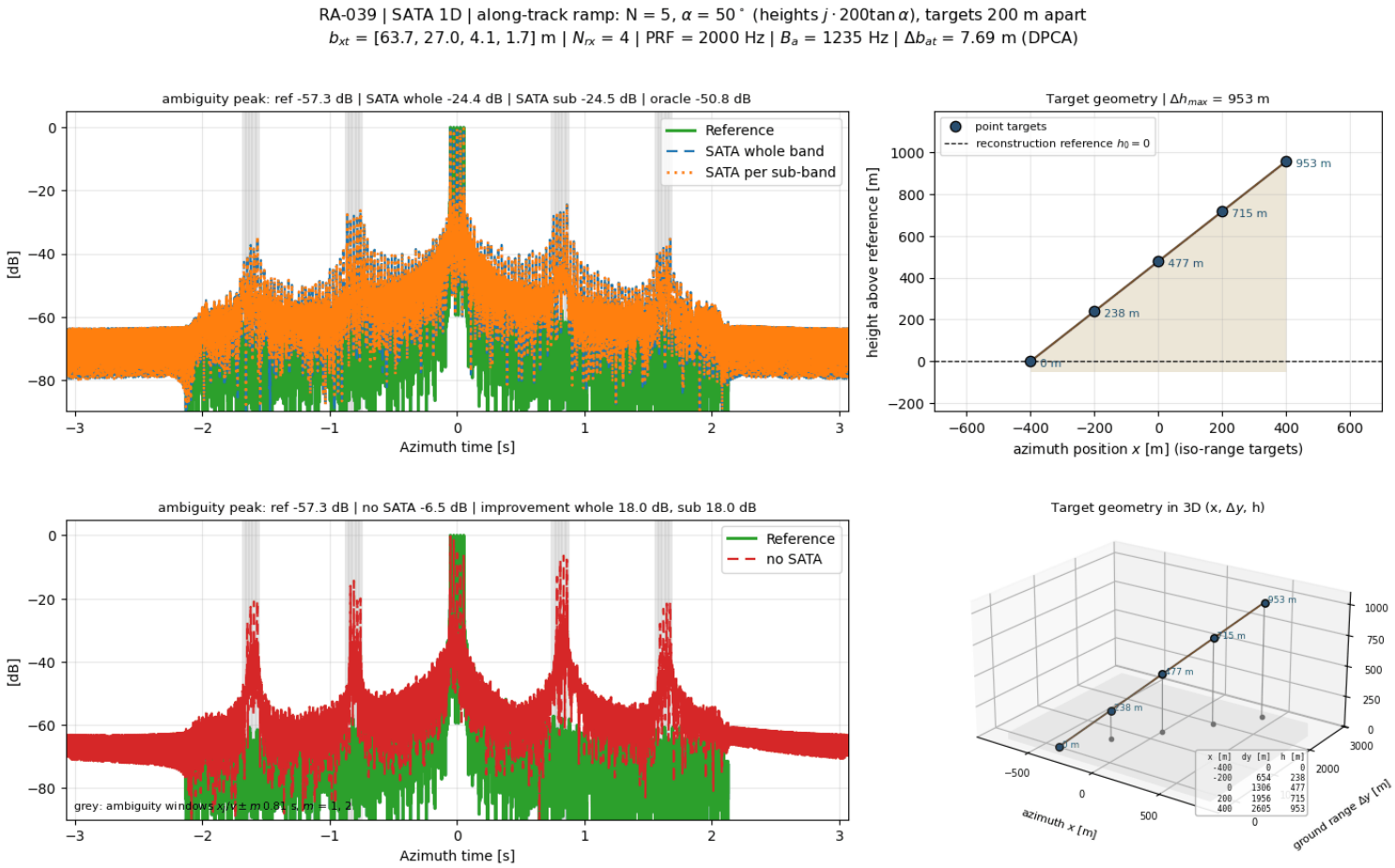


Figure 43: Targets 200 m apart: RA-039, $N = 5$, $\alpha = 50°$. Ambiguity peak: no SATA −6.5, SATA whole −24.4, per sub-band −24.5, oracle −50.8 dB (improvement 18.0 dB).

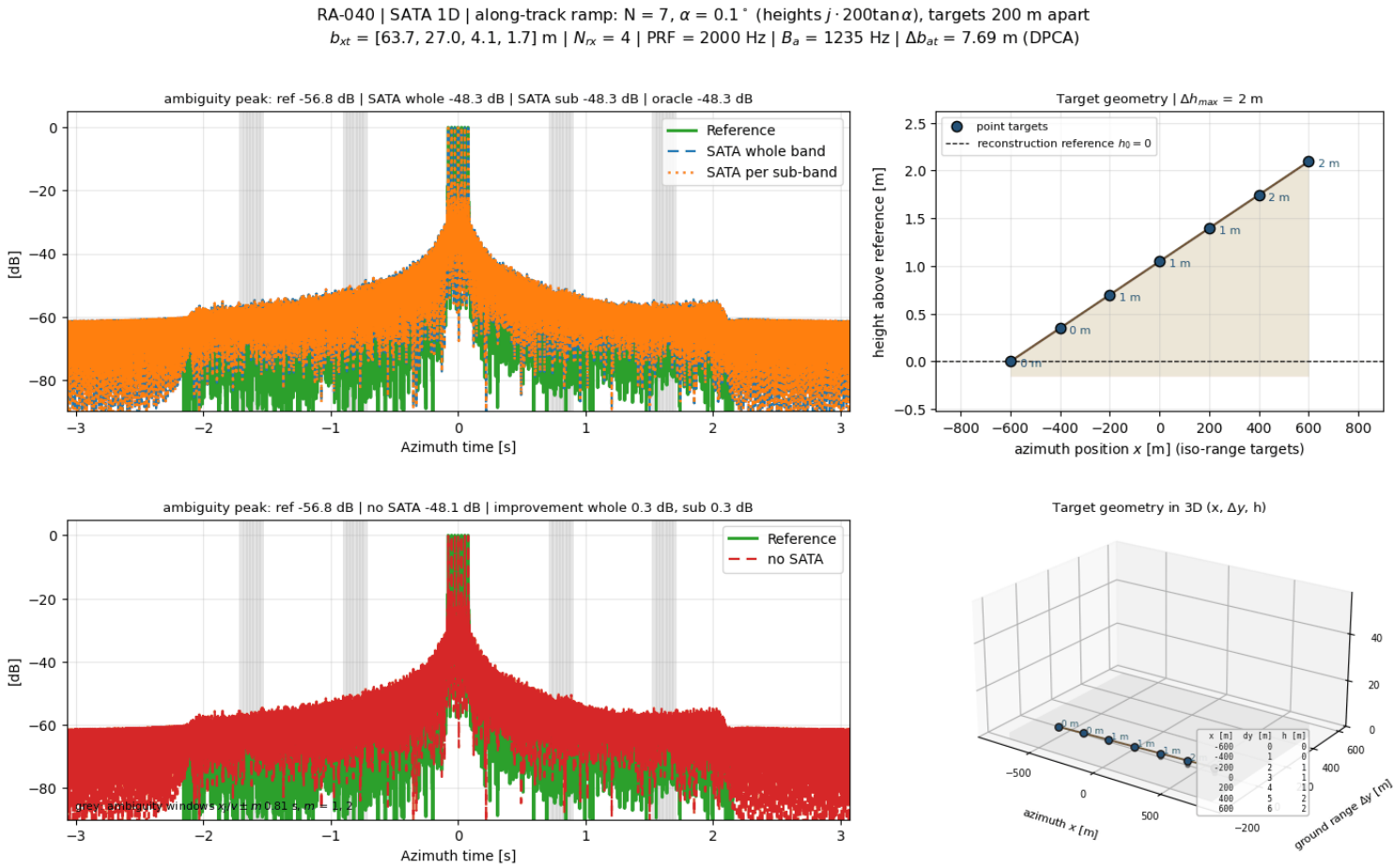


Figure 44: Targets 200 m apart: RA-040, $N = 7$, $\alpha = 0.1°$. Ambiguity peak: no SATA −48.1, SATA whole −48.3, per sub-band −48.3, oracle −48.3 dB (improvement 0.3 dB).

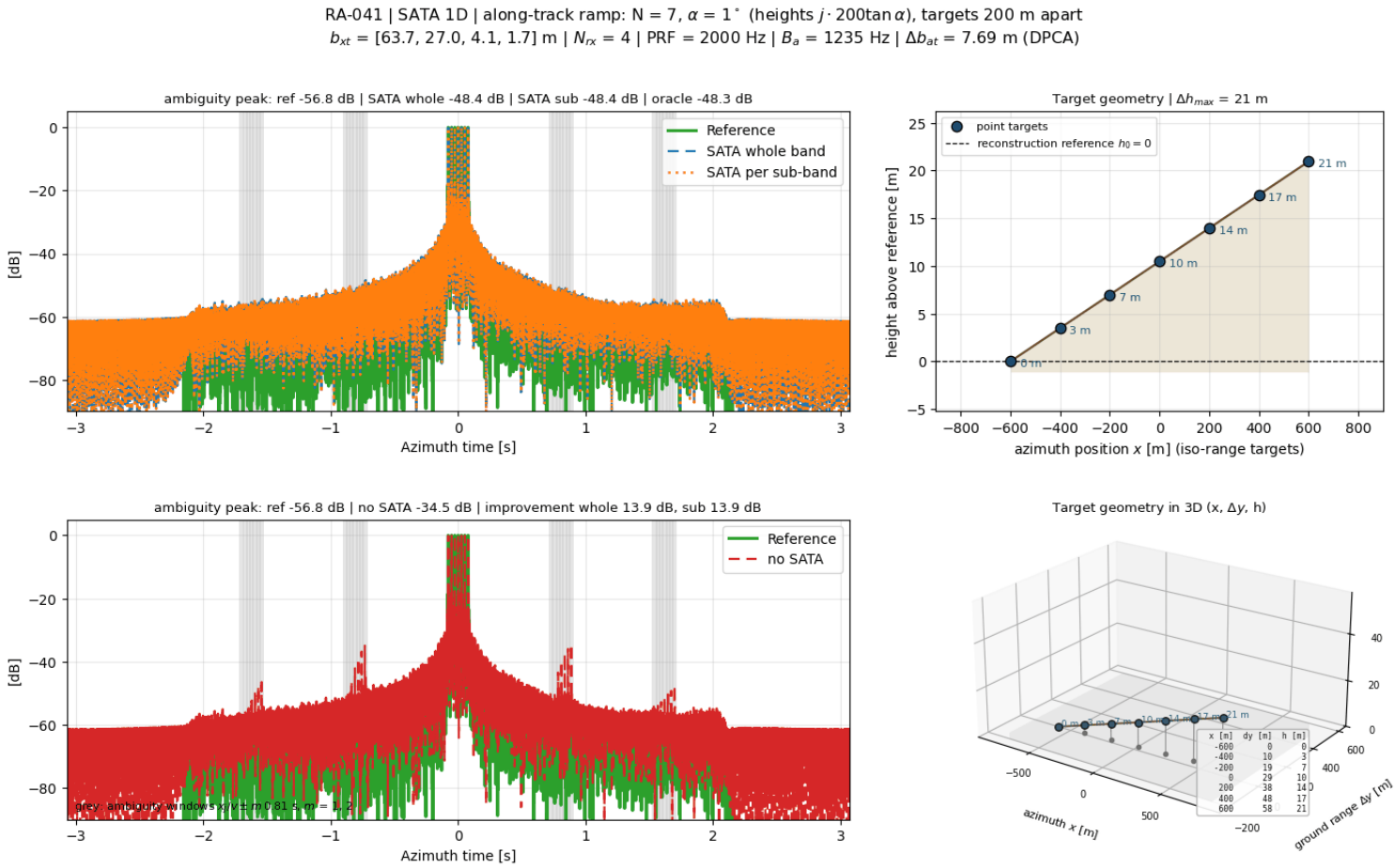


Figure 45: Targets 200 m apart: RA-041, $N = 7$, $\alpha = 1°$. Ambiguity peak: no SATA -34.5, SATA whole -48.4, per sub-band -48.4, oracle -48.3 dB (improvement 13.9 dB).

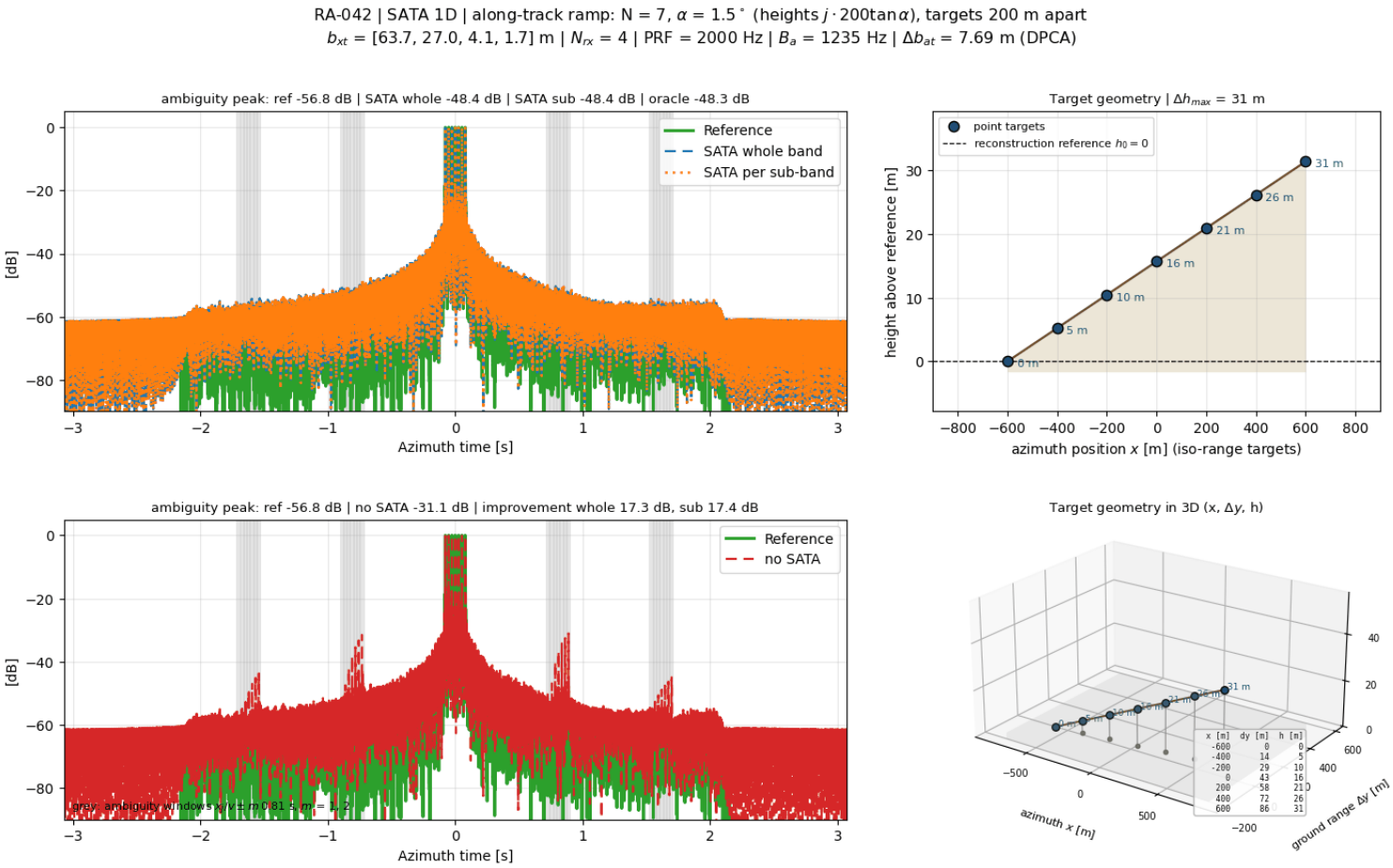


Figure 46: Targets 200 m apart: RA-042, $N = 7$, $\alpha = 1.5°$. Ambiguity peak: no SATA -31.1, SATA whole -48.4, per sub-band -48.4, oracle -48.3 dB (improvement 17.3 dB).

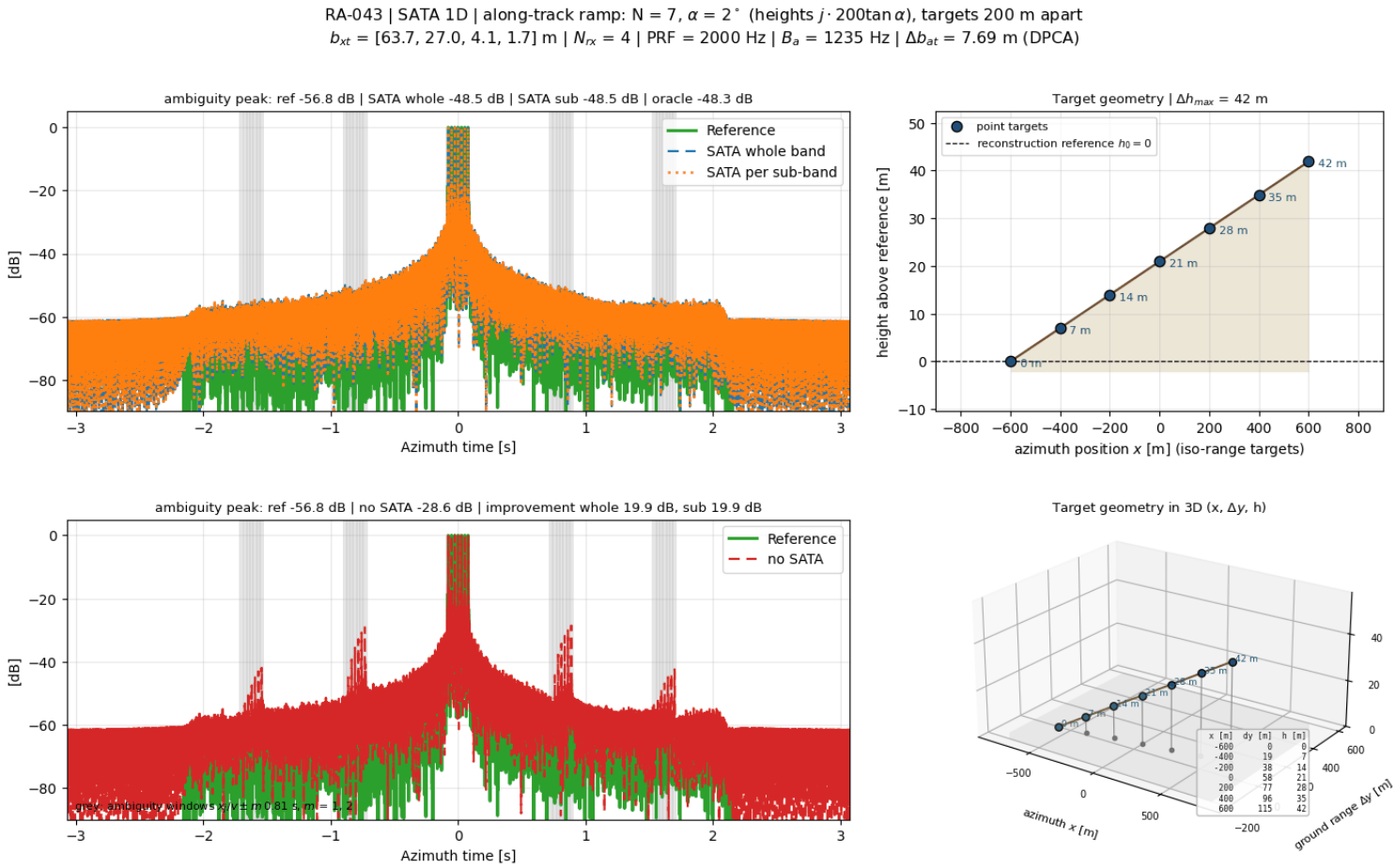


Figure 47: Targets 200 m apart: RA-043, $N = 7$, $\alpha = 2°$. Ambiguity peak: no SATA −28.6, SATA whole −48.5, per sub-band −48.5, oracle −48.3 dB (improvement 19.9 dB).

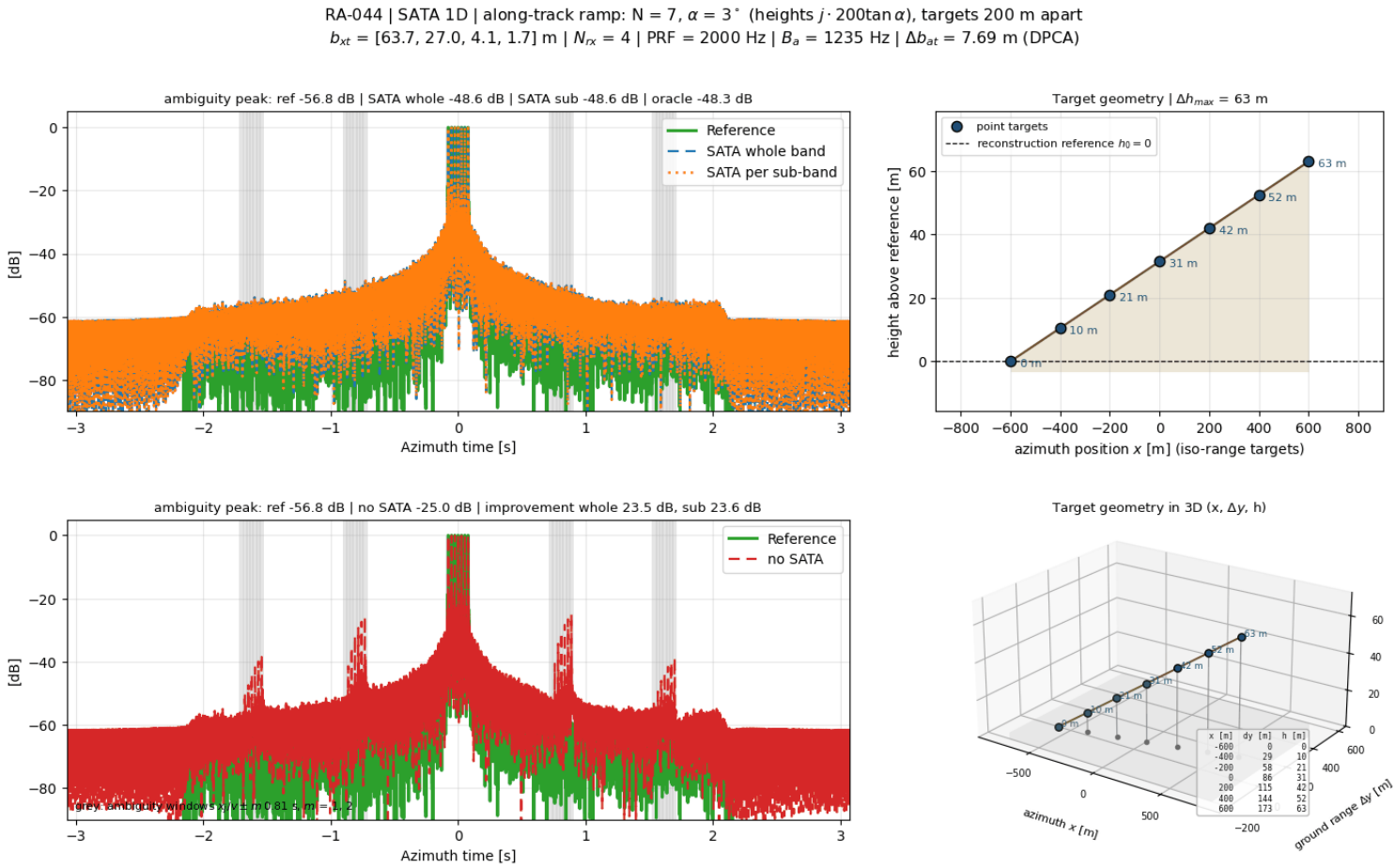


Figure 48: Targets 200 m apart: RA-044, $N = 7$, $\alpha = 3°$. Ambiguity peak: no SATA -25.0, SATA whole -48.6, per sub-band -48.6, oracle -48.3 dB (improvement 23.5 dB).

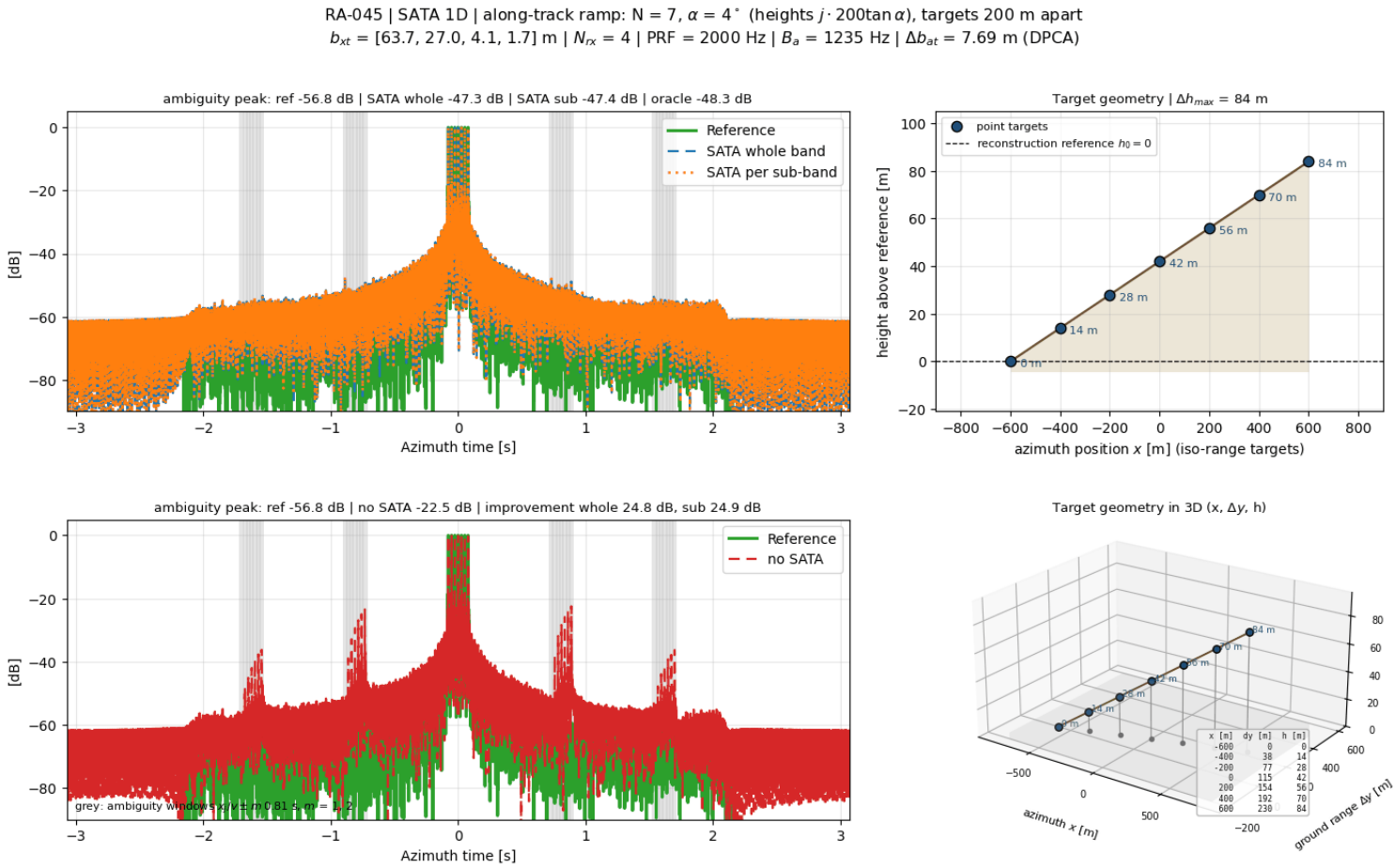


Figure 49: Targets 200 m apart: RA-045, $N = 7$, $\alpha = 4°$. Ambiguity peak: no SATA -22.5, SATA whole -47.3, per sub-band -47.4, oracle -48.3 dB (improvement 24.8 dB).

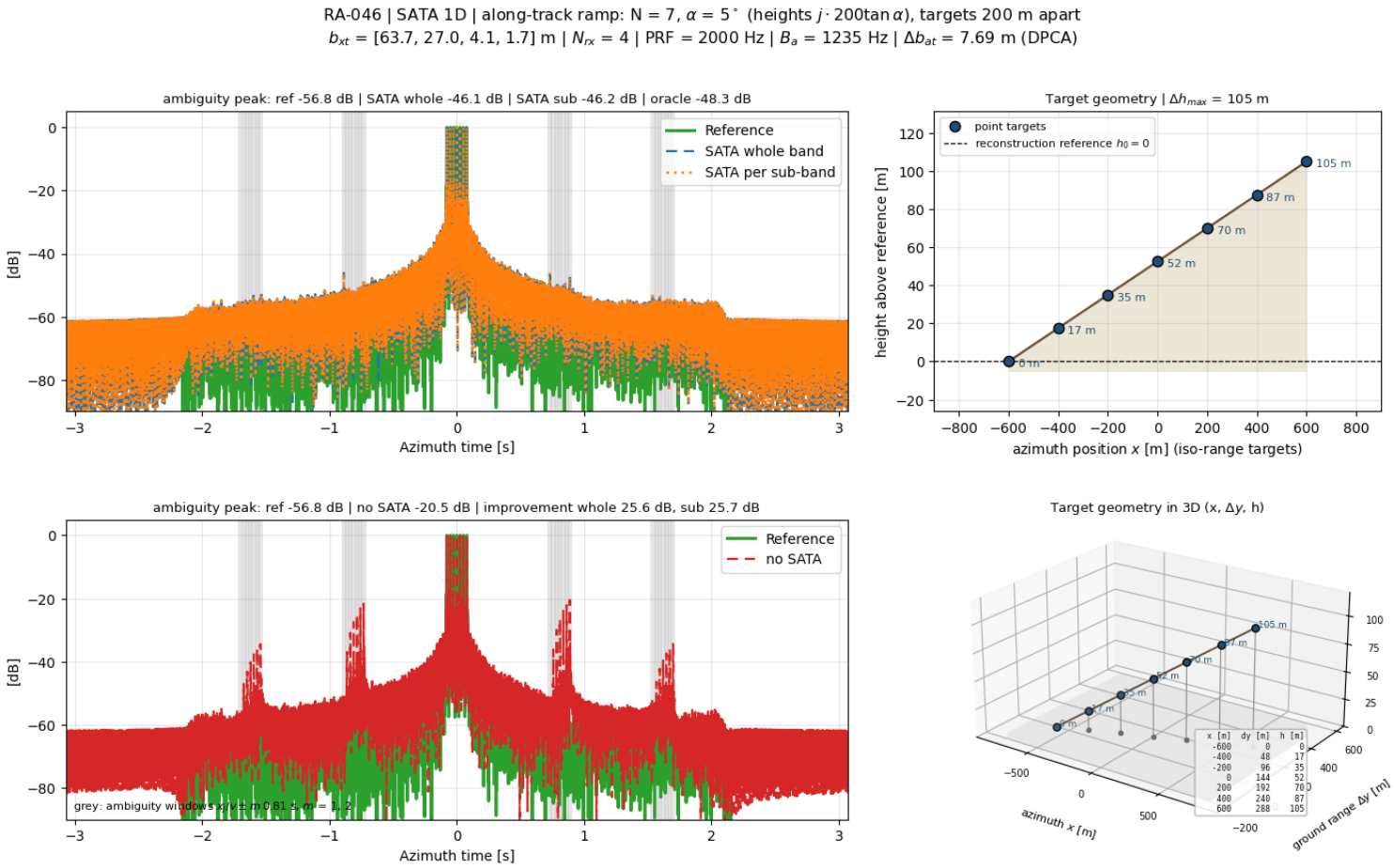


Figure 50: Targets 200 m apart: RA-046, $N = 7$, $\alpha = 5°$. Ambiguity peak: no SATA -20.5, SATA whole -46.1, per sub-band -46.2, oracle -48.3 dB (improvement 25.6 dB).

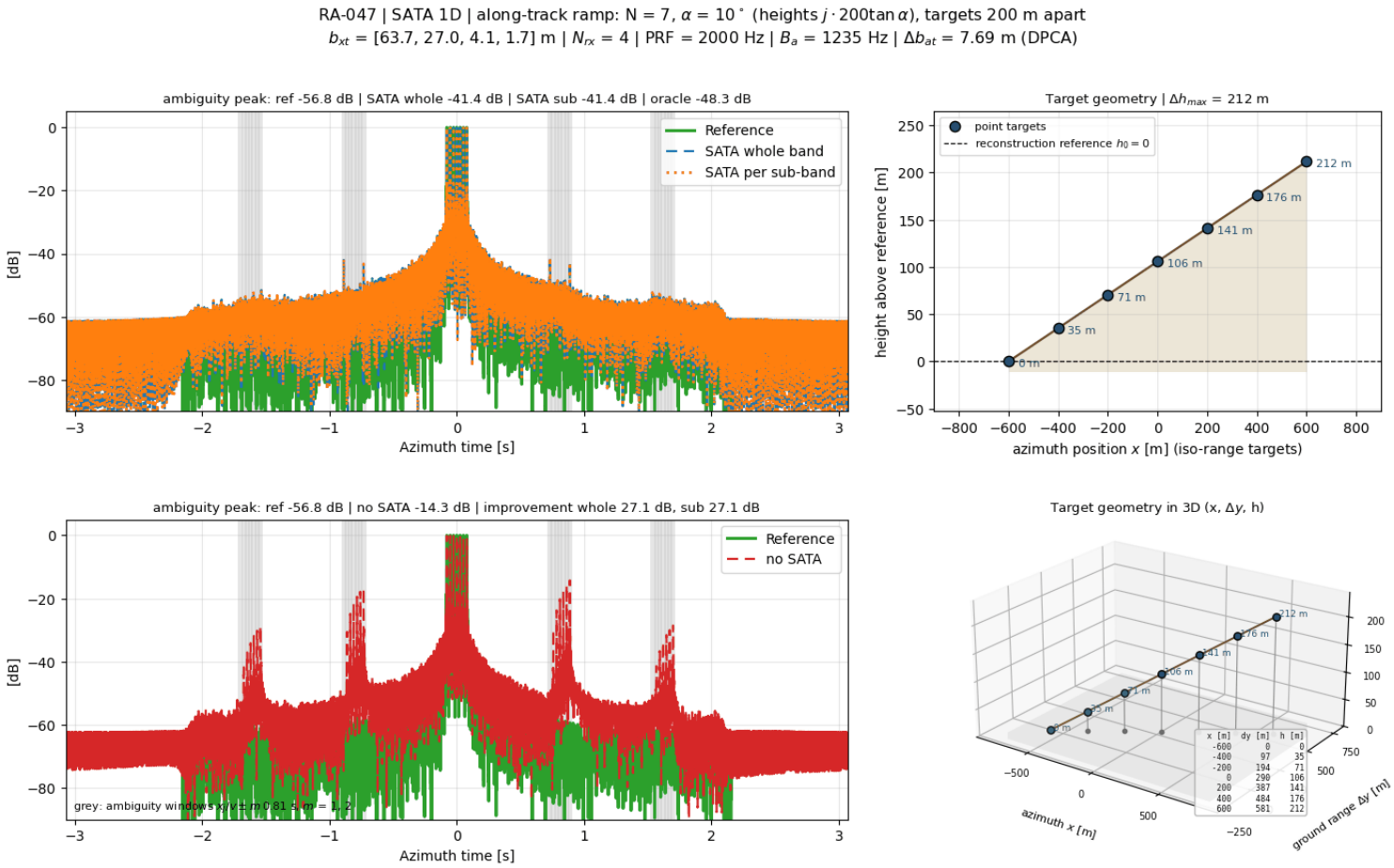


Figure 51: Targets 200 m apart: RA-047, $N = 7$, $\alpha = 10°$. Ambiguity peak: no SATA -14.3, SATA whole -41.4, per sub-band -41.4, oracle -48.3 dB (improvement 27.1 dB).

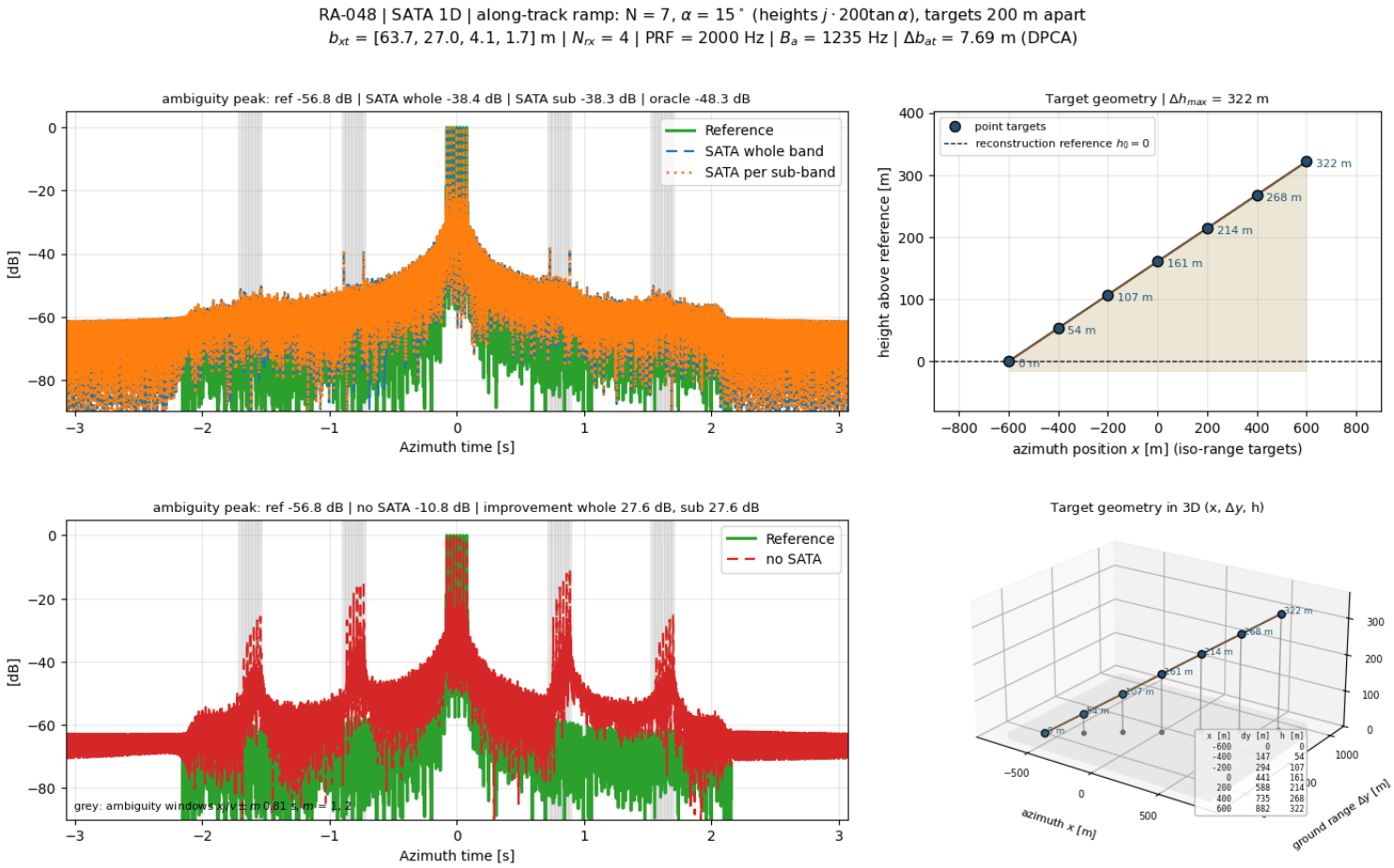


Figure 52: Targets 200 m apart: RA-048, $N = 7$, $\alpha = 15°$. Ambiguity peak: no SATA -10.8, SATA whole -38.4, per sub-band -38.3, oracle -48.3 dB (improvement 27.6 dB).

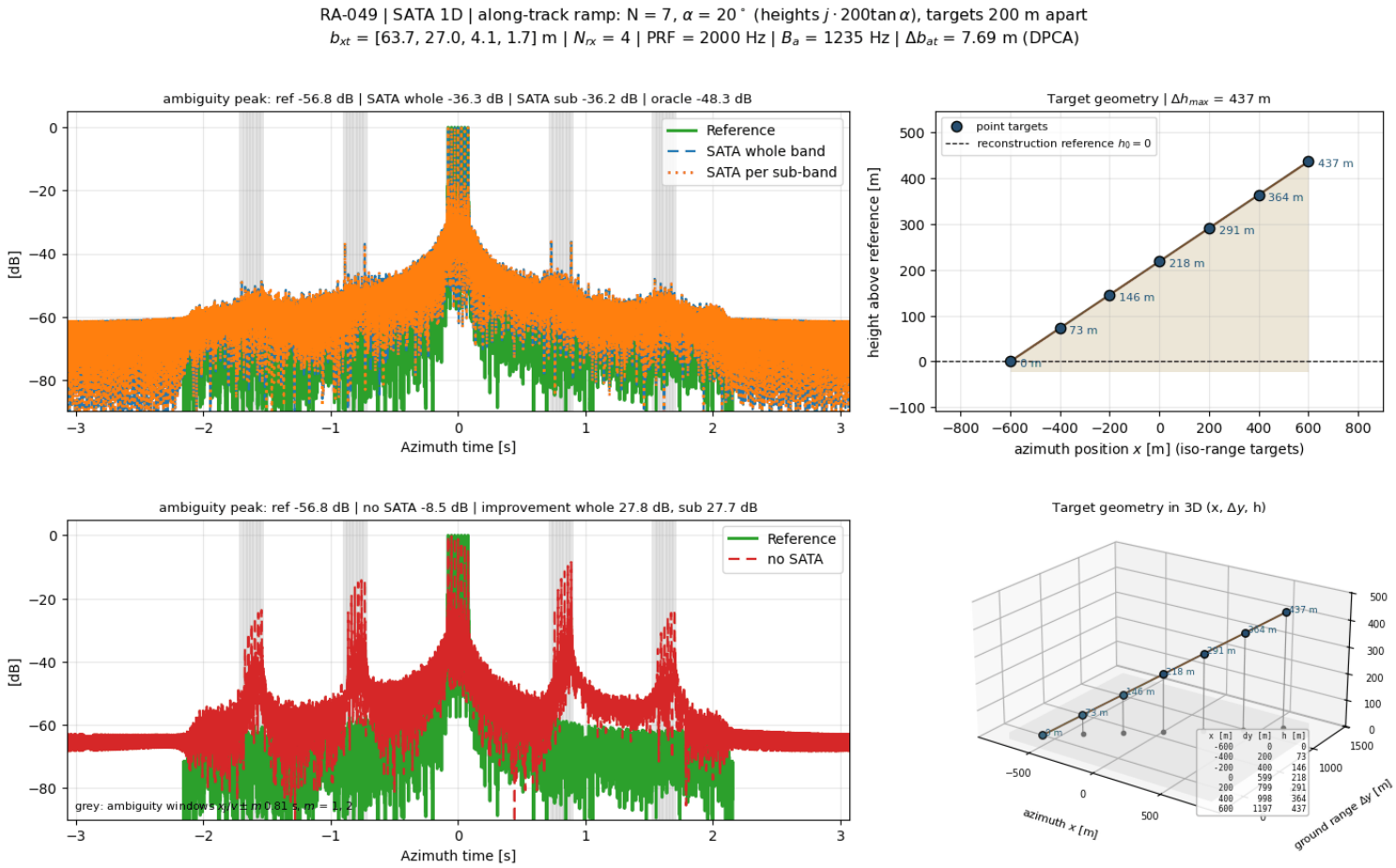


Figure 53: Targets 200 m apart: RA-049, $N = 7$, $\alpha = 20°$. Ambiguity peak: no SATA −8.5, SATA whole −36.3, per sub-band −36.2, oracle −48.3 dB (improvement 27.8 dB).

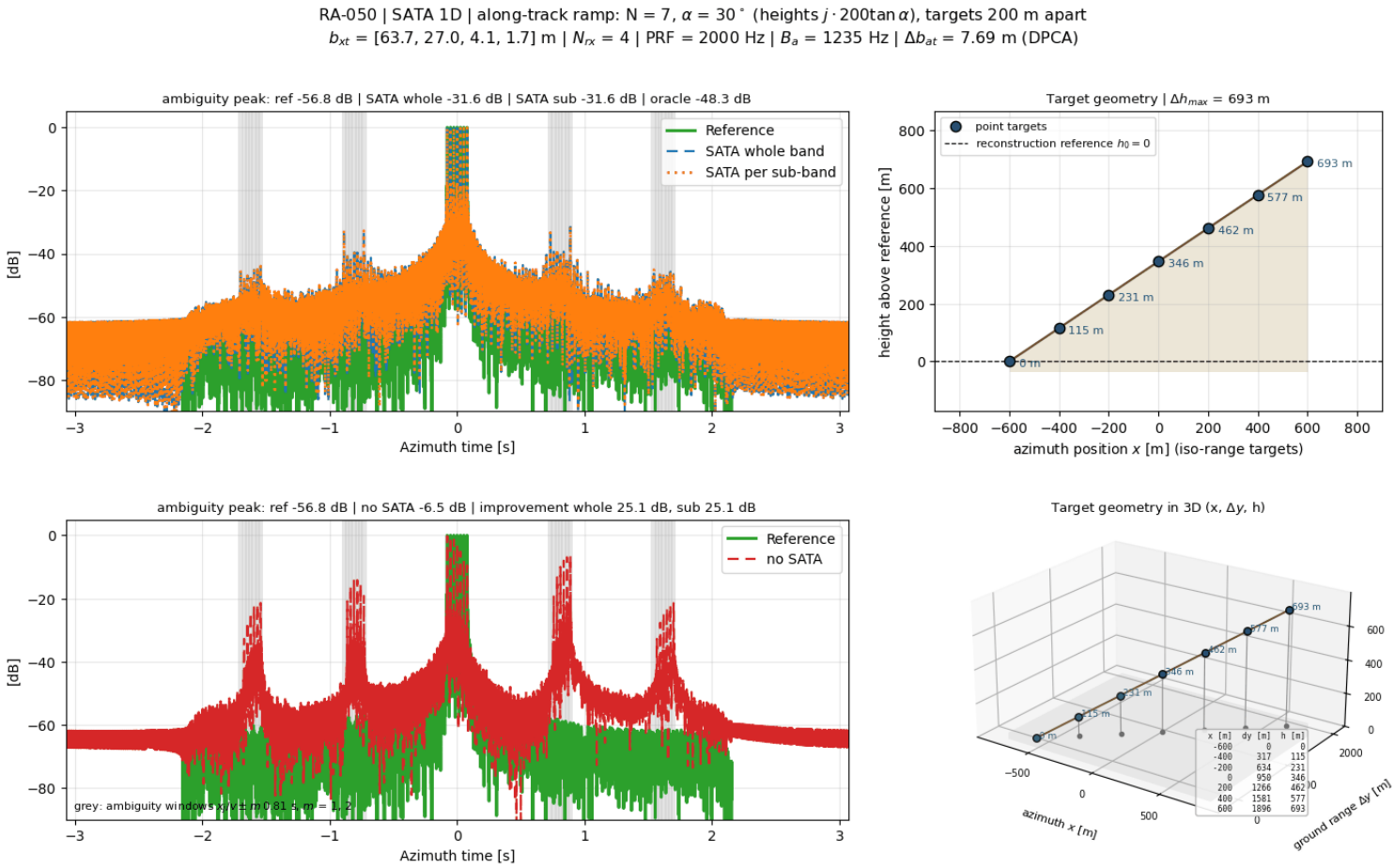


Figure 54: Targets 200 m apart: RA-050, $N = 7$, $\alpha = 30°$. Ambiguity peak: no SATA -6.5, SATA whole -31.6, per sub-band -31.6, oracle -48.3 dB (improvement 25.1 dB).

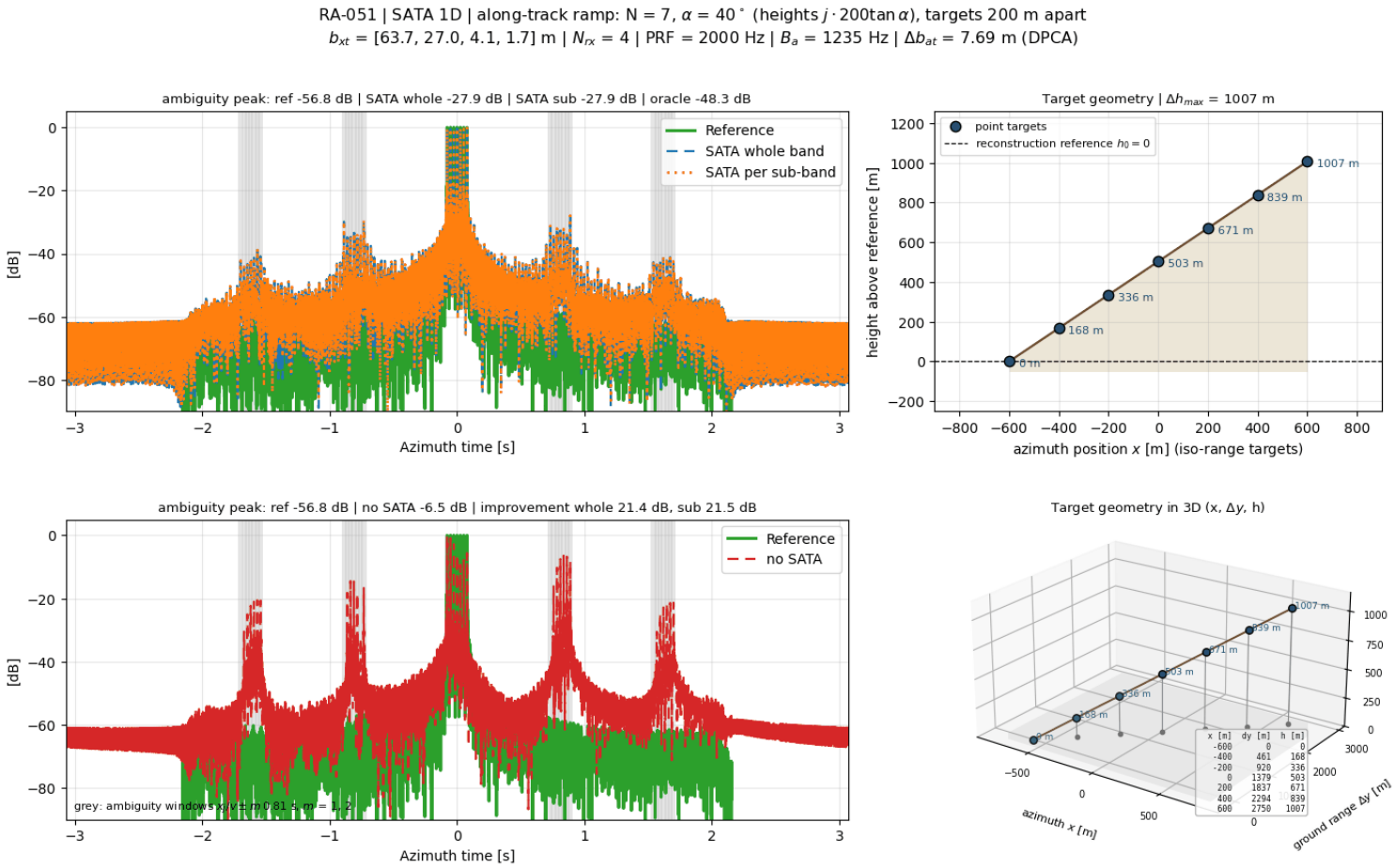


Figure 55: Targets 200 m apart: RA-051, $N = 7$, $\alpha = 40°$. Ambiguity peak: no SATA -6.5, SATA whole -27.9, per sub-band -27.9, oracle -48.3 dB (improvement 21.4 dB).

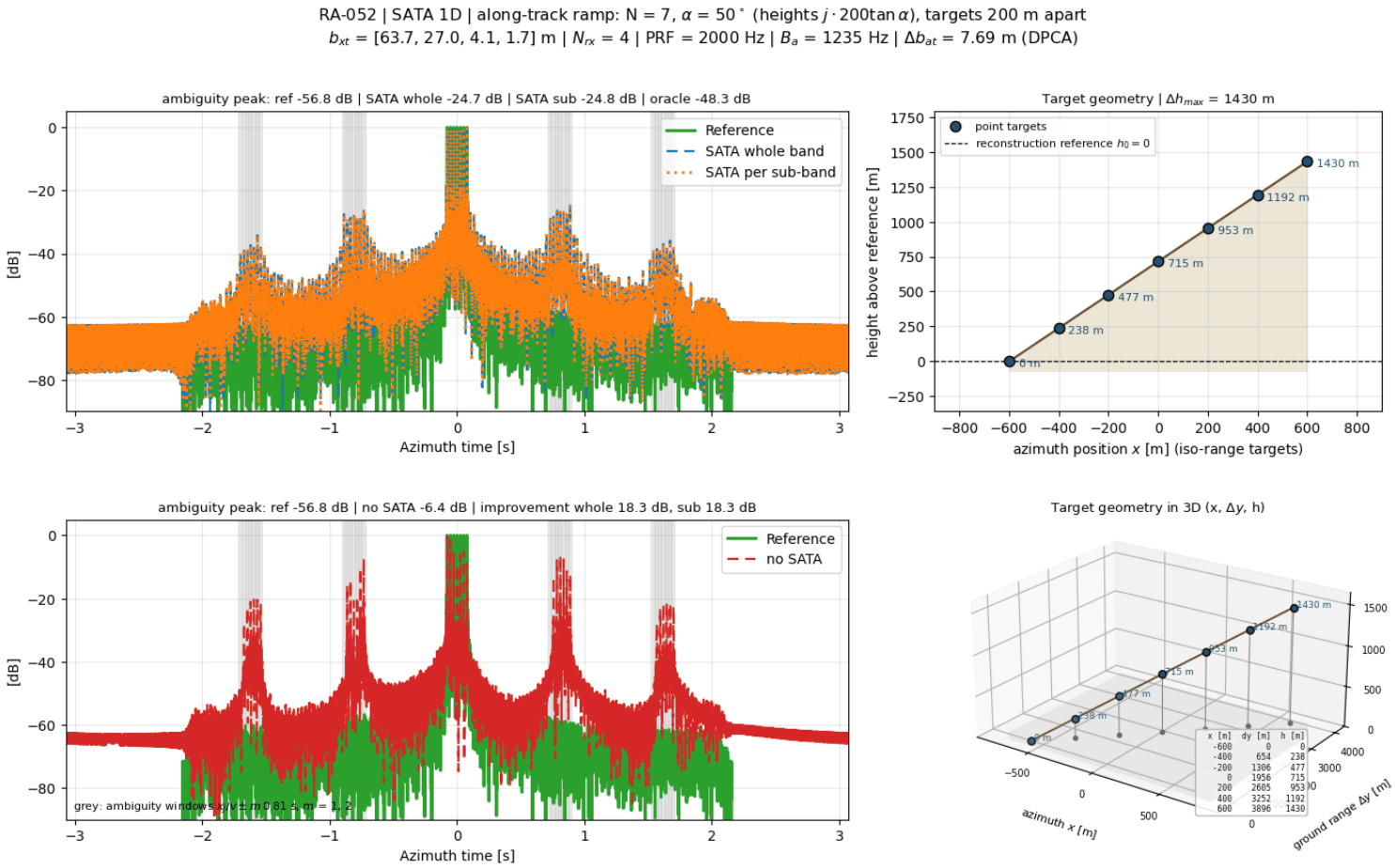


Figure 56: Targets 200 m apart: RA-052, $N = 7$, $\alpha = 50°$. Ambiguity peak: no SATA -6.4, SATA whole -24.7, per sub-band -24.8, oracle -48.3 dB (improvement 18.3 dB).

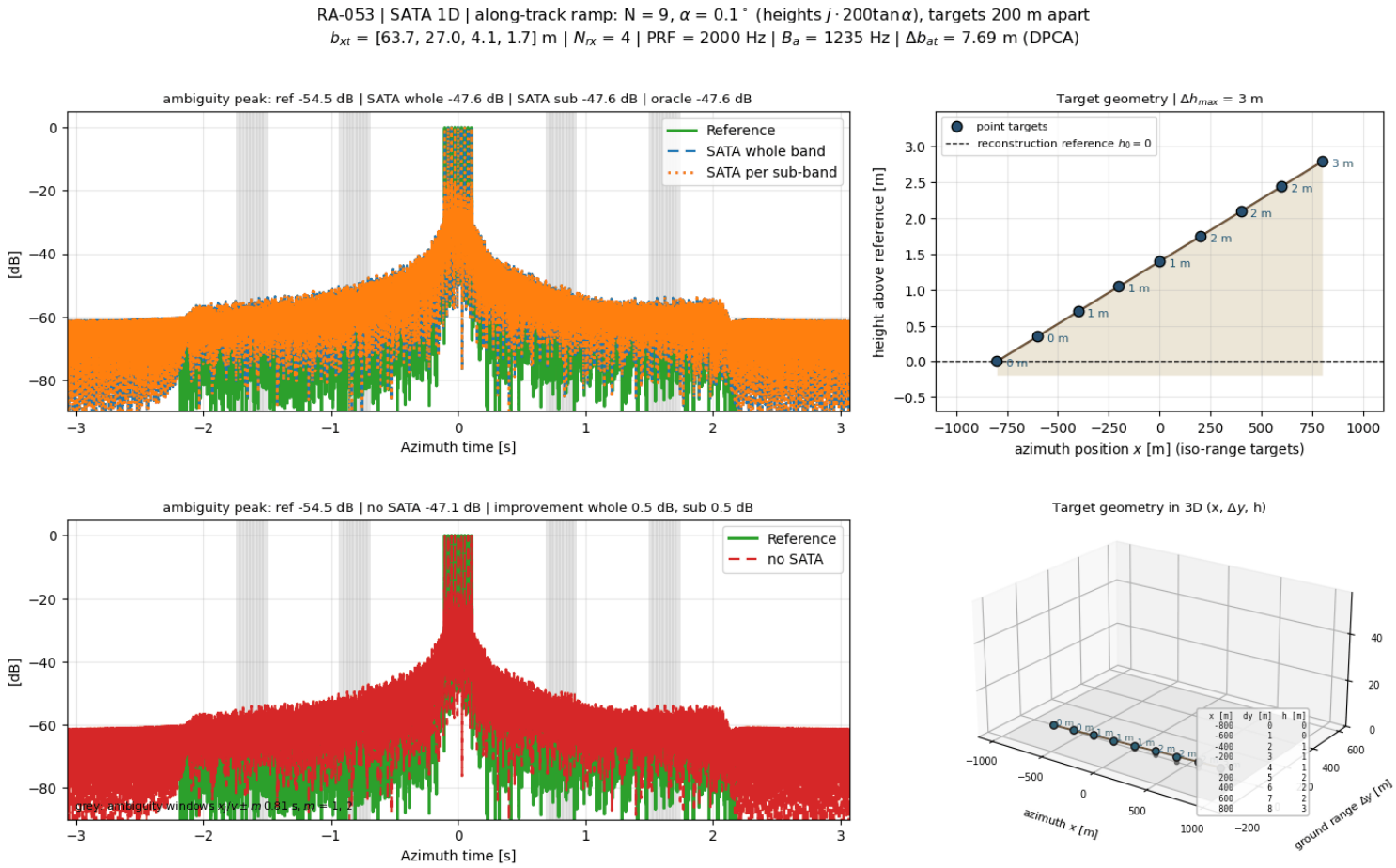


Figure 57: Targets 200 m apart: RA-053, $N = 9$, $\alpha = 0.1°$. Ambiguity peak: no SATA −47.1, SATA whole −47.6, per sub-band −47.6, oracle −47.6 dB (improvement 0.5 dB).

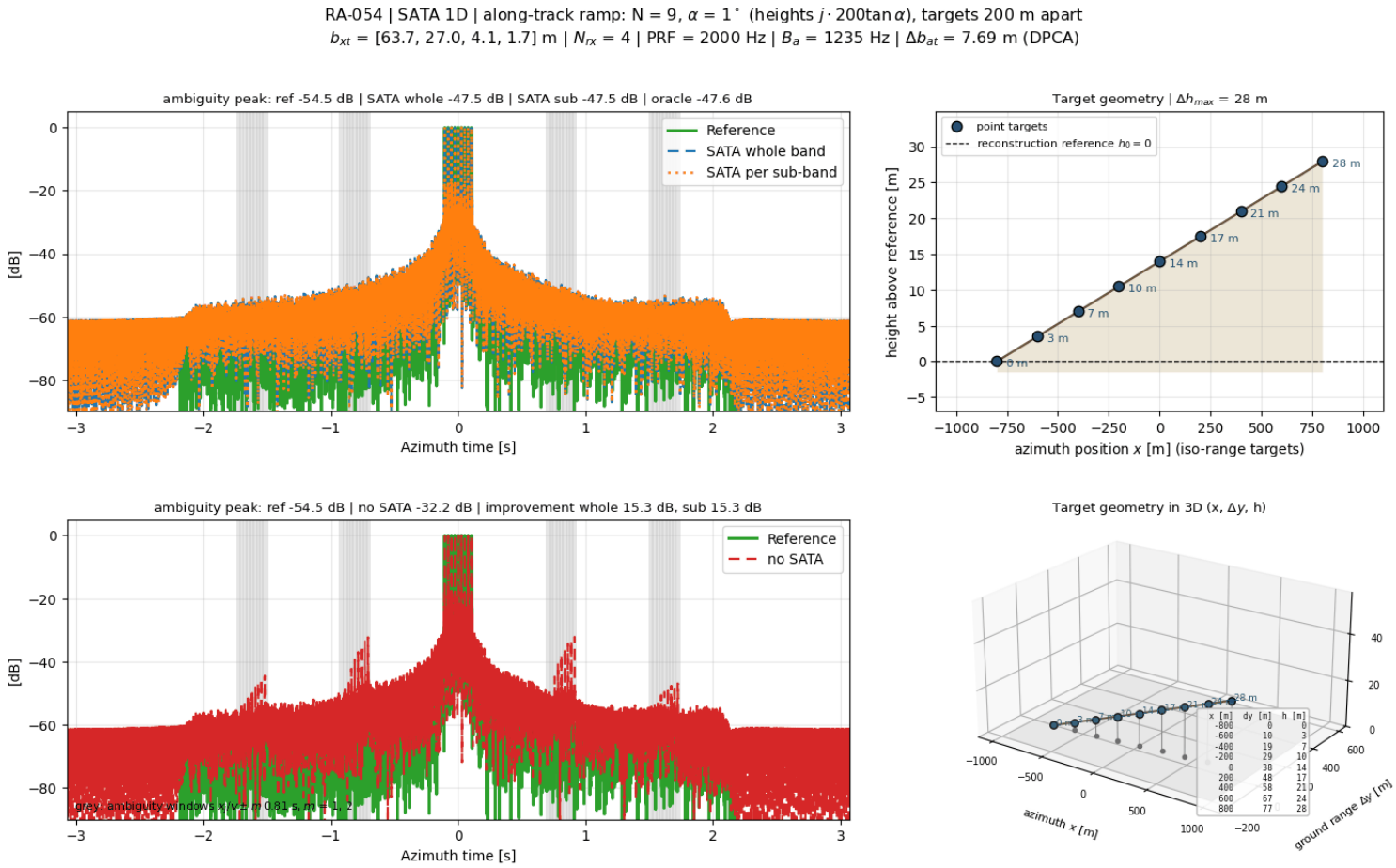


Figure 58: Targets 200 m apart: RA-054, $N = 9$, $\alpha = 1°$. Ambiguity peak: no SATA -32.2, SATA whole -47.5, per sub-band -47.5, oracle -47.6 dB (improvement 15.3 dB).

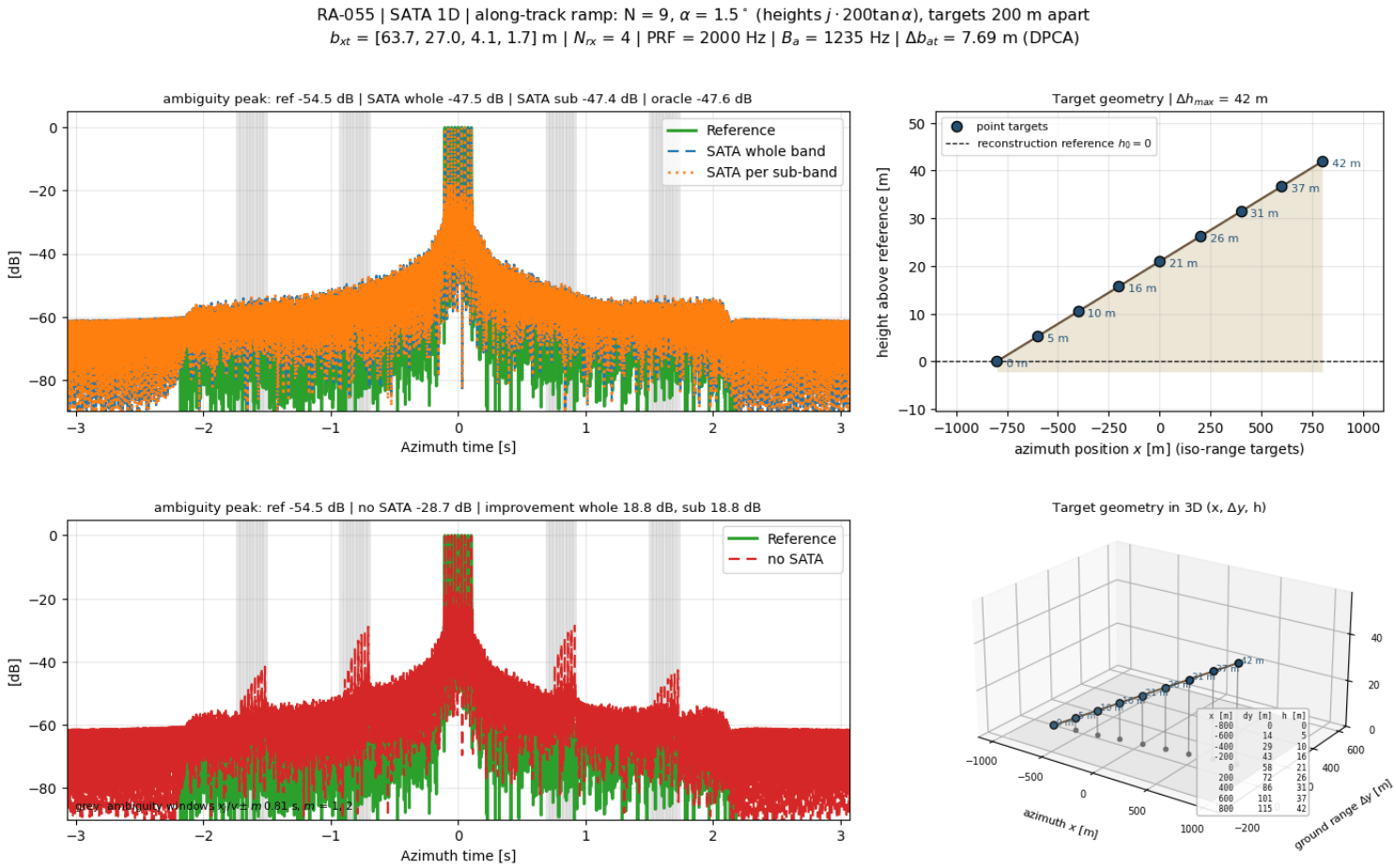


Figure 59: Targets 200 m apart: RA-055, $N = 9$, $\alpha = 1.5°$. Ambiguity peak: no SATA -28.7, SATA whole -47.5, per sub-band -47.4, oracle -47.6 dB (improvement 18.8 dB).

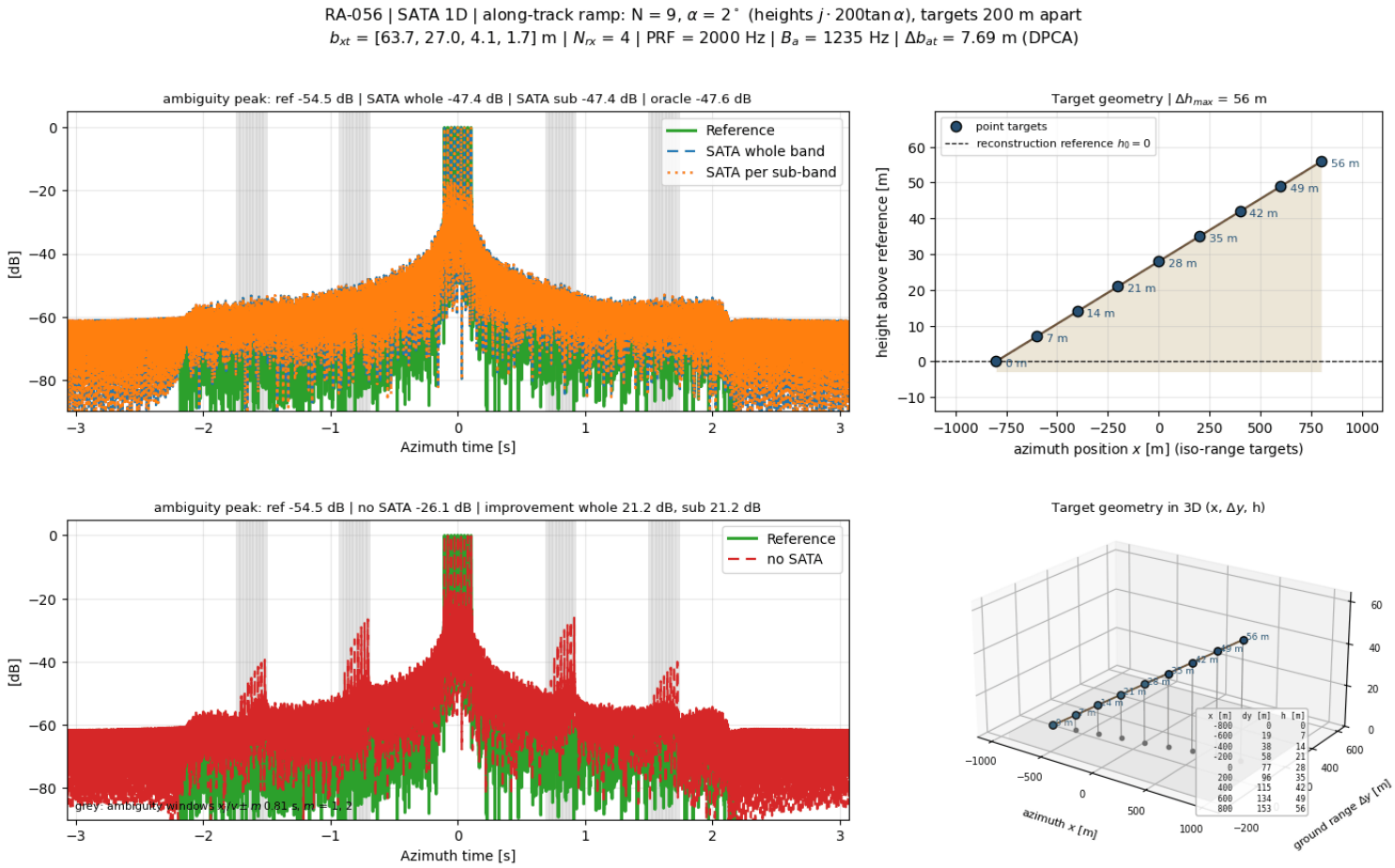


Figure 60: Targets 200 m apart: RA-056, $N = 9$, $\alpha = 2°$. Ambiguity peak: no SATA -26.1, SATA whole -47.4, per sub-band -47.4, oracle -47.6 dB (improvement 21.2 dB).

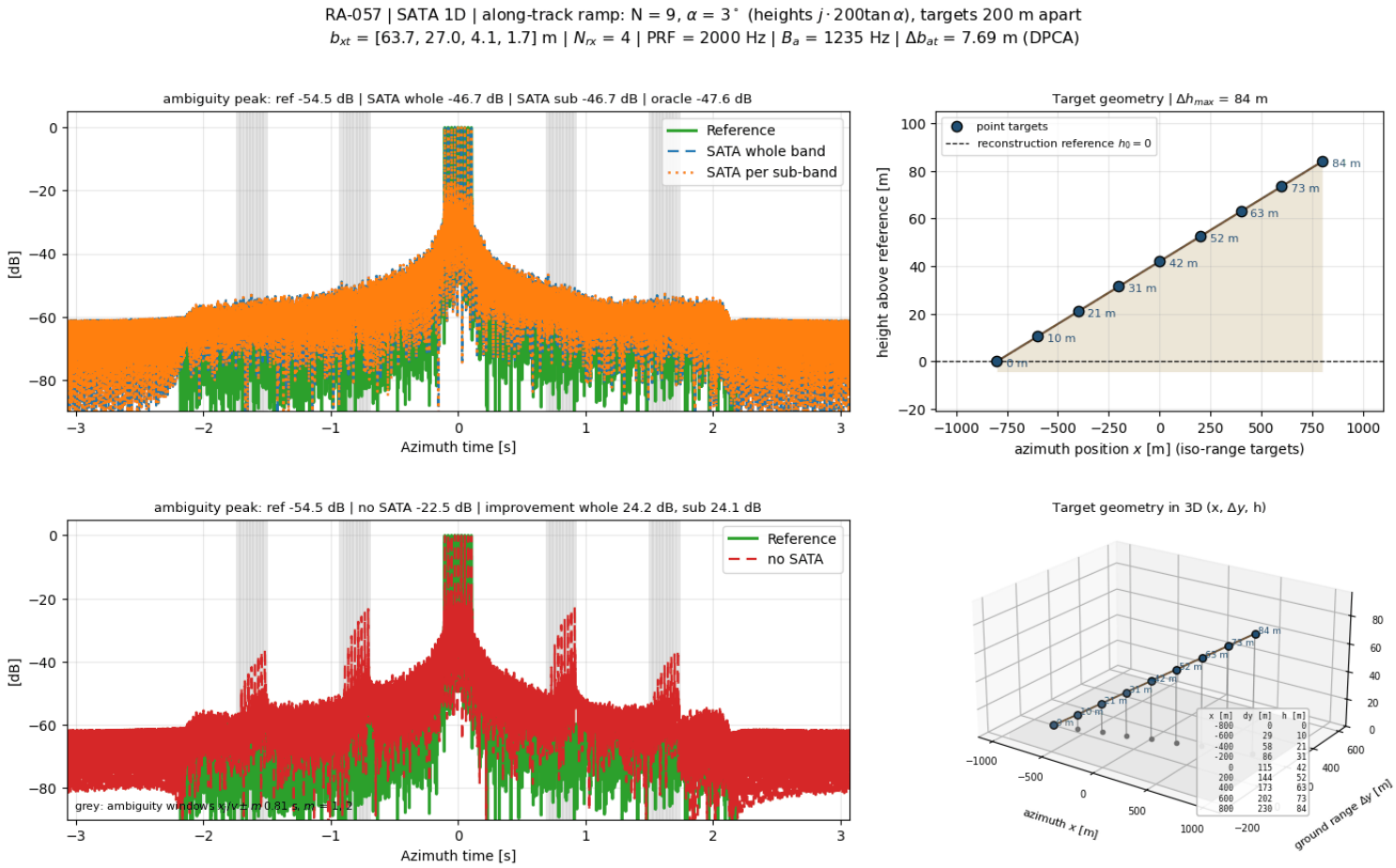


Figure 61: Targets 200 m apart: RA-057, $N = 9$, $\alpha = 3°$. Ambiguity peak: no SATA -22.5, SATA whole -46.7, per sub-band -46.7, oracle -47.6 dB (improvement 24.2 dB).

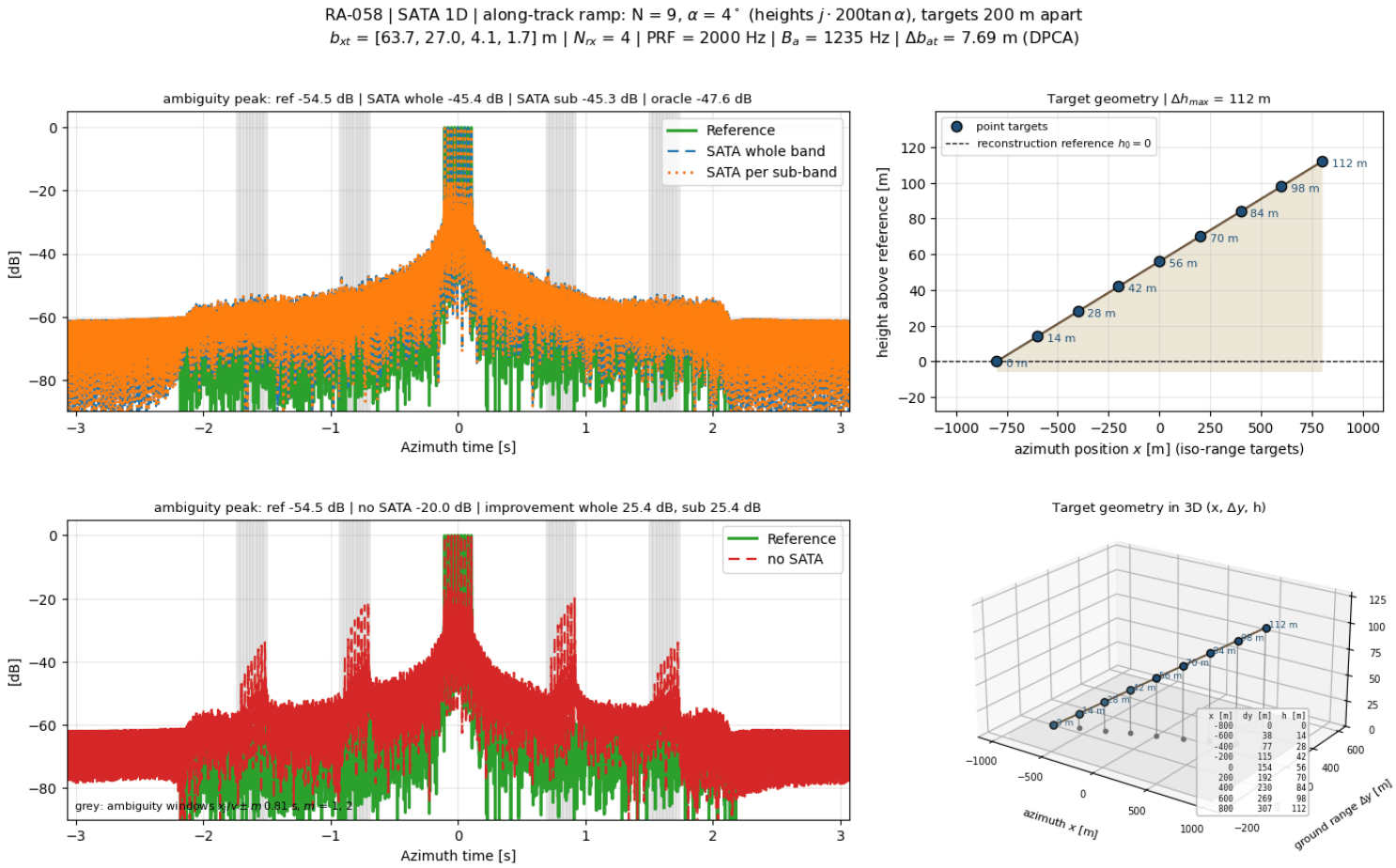


Figure 62: Targets 200 m apart: RA-058, $N = 9$, $\alpha = 4°$. Ambiguity peak: no SATA -20.0, SATA whole -45.4, per sub-band -45.3, oracle -47.6 dB (improvement 25.4 dB).

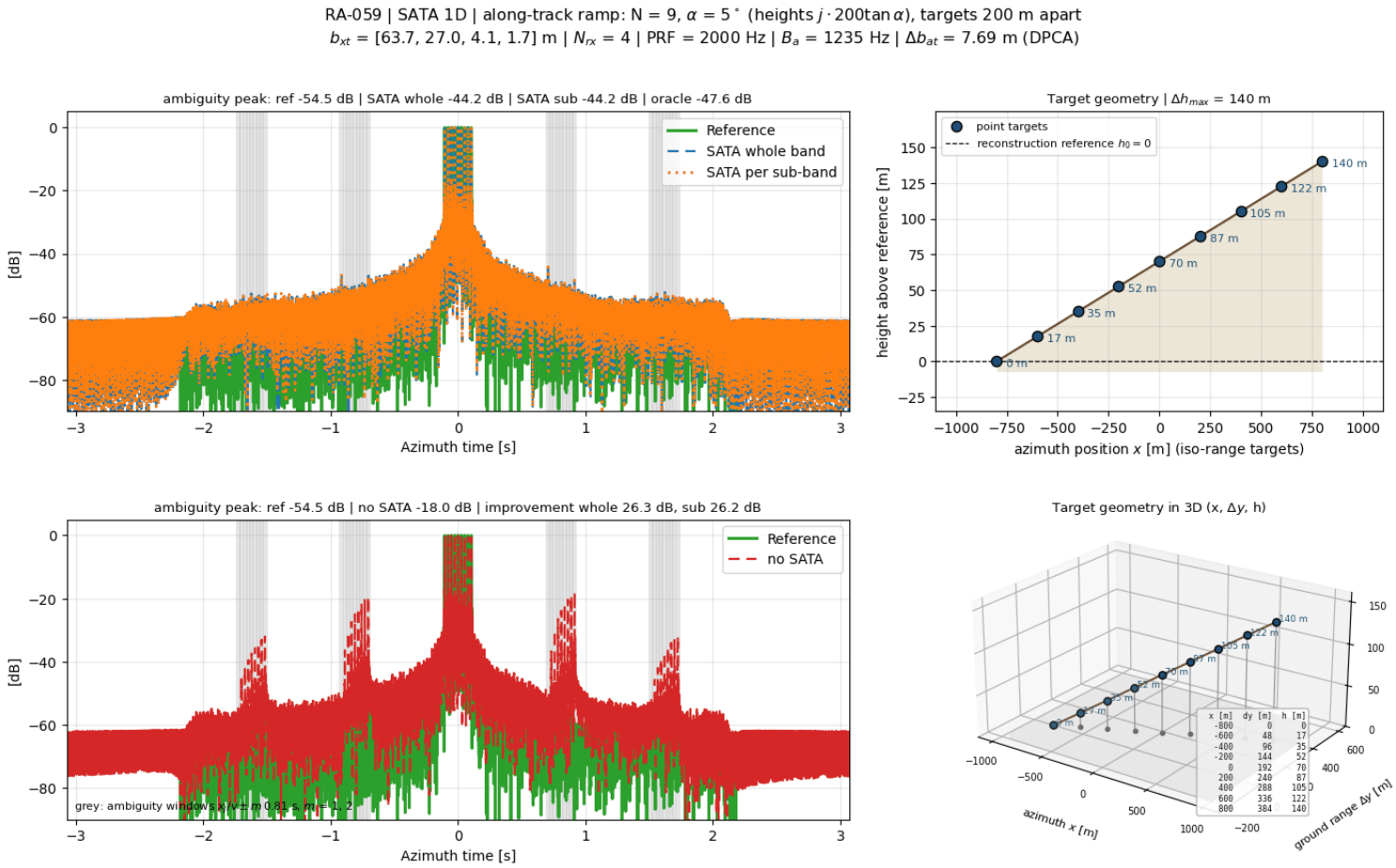


Figure 63: Targets 200 m apart: RA-059, $N = 9$, $\alpha = 5°$. Ambiguity peak: no SATA -18.0, SATA whole -44.2, per sub-band -44.2, oracle -47.6 dB (improvement 26.3 dB).

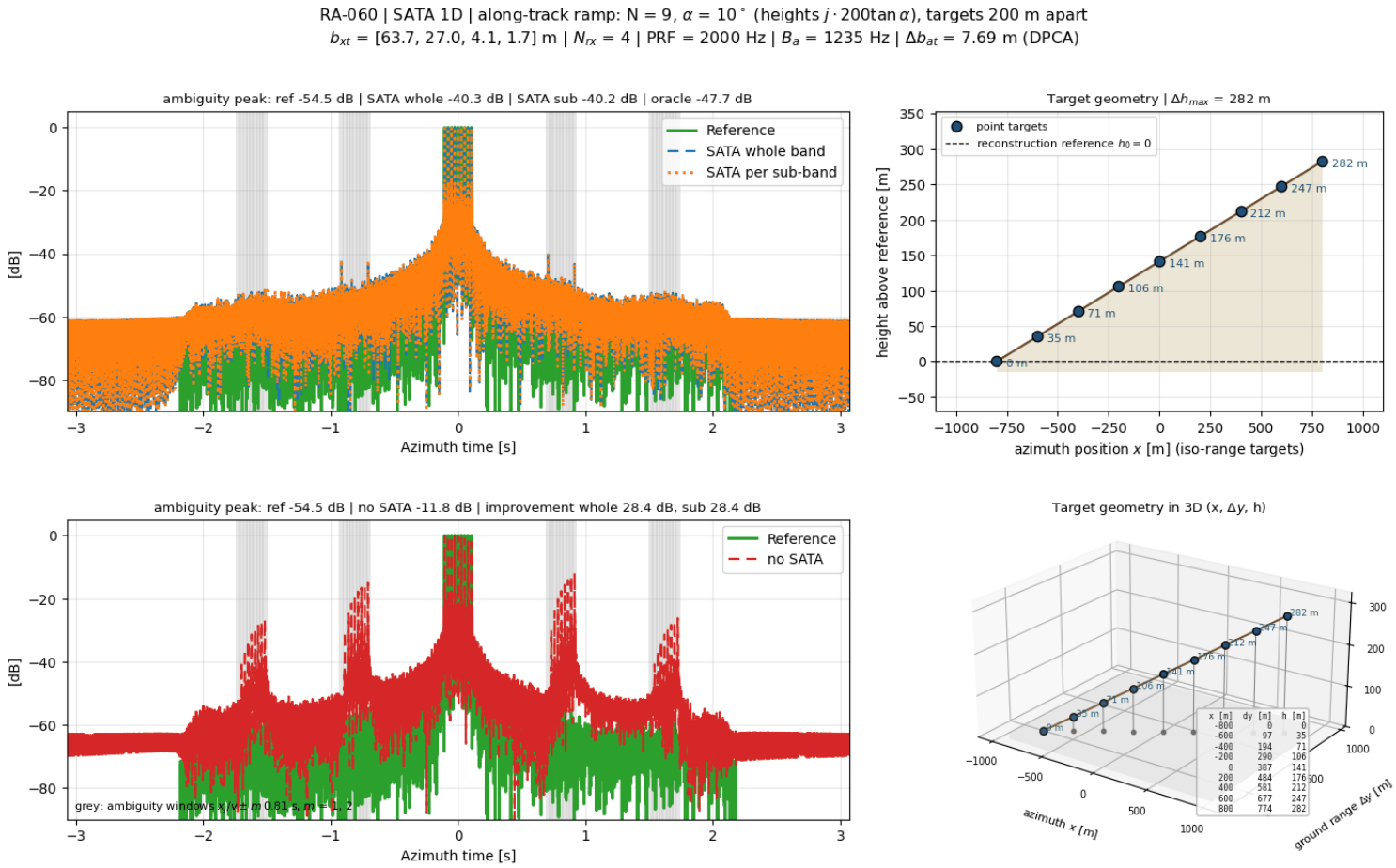


Figure 64: Targets 200 m apart: RA-060, $N = 9$, $\alpha = 10°$. Ambiguity peak: no SATA -11.8, SATA whole -40.3, per sub-band -40.2, oracle -47.7 dB (improvement 28.4 dB).

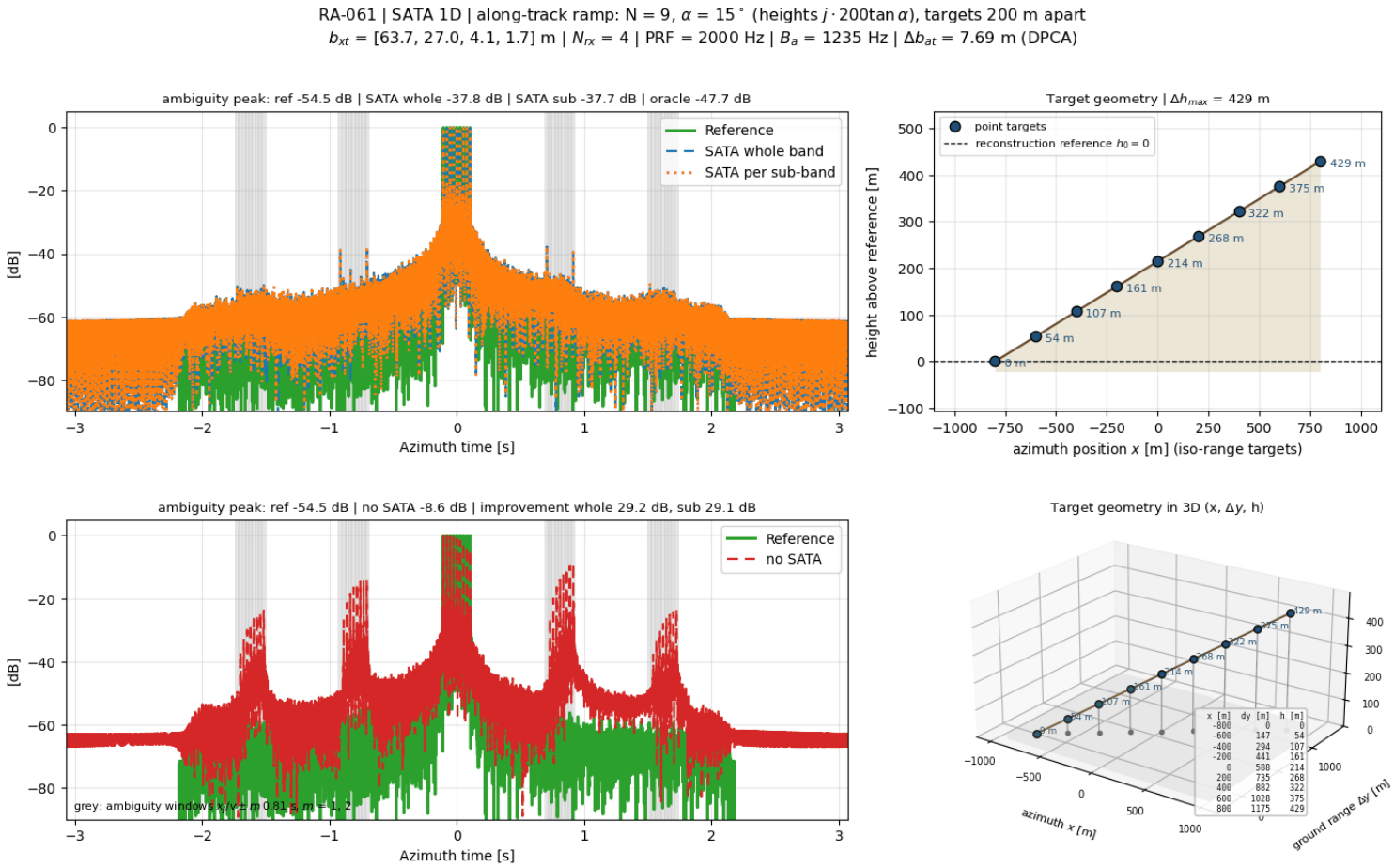


Figure 65: Targets 200 m apart: RA-061, $N = 9$, $\alpha = 15°$. Ambiguity peak: no SATA -8.6, SATA whole -37.8, per sub-band -37.7, oracle -47.7 dB (improvement 29.2 dB).

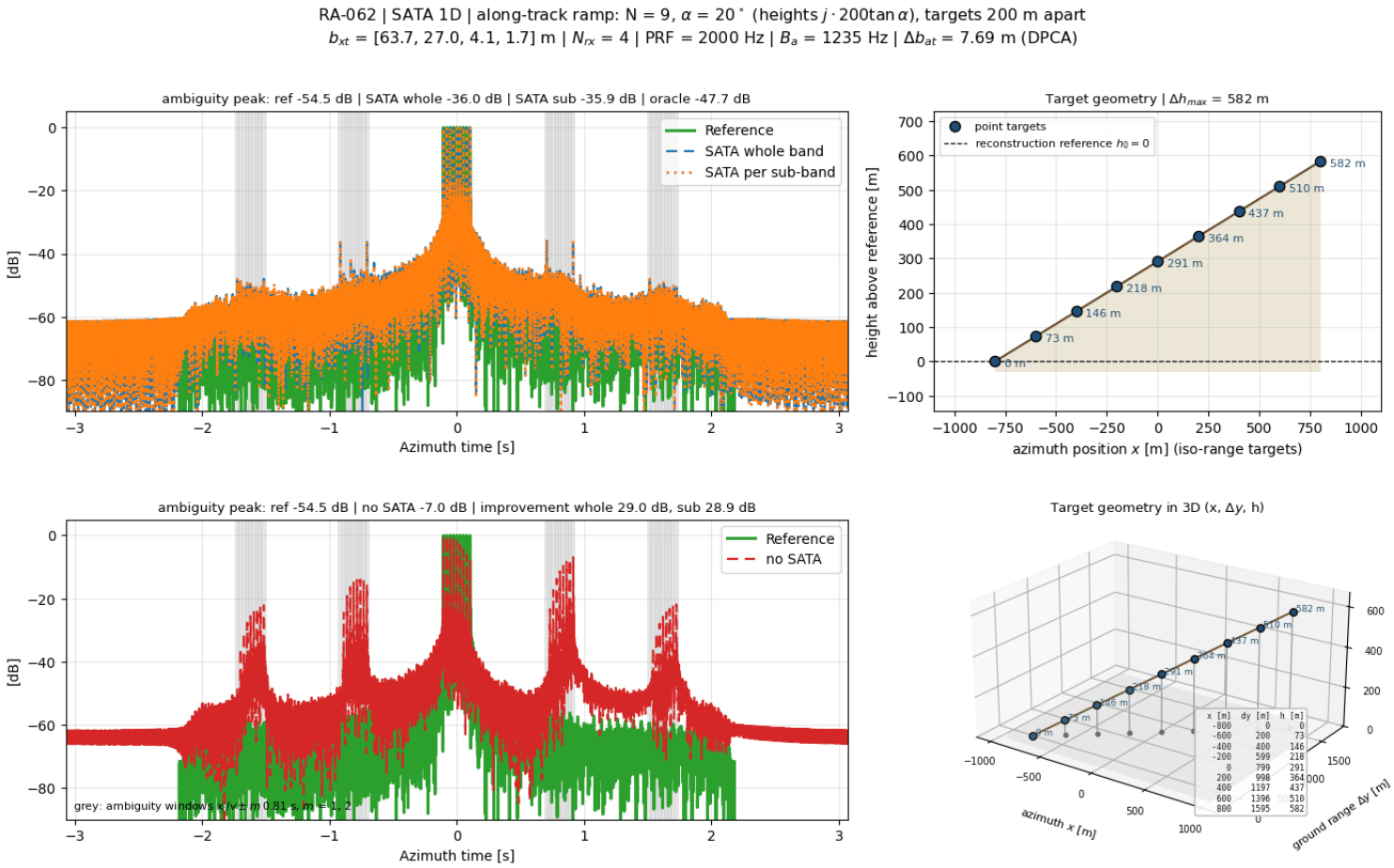


Figure 66: Targets 200 m apart: RA-062, $N = 9$, $\alpha = 20°$. Ambiguity peak: no SATA -7.0, SATA whole -36.0, per sub-band -35.9, oracle -47.7 dB (improvement 29.0 dB).

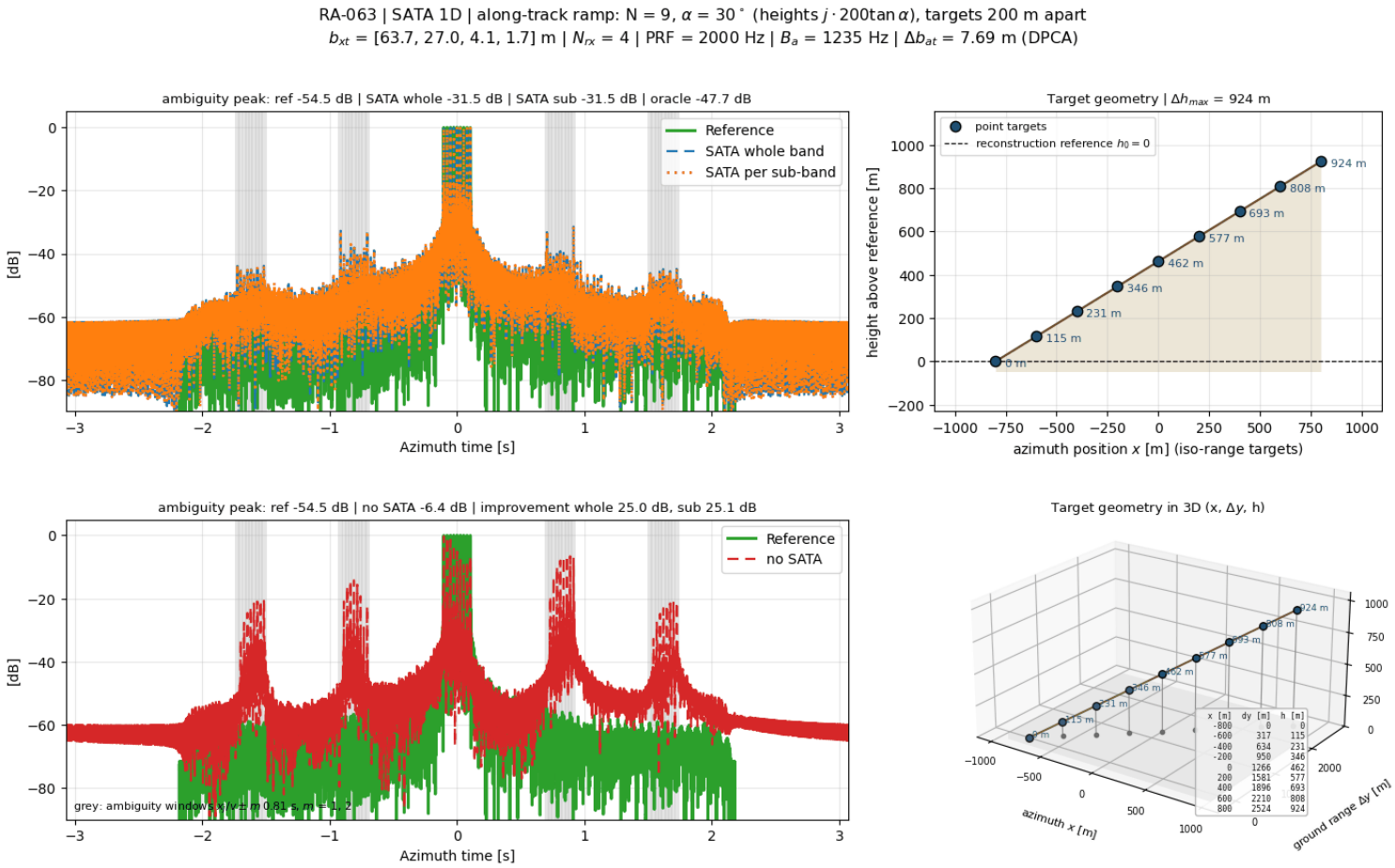


Figure 67: Targets 200 m apart: RA-063, $N = 9$, $\alpha = 30°$. Ambiguity peak: no SATA -6.4, SATA whole -31.5, per sub-band -31.5, oracle -47.7 dB (improvement 25.0 dB).

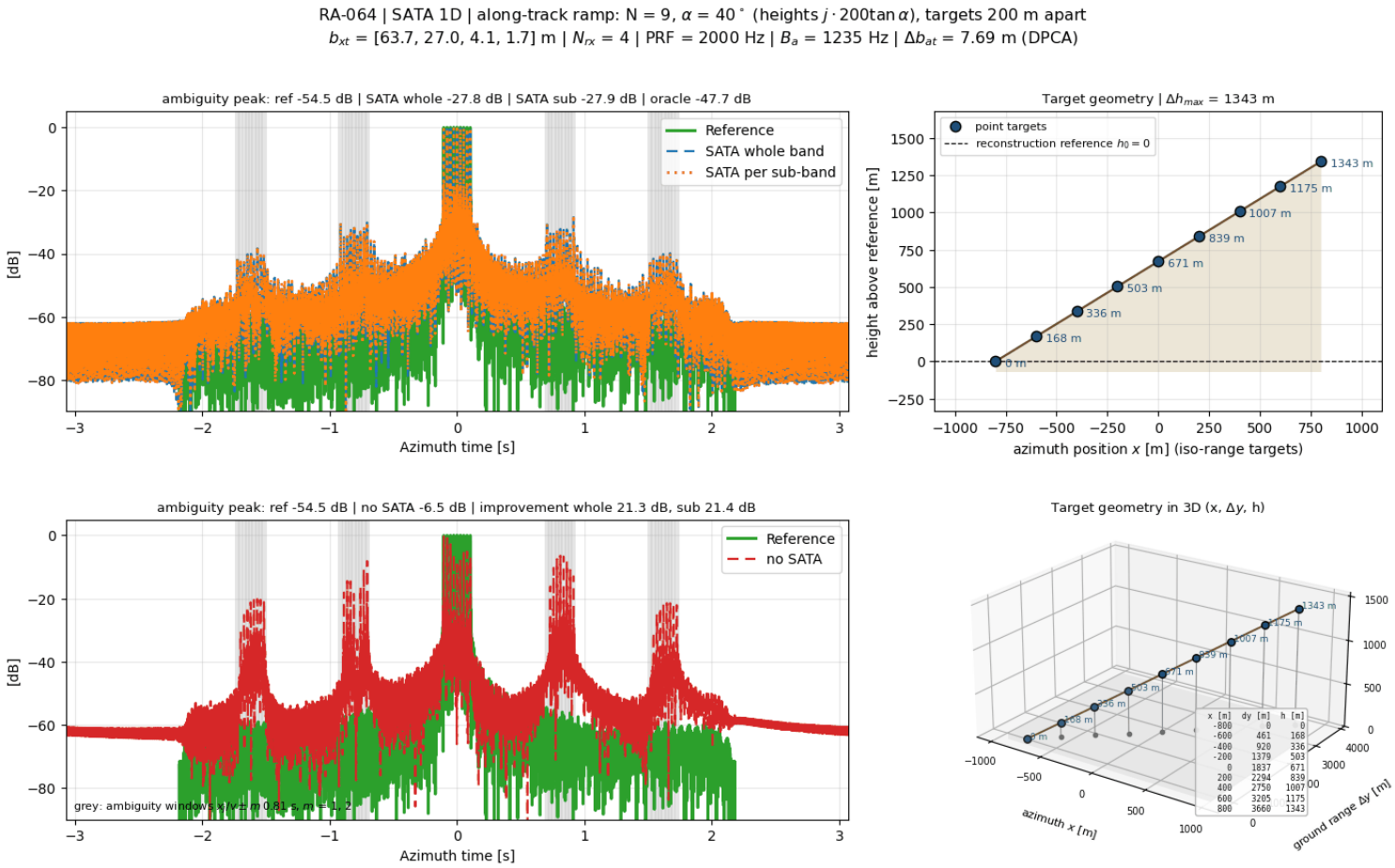


Figure 68: Targets 200 m apart: RA-064, $N = 9$, $\alpha = 40°$. Ambiguity peak: no SATA -6.5, SATA whole -27.8, per sub-band -27.9, oracle -47.7 dB (improvement 21.3 dB).

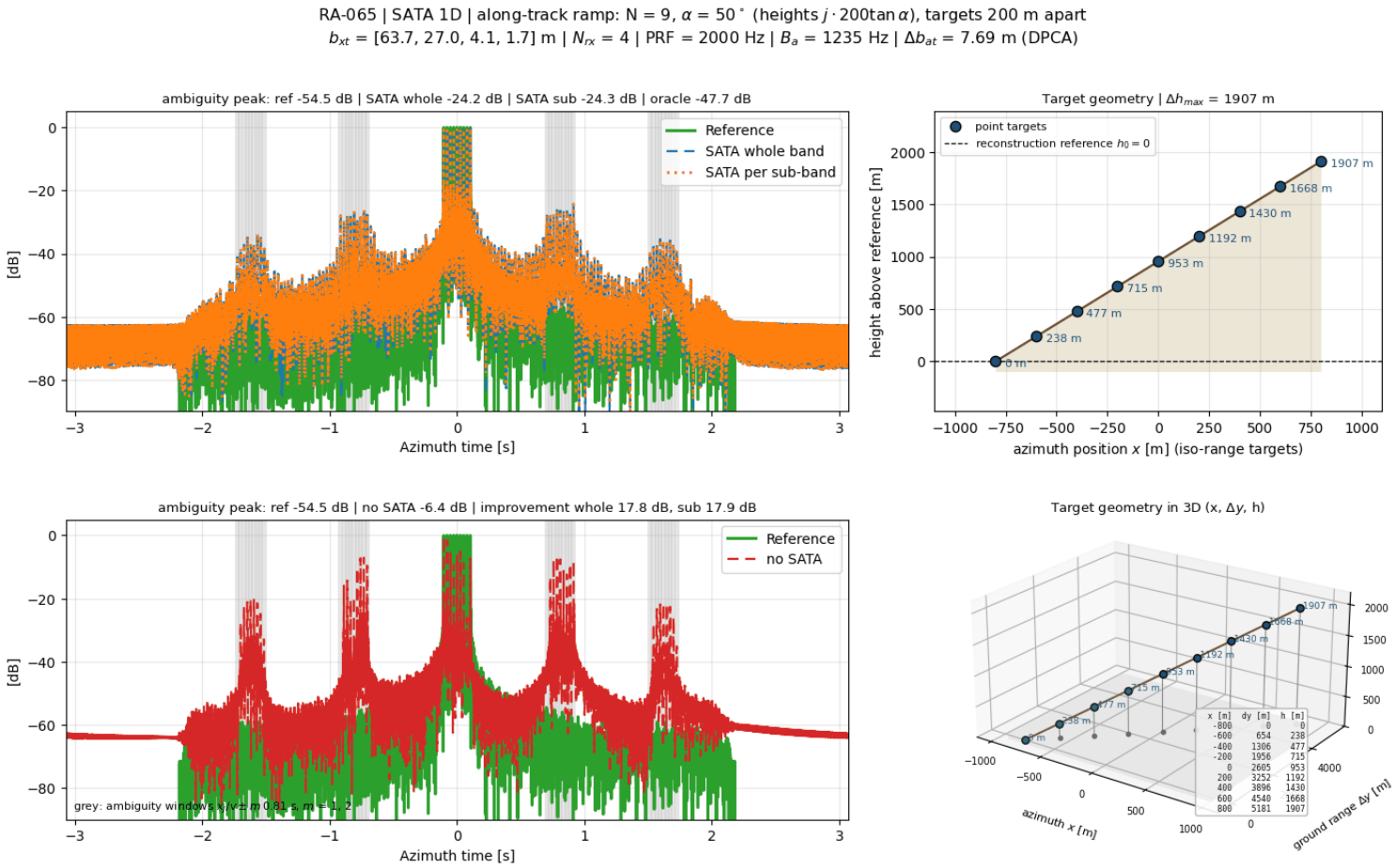


Figure 69: Targets 200 m apart: RA-065, $N = 9$, $\alpha = 50°$. Ambiguity peak: no SATA -6.4, SATA whole -24.2, per sub-band -24.3, oracle -47.7 dB (improvement 17.8 dB).

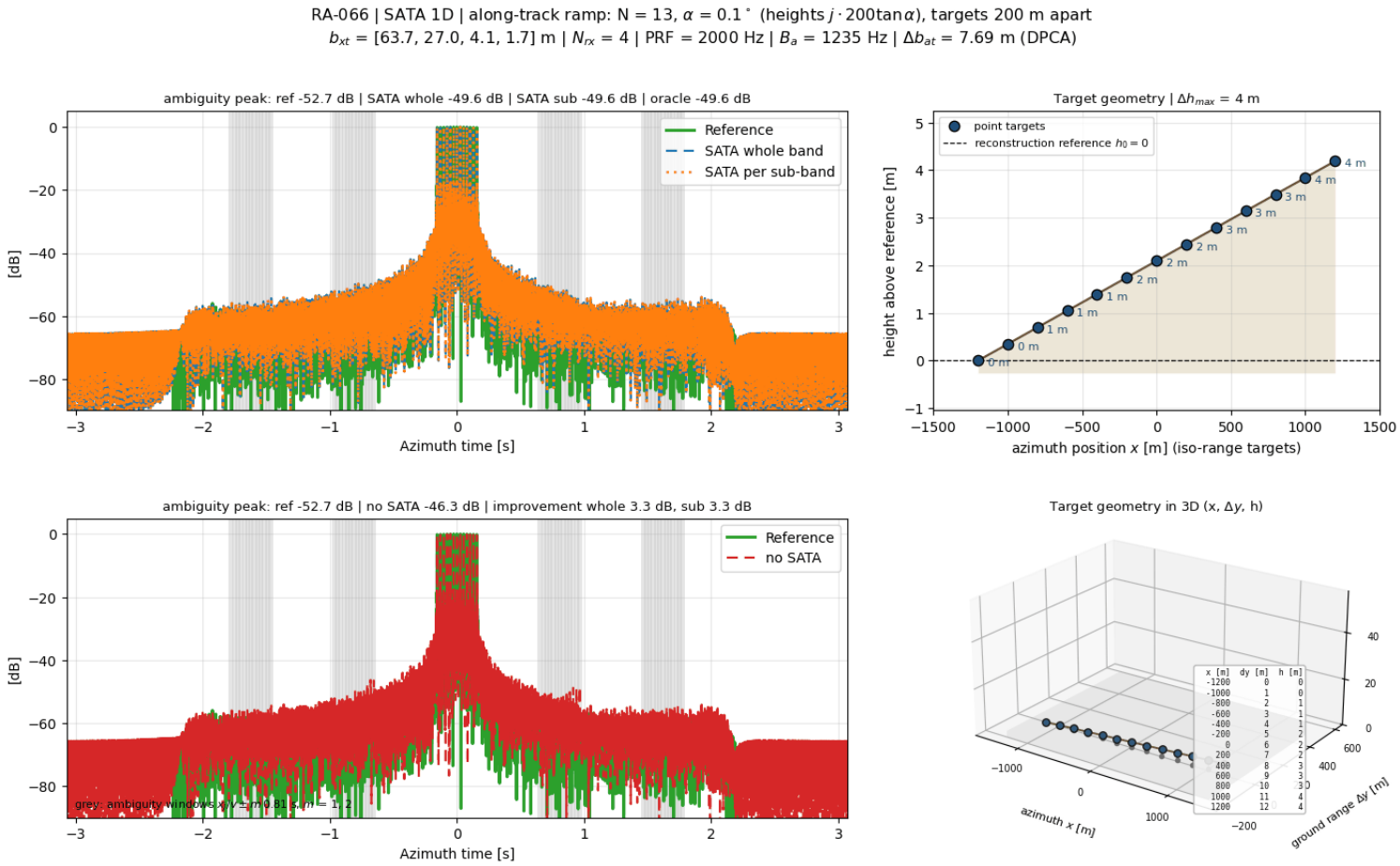


Figure 70: Targets 200 m apart: RA-066, $N = 13$, $\alpha = 0.1°$. Ambiguity peak: no SATA -46.3, SATA whole -49.6, per sub-band -49.6, oracle -49.6 dB (improvement 3.3 dB).

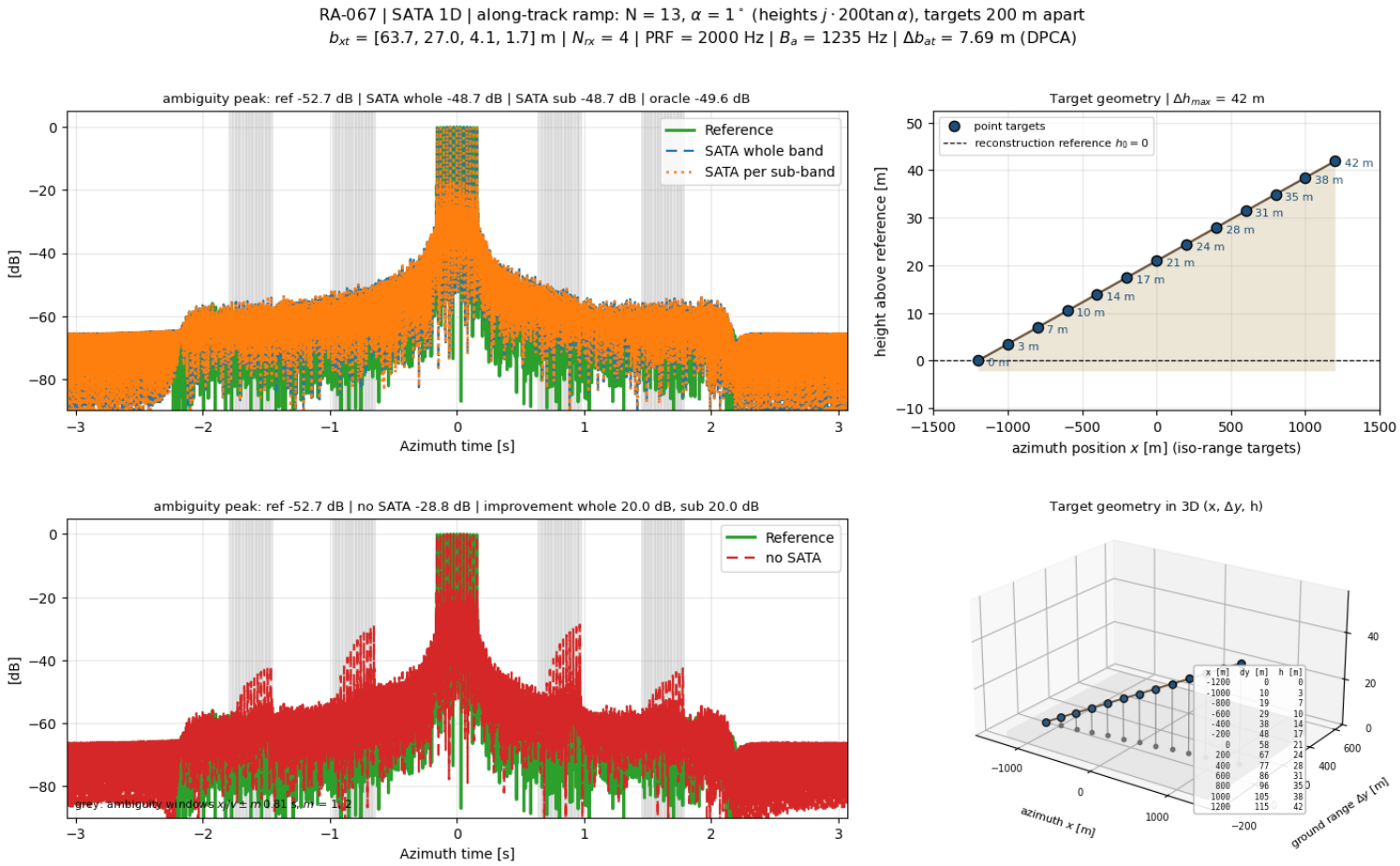


Figure 71: Targets 200 m apart: RA-067, $N = 13$, $\alpha = 1°$. Ambiguity peak: no SATA −28.8, SATA whole −48.7, per sub-band −48.7, oracle −49.6 dB (improvement 20.0 dB).

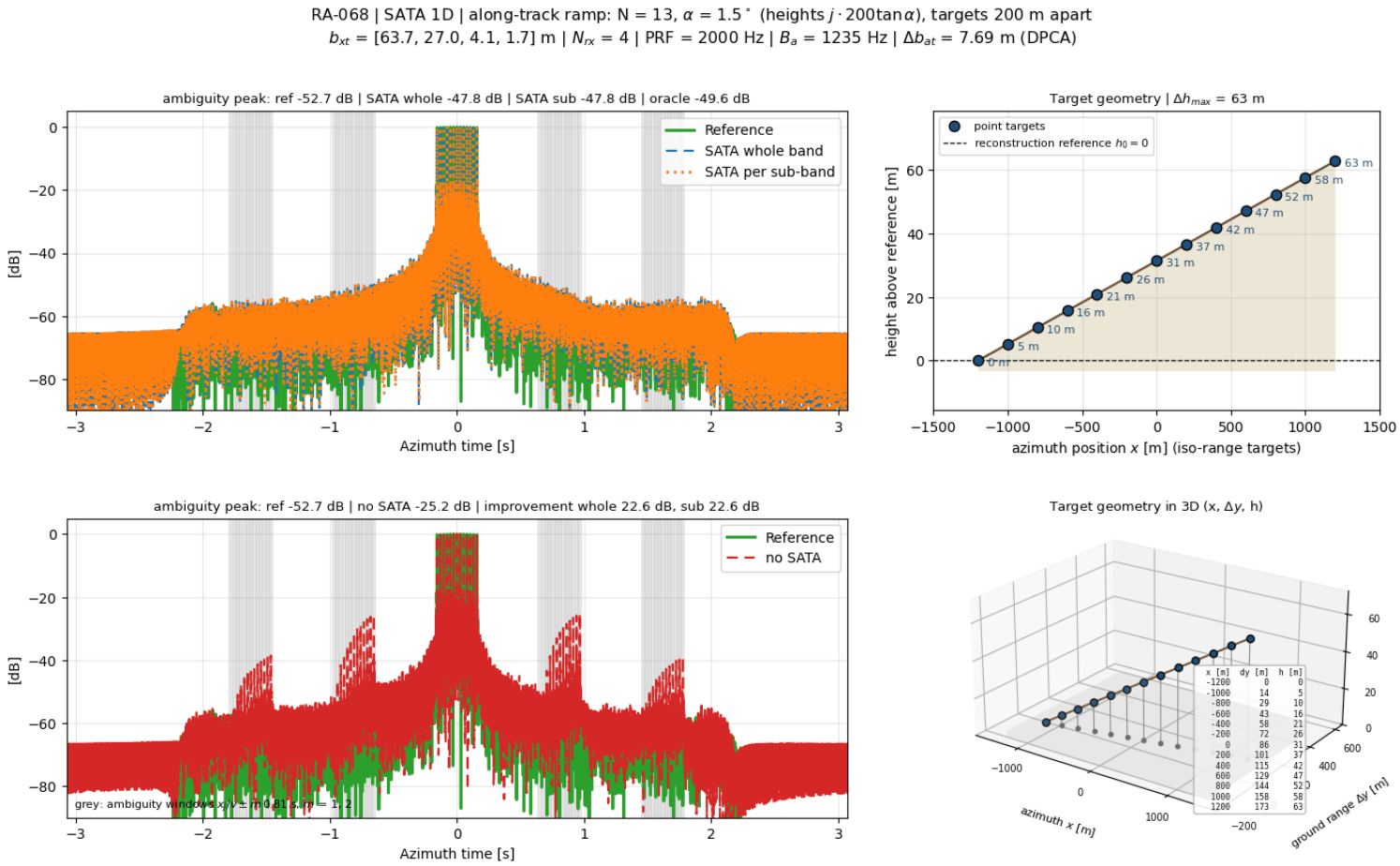


Figure 72: Targets 200 m apart: RA-068, $N = 13$, $\alpha = 1.5°$. Ambiguity peak: no SATA -25.2, SATA whole -47.8, per sub-band -47.8, oracle -49.6 dB (improvement 22.6 dB).

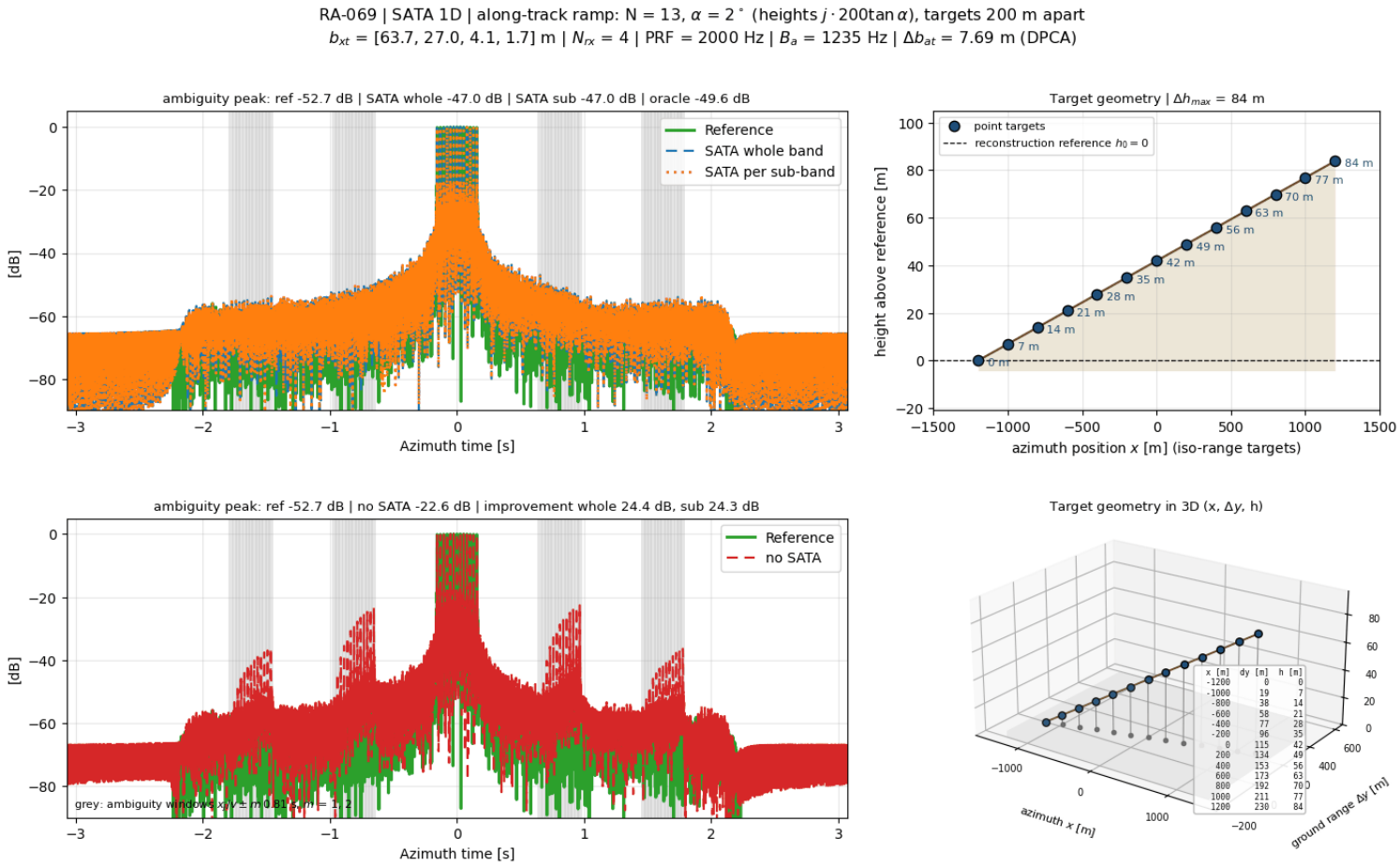


Figure 73: Targets 200 m apart: RA-069, $N = 13$, $\alpha = 2°$. Ambiguity peak: no SATA −22.6, SATA whole −47.0, per sub-band −47.0, oracle −49.6 dB (improvement 24.4 dB).

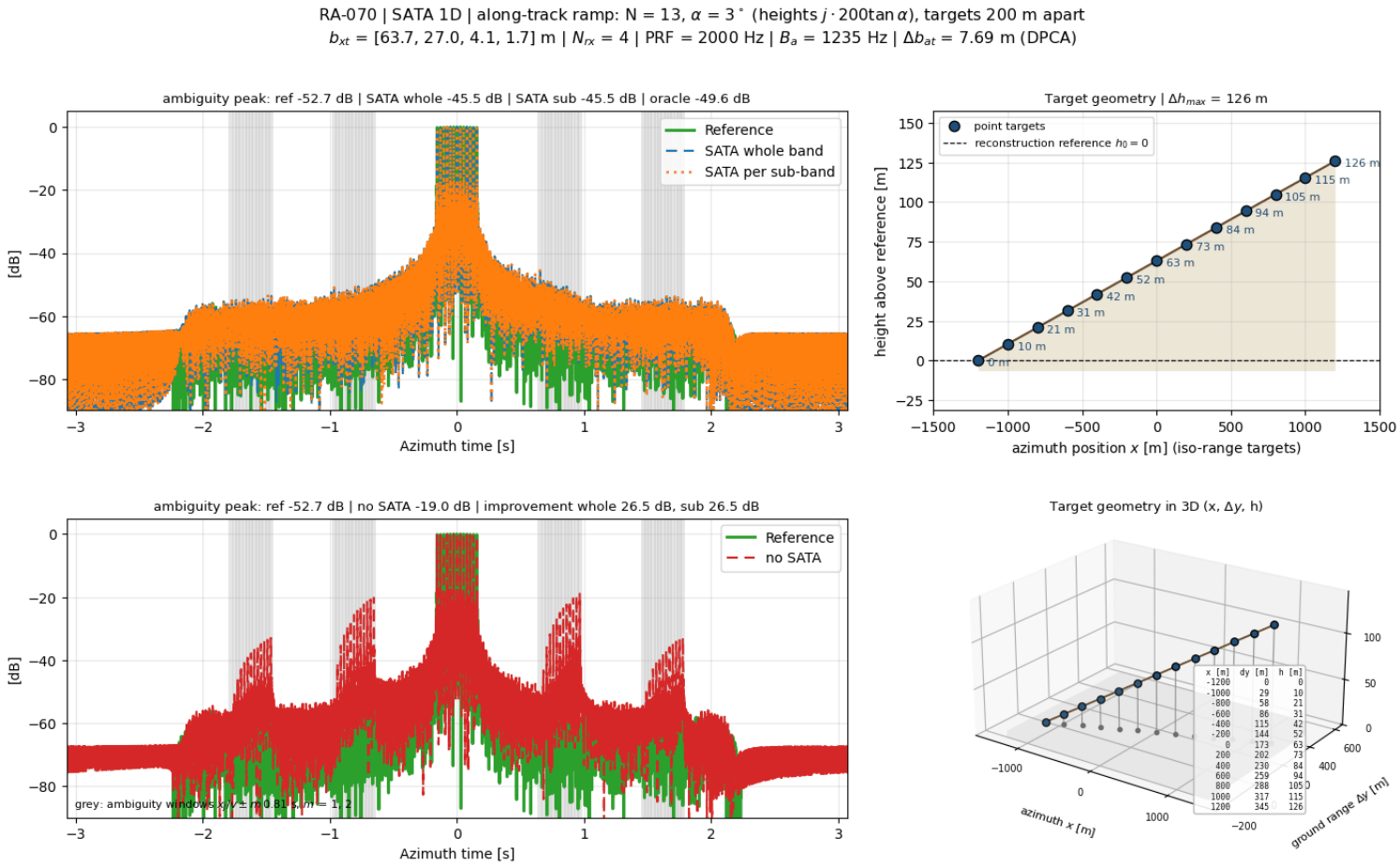


Figure 74: Targets 200 m apart: RA-070, $N = 13$, $\alpha = 3°$. Ambiguity peak: no SATA −19.0, SATA whole −45.5, per sub-band −45.5, oracle −49.6 dB (improvement 26.5 dB).

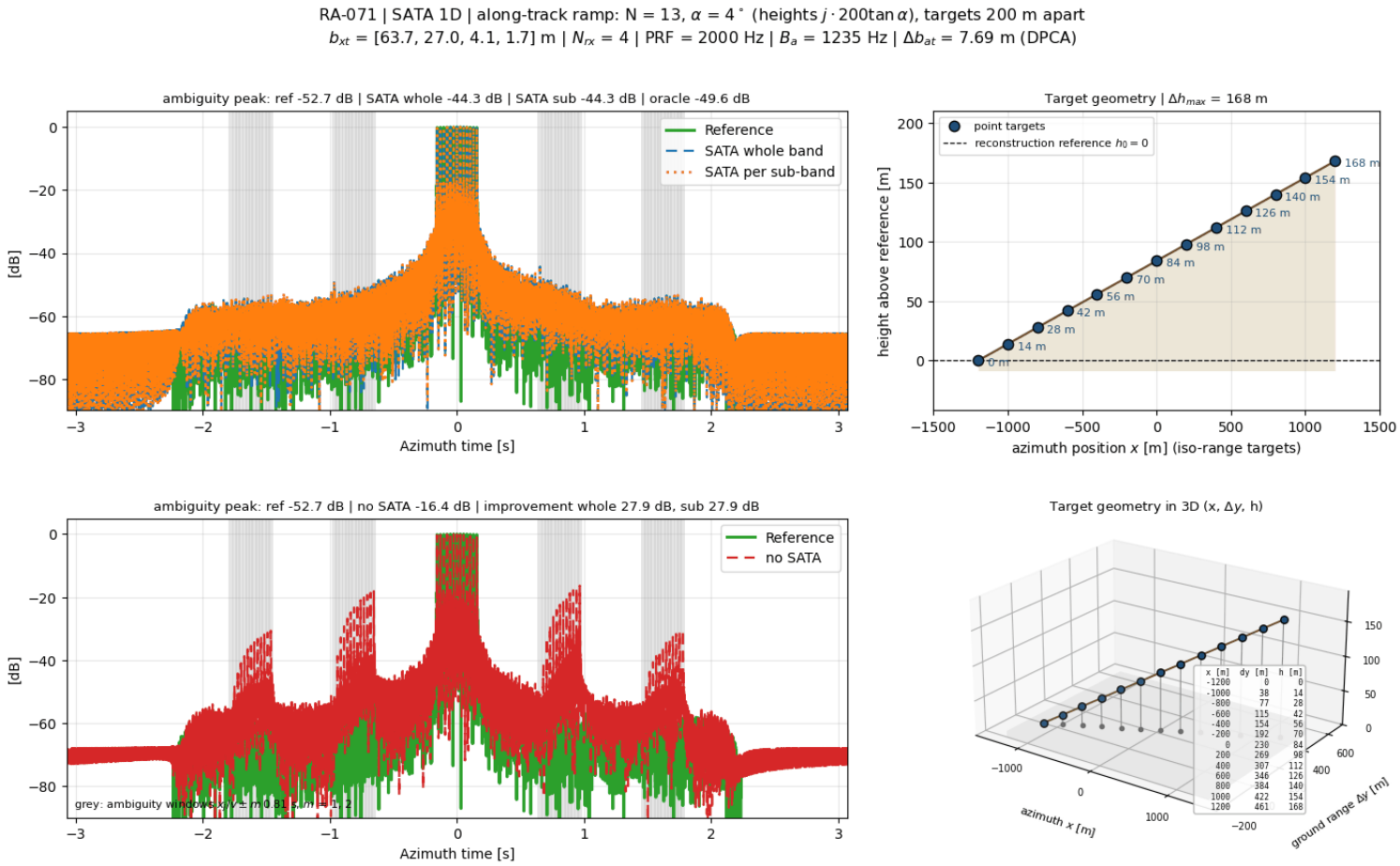


Figure 75: Targets 200 m apart: RA-071, $N = 13$, $\alpha = 4°$. Ambiguity peak: no SATA −16.4, SATA whole −44.3, per sub-band −44.3, oracle −49.6 dB (improvement 27.9 dB).

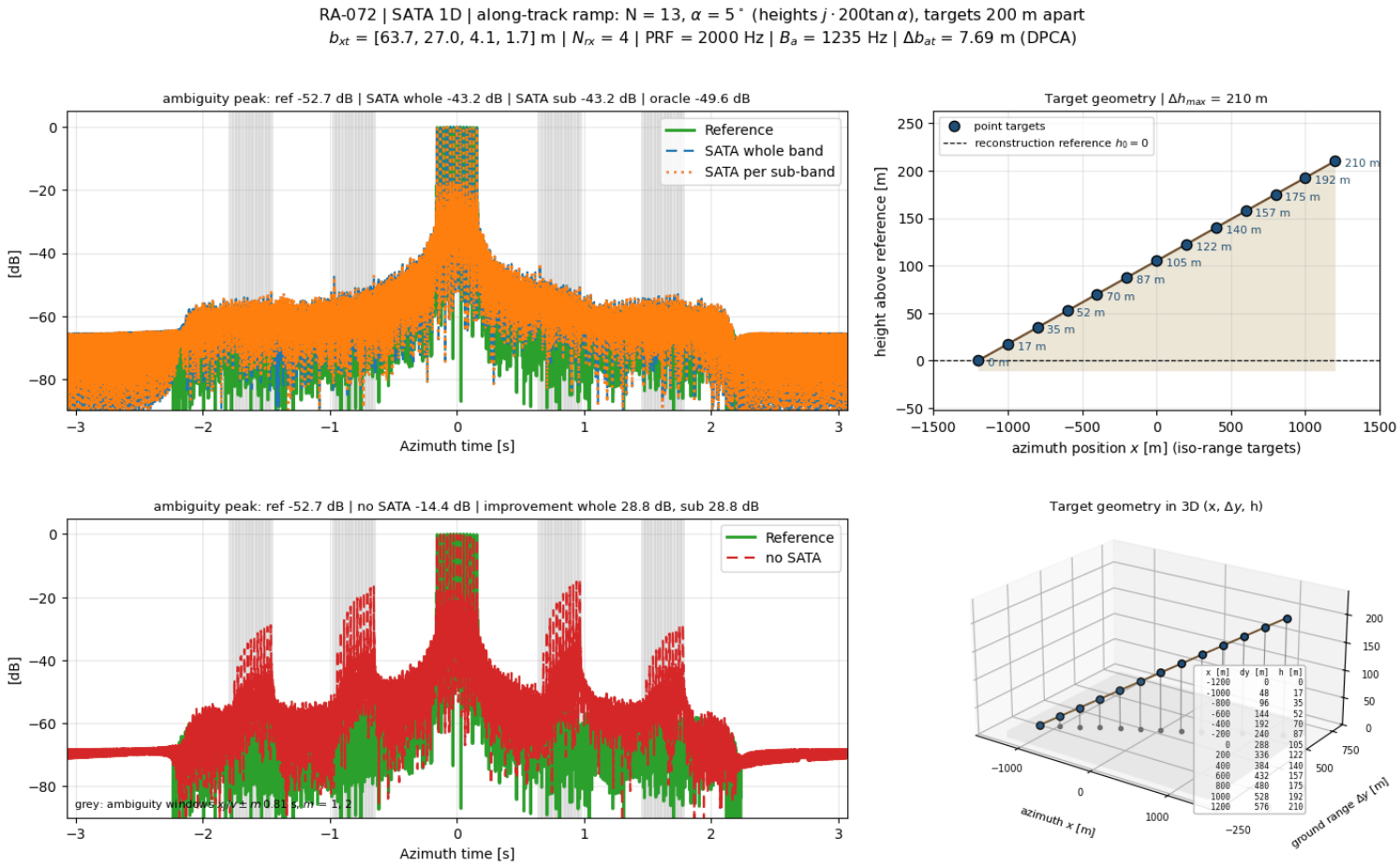


Figure 76: Targets 200 m apart: RA-072, $N = 13$, $\alpha = 5°$. Ambiguity peak: no SATA −14.4, SATA whole −43.2, per sub-band −43.2, oracle −49.6 dB (improvement 28.8 dB).

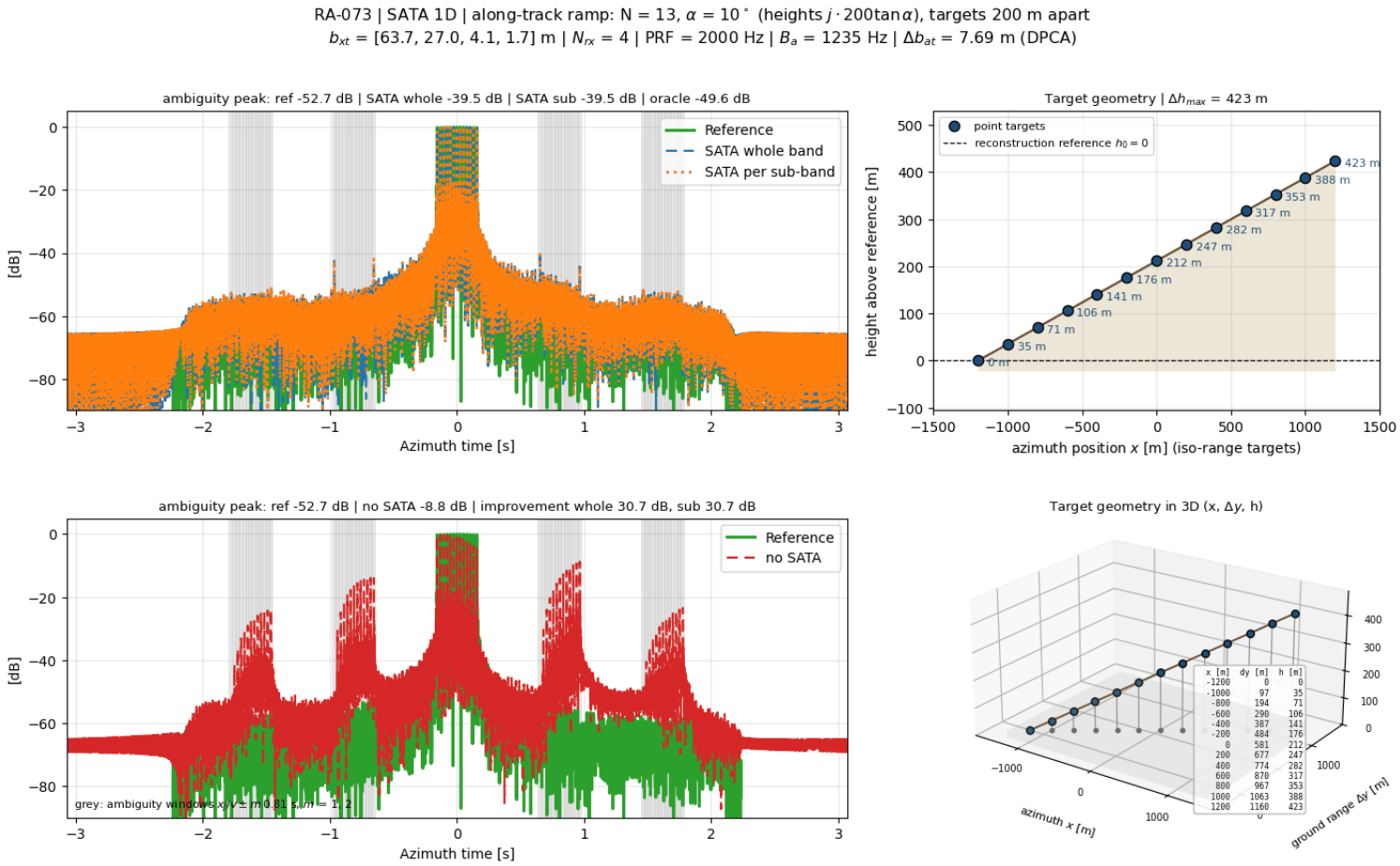


Figure 77: Targets 200 m apart: RA-073, $N = 13$, $\alpha = 10°$. Ambiguity peak: no SATA -8.8, SATA whole -39.5, per sub-band -39.5, oracle -49.6 dB (improvement 30.7 dB).

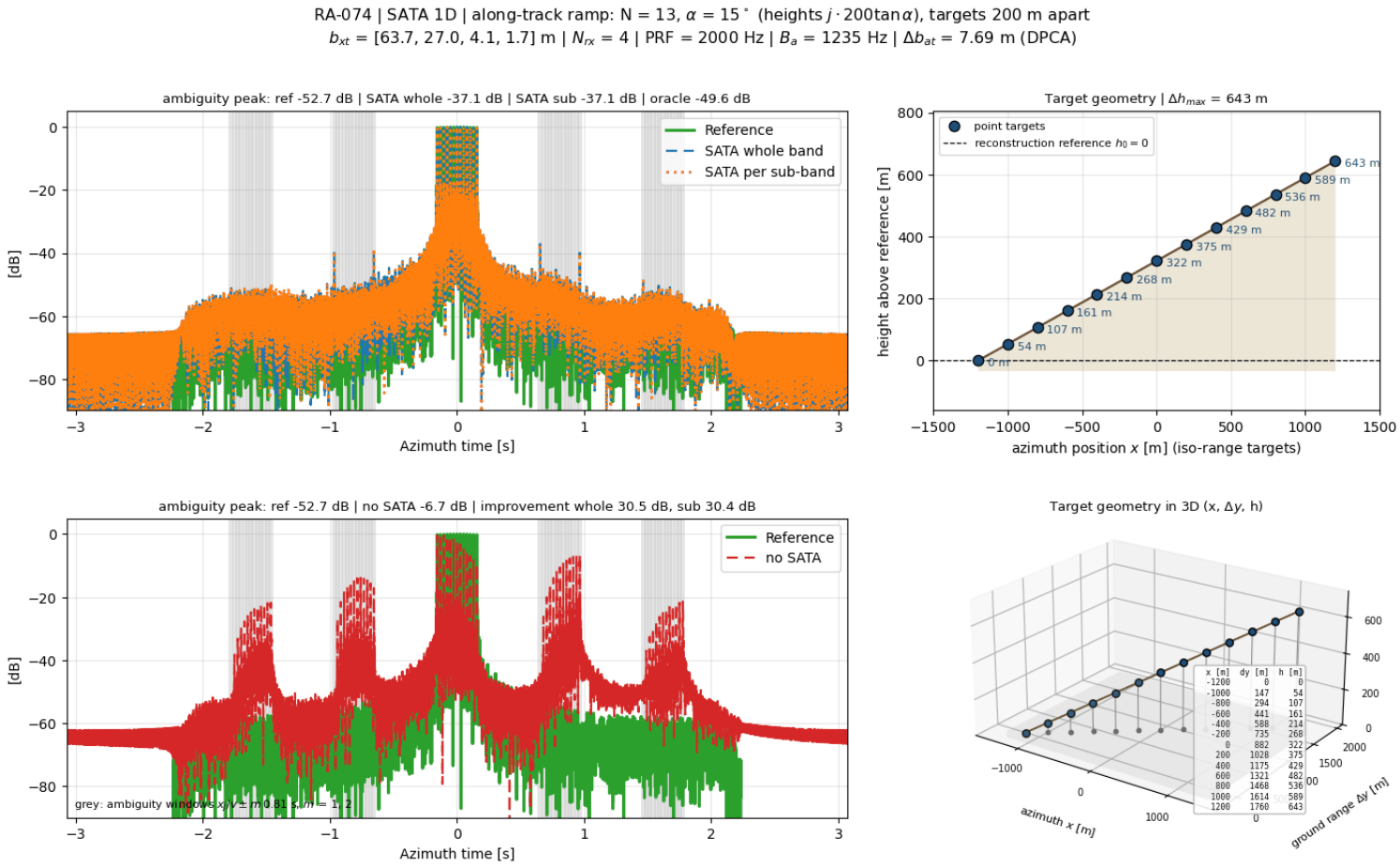


Figure 78: Targets 200 m apart: RA-074, $N = 13$, $\alpha = 15°$. Ambiguity peak: no SATA −6.7, SATA whole −37.1, per sub-band −37.1, oracle −49.6 dB (improvement 30.5 dB).

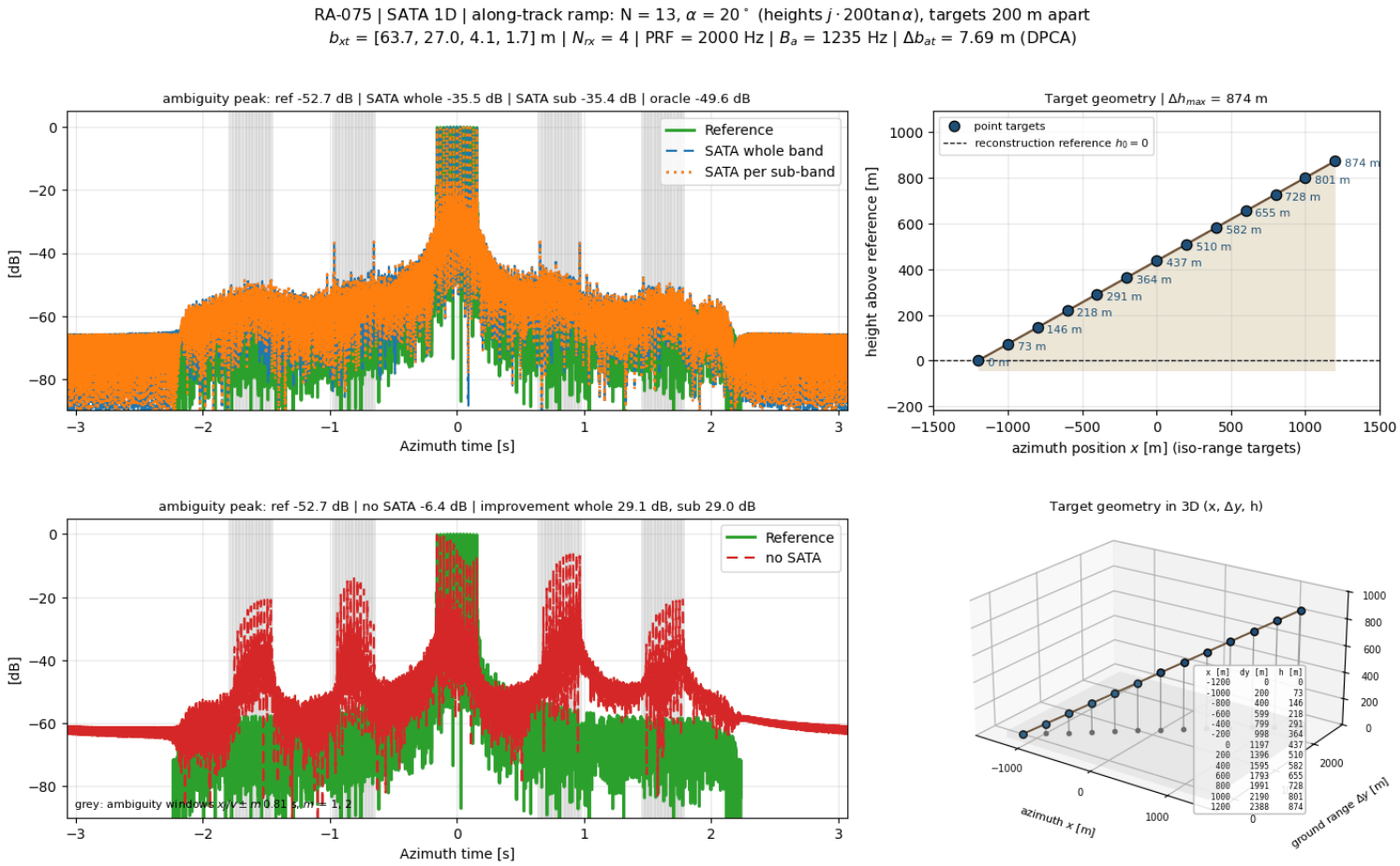


Figure 79: Targets 200 m apart: RA-075, $N = 13$, $\alpha = 20°$. Ambiguity peak: no SATA −6.4, SATA whole −35.5, per sub-band −35.4, oracle −49.6 dB (improvement 29.1 dB).

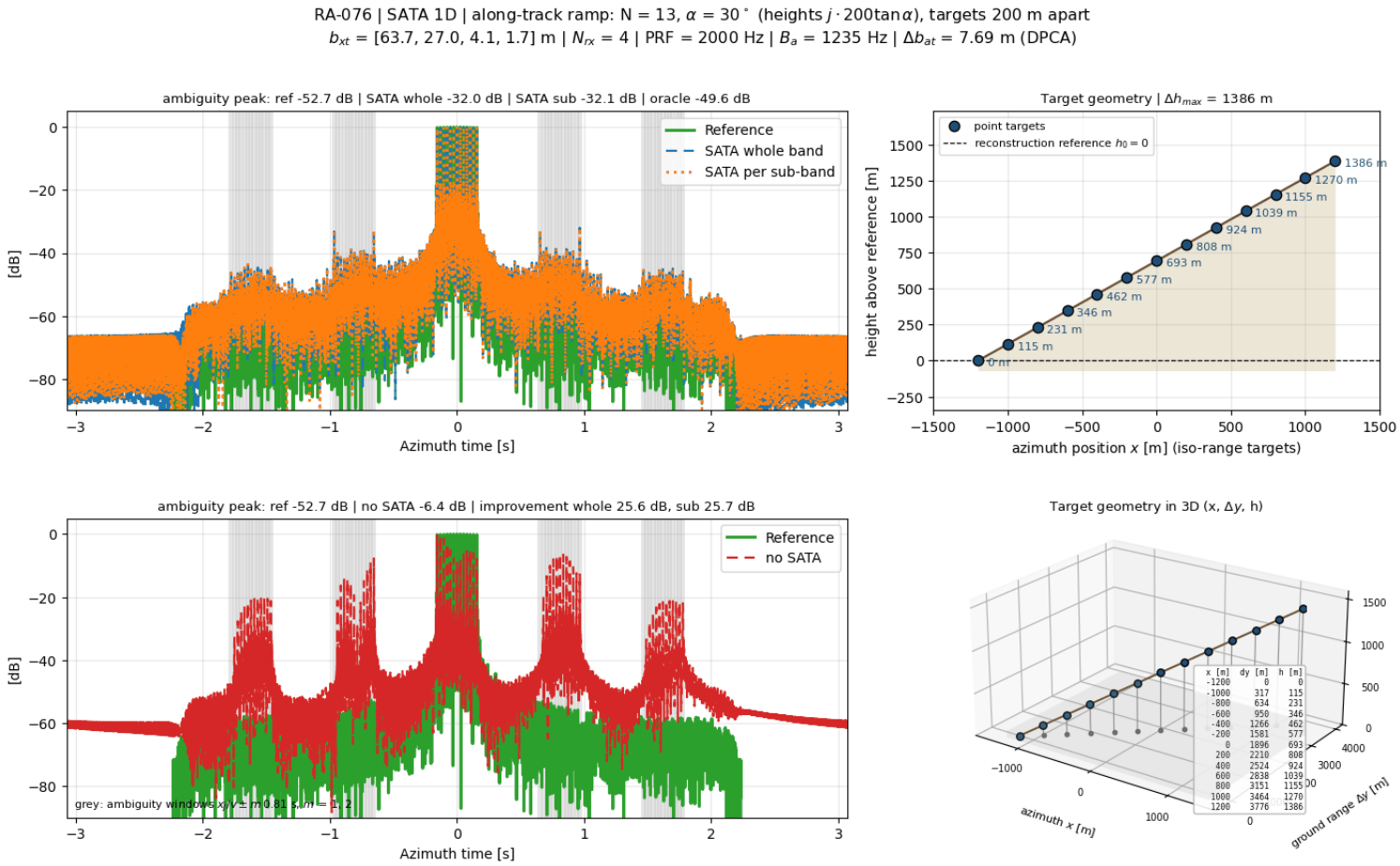


Figure 80: Targets 200 m apart: RA-076, $N = 13$, $\alpha = 30°$. Ambiguity peak: no SATA -6.4, SATA whole -32.0, per sub-band -32.1, oracle -49.6 dB (improvement 25.6 dB).

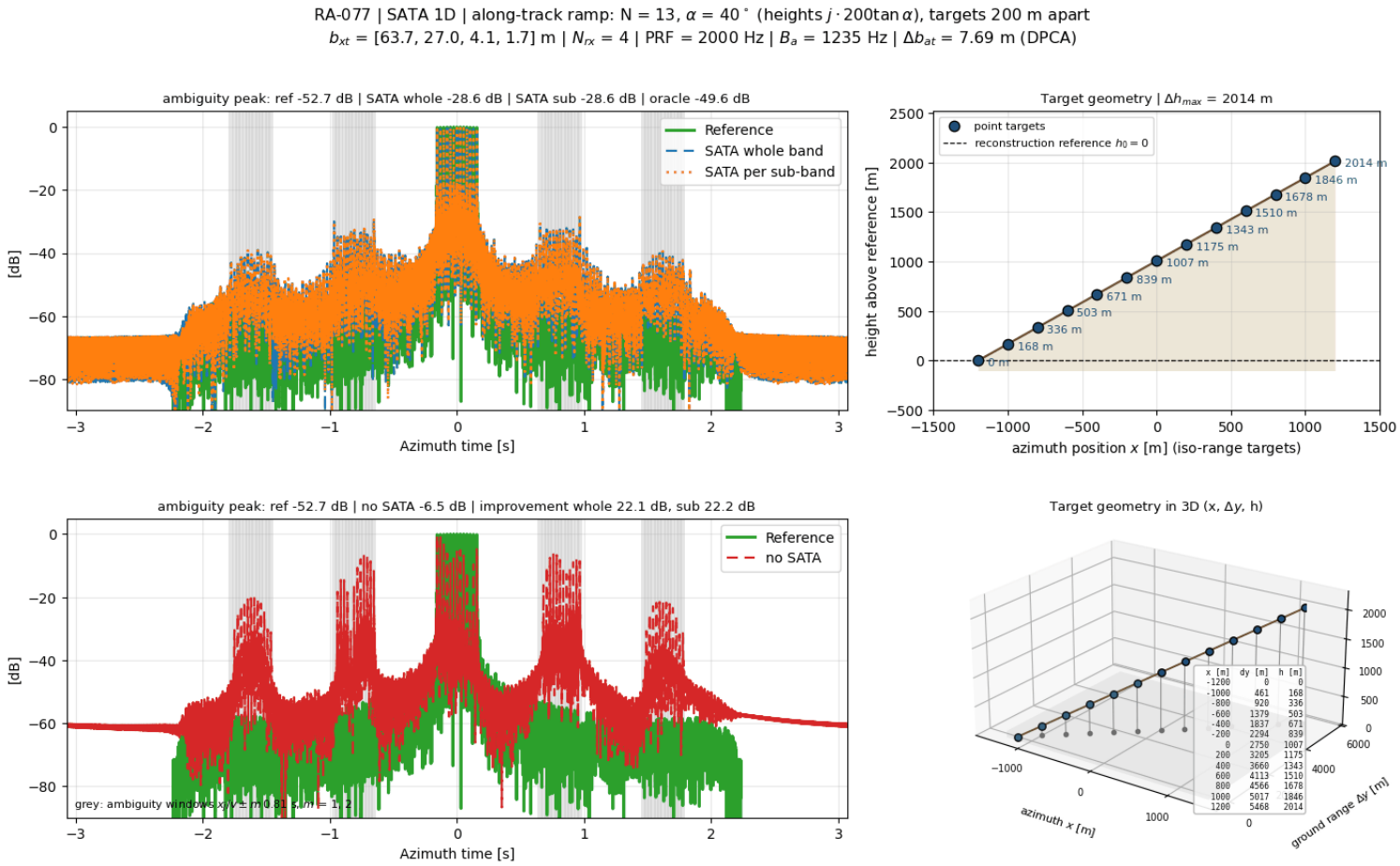


Figure 81: Targets 200 m apart: RA-077, $N = 13$, $\alpha = 40°$. Ambiguity peak: no SATA −6.5, SATA whole −28.6, per sub-band −28.6, oracle −49.6 dB (improvement 22.1 dB).

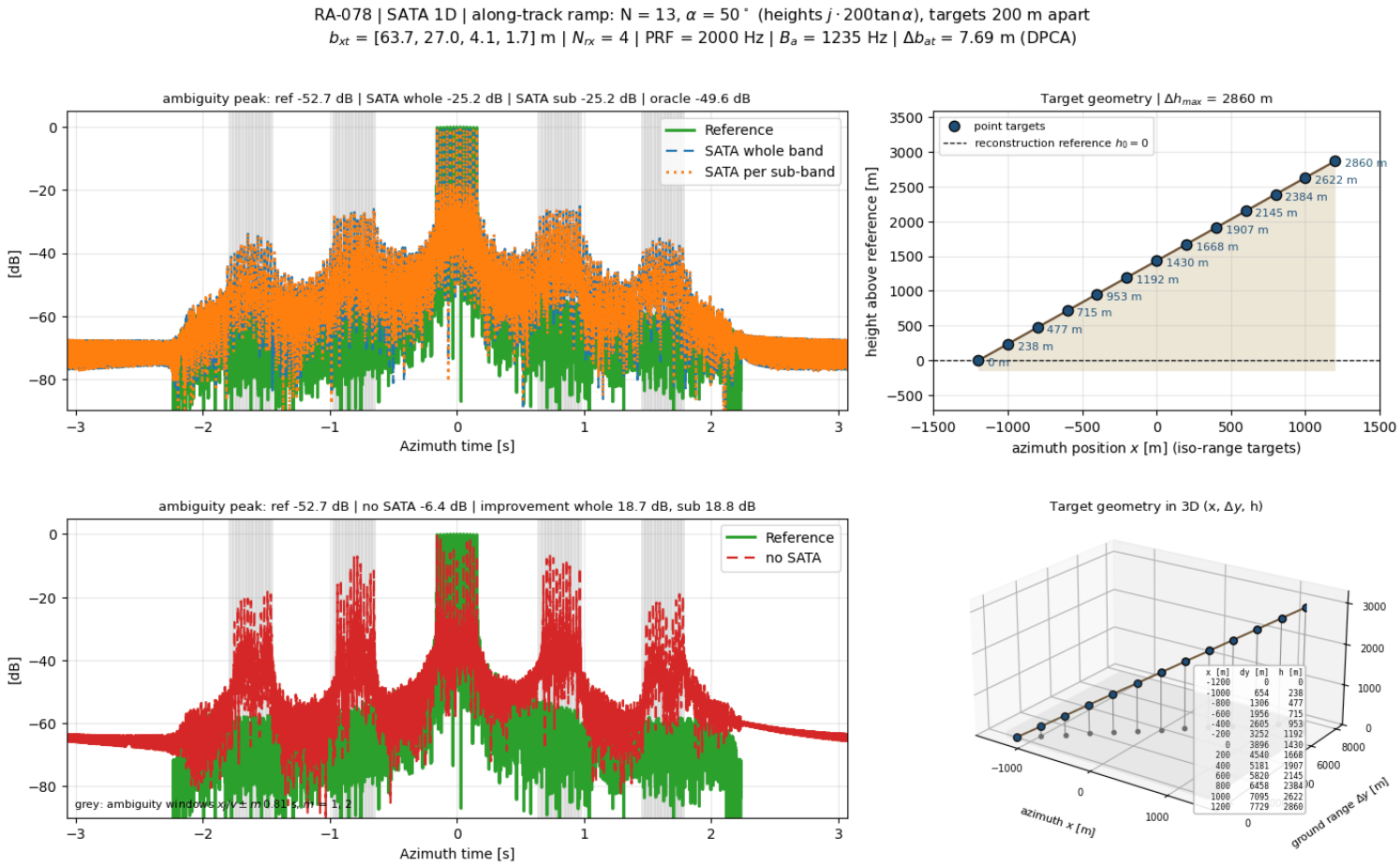


Figure 82: Targets 200 m apart: RA-078, $N = 13$, $\alpha = 50°$. Ambiguity peak: no SATA -6.4, SATA whole -25.2, per sub-band -25.2, oracle -49.6 dB (improvement 18.7 dB).

4 IRFs, targets 2.6 km apart (same heights)

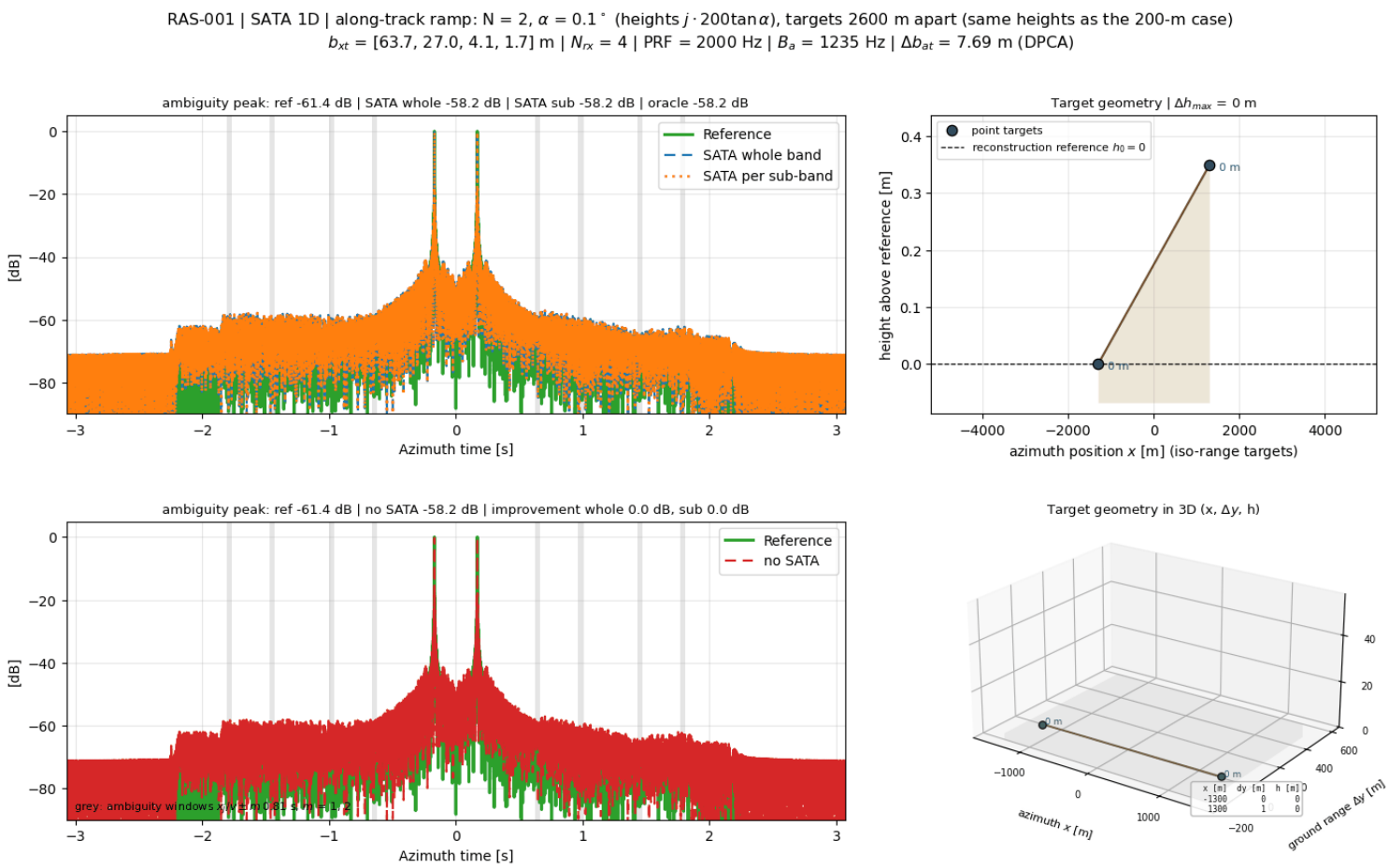


Figure 83: Targets 2.6 km apart, same heights: RAS-001, $N = 2$, $\alpha = 0.1°$. Ambiguity peak: no SATA −58.2, SATA whole −58.2, per sub-band −58.2, oracle −58.2 dB (improvement 0.0 dB).

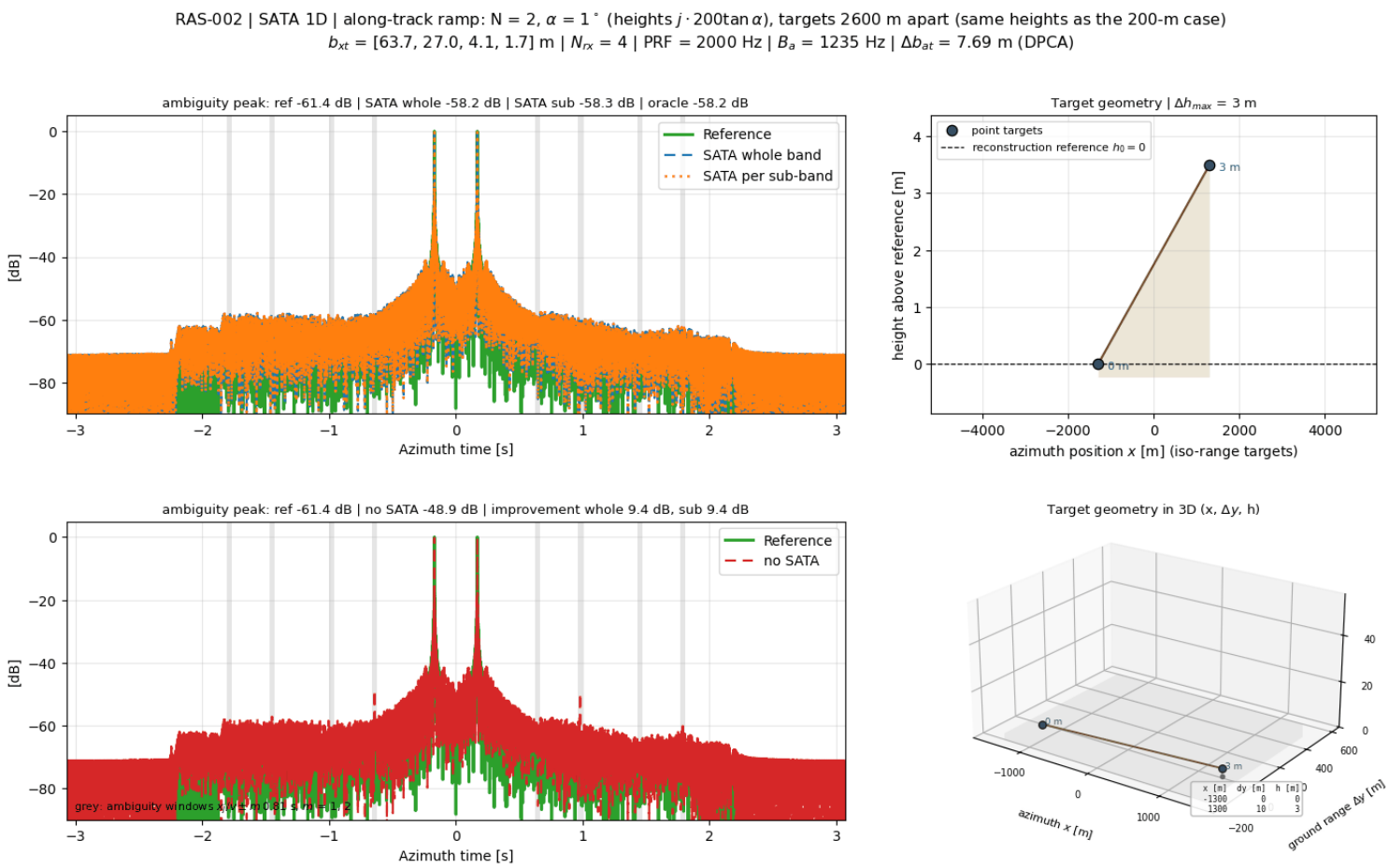


Figure 84: Targets 2.6 km apart, same heights: RAS-002, $N = 2$, $\alpha = 1°$. Ambiguity peak: no SATA −48.9, SATA whole −58.2, per sub-band −58.3, oracle −58.2 dB (improvement 9.4 dB).

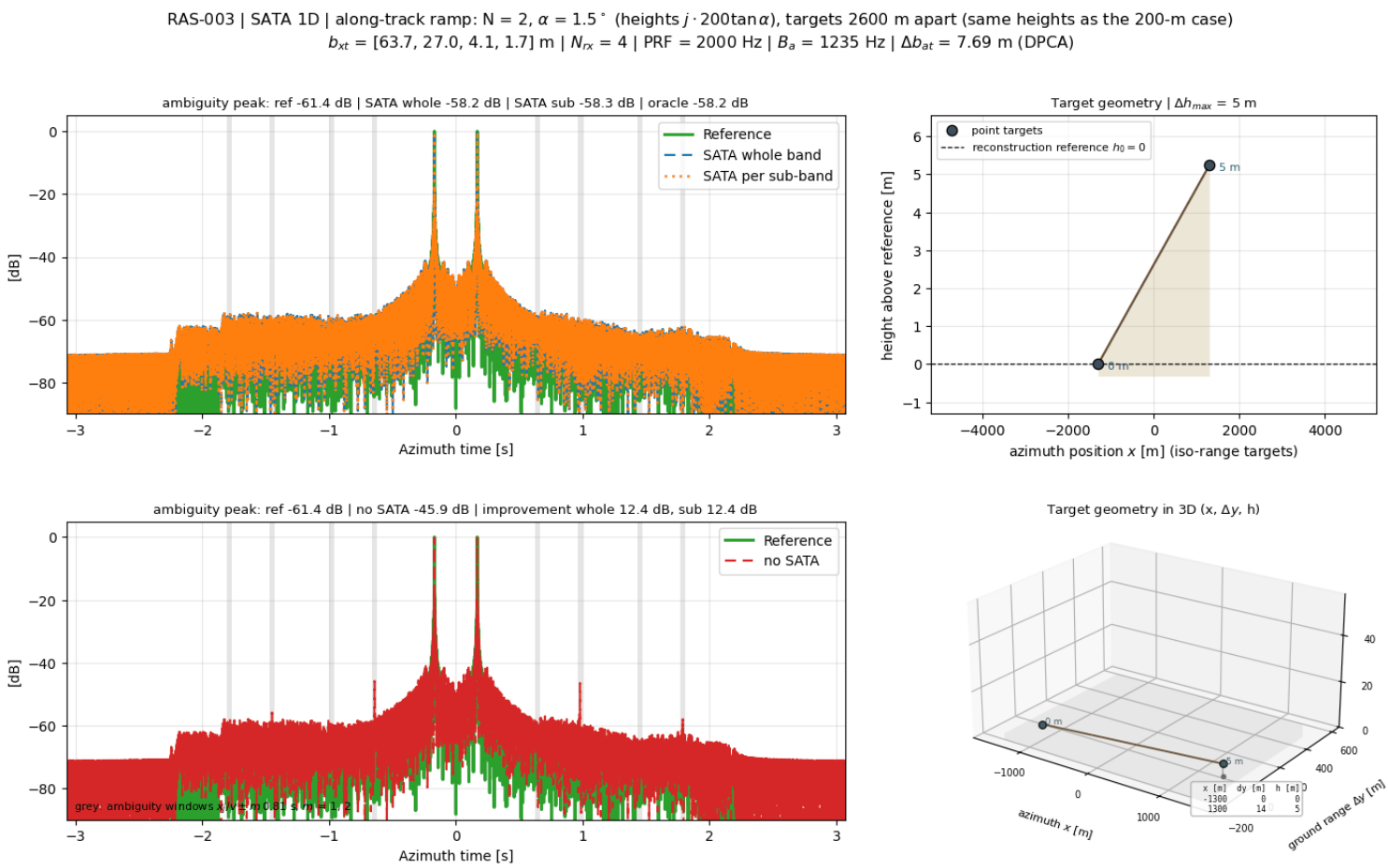


Figure 85: Targets 2.6 km apart, same heights: RAS-003, $N = 2$, $\alpha = 1.5°$. Ambiguity peak: no SATA -45.9, SATA whole -58.2, per sub-band -58.3, oracle -58.2 dB (improvement 12.4 dB).

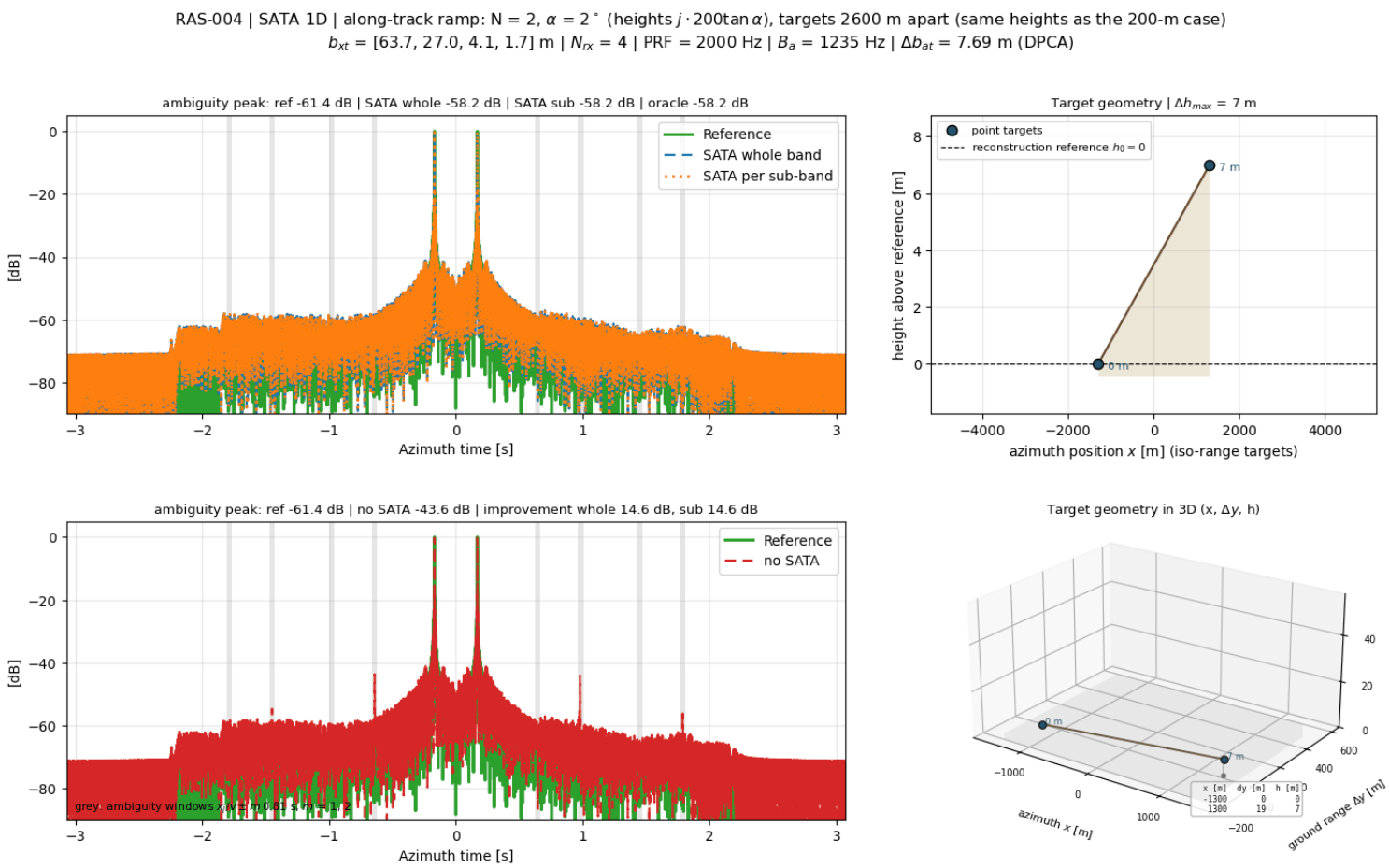


Figure 86: Targets 2.6 km apart, same heights: RAS-004, $N = 2$, $\alpha = 2°$. Ambiguity peak: no SATA −43.6, SATA whole −58.2, per sub-band −58.2, oracle −58.2 dB (improvement 14.6 dB).

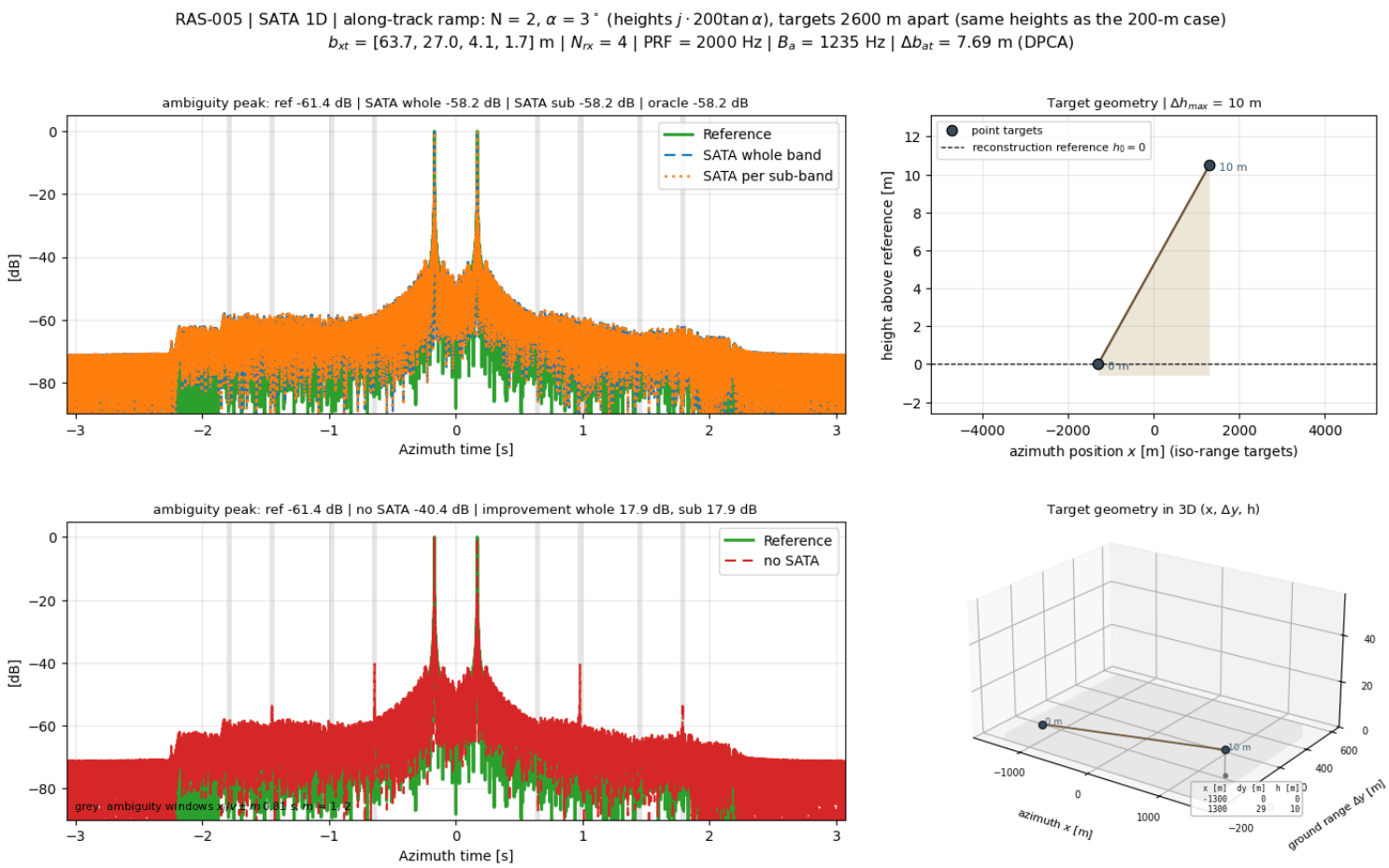


Figure 87: Targets 2.6 km apart, same heights: RAS-005, $N = 2$, $\alpha = 3°$. Ambiguity peak: no SATA −40.4, SATA whole −58.2, per sub-band −58.2, oracle −58.2 dB (improvement 17.9 dB).

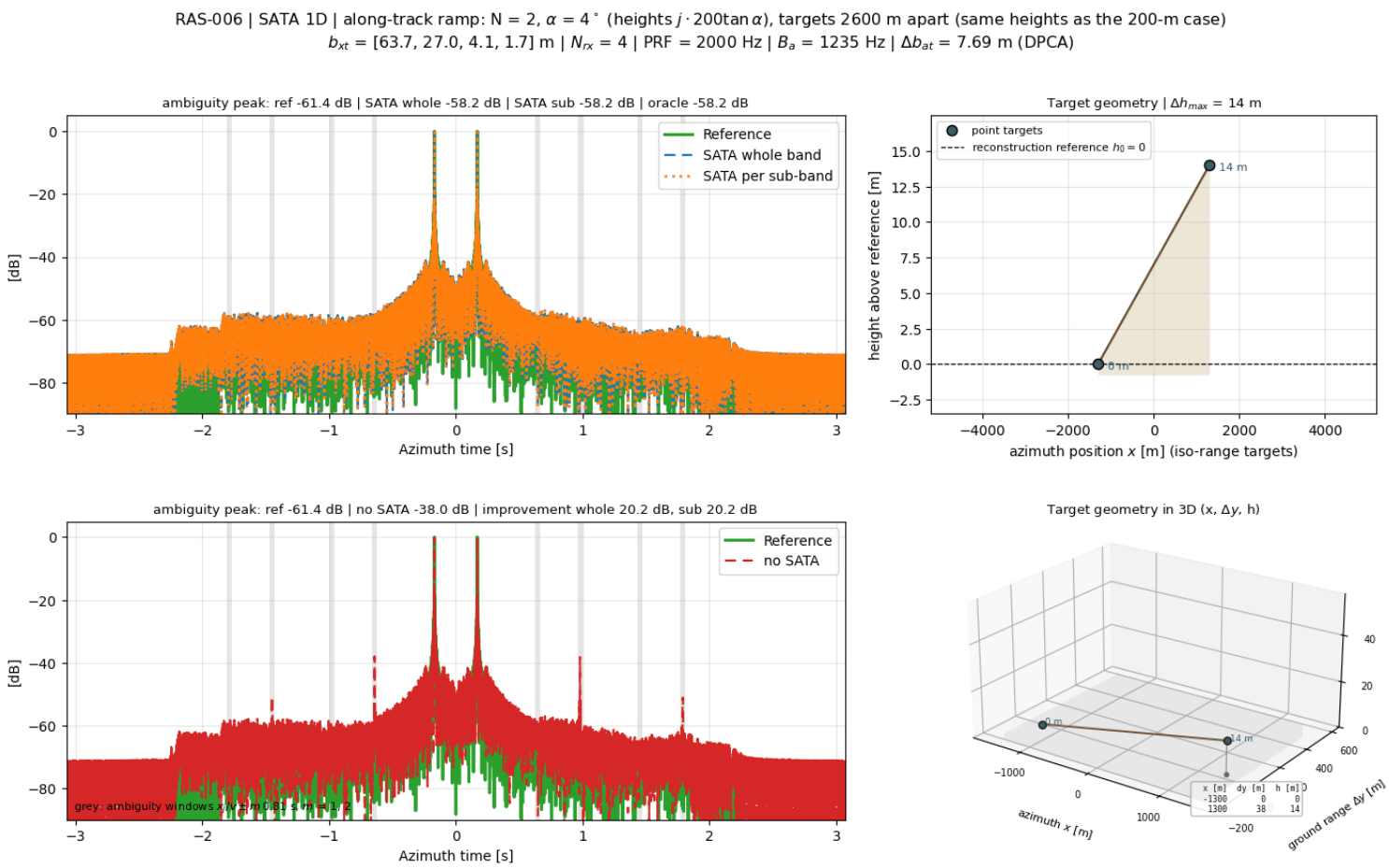


Figure 88: Targets 2.6 km apart, same heights: RAS-006, $N = 2$, $\alpha = 4°$. Ambiguity peak: no SATA −38.0, SATA whole −58.2, per sub-band −58.2, oracle −58.2 dB (improvement 20.2 dB).

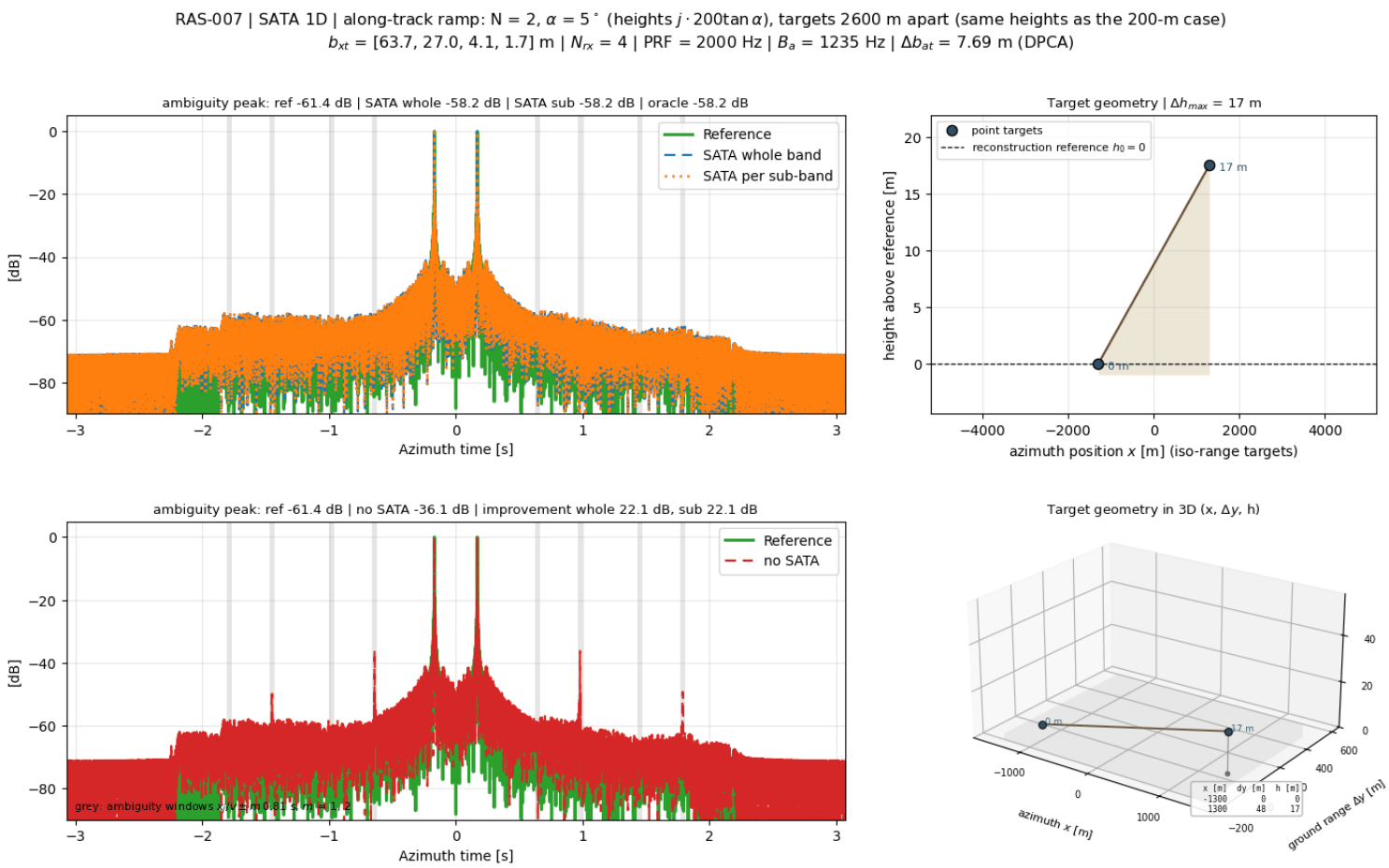


Figure 89: Targets 2.6 km apart, same heights: RAS-007, $N = 2$, $\alpha = 5°$. Ambiguity peak: no SATA −36.1, SATA whole −58.2, per sub-band −58.2, oracle −58.2 dB (improvement 22.1 dB).

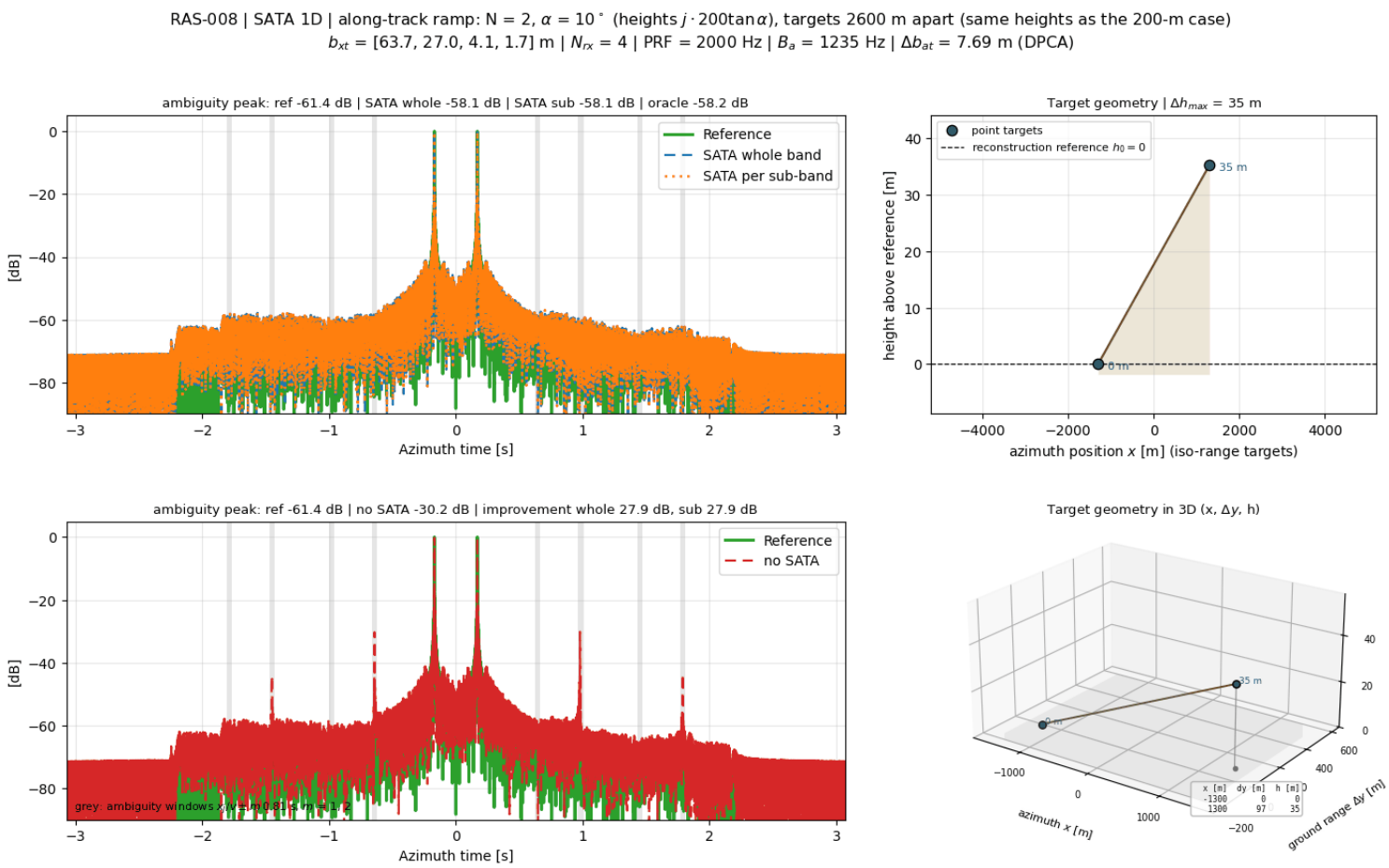


Figure 90: Targets 2.6 km apart, same heights: RAS-008, $N = 2$, $\alpha = 10°$. Ambiguity peak: no SATA -30.2, SATA whole -58.1, per sub-band -58.1, oracle -58.2 dB (improvement 27.9 dB).

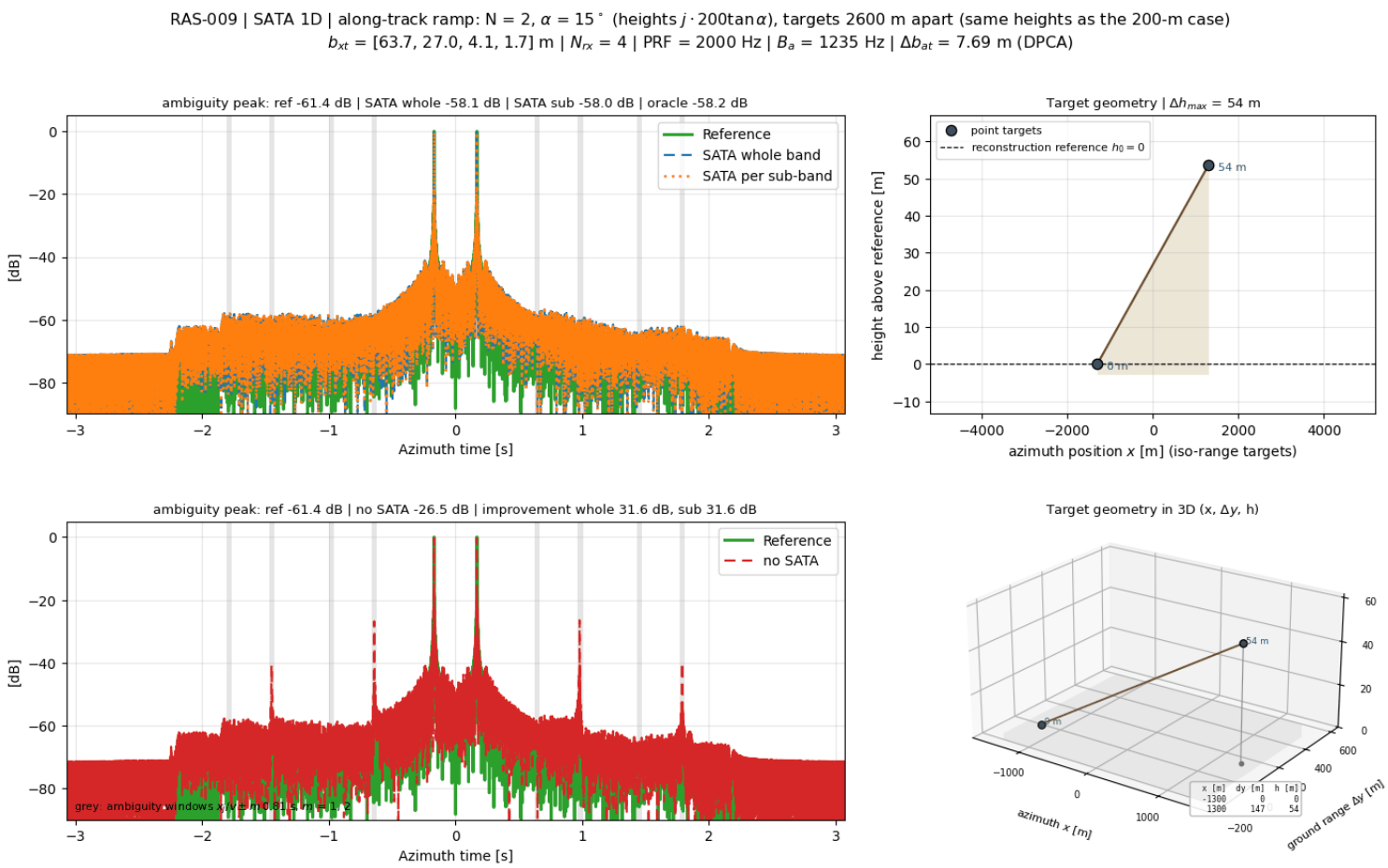


Figure 91: Targets 2.6 km apart, same heights: RAS-009, $N = 2$, $\alpha = 15°$. Ambiguity peak: no SATA −26.5, SATA whole −58.1, per sub-band −58.0, oracle −58.2 dB (improvement 31.6 dB).

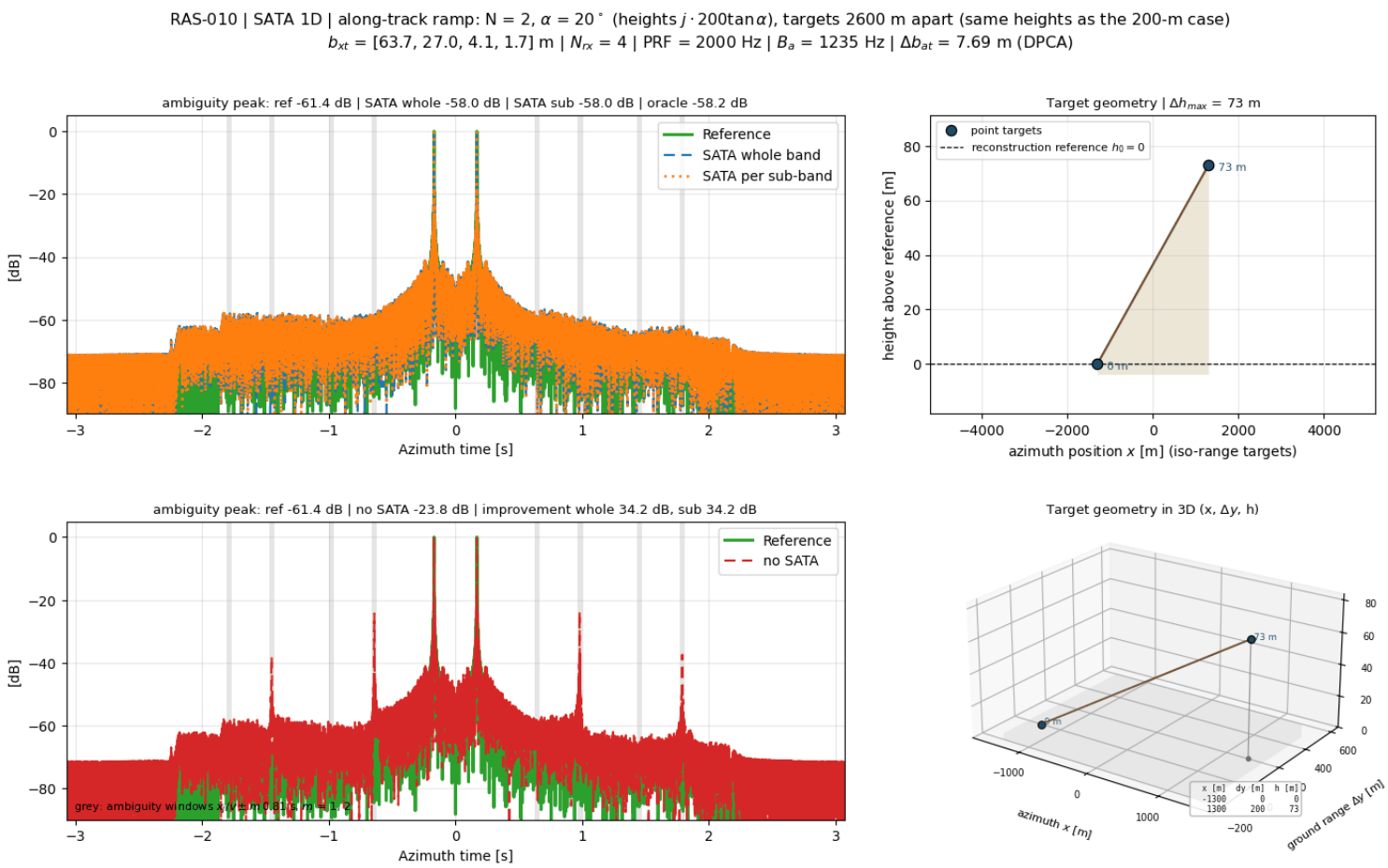


Figure 92: Targets 2.6 km apart, same heights: RAS-010, $N = 2$, $\alpha = 20°$. Ambiguity peak: no SATA -23.8, SATA whole -58.0, per sub-band -58.0, oracle -58.2 dB (improvement 34.2 dB).

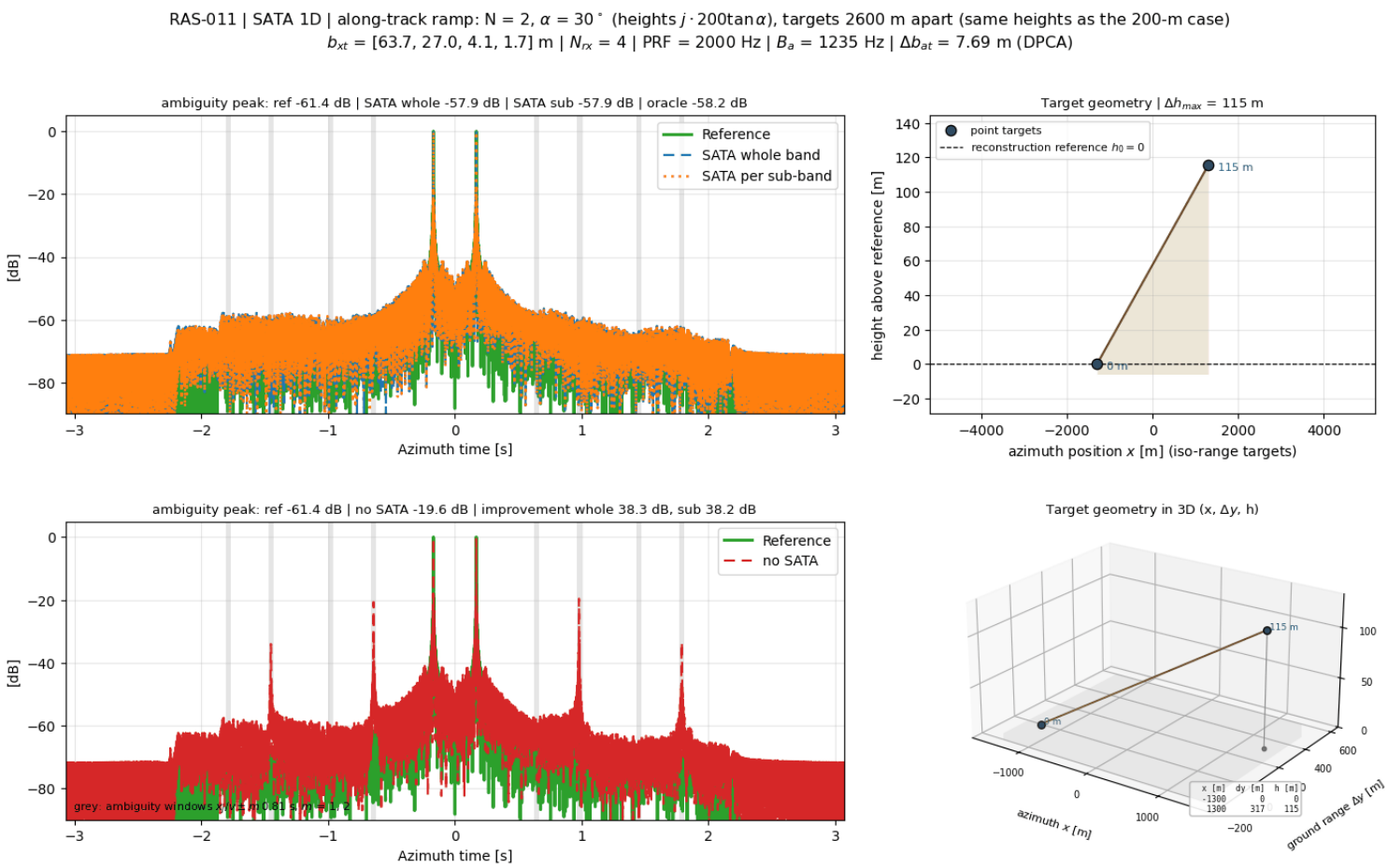


Figure 93: Targets 2.6 km apart, same heights: RAS-011, $N = 2$, $\alpha = 30°$. Ambiguity peak: no SATA -19.6, SATA whole -57.9, per sub-band -57.9, oracle -58.2 dB (improvement 38.3 dB).

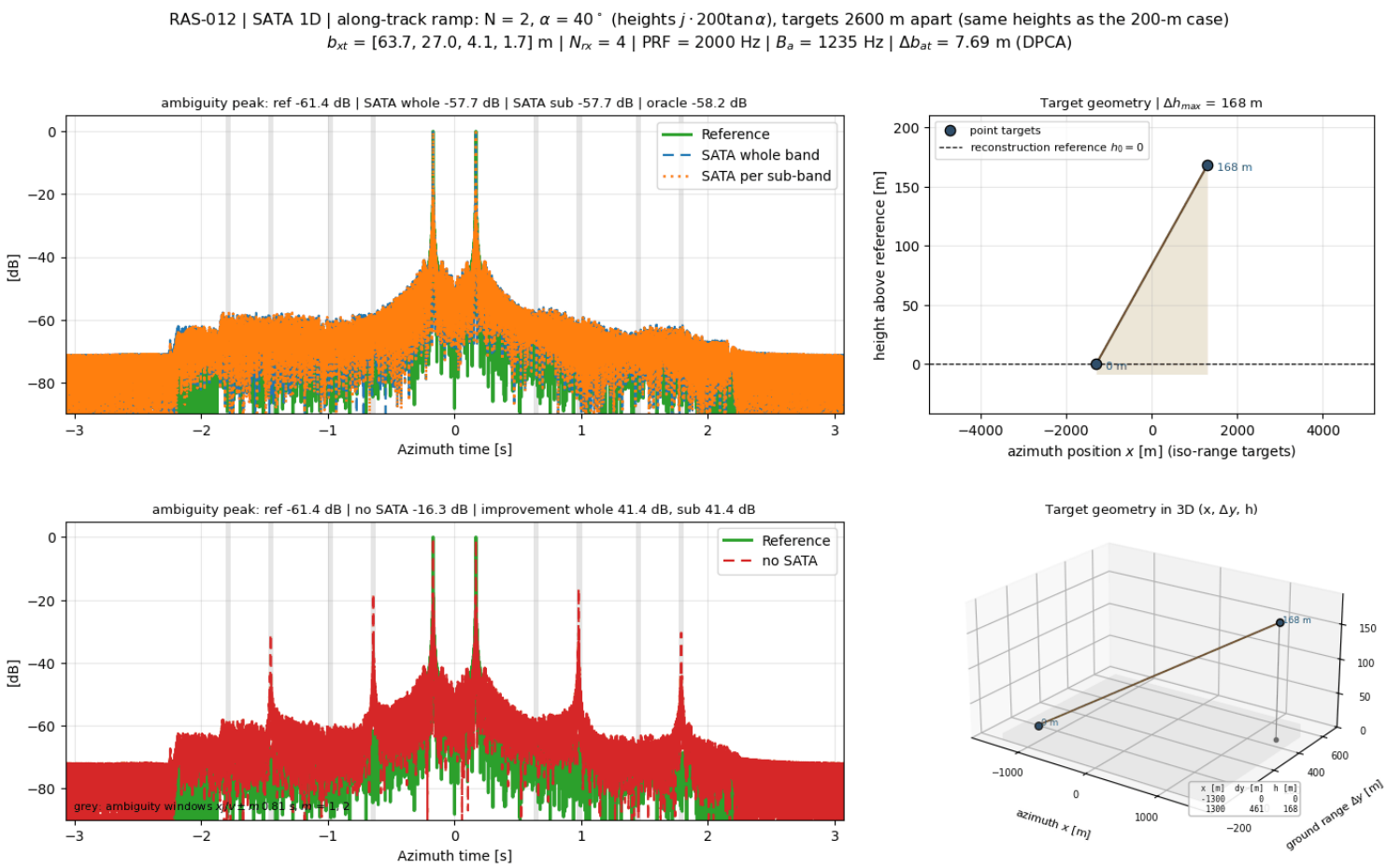


Figure 94: Targets 2.6 km apart, same heights: RAS-012, $N = 2$, $\alpha = 40°$. Ambiguity peak: no SATA -16.3, SATA whole -57.7, per sub-band -57.7, oracle -58.2 dB (improvement 41.4 dB).

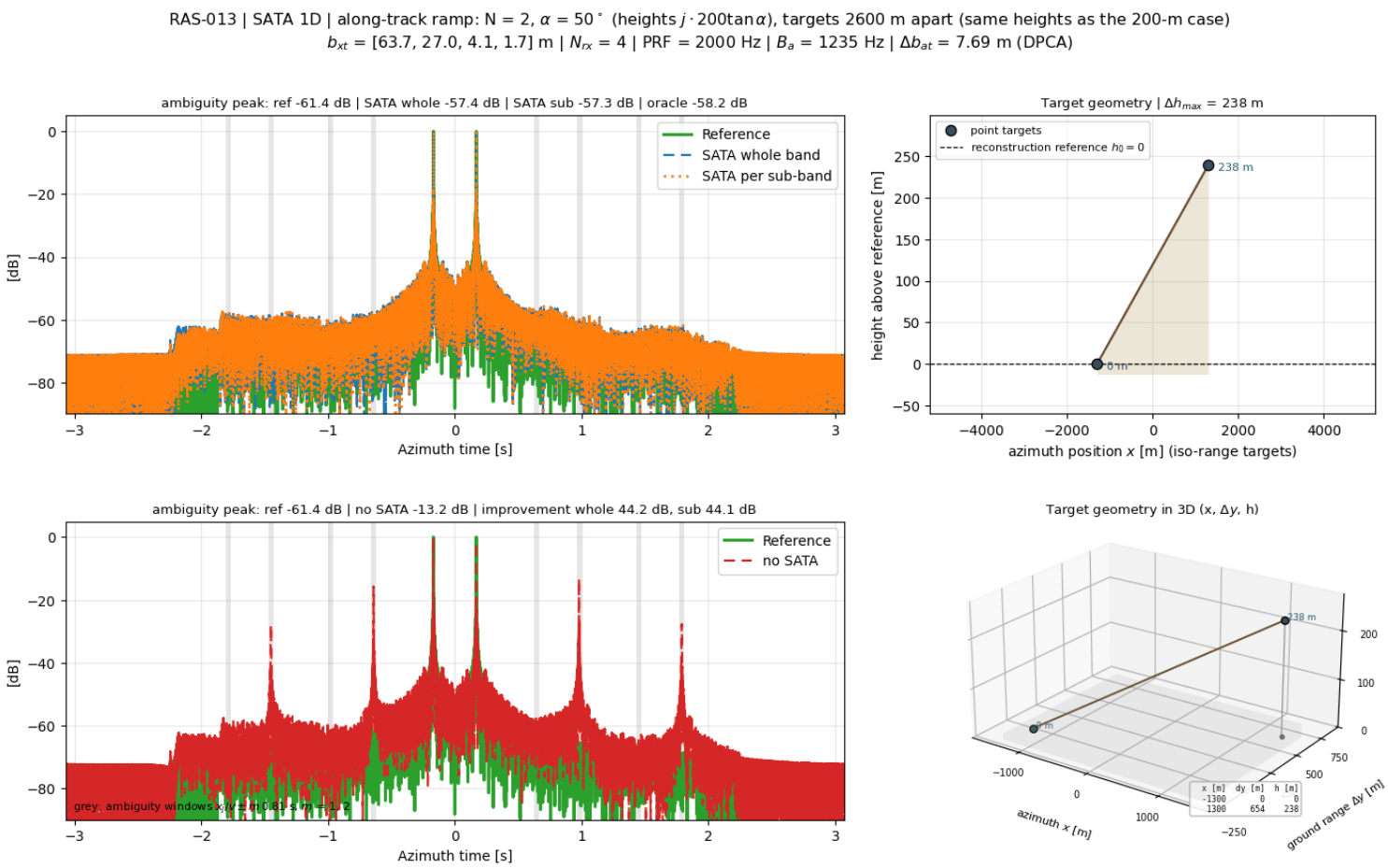


Figure 95: Targets 2.6 km apart, same heights: RAS-013, $N = 2$, $\alpha = 50°$. Ambiguity peak: no SATA -13.2, SATA whole -57.4, per sub-band -57.3, oracle -58.2 dB (improvement 44.2 dB).

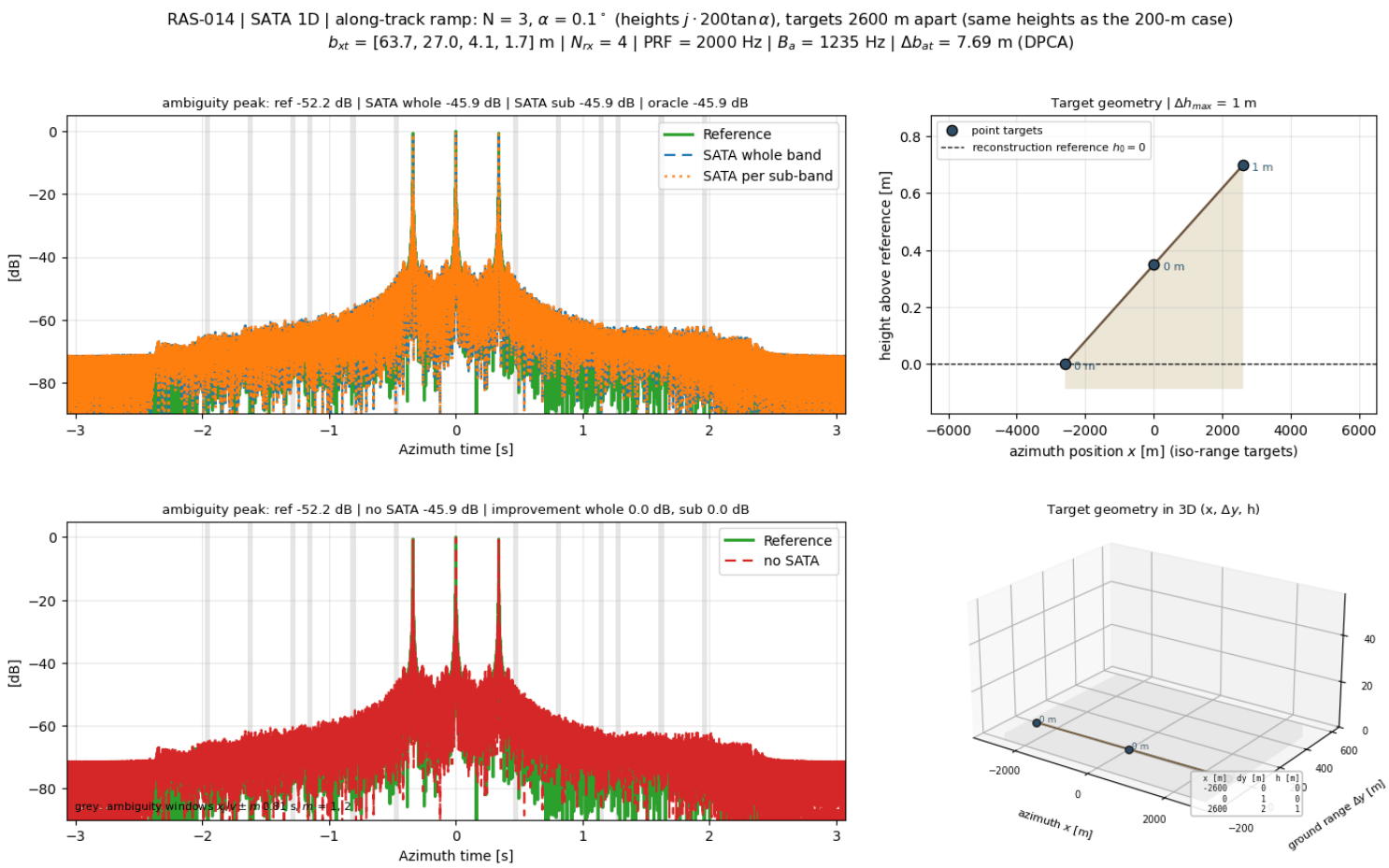


Figure 96: Targets 2.6 km apart, same heights: RAS-014, $N = 3$, $\alpha = 0.1°$. Ambiguity peak: no SATA -45.9, SATA whole -45.9, per sub-band -45.9, oracle -45.9 dB (improvement 0.0 dB).

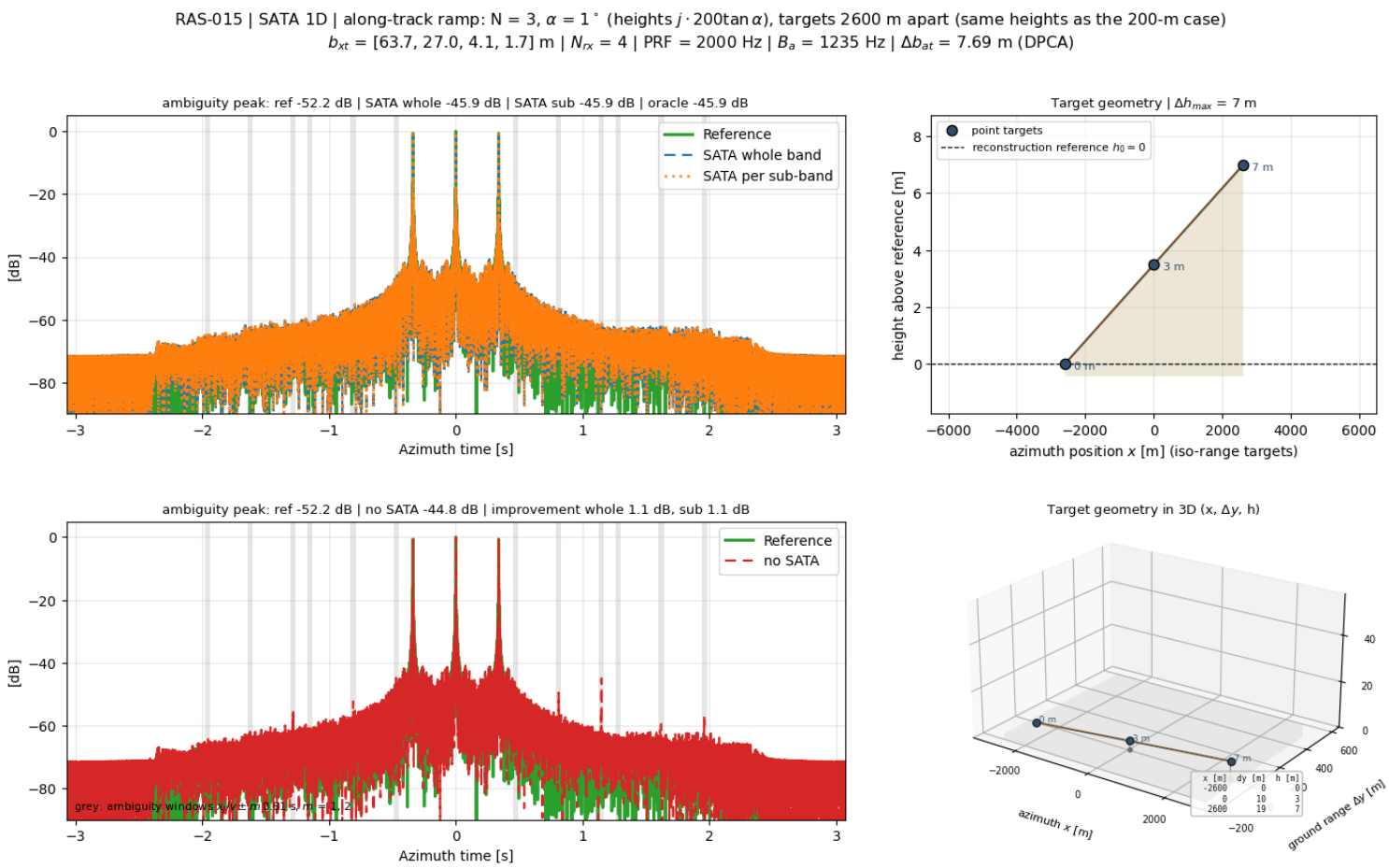


Figure 97: Targets 2.6 km apart, same heights: RAS-015, $N = 3$, $\alpha = 1°$. Ambiguity peak: no SATA -44.8, SATA whole -45.9, per sub-band -45.9, oracle -45.9 dB (improvement 1.1 dB).

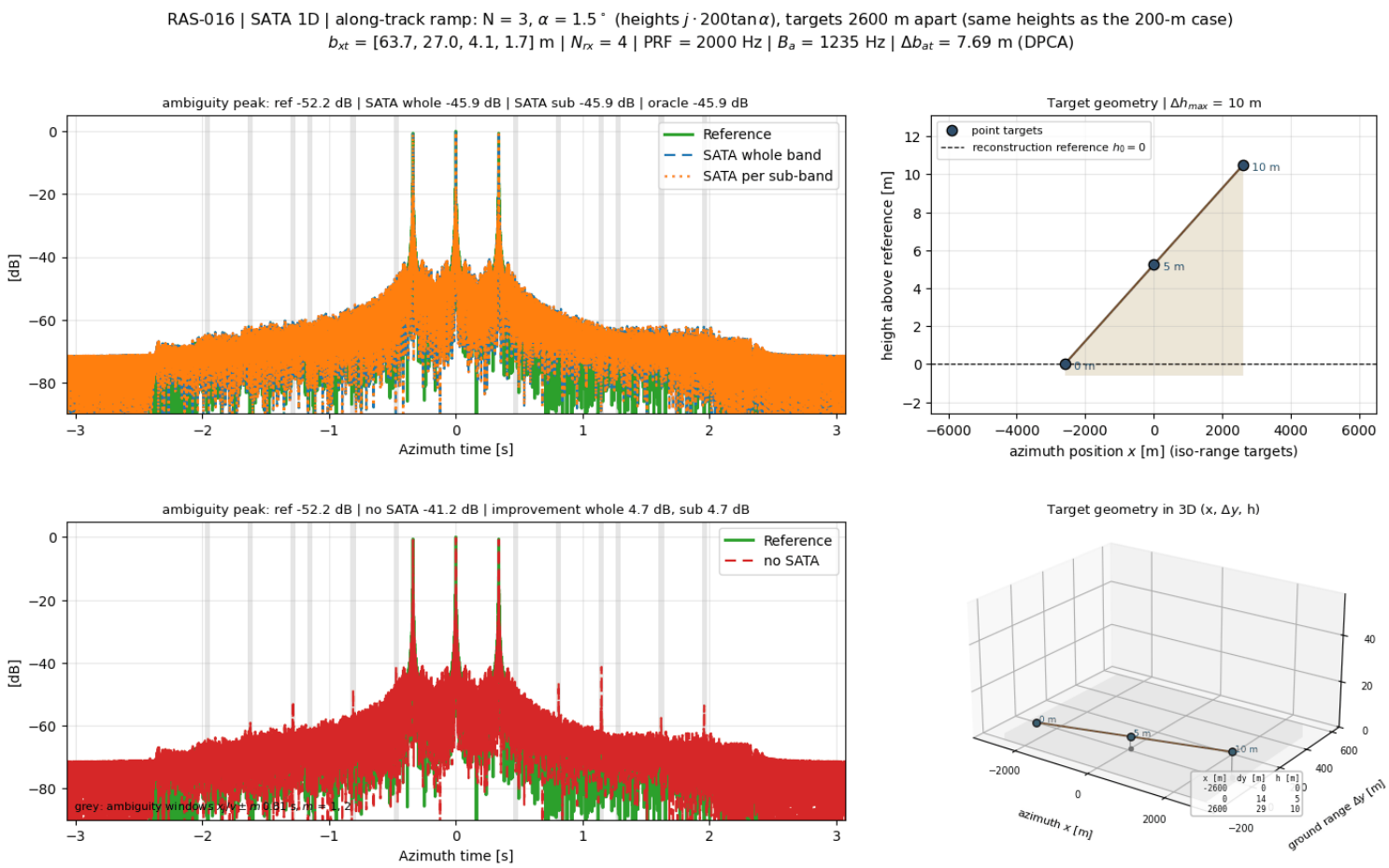


Figure 98: Targets 2.6 km apart, same heights: RAS-016, $N = 3$, $\alpha = 1.5°$. Ambiguity peak: no SATA -41.2, SATA whole -45.9, per sub-band -45.9, oracle -45.9 dB (improvement 4.7 dB).

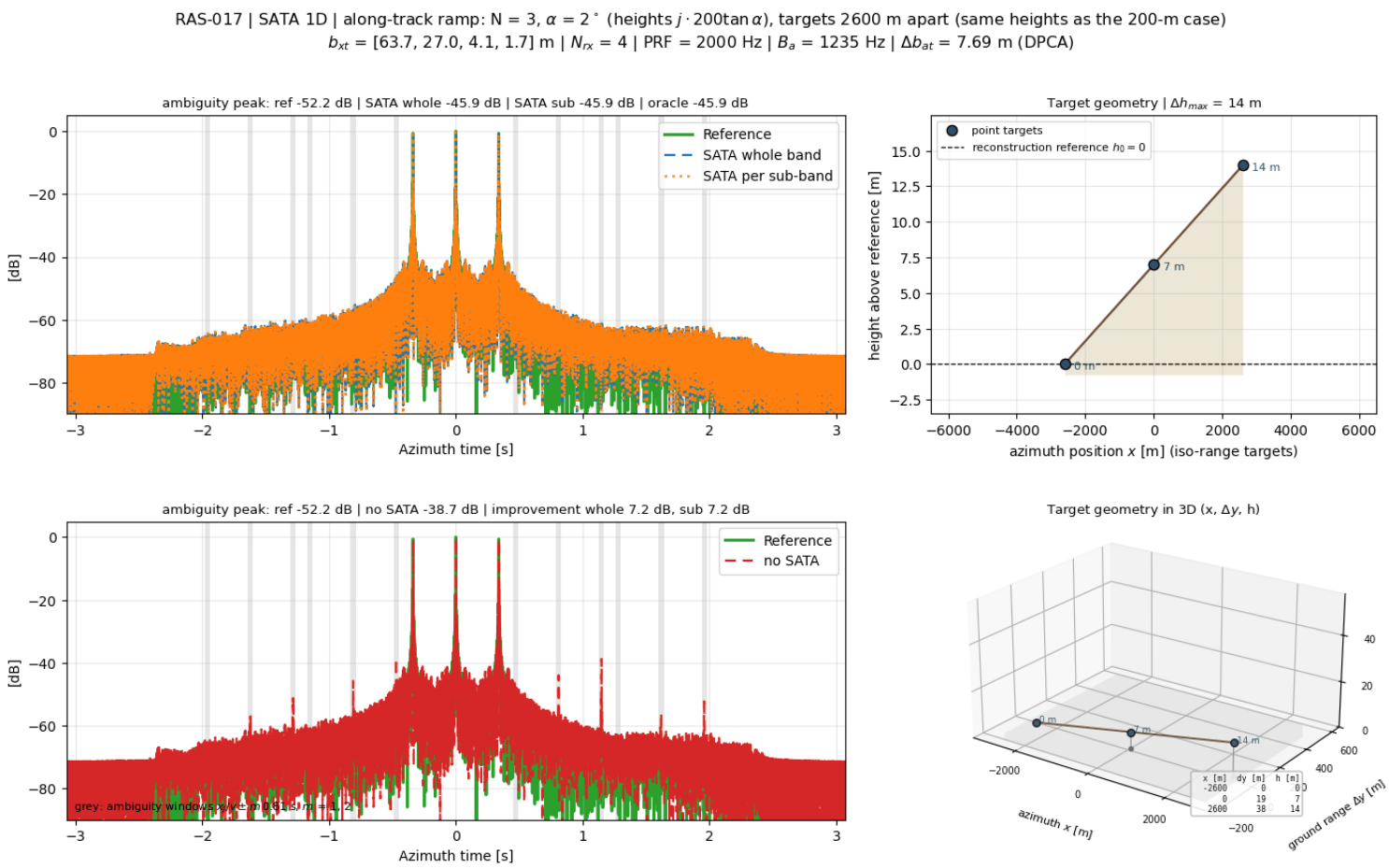


Figure 99: Targets 2.6 km apart, same heights: RAS-017, $N = 3$, $\alpha = 2°$. Ambiguity peak: no SATA −38.7, SATA whole −45.9, per sub-band −45.9, oracle −45.9 dB (improvement 7.2 dB).

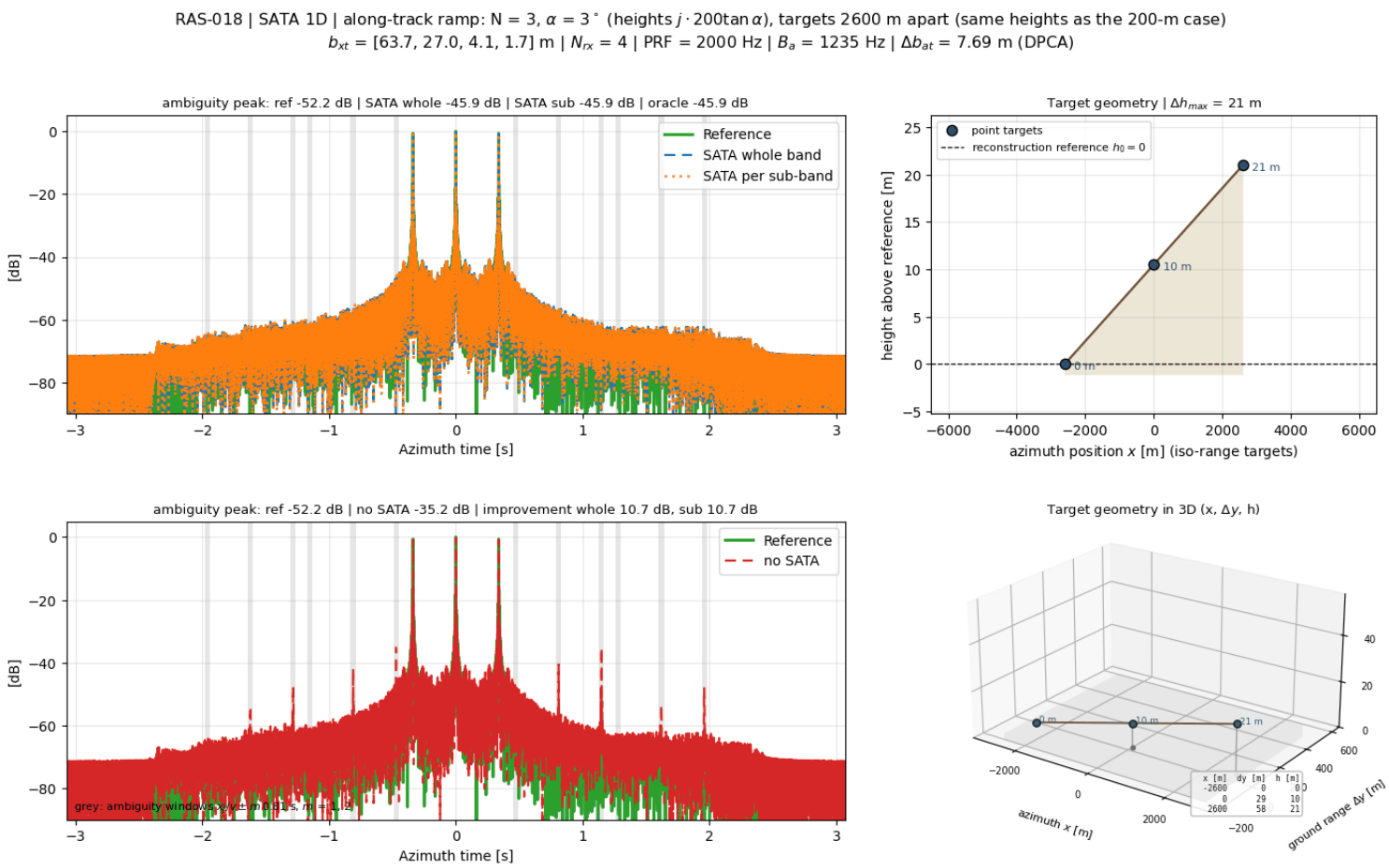


Figure 100: Targets 2.6 km apart, same heights: RAS-018, $N = 3$, $\alpha = 3°$. Ambiguity peak: no SATA -35.2, SATA whole -45.9, per sub-band -45.9, oracle -45.9 dB (improvement 10.7 dB).

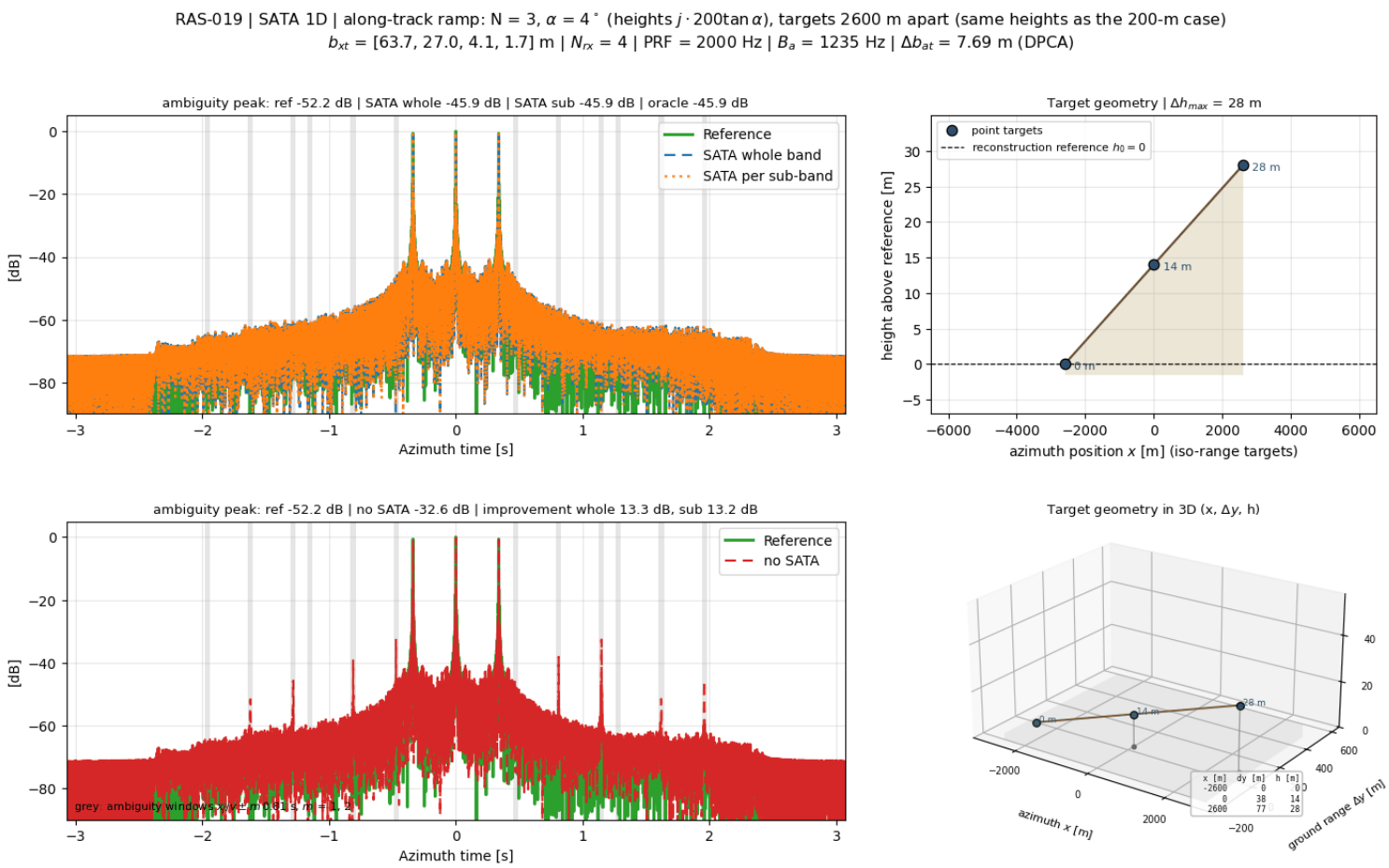


Figure 101: Targets 2.6 km apart, same heights: RAS-019, $N = 3$, $\alpha = 4°$. Ambiguity peak: no SATA -32.6, SATA whole -45.9, per sub-band -45.9, oracle -45.9 dB (improvement 13.3 dB).

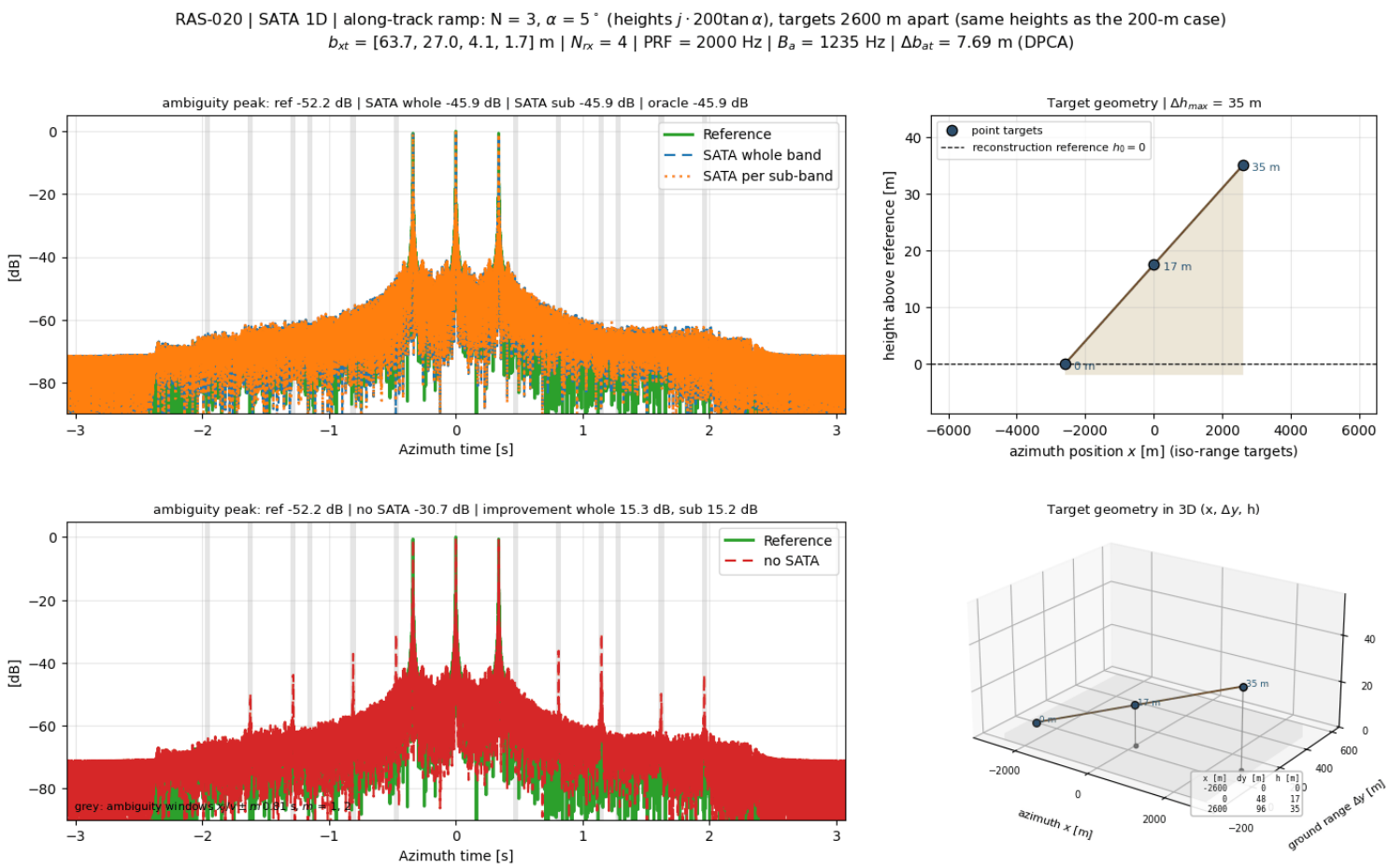


Figure 102: Targets 2.6 km apart, same heights: RAS-020, $N = 3$, $\alpha = 5°$. Ambiguity peak: no SATA −30.7, SATA whole −45.9, per sub-band −45.9, oracle −45.9 dB (improvement 15.3 dB).

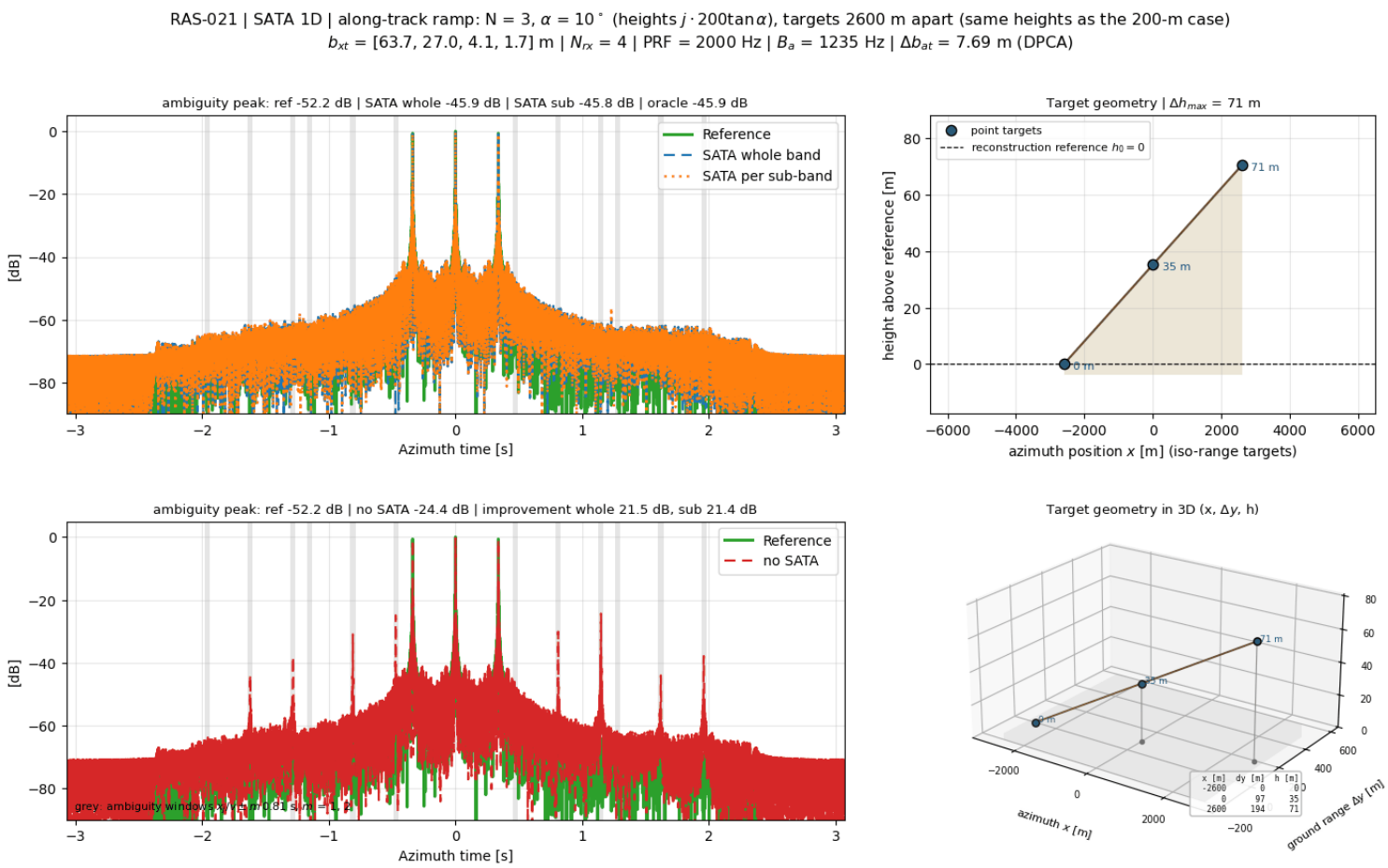


Figure 103: Targets 2.6 km apart, same heights: RAS-021, $N = 3$, $\alpha = 10°$. Ambiguity peak: no SATA -24.4, SATA whole -45.9, per sub-band -45.8, oracle -45.9 dB (improvement 21.5 dB).

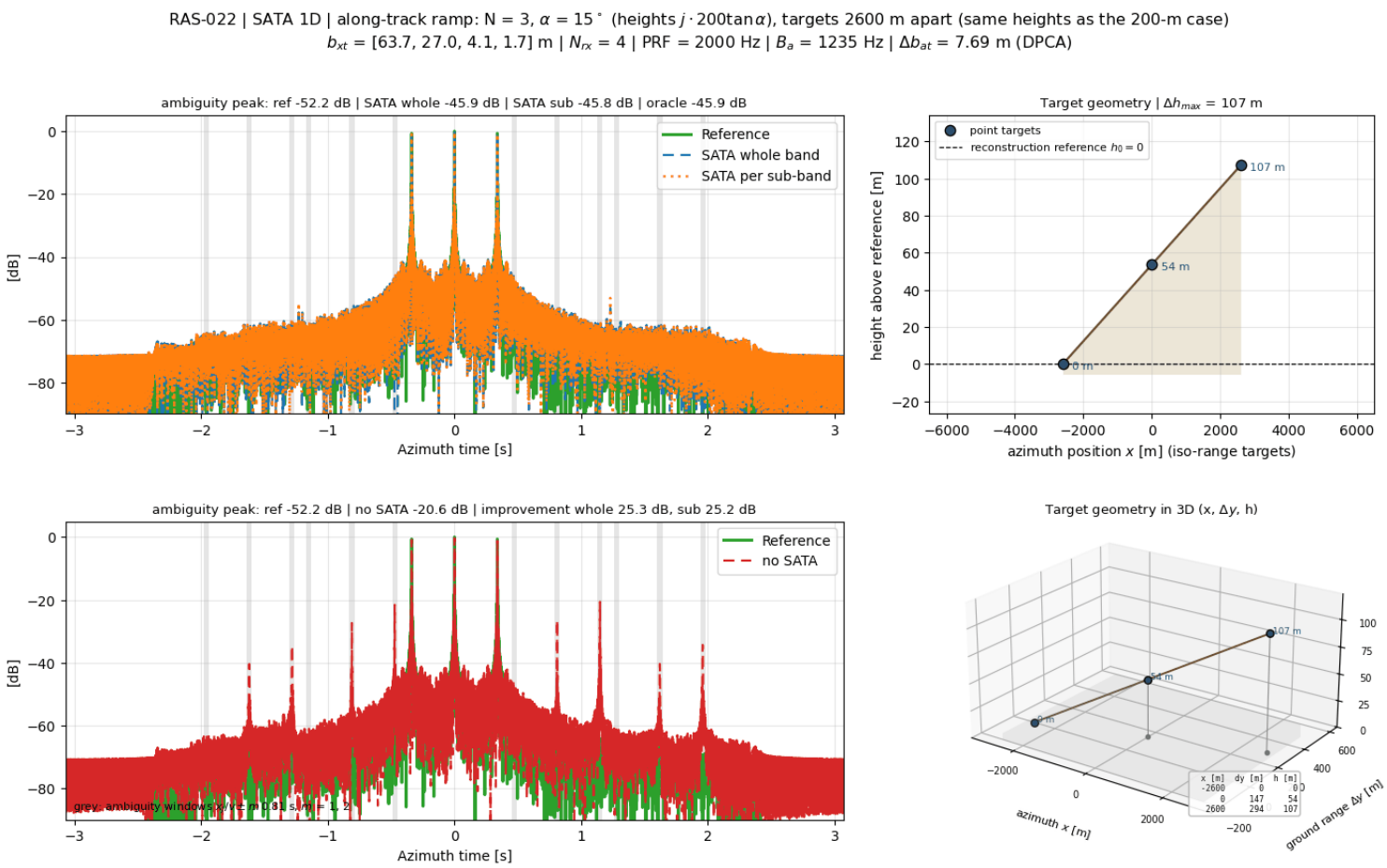


Figure 104: Targets 2.6 km apart, same heights: RAS-022, $N = 3$, $\alpha = 15°$. Ambiguity peak: no SATA -20.6, SATA whole -45.9, per sub-band -45.8, oracle -45.9 dB (improvement 25.3 dB).

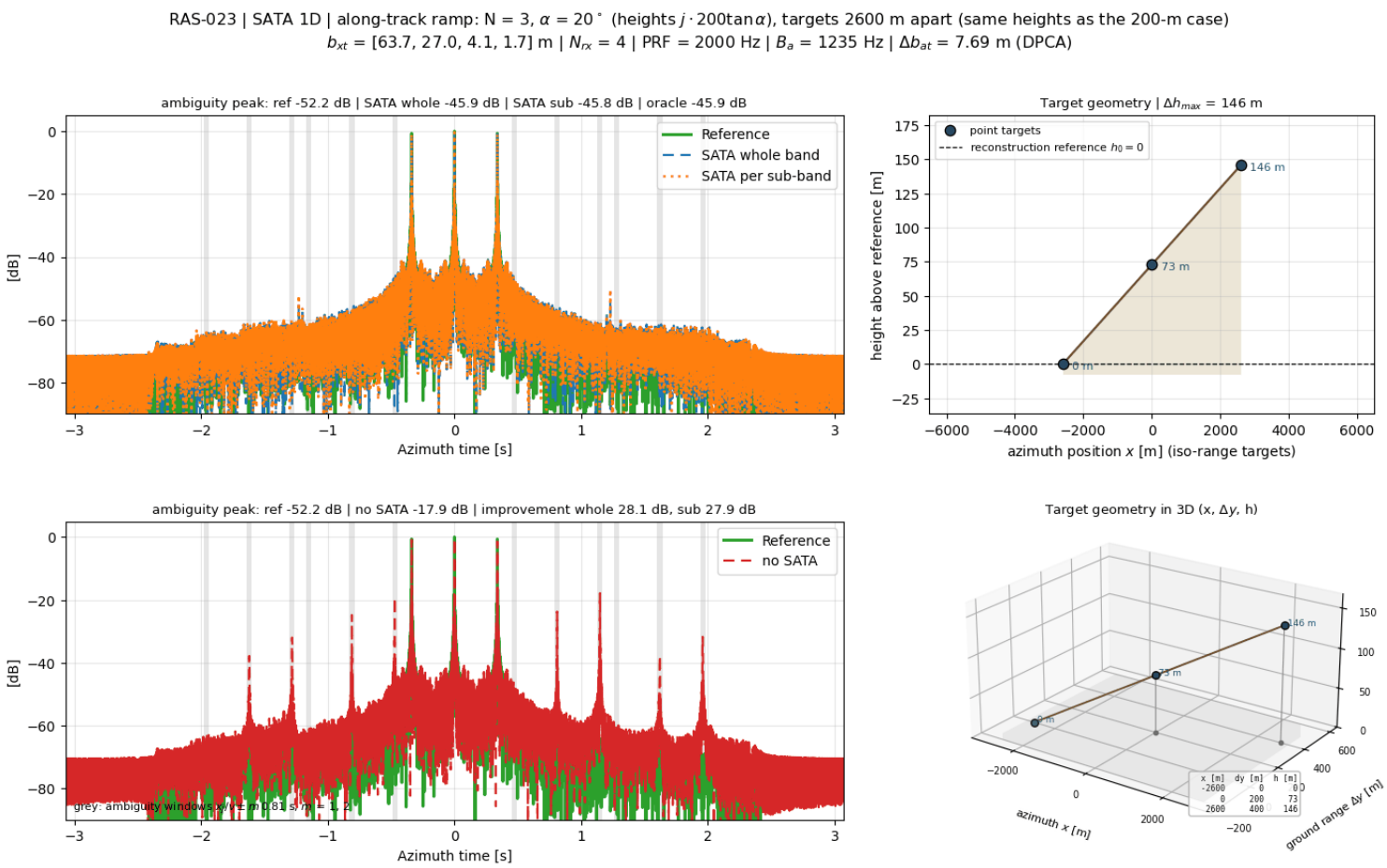


Figure 105: Targets 2.6 km apart, same heights: RAS-023, $N = 3$, $\alpha = 20°$. Ambiguity peak: no SATA −17.9, SATA whole −45.9, per sub-band −45.8, oracle −45.9 dB (improvement 28.1 dB).

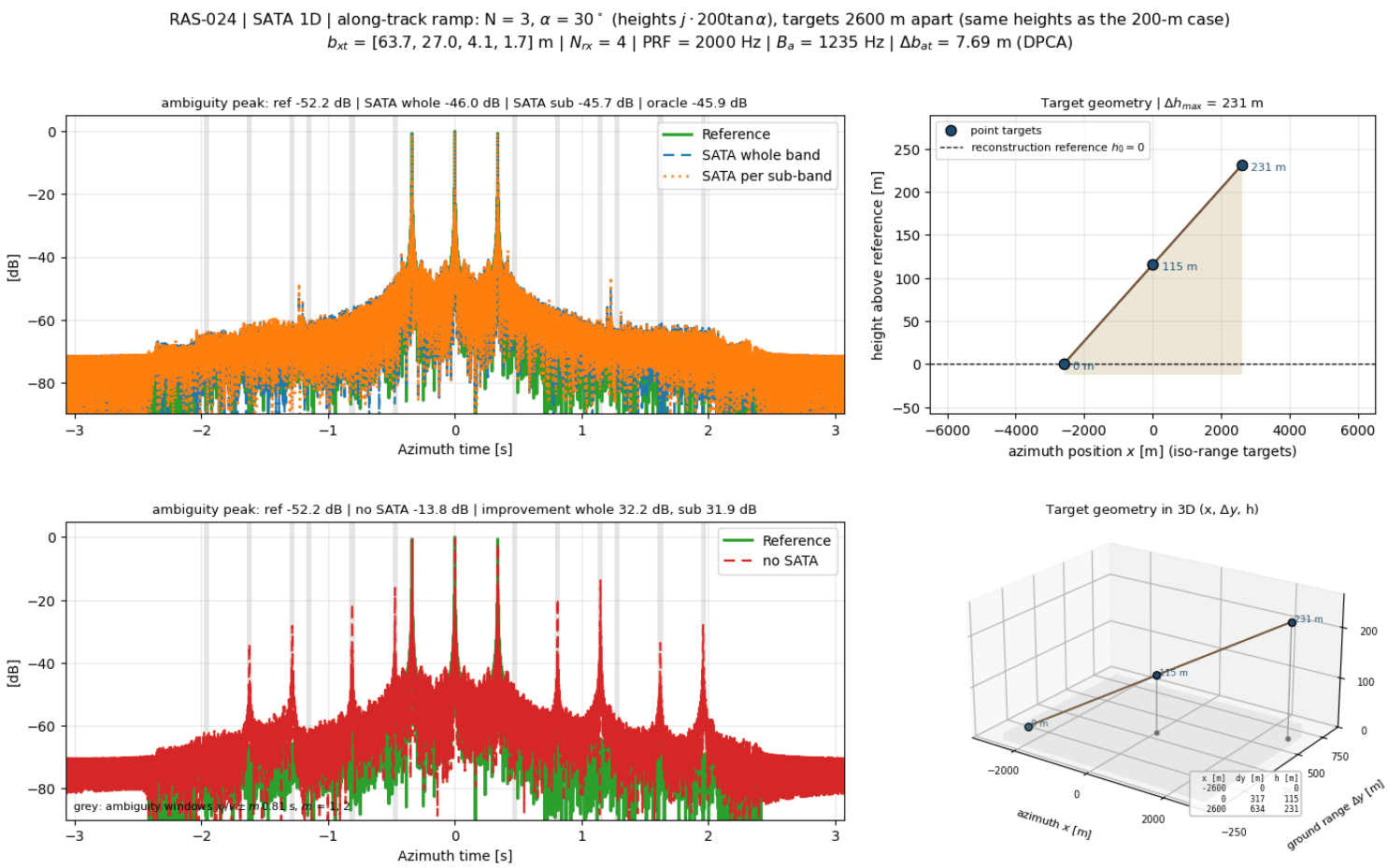


Figure 106: Targets 2.6 km apart, same heights: RAS-024, $N = 3$, $\alpha = 30°$. Ambiguity peak: no SATA -13.8, SATA whole -46.0, per sub-band -45.7, oracle -45.9 dB (improvement 32.2 dB).

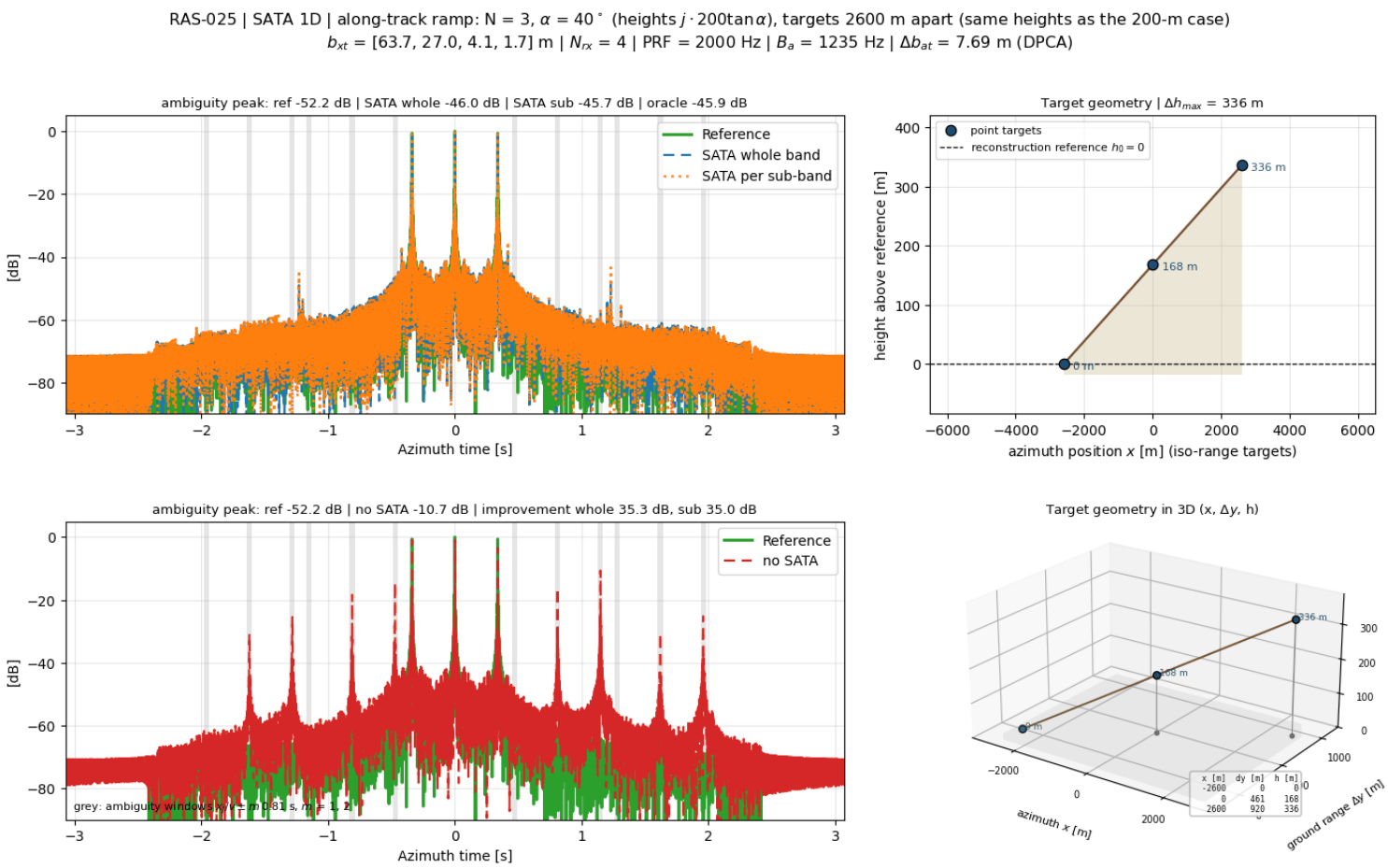


Figure 107: Targets 2.6 km apart, same heights: RAS-025, $N = 3$, $\alpha = 40°$. Ambiguity peak: no SATA -10.7, SATA whole -46.0, per sub-band -45.7, oracle -45.9 dB (improvement 35.3 dB).

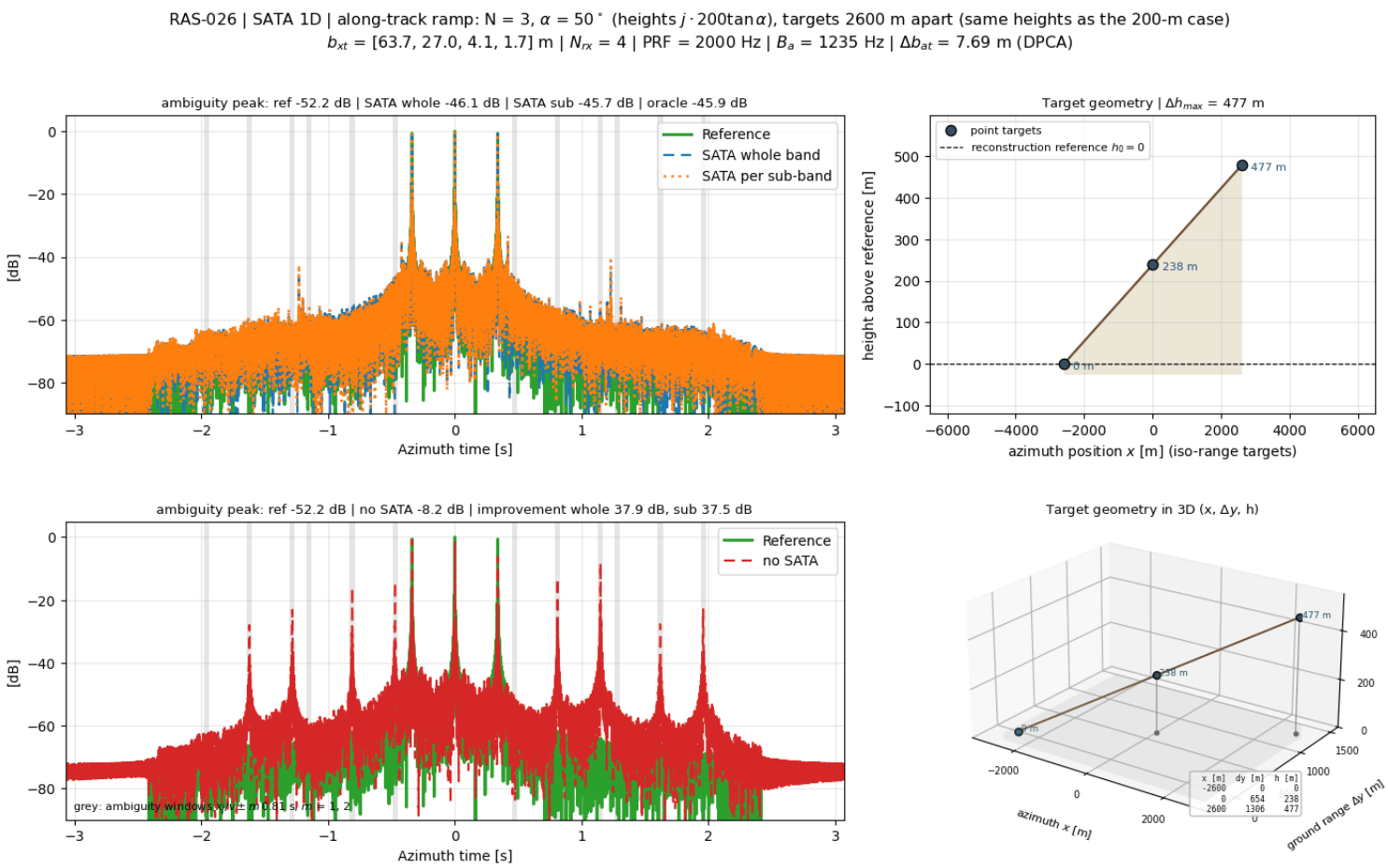


Figure 108: Targets 2.6 km apart, same heights: RAS-026, $N = 3$, $\alpha = 50°$. Ambiguity peak: no SATA −8.2, SATA whole −46.1, per sub-band −45.7, oracle −45.9 dB (improvement 37.9 dB).

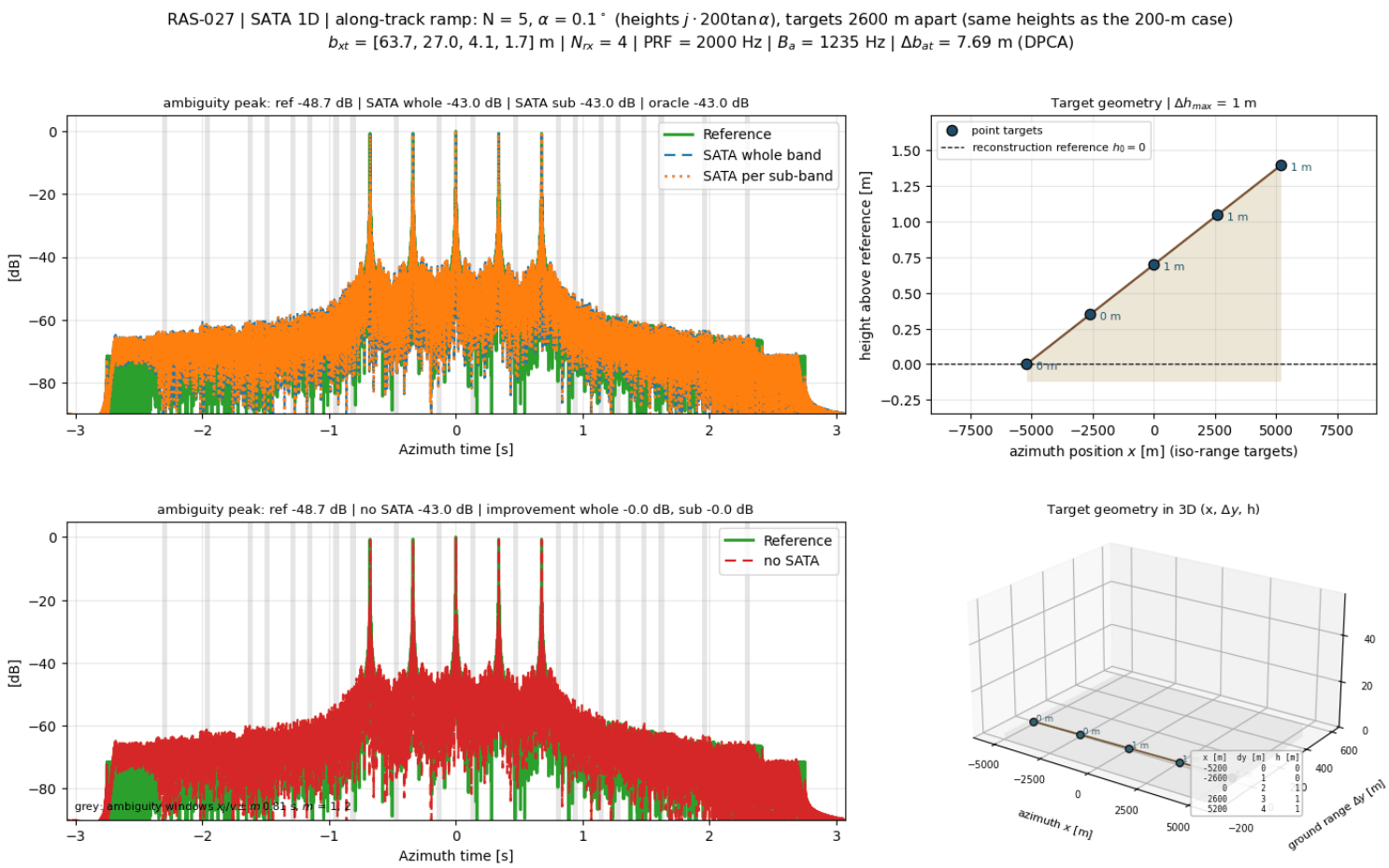


Figure 109: Targets 2.6 km apart, same heights: RAS-027, $N = 5$, $\alpha = 0.1°$. Ambiguity peak: no SATA −43.0, SATA whole −43.0, per sub-band −43.0, oracle −43.0 dB (improvement -0.0 dB).

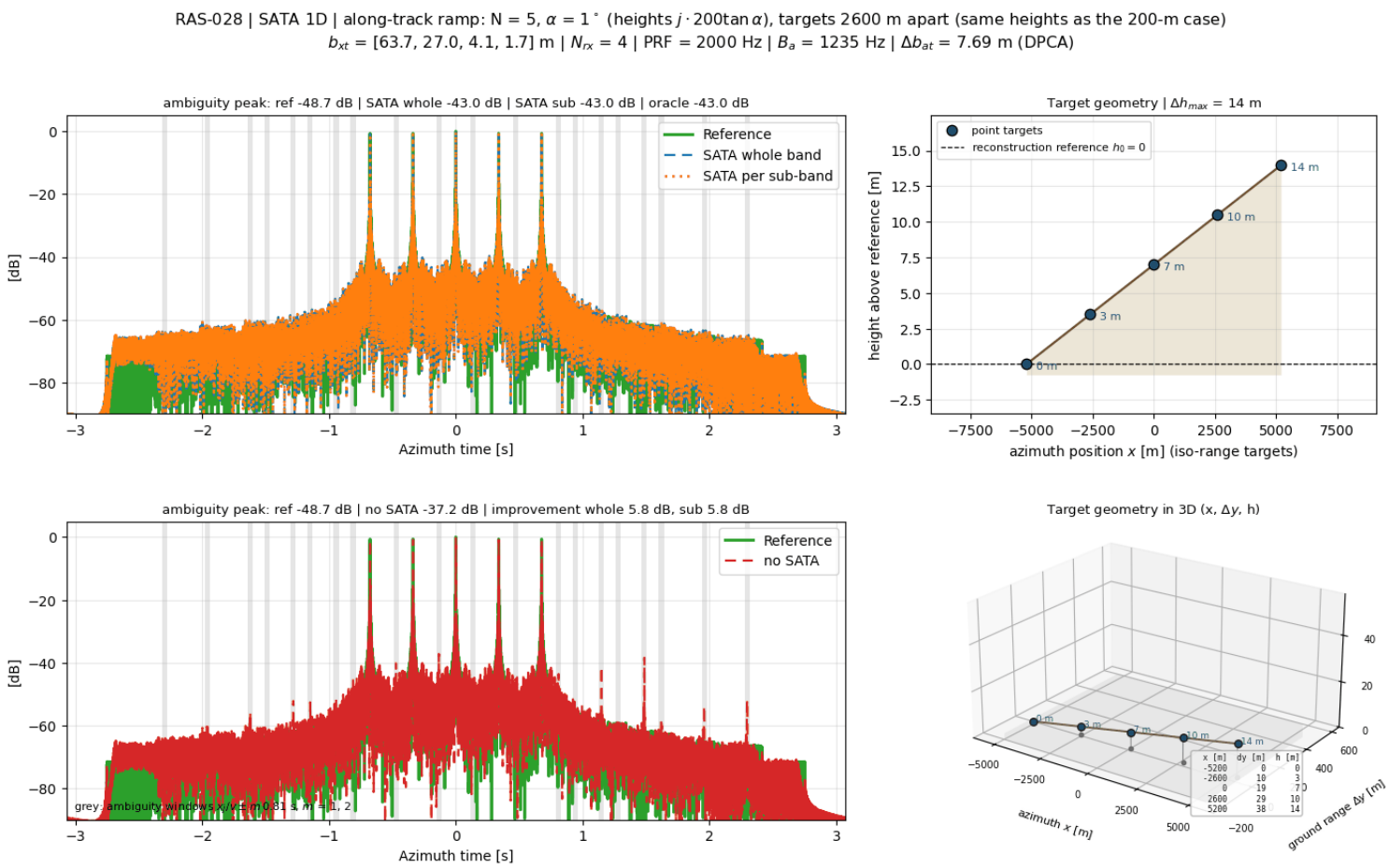


Figure 110: Targets 2.6 km apart, same heights: RAS-028, $N = 5$, $\alpha = 1°$. Ambiguity peak: no SATA -37.2, SATA whole -43.0, per sub-band -43.0, oracle -43.0 dB (improvement 5.8 dB).

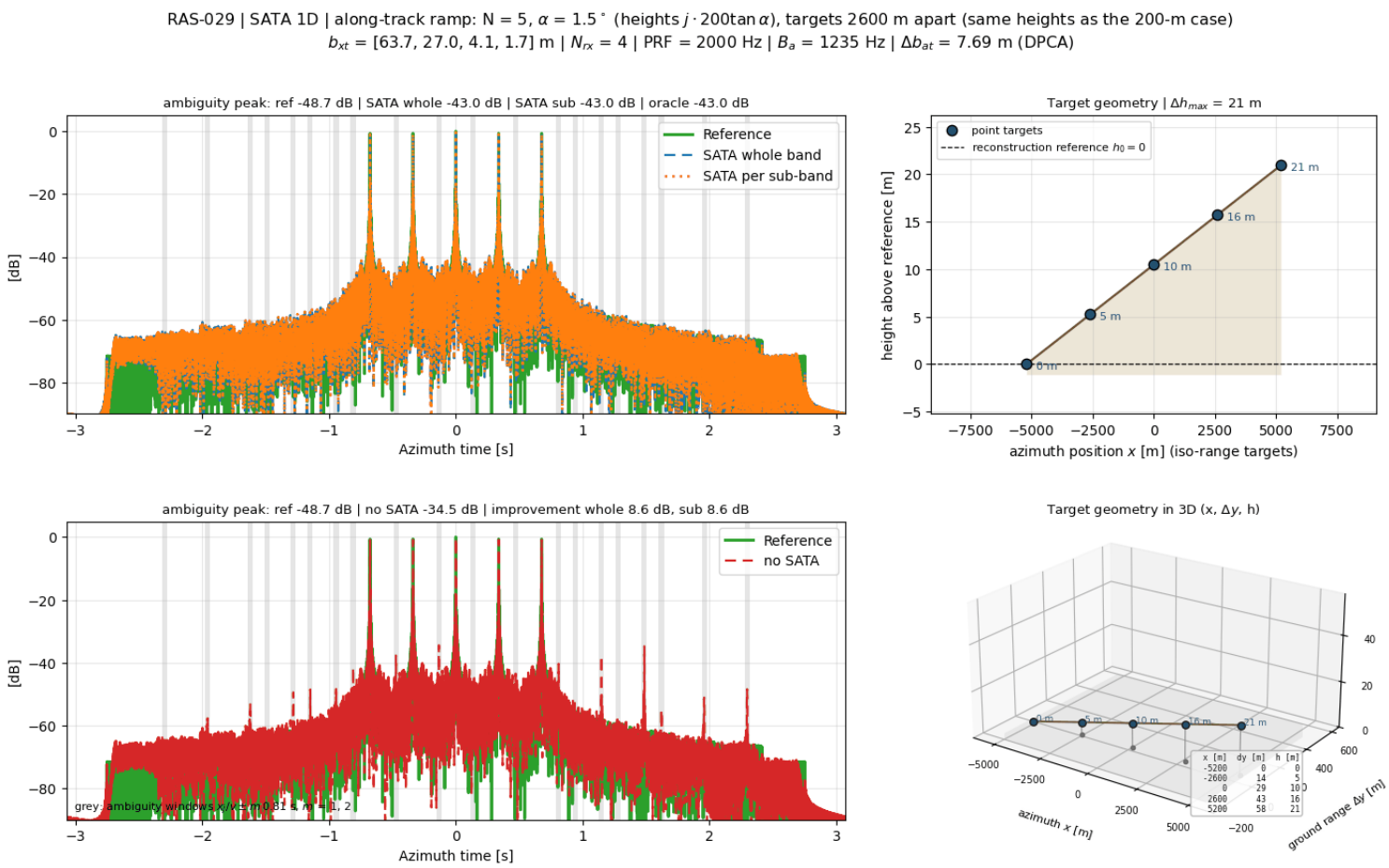


Figure 111: Targets 2.6 km apart, same heights: RAS-029, $N = 5$, $\alpha = 1.5°$. Ambiguity peak: no SATA -34.5, SATA whole -43.0, per sub-band -43.0, oracle -43.0 dB (improvement 8.6 dB).

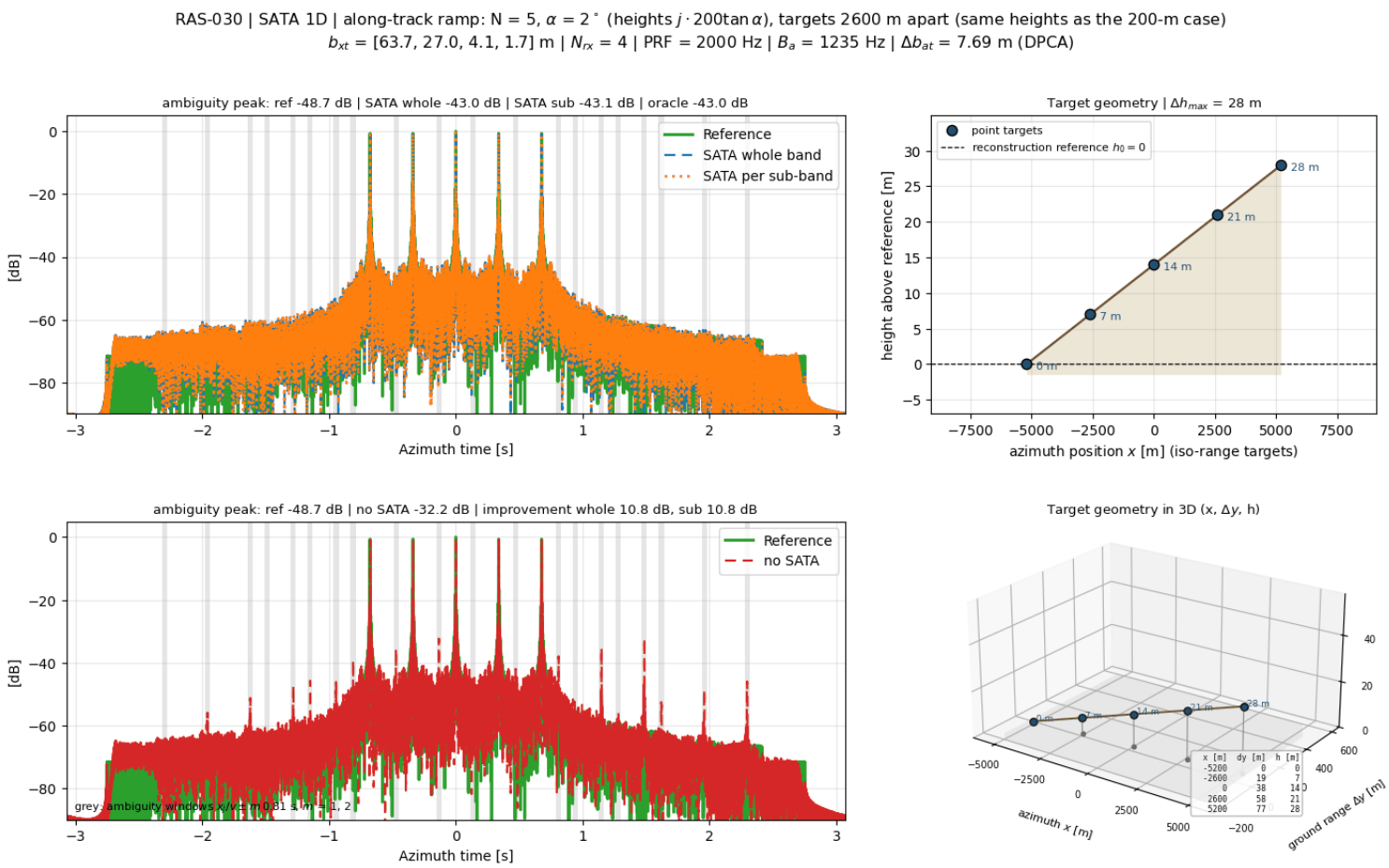


Figure 112: Targets 2.6 km apart, same heights: RAS-030, $N = 5$, $\alpha = 2°$. Ambiguity peak: no SATA -32.2, SATA whole -43.0, per sub-band -43.1, oracle -43.0 dB (improvement 10.8 dB).

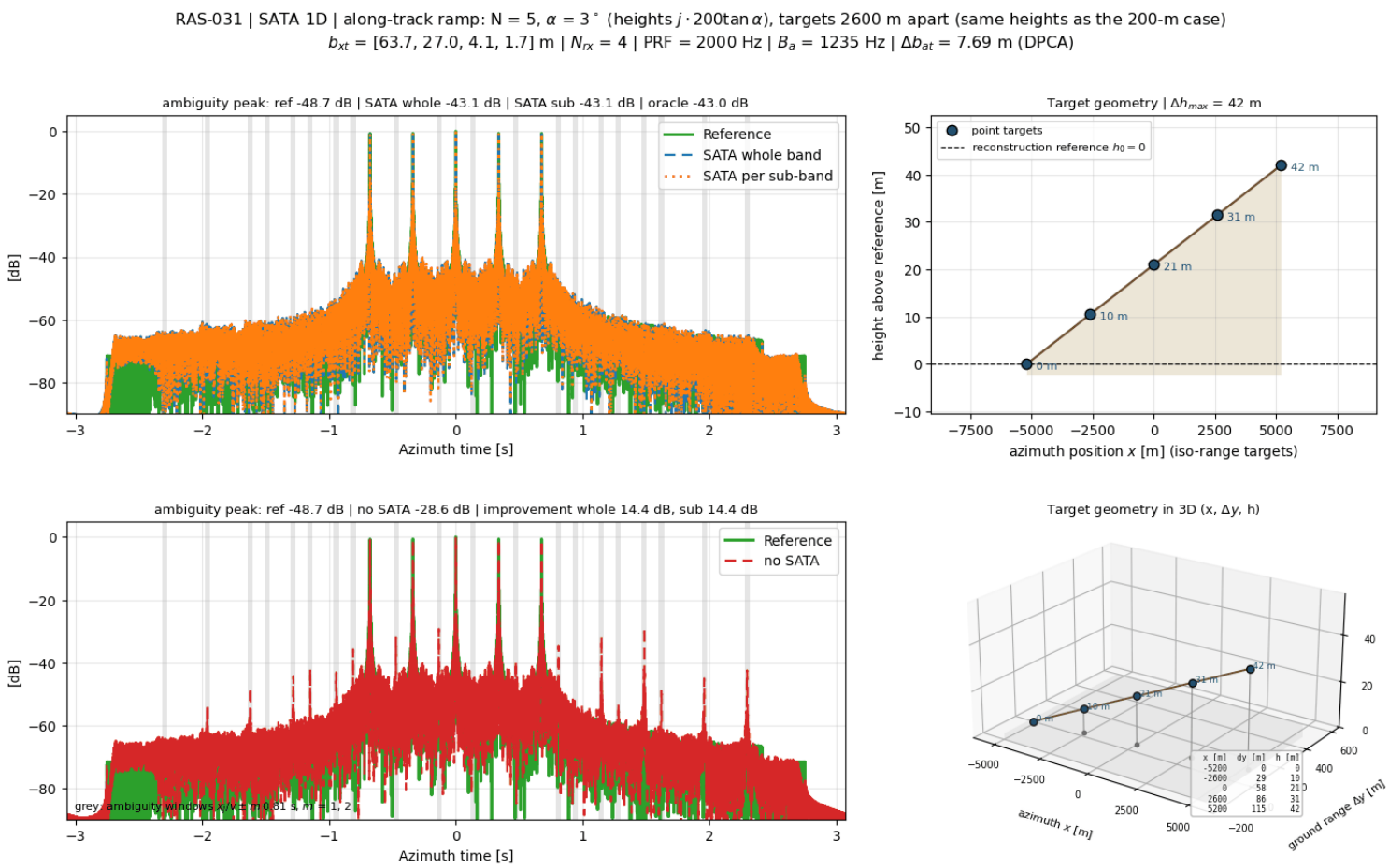


Figure 113: Targets 2.6 km apart, same heights: RAS-031, $N = 5$, $\alpha = 3°$. Ambiguity peak: no SATA −28.6, SATA whole −43.1, per sub-band −43.1, oracle −43.0 dB (improvement 14.4 dB).

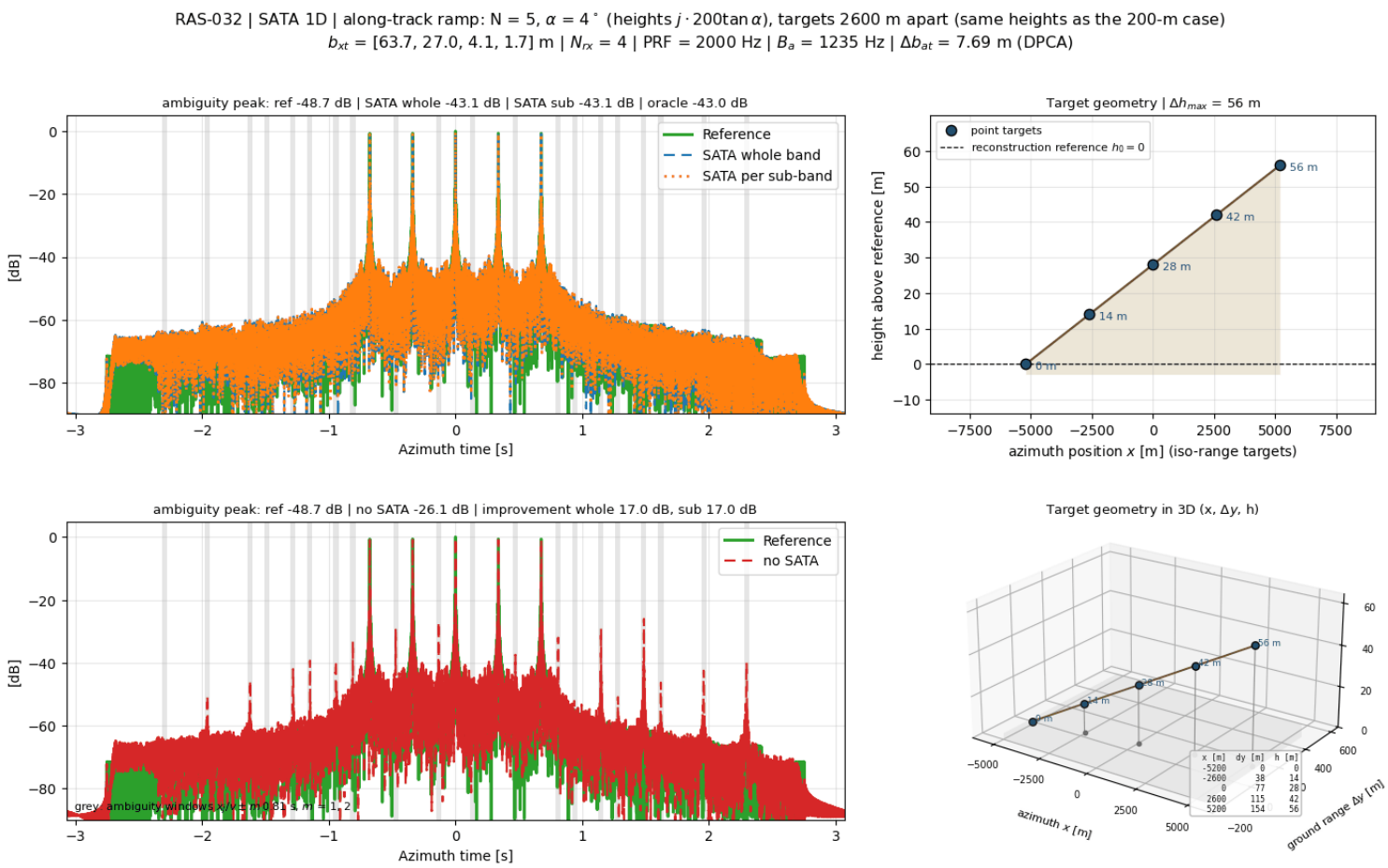


Figure 114: Targets 2.6 km apart, same heights: RAS-032, $N = 5$, $\alpha = 4°$. Ambiguity peak: no SATA −26.1, SATA whole −43.1, per sub-band −43.1, oracle −43.0 dB (improvement 17.0 dB).

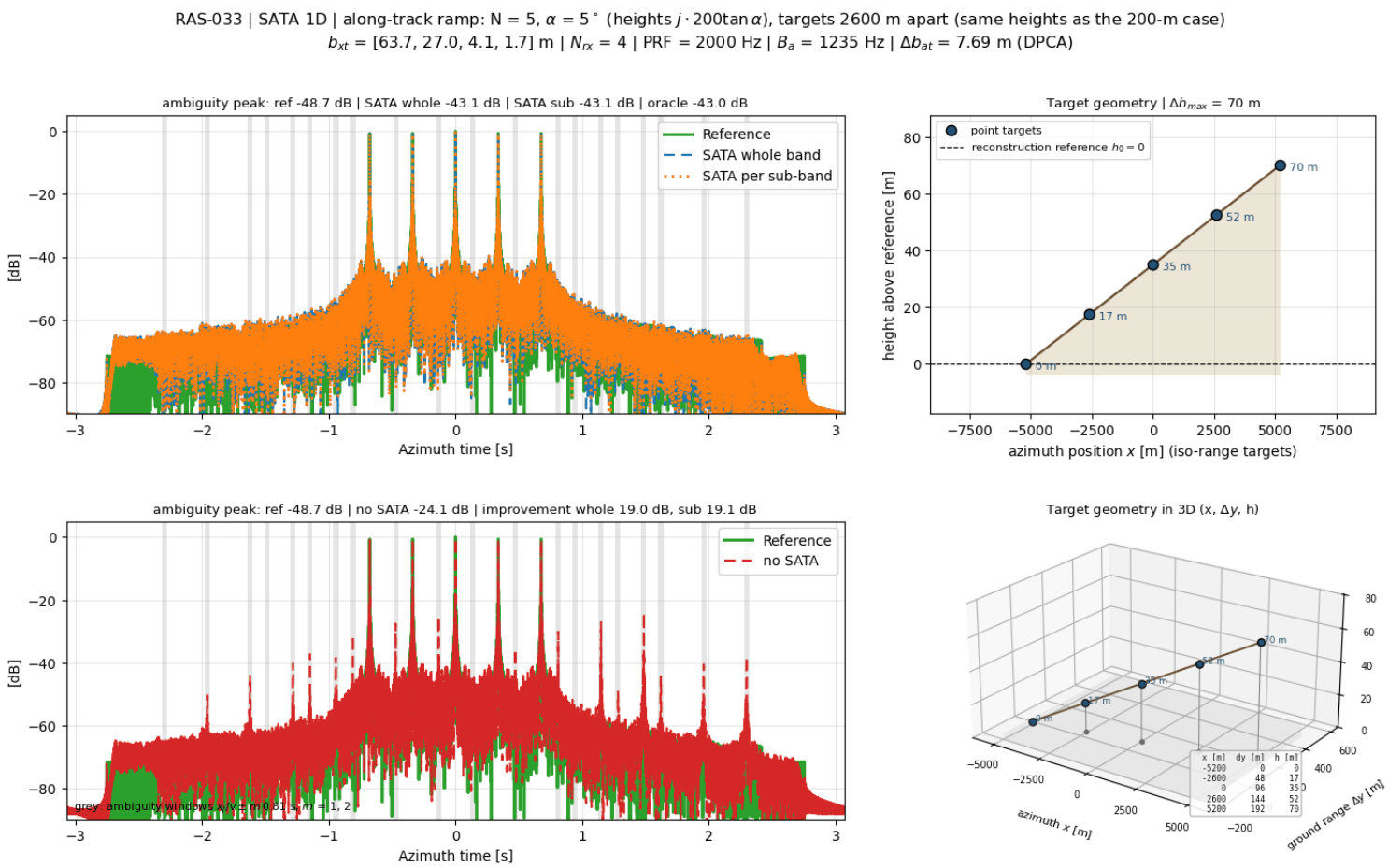


Figure 115: Targets 2.6 km apart, same heights: RAS-033, $N = 5$, $\alpha = 5°$. Ambiguity peak: no SATA −24.1, SATA whole −43.1, per sub-band −43.1, oracle −43.0 dB (improvement 19.0 dB).

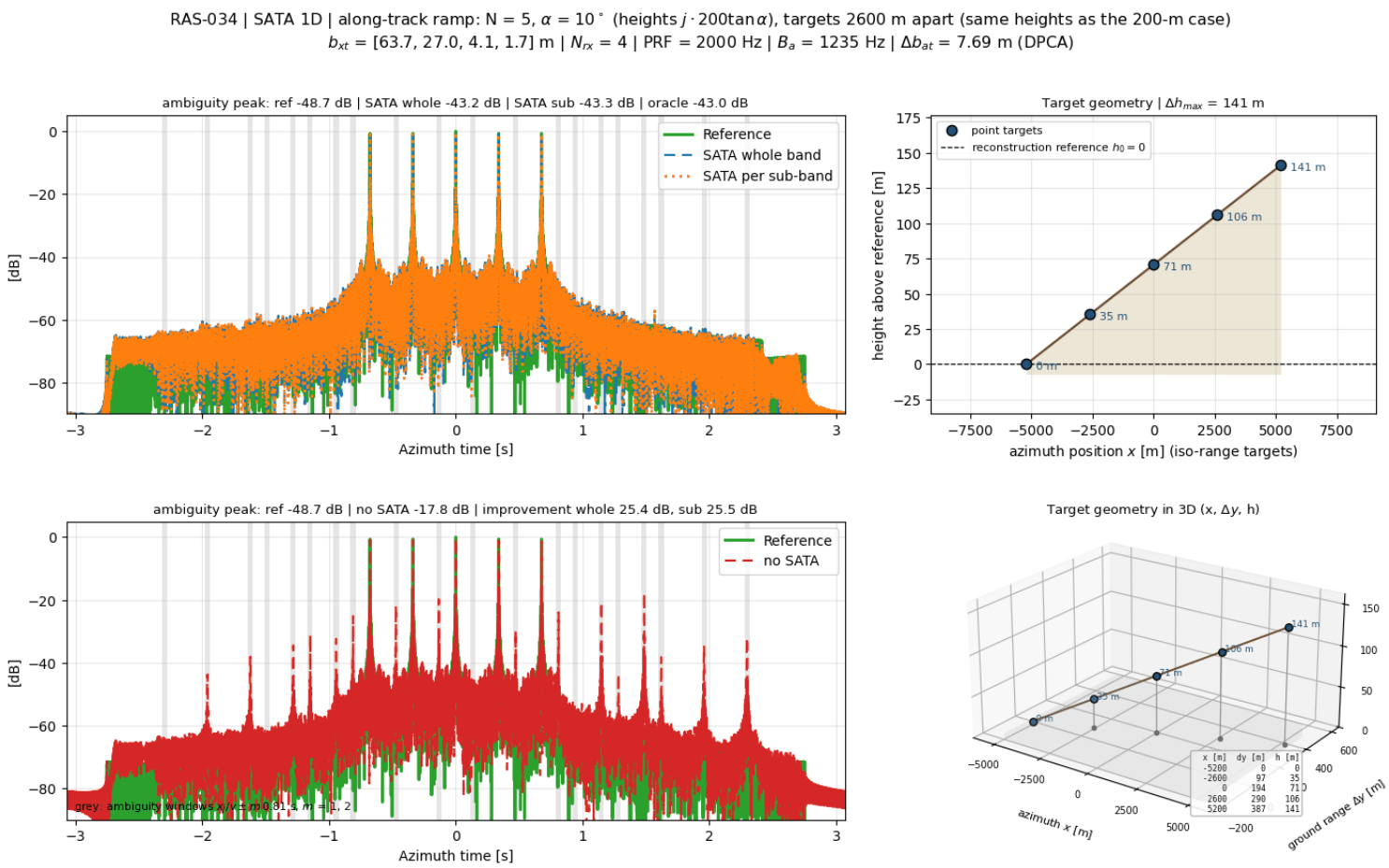


Figure 116: Targets 2.6 km apart, same heights: RAS-034, $N = 5$, $\alpha = 10°$. Ambiguity peak: no SATA −17.8, SATA whole −43.2, per sub-band −43.3, oracle −43.0 dB (improvement 25.4 dB).

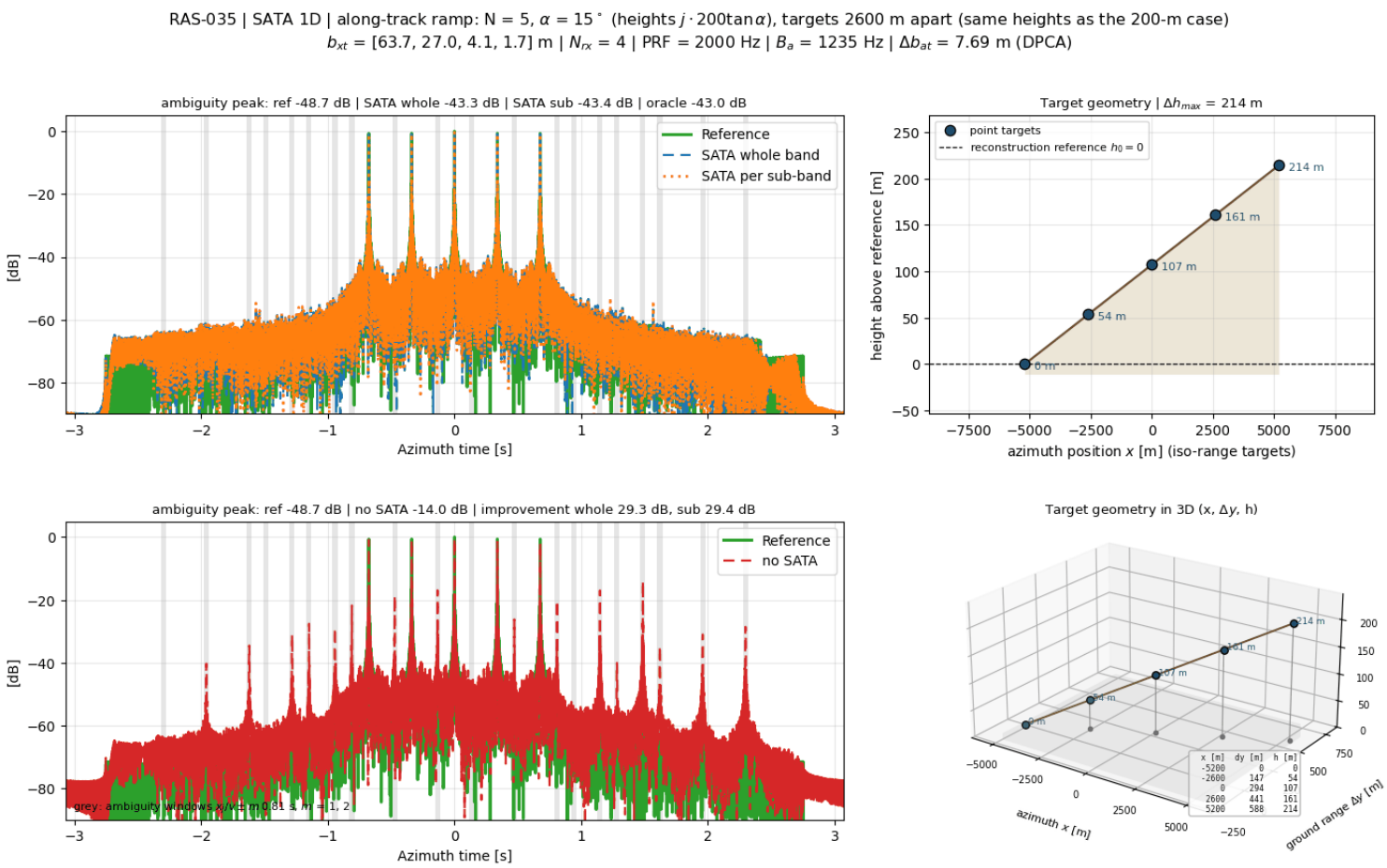


Figure 117: Targets 2.6 km apart, same heights: RAS-035, $N = 5$, $\alpha = 15°$. Ambiguity peak: no SATA -14.0, SATA whole -43.3, per sub-band -43.4, oracle -43.0 dB (improvement 29.3 dB).

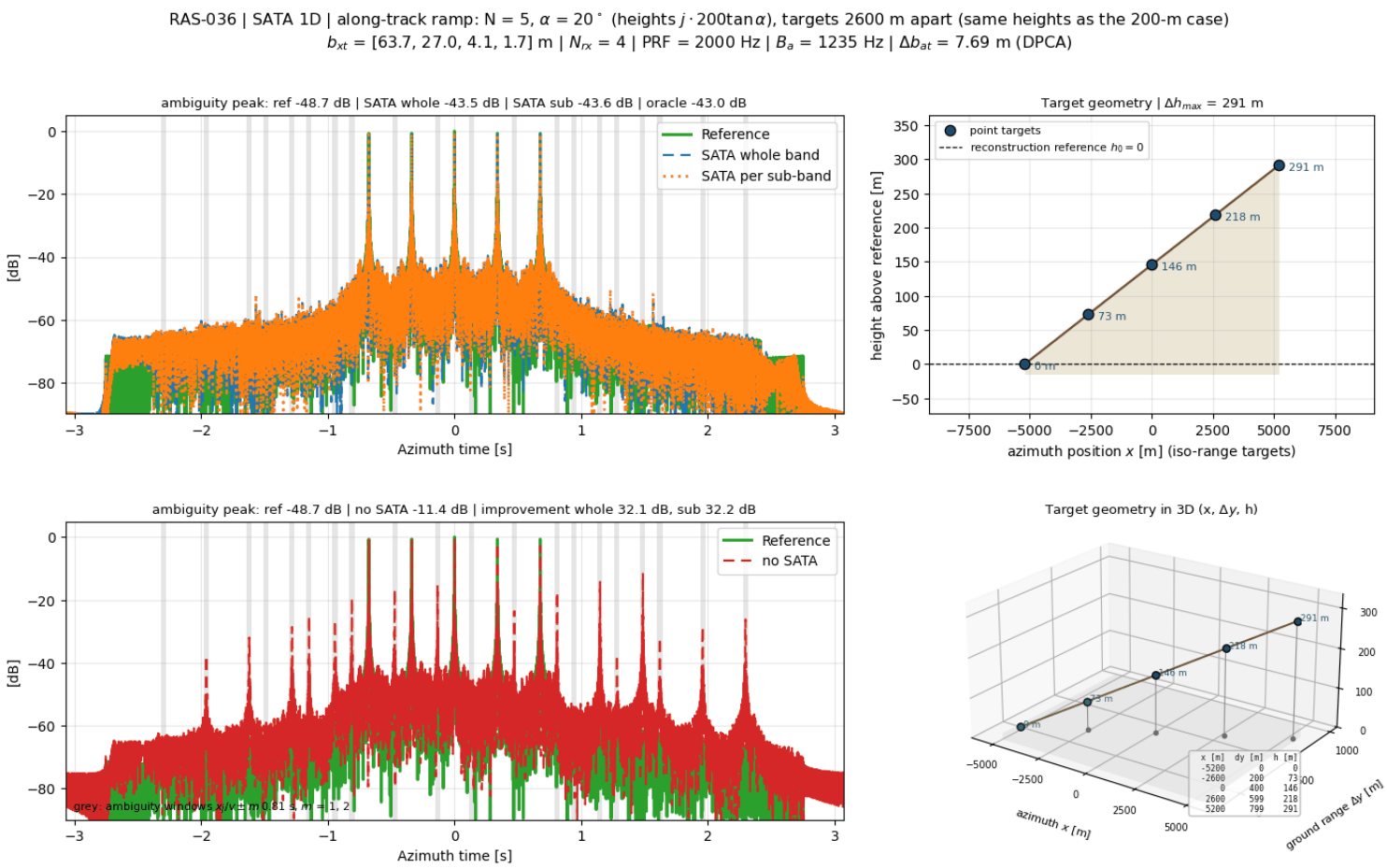


Figure 118: Targets 2.6 km apart, same heights: RAS-036, $N = 5$, $\alpha = 20°$. Ambiguity peak: no SATA −11.4, SATA whole −43.5, per sub-band −43.6, oracle −43.0 dB (improvement 32.1 dB).

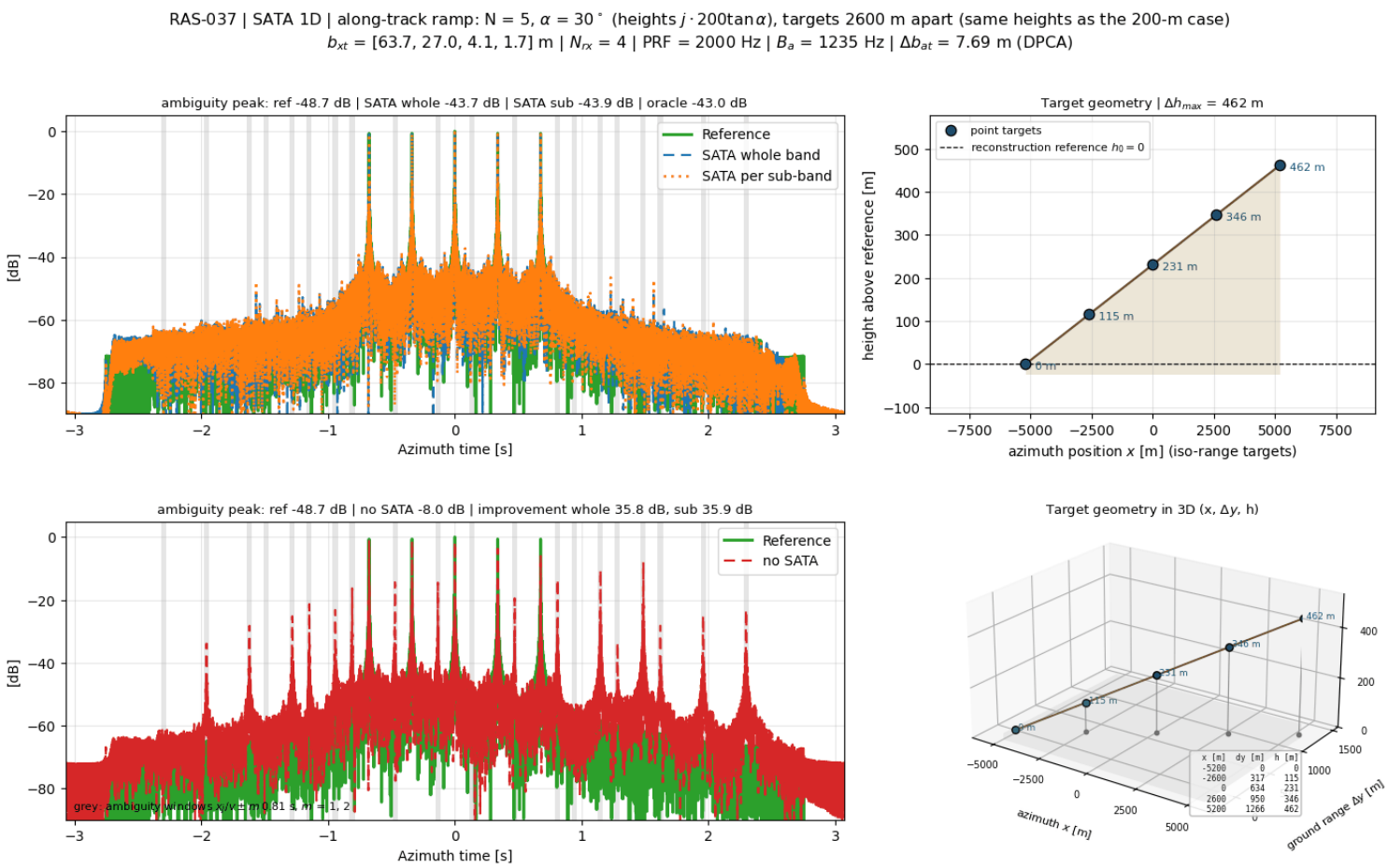


Figure 119: Targets 2.6 km apart, same heights: RAS-037, $N = 5$, $\alpha = 30°$. Ambiguity peak: no SATA −8.0, SATA whole −43.7, per sub-band −43.9, oracle −43.0 dB (improvement 35.8 dB).

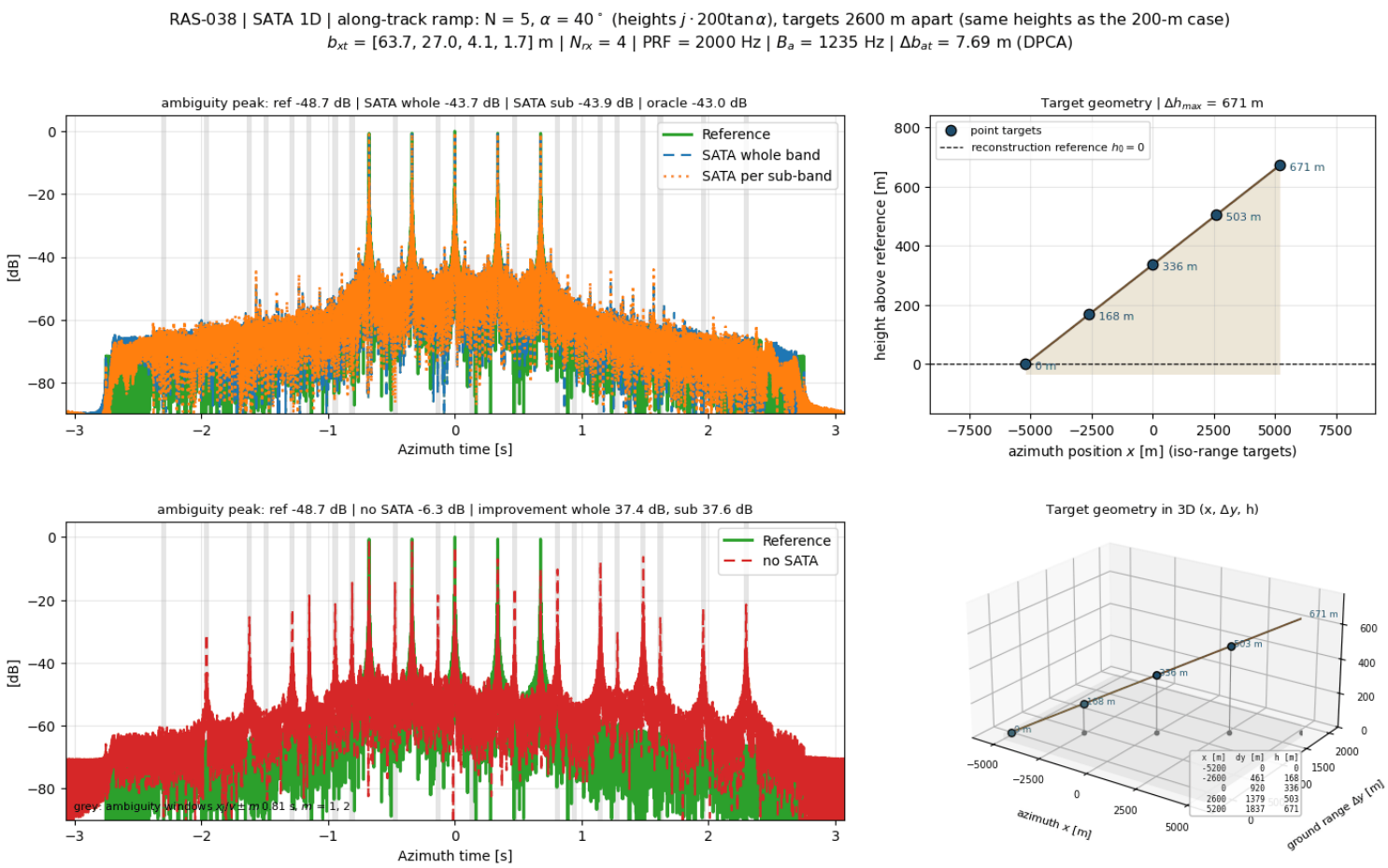


Figure 120: Targets 2.6 km apart, same heights: RAS-038, $N = 5$, $\alpha = 40°$. Ambiguity peak: no SATA -6.3, SATA whole -43.7, per sub-band -43.9, oracle -43.0 dB (improvement 37.4 dB).

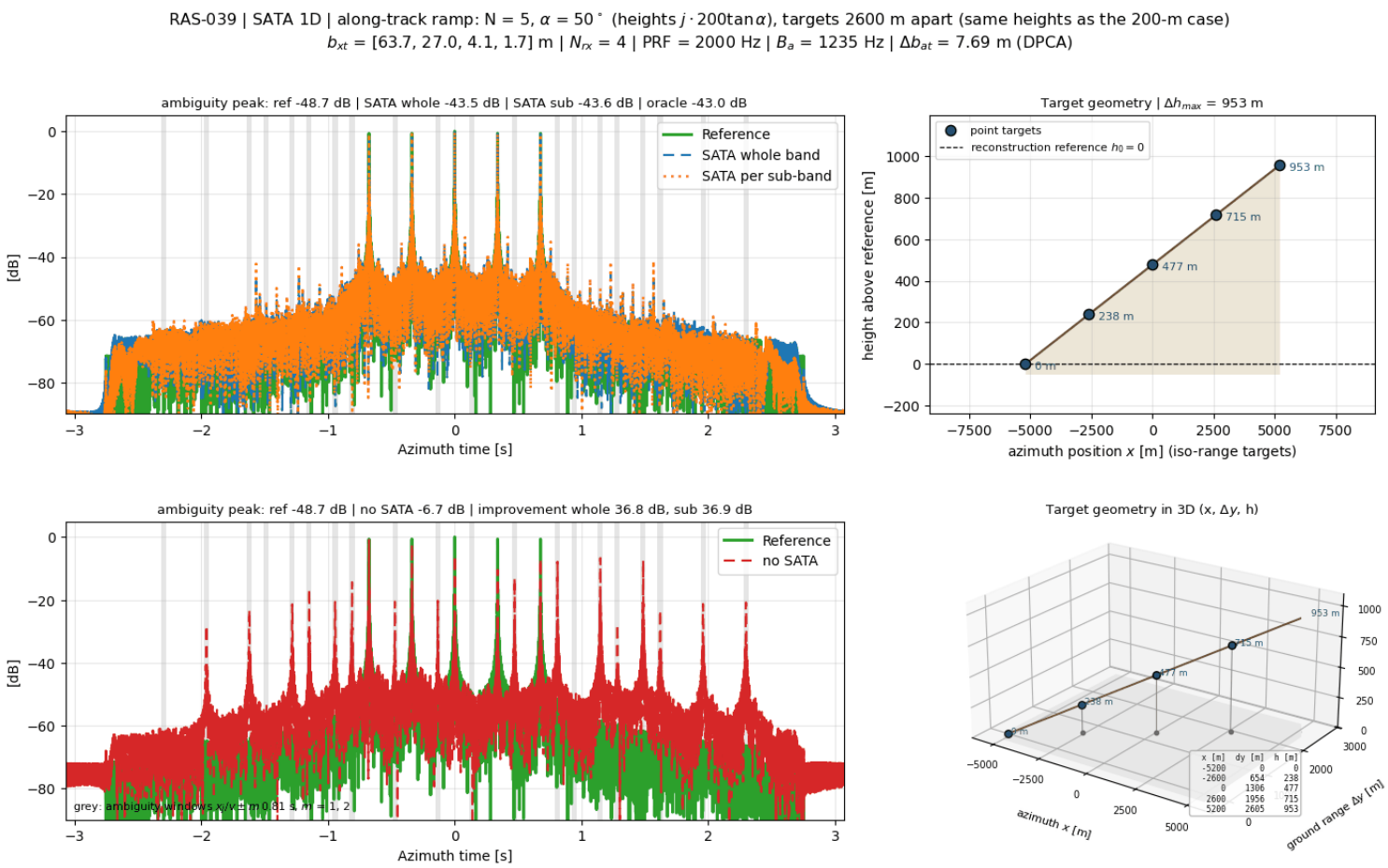


Figure 121: Targets 2.6 km apart, same heights: RAS-039, $N = 5$, $\alpha = 50°$. Ambiguity peak: no SATA -6.7, SATA whole -43.5, per sub-band -43.6, oracle -43.0 dB (improvement 36.8 dB).